

Module 4: Physical Chemistry

- **Ideal Gas**
- **Real Gas**
- **Atomic Structure**
- **Chemical Equilibrium**

IDEAL GAS

1. INTRODUCTION

Matter, as we know, broadly exist in three states - solid, liquid and gas.

There are always two opposite tendencies between particles of matter which determine the state of matter :

- Intermolecular forces.
- The molecular motion / random motion (energy of particles)

Intermolecular forces are the forces of attraction and repulsion between atoms or molecules. Attractive intermolecular forces are known as ***vander Waals forces***. These are dispersion forces, dipole-dipole forces & dipole induced forces. When two molecules are brought very close, they will exert repulsive forces. Magnitude of the repulsion rises very rapidly as the distance separating the molecules decreases.

This is the reason that liquid & solids are hard to compress. In these states, molecules are already in close contact, therefore they resist further compression (in that case repulsive interaction will increase)

Thermal energy is the energy of a body arising from motion of its atoms or molecules. It is directly proportional to the temperature of the substance. It is the measure of average kinetic energy of the particles of the matter and is thus responsible for movement of particles. This movement of particles is called ***thermal motion***. Intermolecular forces tend to keep the molecules together but thermal energy of the molecules tends to keep them apart. Three states of matter are the result of balance between intermolecular forces and the thermal energy of the molecules.

2. GENERAL CHARACTERISTICS OF SOLID, LIQUID & GAS

Each physical state of matter possesses characteristics properties of its own. For example,

❖ Solids :

All solids show the following characteristics :

- (i) Solids are rigid and incompressible.
- (ii) Solids have fixed shape and definite volume.
- (iii) Solids have their melting and boiling points above room temperature.
- (iv) Density of solid is high.

❖ Liquids :

All liquids show the following characteristics :

- (i) Liquids are almost incompressible but less incompressible than solids.
- (ii) Liquids have fixed volume but no fixed shape.
- (iii) Liquids have their melting points below room temperature and boiling points above room temperature, under normal conditions.
- (iv) Density of liquids is lower than that of solids but much higher than that of gases.

❖ **Gases :**

All gases show the following characteristics :

- (i) Gases are highly compressible, i.e. gases can be compressed easily by applying pressure.
- (ii) Gases have no fixed volume and shape. Gases fill the container of any size and shape completely.
- (iii) Gases can diffuse into each other rapidly.
- (iv) Gases have their melting and boiling points both below room temperature.
- (v) Gases generally have low density.

3. MEASURABLE PROPERTIES OF GASES

The characteristics of gases are described fully in terms of four parameters or measurable properties :

- (I) Amount of the gas (i.e., mass or number of moles).
- (II) Volume (V) of the gas.
- (III) Temperature (T)
- (IV) Pressure (P)

I. Amount of the gas :

- (i) The mass of a gas can be determined by weighing the container in which the gas is enclosed and again weighing the container after removing the gas. The difference between the two masses gives the mass of the gas.

$$\text{Mass of gas (m)} = \text{Mass of filled container} - \text{mass of empty container}$$

- (ii) The mass of the gas is related to the number of moles of the gas as

$$\text{Moles of gas (n)} = \frac{\text{Mass (in gm)}}{\text{Molar mass}} = \frac{m}{M}$$

- (iii) Mass is expressed in gram or kg.

II. Gas volume :

- (i) Since gases occupy the entire space available to them, the measurement of volume of a gas only requires a measurement of the capacity of container confining the gas.
- (ii) Volume is expressed in litres (L), millilitres (mL), cubic centimeters (cm³), cubic meters (m³), etc.
- (iii) $1 \text{ L} = 1000 \text{ mL} = 1 \text{ dm}^3 = 10^{-3} \text{ m}^3$
 $1 \text{ m}^3 = 10^3 \text{ dm}^3 = 10^6 \text{ cm}^3 = 10^6 \text{ mL} = 10^3 \text{ L}$
 $1 \text{ mL} = 1 \text{ cc} = 1 \text{ cm}^3$

III. Temperature :

- (ii) Temperature is measured in degree Celsius (°C), degree Fahrenheit (°F), Kelvin (K), etc.
- (iii) S.I. unit of temperature is kelvin (K)
 $K = ^\circ\text{C} + 273.15$
- (iv) Relation between °F and °C is

$$\frac{^\circ\text{C}}{5} = \frac{^\circ\text{F} - 32}{9}$$

IV. Pressure :

Pressure is the force exerted by the gas per unit area on the walls of the container in all directions. Thus,

$$\text{Pressure (P)} = \text{Force (F)} / \text{Area (A)}$$

❖ Atmospheric pressure :

The pressure exerted by atmosphere on earth's surface at sea level is called atmospheric pressure. Generally it is expressed in atm.

$$1 \text{ atm} = 1.013 \text{ bar} = 1.013 \times 10^5 \text{ N/m}^2 = 1.013 \times 10^5 \text{ Pa} = 1.013 \times 10^6 \text{ dyne/cm}^2$$

$$1 \text{ atm} = 76 \text{ cm of Hg} = 760 \text{ mm of Hg} = 760 \text{ torr}$$

$$1 \text{ bar} = 75 \text{ cm Hg.}$$

4. GAS LAWS

The behaviour of the gases is governed by same general laws, which were discovered as a result of their experimental studies. These laws are relationships between measurable properties of gases. Some of these properties like pressure, volume, temperature and mass are very important because relationships between these variables describe state of the gas.

The first reliable measurement on properties of gases was made by Anglo-Irish scientist Robert Boyle in 1662. The law which he formulated is known as **Boyle's Law**. Later on attempts to fly in air with the help of hot air balloons motivated Jacques Charles and Joseph Lewis Gay Lussac to discover additional gas laws. Contribution from *Avogadro* and others provided lot of information about gaseous state.

4.1 Boyle's Law :

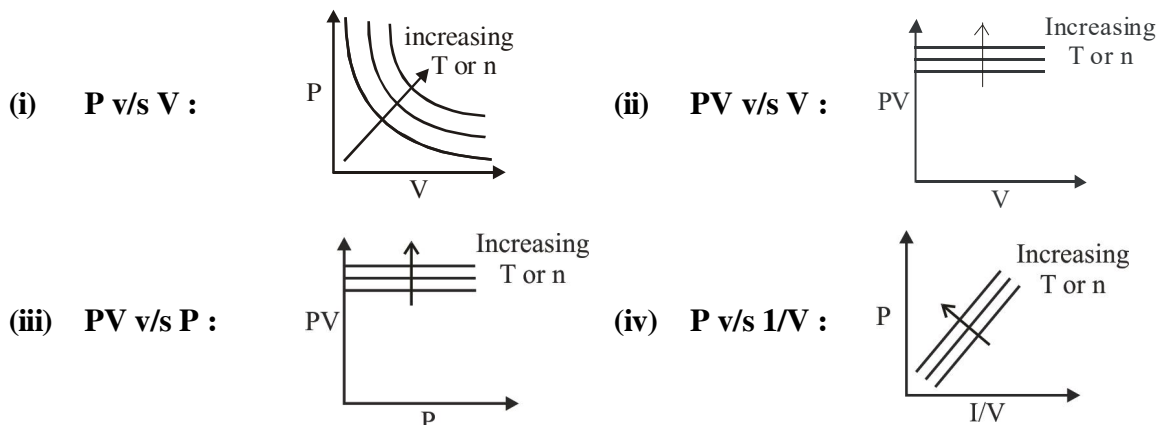
For a fixed amount of gas at constant temperature, the volume occupied by the gas is inversely proportional to the pressure applied on the gas or pressure of the gas.

$$\Rightarrow V \propto \frac{1}{P}$$

$$\Rightarrow PV = \text{constant}$$

$$\Rightarrow \text{Working formula : } P_1 V_1 = P_2 V_2$$

❖ Graphical representation of Boyle's law (Isotherms) :



Ex.1 A sample of gas occupies 100 dm^3 at 1 bar pressure and at $T^\circ\text{C}$. If the volume of the gas is reduced to 5 dm^3 at the same temperature, what additional pressure must be applied ?

Sol. From the given data :

$$P_1 = 1 \text{ bar} \quad P_2 = ?$$

$$V_1 = 100 \text{ dm}^3 \quad V_2 = 5 \text{ dm}^3$$

Since the temperature is constant, Boyle's law can be applied

$$P_1 V_1 = P_2 V_2 = P_2 = \frac{P_1 V_1}{V_2}$$

$$P_2 = \frac{(1 \text{ bar}) \times (100 \text{ dm}^3)}{(5 \text{ dm}^3)} = 20 \text{ bar}$$

$\therefore$ Additional pressure applied = $20 - 1 = 19 \text{ bar}$

4.2. Charle's Law :

For a fixed amount of gas at constant pressure, volume occupied by the gas is directly proportional to temperature of the gas on absolute scale of temperature.

$$\Rightarrow V \propto T$$

$$\boxed{\frac{V}{T} = \text{constant}}$$

T = Temperature on absolute scale, kelvin scale or ideal gas scale.

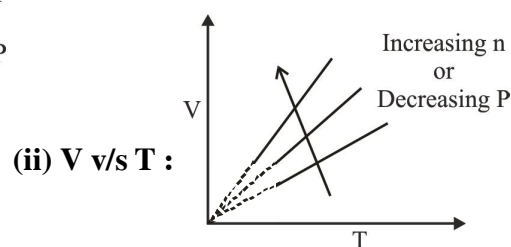
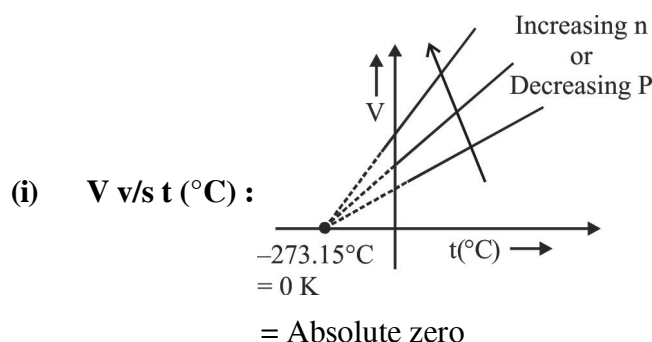
Working formula : $\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$

$$* \quad V = V_0 \left(1 + \frac{t}{273.15} \right)$$

t = temperature on celcius scale.

V_0 = volume of gas at 0°C .

❖ Graphical representation of Charle's Law (Isobars) :



❖ Important Points :

- Absolute zero ($0 \text{ K} = -273.15^\circ\text{C}$) is theoretically the minimum possible temperature.
- Absolute zero can never be achieved.

Ex.2 A certain amount of a gas at 27°C and 1 bar pressure occupies a volume of 25 m^3 . If the pressure is kept constant and the temperature is raised to 77°C , what will be the volume of the gas ?

Sol. From the available data :

$$V_1 = 25\text{ m}^3 \quad T_1 = 27 + 273 = 300\text{ K}$$

$$V_2 = ? \quad T_2 = 77 + 273 = 350\text{ K}$$

Since the pressure of the gas is constant, Charles law is applicable

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$V_2 = \frac{V_1 \times T_2}{T_1} = \frac{(25\text{ m}^3) \times (350\text{ K})}{(300\text{ K})} = 29.17\text{ m}^3$$

Ex.3 A balloon is inflated to $\frac{7}{8}$ of its maximum volume at 27°C then calculate the minimum temperature above which it will burst.

Sol. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$$\frac{7V}{8 \times 300} = \frac{V}{T}$$

$$T = 342.8\text{ K}$$

4.3. Gay-lussac's law or Pressure law :

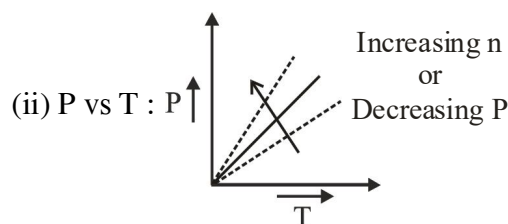
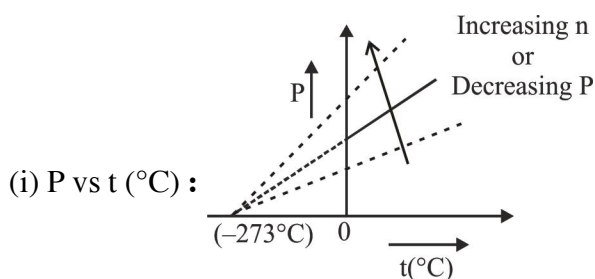
For a fixed amount of gas at constant volume, pressure of the gas is directly proportional to temperature of the gas on absolute scale of temperature.

$$\Rightarrow P \propto T$$

$$\frac{P}{T} = \text{constant}$$

Working formula $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$$* \quad P = P_0 \left(1 + \frac{t}{273.15} \right)$$



Ex.4 A gas cylinder containing cooking gas can with stand a pressure of 18 atm. The pressure gauge of cylinder indicates 12 atm at 27°C. Due to sudden fire in building the temperature start rising at what temperature will the cylinder explode.

Sol. $\frac{P_1}{T_2} = \frac{P_2}{T_1}$

$$\frac{12}{300} = \frac{18}{T}$$
$$T = 450 \text{ K}$$

4.4. Avogadro's law :

Equal volumes of all the gases under similar conditions of temperature and pressure contains equal number of molecules or moles of molecules (not atoms).

$$V \propto N \quad (\text{Temperature and pressure constant})$$

$$V \propto n \quad (\text{Temperature and pressure constant})$$

Where, N = number of molecules, n = number of moles of molecules

$$\frac{V_1}{N_1} = \frac{V_2}{N_2} \text{ or } \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

Since volume of a gas is directly proportional to the number of moles; one mole of each gas at standard temperature and pressure (STP) will have same volume.

Standard temperature and pressure means 273.15 K (0°C) temperature and 1 bar (i.e., exactly 10^5 pascal) pressure. **At STP, molar volume of an ideal gas or a combination of ideal gases is 22.71098 L mol⁻¹, normally taken as 22.7 L mol⁻¹.**

* The previous standard is still often used, and applies to all chemistry data more than decade old. In this definition STP denotes the same temperature of 0°C (273.15 K), but a slightly higher pressure of 1 atm (101.325 kPa). One mole of any gas or a combination of gases occupies 22.413996 L of volume at STP, normally taken as 22.4 L.

Standard ambient temperature and pressure (SATP), conditions are also used in some scientific works. SATP conditions means 298.15 K and 1 bar (i.e., exactly 10^5 Pa). **At SATP (1 bar and 298.15 K), the molar volume of an ideal gas is 24.789 L mol⁻¹.**

Molar volume in litres per mole of some gases at 273.15 K and 1 bar (STP).	
Argon	22.37
Carbon dioxide	22.54
Dinitrogen	22.69
Dioxygen	22.69
Dihydrogen	22.72
Ideal gas	22.71

Ex.5 A toy balloon originally held 1.0 gm of He gas and had a radius 10 cm. During the night, 0.875 gm of the gas effused from the balloon. Assuming ideal gas behaviour, under these constant P and T conditions, what was the radius of the balloon the next morning.

Ans. 5 cm

Sol. $\frac{V_1}{n_1} = \frac{V_2}{n_2} \Rightarrow \frac{V_1}{m_1} = \frac{V_2}{m_2}$

$$\Rightarrow \frac{\frac{4}{3}\pi(10)^3}{1.0} = \frac{\frac{4}{3}\pi r^3}{1.0 - 0.875}$$

$$\therefore r = 5 \text{ cm.}$$

5. IDEAL GAS EQUATION

A single equation which is combination of Boyle's, Charle's & Avogadro's Law is known as **Ideal gas equation**, or we can say a single equation which describe the simultaneous effects of the change in temperature and pressure on volume of the given amount of the gas is called the **Ideal gas equation**.

According to Boyle's law, $V \propto \frac{1}{P}$ (at constant T and n)

According to Charle's law, $V \propto T$ (at constant P and n)

According to Avogadro's law, $V \propto n$ (at constant T and P)

Combining these three laws, $V \propto \frac{nT}{P}$ or $PV \propto nT$

or **$PV = nRT$** [Equation of state for ideal gas or combined gas equation or ideal gas equation]

where R is a constant called **Universal gas constant**, which does not depend on variables (P , V , n , T) and nature of gas.

5.1 Molar volume is the volume of 1 mole of gas.

$$\text{Molar volume } (V_m) = \frac{\text{Volume}}{\text{mole}}$$

$$PV_m = RT$$

Volume of 1 mole of an ideal gas under STP conditions (273.15 K, normally taken as 273 K, and 1 bar pressure) is 22.7 L mol^{-1} .

5.2 Dimension of R :

$$R = \frac{PV}{nT} = \frac{\text{Pressure} \times \text{Volume}}{\text{Mole} \times \text{Temperature}} = \frac{(\text{Force} / \text{Area}) \times (\text{Area} \times \text{Length})}{\text{Mole} \times \text{Temperature (K)}}$$

$$= \frac{\text{Force} \times \text{Length}}{\text{Mole} \times \text{Temperature (K)}} = \frac{\text{Work or energy}}{\text{Mole} \times \text{Temperature (K)}}$$

5.3 Physical significance of R :

The dimensions of R is energy per mole per kelvin and hence it represents the amount of work (or energy) that can be obtained from one mole of a gas when its temperature is raised by 1 K, isobarically.

5.4 Value of R in different units :

$$(i) \quad \text{In L-atm :} \quad R = \frac{1 \text{ atm} \times 22.4 \text{ lit.}}{1 \text{ mol} \times 273 \text{ K}} = \mathbf{0.082 \text{ L-atm mol}^{-1} \text{K}^{-1}}$$

$$(ii) \quad \text{In L-bar :} \quad R = \frac{1 \text{ bar} \times 22.7 \text{ L}}{1 \text{ mol} \times 273 \text{ K}} = \mathbf{0.083 \text{ L-bar/K-mol}}$$

$$(iii) \quad \text{In C.G.S system :} \quad R = \frac{(1 \times 76 \times 13.6 \times 980 \text{ dyne cm}^{-2}) \times (22400 \text{ cm}^3)}{1 \text{ mol} \times 273 \text{ K}}$$

$$= \mathbf{8.314 \times 10^7 \text{ erg mole}^{-1} \text{K}^{-1}}.$$

$$(iv) \quad \text{In SI units :} \quad R = \mathbf{8.314 \text{ Joule mole}^{-1} \text{K}^{-1}}. \quad [10^7 \text{ erg} = 1 \text{ joule}]$$

$$(v) \quad \text{In calories :} \quad R = 1.987 \approx \mathbf{2 \text{ calorie mol}^{-1} \text{K}^{-1}}.$$

5.5 Ideal gas equation in terms of density of gas :

$$PV = nRT = \frac{W}{M} RT$$

$$\text{or } PM = \frac{W}{V} \cdot RT$$

$$\therefore \mathbf{PM = dRT}$$

Ex.6 The best vacuum so far attained in laboratory is 10^{-10} mm of Hg. What is the number of molecules of gas remain per cm^3 at 20°C in this vacuum ?

Sol. Given conditions -

$$P = 10^{-10} \text{ mmHg} = \frac{10^{-10}}{760} \text{ atm} ; V = 1 \text{ cm}^3 = \frac{1}{1000} \text{ L} ; T = 20^\circ\text{C} = 293 \text{ K}$$

No. of molecules, $N = ?$

Now, applying

$$PV = nRT$$

$$\therefore \text{ number of molecules per cm}^3, N = \frac{N_A \times P \times V}{RT}$$

$$\Rightarrow N = \frac{6.022 \times 10^{23} \times (10^{-10}/760) \times (1/1000)}{(0.082)(293)}$$

$$\mathbf{N = 3.29 \times 10^6 \text{ molecules}}$$

Ex.7 The density of a gas is found to be 1.56 g dm^{-3} at 0.98 bar pressure and 65°C . Calculate the molar mass of the gas.

Use $R = 0.083 \text{ dm}^3 \text{ bar K}^{-1} \text{ mol}^{-1}$

Sol. We know that :

$$M = \frac{dRT}{P} = \frac{(1.56 \text{ g dm}^{-3}) \times (0.083 \text{ dm}^3 \text{ bar K}^{-1} \text{ mol}^{-1}) \times 338 \text{ K}}{(0.98 \text{ bar})} = 44.66 \text{ g mol}^{-1}$$

Ex.8 A tyre tube of maximum volume 8.2 lit. can withstand a pressure of 10 atm . Initially the tube is empty.

(i) Calculate the number of moles required to inflate completely the tube upto a pressure of 1 atm & 300 K temperature.

(ii) Calculate the minimum number of moles required to burst the tyre tube at 300 K .

Sol. (i) $PV = nRT$

$$1 \times 8.2 = n_1 \times 0.082 \times 300$$

$$n_1 = \frac{1}{3}$$

(ii) $PV = nRT$

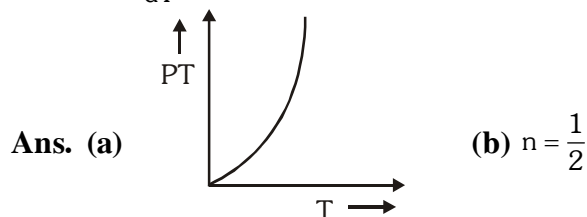
$$10 \times 8.2 = n_2 \times 0.082 \times 300$$

$$n_2 = \frac{10}{3}$$

Ex.9 (a) Plot the curve between PT vs T at const V & constant no. of moles.

(b) Find the number of moles of gas taken when the volume of the vessel is 82.0 ml and

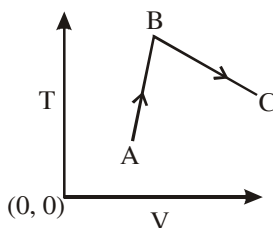
$\frac{d}{dT}[PT]$ at $300 \text{ K} = 300$ for the given curve.



Sol. (b) $\frac{d(PT)}{dT} = \frac{d}{dT} \left[\frac{nRT}{V} \cdot T \right] = \frac{2nRT}{V} = 300$

$$\therefore n = \frac{300 \times (82 \times 10^{-3})}{2 \times 0.082 \times 300} = \frac{1}{2}$$

Ex.10 In the given isobaric process shown by graph between T & V .



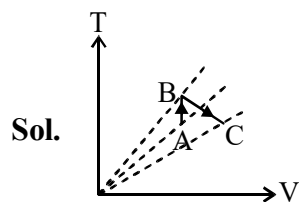
(A) Moles decreases throughout

(B) Moles first increases then decreases

(C) Moles first decreases then increases

(D) Moles cannot be predicted from given data

Ans. (C)

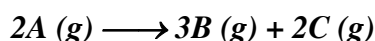


$$\frac{dT}{dV} = \frac{P}{nR}$$

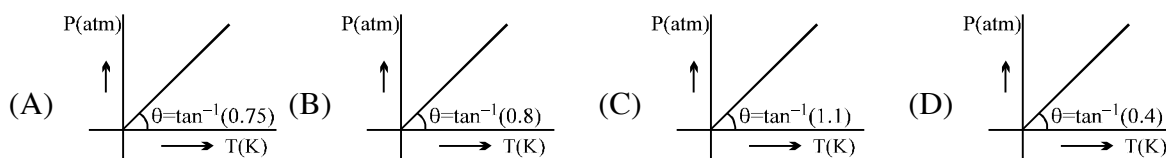
At constant pressures slope is higher for smaller moles of gas.

Hence, from A to B, mole of gas decreases and for B to C, moles increases.

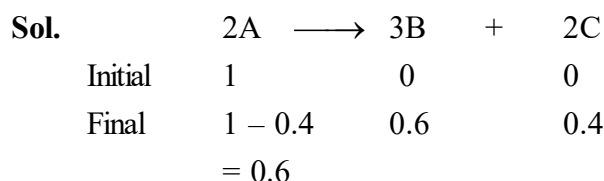
Ex.11 Gas A (1 mol) dissociates in a closed rigid container of volume 0.16 lit. as per following reaction.



If degree of dissociation of A is 0.4 and remains constant in entire range of temperature, then the correct P vs T graph is [Given $R = 0.08 \text{ lit-atm/mol/K}$]



Ans. (B)



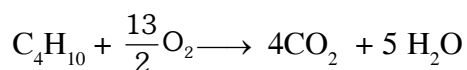
Total moles = $0.6 + 0.6 + 0.4 = 1.6$

$$\text{Now, } \frac{dP}{dT} = \frac{nR}{V} = \frac{1.6 \times 0.08}{0.16} = 0.8 \tan \theta$$

$$\therefore \theta = \tan^{-1}(0.8)$$

Ex.12 LPG is a mixture of n-butane & iso-butane. What is the volume of oxygen needed to burn 1 kg of LPG at 1 atm, 273 K?

Sol. During the burning of LPG following reaction takes place -



Now, for complete combustion of 1 mole of LPG $\longrightarrow \frac{13}{2}$ moles of O_2 are required.

$$\therefore \text{ for } \frac{1000}{58} \text{ moles of LPG } \longrightarrow \frac{13}{2} \times \frac{1000}{58} \text{ moles of } O_2 \text{ are required.}$$

Thus, the volume of oxygen needed to burn 1 kg of LPG at 1 atm & 273K would be

$$\text{Vol. of } O_2 = \text{Moles of } O_2 \times 22.4 \text{ L} = \frac{13}{2} \times \frac{1000}{58} \times 22.4 = \mathbf{2510 \text{ L}}$$

Ex.13 An open flask contains air at 27°C. Calculate the temperature at which it should be heated so that

- (a) $1/3^{\text{rd}}$ of air measured in the container at 27°C escape out.
 (b) $1/3^{\text{rd}}$ of air measured in the container at final temperature escape out.

Sol.

(a) $n_{\text{initial}} - n_{\text{final}} = n_{\text{expelled}}$

$$\frac{PV}{R \times 300} - \frac{PV}{R \times T} = \frac{1}{3} \times \frac{PV}{R \times 300} = \frac{1}{300} - \frac{1}{T} = \frac{1}{900}$$

$$\frac{1}{T} = \frac{1}{300} - \frac{1}{900} = \frac{3-1}{900} = \frac{2}{900} \Rightarrow T = 450 \text{ K} = 177^\circ\text{C}$$

(b) $n_{\text{initial}} - n_{\text{final}} = n_{\text{expelled}}$

$$\frac{PV}{R \times 300} - \frac{PV}{R \times T} = \frac{1}{3} \times \frac{PV}{R \times T}$$

$$\frac{1}{300} - \frac{1}{T} = \frac{1}{3 \times T}$$

$$\Rightarrow T = 400 \text{ K} = 127^\circ\text{C}$$

Ex.14 A bulb of unknown volume containing air is heated from 27°C to 227°C at constant pressure. The expelled air is measured at different temperature of determine volume of container. What will be volume of container if -

- (a) 200 ml of air measured at 227°C was expelled.
 (b) 200 ml of air measured at 27°C was expelled.
 (c) 200 ml of air measured at 127°C was expelled.

Sol.

(a) $n_{\text{initial}} - n_{\text{final}} = n_{\text{expelled}}$

$$\frac{PV}{R \times 300} - \frac{PV}{R \times 500} = \frac{P \times 200}{R \times 500}$$

$$\frac{V}{3} - \frac{V}{5} = \frac{200}{5}$$

$$\frac{5V - 3V}{15} = \frac{200}{5}$$

$$\Rightarrow 2V = \frac{200 \times 15}{5} \Rightarrow V = 300 \text{ ml}$$

(b) $\frac{PV}{R \times 300} - \frac{PV}{R \times 500} = \frac{P \times 200}{R \times 300} = \frac{2V}{15} = \frac{200}{3} \Rightarrow V = \frac{100 \times 15}{2 \times 3} = 500 \text{ ml}$

(c) $\frac{PV}{R \times 300} - \frac{PV}{R \times 500} = \frac{P \times 200}{R \times 400}$

$$\frac{2V}{15} = \frac{200}{4} \Rightarrow V = \frac{200 \times 15}{2 \times 4} = 375 \text{ ml}$$

Ex.15 A 10 litre container consist of 1 mole of gas at 300 K. It is connected to another container having volume 40 litre and is initially at 300 K. The nozzle connecting two containers is opened for a long time and once the movement of gas stopped, the larger container was heated to a temperature of 600 K. Calculate

- (a) Moles and pressure of gas in both the containers before heating.
 (b) Moles and pressure in two containers after heating.
 (Assume that initially the larger container is completely evacuated.)

Sol. (a) Before heating :

$$P_I = P_{II}$$

$$\frac{(1-x)R \times 300}{10} = \frac{x \times R \times 300}{40}$$

$$x = 0.8 \text{ moles}$$

$$n_I = 1 - x = 0.2 \text{ mole}$$

$$n_{II} = x = 0.8 \text{ mole}$$

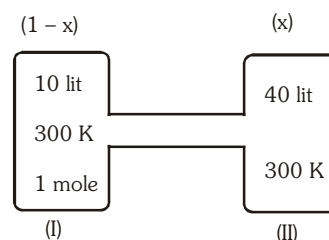
$$\text{Pressure} = \frac{x \times R \times T}{V} = \frac{0.8 \times R \times 300}{40} = 0.492 \text{ atm}$$

(b) After heating :

$$\frac{(1-x_1)R \times 300}{10} = \frac{x_1 \times R \times 600}{40}$$

$$x_1 = 0.67 \text{ moles, Given } T_1 = 600 \text{ K}$$

$$\text{Pressure} = \frac{x_1 \times R \times T_1}{V} = \frac{0.67 \times 0.082 \times 600}{40} = 0.82 \text{ atm}$$



6. DALTON'S LAW OF PARTIAL PRESSURES

6.1 Partial pressure :

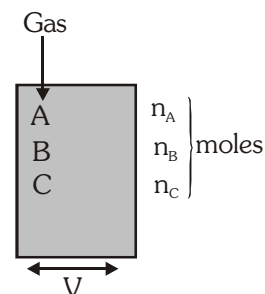
In a mixture of non-reacting gases, partial pressure of any component gas is defined as the pressure exerted by the individual gas if the entire volume of mixture had been occupied by this component only.

Partial pressure of component gases are -

$$P_A = \frac{n_A RT}{V} = \text{partial pressure of A}$$

$$P_B = \frac{n_B RT}{V} = \text{partial pressure of B}$$

$$P_C = \frac{n_C RT}{V} = \text{partial pressure of C}$$



6.2 Dalton's Law :

Dalton's law of partial pressure states "at a given temperature, the total pressure exerted by two or more non-reacting gases occupying a definite volume is equal to the sum of the partial pressures of the component gases."

Ex.17 (a) Find the total pressure and partial pressure of each component if a container of volume 8.2 lit. contains 2 moles of A and 3 mole of B at 300K.

(b) What will be the final pressure and partial pressure of each component if 5 moles of C is also added to the container at same temperature.

Sol. (a) $P_A = \frac{n_A RT}{V} = \frac{2 \times 0.082 \times 300}{8.22} = 6 \text{ atm}$

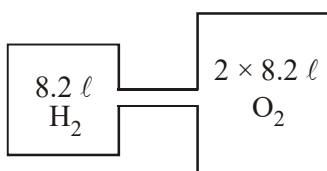
$$P_B = \frac{n_B RT}{V} = \frac{3 \times 0.082 \times 300}{8.22} = 9 \text{ atm}$$

$$\text{Total pressure } P_T = P_A + P_B = 6 + 9 = 15 \text{ atm}$$

(b) $P_C = \frac{n_C RT}{V} = \frac{5 \times 0.082 \times 300}{8.22} = 15 \text{ atm}$

$$\text{and total pressure} = 15 + 15 = 30 \text{ atm.}$$

Ex.18 A container of 8.2 lit. capacity is filled with 1 mole of H_2 at 300 K and it is connected to another container of capacity 2×8.2 lit. containing 4 moles of O_2 at 300 K, then find the final pressure & partial pressure of each gas.

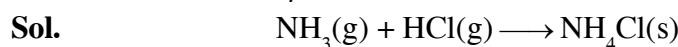


Sol. $P_f V_f = n_f RT$
 $P_f (3 \times 8.2) = 5 \times 0.082 \times 300$
 $P_f = 5 \text{ atm}$

$$P_{H_2} = x_{H_2} P_f = \frac{1}{1+4} \times 5 = 1 \text{ atm}$$

$$P_{O_2} = x_{O_2} P_f = \frac{4}{1+4} \times 5 = 4 \text{ atm}$$

Ex.19 2 moles of $NH_3(g)$ and 1 mole of $HCl(g)$ are taken in a container of capacity 8.2 lit at 300K to produce $NH_4Cl(s)$. Find the total pressure after the reaction.



initial	2	1
---------	---	---

after rxn	1	0
-----------	---	---

In this reaction HCl is L.R. so it will be completely consumed. We don't consider pressure due to solid.

$$\therefore P_T = \frac{nRT}{V} = \frac{1 \times 0.082 \times 300}{8.2} = 3 \text{ atm}$$

Ex.20 A closed container containing O_2 and some liquid water was found to exert 740 mm pressure at $27^\circ C$.

(a) Then calculate the pressure exerted by O_2 if aqueous tension at $27^\circ C$ is 20 mm.

(b) What will be the final pressure if volume is reduced to half.
 (consider volume of liquid water negligible)

(c) What will be the final pressure if volume is doubled.

Sol. (a) $P_T = P_{\text{dry gas}} + P_{\text{aq. tension}}$

$$740 = P_{O_2} + 20$$

$$P_{O_2} = 740 - 20 = 720 \text{ mm}$$

(b) $(P_{O_2} V_{O_2})_{\text{initial}} = (P_{O_2} V_{O_2})_{\text{final}}$ (Boyle's law)

$$720 \times V = P_{O_2} \times \frac{V}{2}$$

$$P_{O_2} = 1440 \text{ mm}$$

$$P_T = P_T = P_{O_2} + P_{\text{aq.}}$$

$$= 1440 + 20 = 1460 \text{ mm}$$

(c) $(P_{O_2} V_{O_2})_{\text{initial}} = (P_{O_2} V_{O_2})_{\text{final}}$

$$720 \times V = P_{O_2} \times 2V$$

$$P_{O_2} = 360 \text{ mm}$$

$$P_T = P_{O_2} + P_{\text{aq.}}$$

$$= 360 + 20 = \mathbf{380 \text{ mm}}$$

7. AMAGAT'S LAW OF PARTIAL VOLUME

7.1 Partial Volume :

Partial volume of any component is defined as the volume occupied by that particular component when it is kept at same total pressure and temperature as of the mixture.

7.2 Amagat's law :

According to this law at constant temperature and pressure, the total volume of mixture of non-reacting gases is equal to the sum of partial volumes of each component present in mixture.

$$V_T = V_1 + V_2 + V_3 + \dots$$

Let us consider three non-reacting gases A, B and C are present in a container which have no. of moles n_A , n_B and n_C respectively. For each gas partial volume is

$$V_A = n_A \left(\frac{RT}{P} \right) = \text{partial volume of A}$$

$$V_B = n_B \left(\frac{RT}{P} \right) = \text{partial volume of B}$$

$$V_C = n_C \left(\frac{RT}{P} \right) = \text{partial volume of C}$$

Now, total volume

$$V_T = V_A + V_B + V_C = (n_A + n_B + n_C) \left(\frac{RT}{P} \right) = n_T \left(\frac{RT}{P} \right)$$

$$\frac{V_A}{V_T} = \frac{n_A}{n_T} = x_A \quad (\text{Mole fraction of gas A})$$

$$\frac{V_B}{V_T} = \frac{n_B}{n_T} = x_B \quad (\text{Mole fraction of gas B})$$

$$\frac{V_C}{V_T} = \frac{n_C}{n_T} = x_C \quad (\text{Mole fraction of gas C})$$

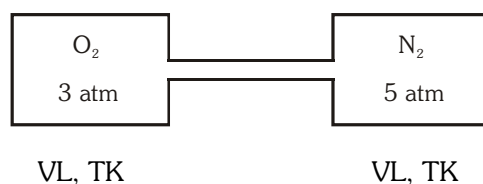
$\therefore$ **Partial volume of a gas = Mole fraction $\times$ Total volume**

8. GRAHAM'S LAW OF DIFFUSION AND EFFUSION

8.1 Diffusion :

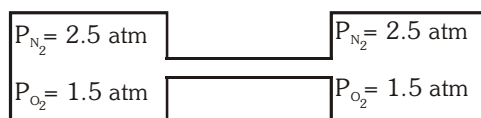
Diffusion is the process of gradual mixing of molecules of one gas with molecules of another gas due to their molecular motion (kinetic energy). The diffusion always proceeds from a region of high concentration to a region of lower concentration (or high partial pressure to low partial pressure). For example, when a bottle of perfume is opened at one end of the room, the person sitting at the other end of the room can smell the perfume because of the diffusion process of perfume molecules.

Note : Initially



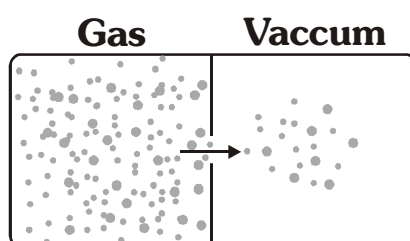
When stop cock is removed flow will be from both sides, N₂ will try to equalise its partial pressure in both the vessels, and so will O₂.

Finally



8.2 Effusion :

The effusion is the process of forcing a gas through a pin hole or small orifice from one compartment to another empty (vacuum) compartment.



8.3 Graham's Law :

Under similar condition of pressure (partial pressure) and temperature, the rate of diffusion or effusion of different gases is inversely proportional to square root of their density.

$$\Rightarrow \text{rate of diffusion, } r \propto \frac{1}{\sqrt{d}}$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{\sqrt{d_2}}{\sqrt{d_1}} = \frac{\sqrt{M_2}}{\sqrt{M_1}} = \frac{\sqrt{V.D_2}}{\sqrt{V.D_1}}$$

where, d = density of gas

$V.D$ = vapour density

M = molar mass of gas

8.4 Under conditions of same temperature but different pressure :

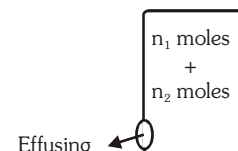
$$r \propto \frac{P}{\sqrt{M}}$$

$$\boxed{\frac{r_1}{r_2} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}}$$

If both gases are present in the same container at same temperature.

$$P \propto n$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{n_1}{n_2} \sqrt{\frac{M_2}{M_1}}$$



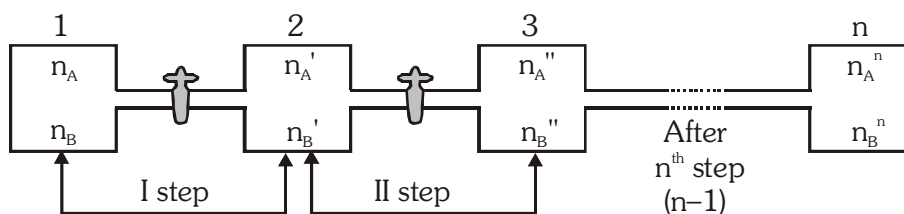
8.5 Rate of diffusion/effusion can be expressed as -

$$r = \frac{\text{volume diffused}}{\text{time taken}} \text{ or } \frac{\text{moles diffused}}{\text{time taken}} \text{ or } \frac{\text{pressure dropped}}{\text{time taken}}$$

$$\text{or } \frac{\text{distance travelled in horizontal tube of uniform cross - section}}{\text{time taken}}$$

8.6 Enrichment factor or Isotopic separation factor :

When two gases present in a container are allowed to diffuse in another container then gas having lower molecular mass will diffuse more and if this process is continued for large number of steps then we can obtain a mixture which is rich in a gas having lower molecular mass. Hence ultimately in the ultimate (last) container amount of lighter gas is larger as compared to heavier gas.



where, $M_A > M_B$ [similar condition of T and V] & $n_B^n \gg n_A^n$

Enrichment factor (f) is defined as the ratio of final ratio of moles in the mixture [after n^{th} step] with initial mole ratio of the mixture.

$$f = \frac{\left(\frac{n_A^n}{n_B^n}\right)}{\left(\frac{n_A}{n_B}\right)} = \left(\sqrt{\frac{M_B}{M_A}}\right)^n = \frac{\text{final molar ratio}}{\text{initial molar ratio}}$$

Ex.21 32 ml of He effuses through a fine orifice in 1 minute. Then what volume of CH₄ will diffuse in 1 minute under the similar condition.

Sol. $\therefore r \propto \frac{1}{\sqrt{M}}$

$$r = \frac{\text{volume diffused}}{\text{time}}$$

$\therefore$ time is same so

$$\frac{V_{\text{CH}_4}}{V_{\text{He}}} = \sqrt{\frac{M_{\text{He}}}{M_{\text{CH}_4}}}$$

$$\frac{V_{\text{CH}_4}}{32} = \sqrt{\frac{4}{16}}$$

$$V_{\text{CH}_4} = \frac{1}{2} \times 32 = 16 \text{ mL}$$

Ex.22 20 dm³ of Ne diffuse through a porous partition in 60 seconds. What volume of SO₃ will diffuse under similar conditions in 30 sec. (Atomic wt. of Ne = 20, S = 32)

Sol. $\frac{r_{\text{Ne}}}{r_{\text{SO}_3}} = \sqrt{\frac{M_{\text{SO}_3}}{M_{\text{Ne}}}} \Rightarrow \frac{V_{\text{Ne}}/t_{\text{Ne}}}{V_{\text{SO}_3}/t_{\text{SO}_3}} = \sqrt{\frac{M_{\text{SO}_3}}{M_{\text{Ne}}}}$

$$\Rightarrow V_{\text{SO}_3} = \frac{V_{\text{Ne}} \times t_{\text{SO}_3}}{t_{\text{Ne}}} \times \sqrt{\frac{M_{\text{Ne}}}{M_{\text{SO}_3}}} = \frac{20 \times 30}{60} \times \sqrt{\frac{20}{80}} = \frac{10}{2} = 5 \text{ dm}^3$$

Ex.23 A gaseous mixture of O₂ and X containing 20% (mole %) of X, diffused through a small hole in 234 seconds while pure O₂ takes 224 seconds to diffuse through the same hole. Molecular weight of X is :

Sol. $\frac{t_{\text{mix}}}{t_{\text{O}_2}} = \sqrt{\frac{M_{\text{mix}}}{M_{\text{O}_2}}}$

$$\frac{234}{224} = \sqrt{\frac{M_{\text{mix}}}{32}}$$

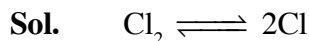
$$\therefore M_{\text{mix}} = 34.921.$$

As the mixture contains 20% (mole %) of X, the molar ratio of O₂ and X may be represented as 0.8n : 0.2n, n being the total no. of moles.

$$\therefore M_{\text{mix}} = \frac{32 \times 0.8n + M_x \times 0.2n}{n} = 34.921$$

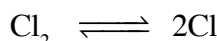
$$\therefore M_x (\text{mol. wt. of X}) = 46.6$$

Ex.24 At 1200°C, mixture of Cl₂ and Cl atoms (both in gaseous state) effuses 1.16 times as fast as krypton effuses under identical conditions. Calculate the percentage of chlorine molecules dissociated into atoms. M (Kr) = 83.8 g mol⁻¹.



$$\frac{r(\text{Cl}_2 \text{ and Cl mix})}{r(\text{Kr})} = 1.16 \sqrt{\frac{M(\text{Kr})}{M_{\text{av}}(\text{Cl}_2 + \text{Cl})}} = \sqrt{\frac{83.8}{M_{\text{av}}}}$$

$$\therefore M_{\text{av}} = \frac{83.8}{(1.16)^2} = 62.28 \text{ g mol}^{-1}$$



Initial mole 1 0

After dissociation (1 - x) 2x

(x = degree of dissociation)

Total moles after dissociation = 1 - x + 2x = (1 + x)

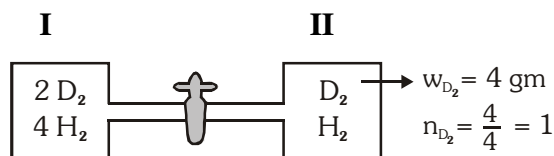
$$\therefore \frac{(1-x)M(\text{Cl}_2) + 2xM(\text{Cl})}{(1+x)} = 62.28 \Rightarrow \frac{(1-x) \times 71 + 2x \times 35.5}{1+x} = 62.28$$

$$x = 0.14$$

$$\therefore \% \text{ dissociation} = 14\%$$

Ex.25 A mixture containing 2 moles of D_2 and 4 moles of H_2 is taken inside a container which is connected to another empty container through a nozzle. The nozzle is opened for certain time and then closed. The second bulb was found to contain 4 gm D_2 . Then find % by moles of the lighter gases in second container.

Sol.



lighter gas = H_2

$$\frac{r_{\text{H}_2}}{r_{\text{D}_2}} = \frac{n_{\text{H}_2}}{n_{\text{D}_2}} \sqrt{\frac{M_{\text{D}_2}}{M_{\text{H}_2}}} = \frac{\dot{n}_{\text{H}_2}/t}{\dot{n}_{\text{D}_2}/t}$$

$$(\% \text{ mole of } \text{H}_2)_{\text{II}} = \left(\frac{\dot{n}_{\text{H}_2}}{\dot{n}_{\text{H}_2} + \dot{n}_{\text{D}_2}} \right) \times 100$$

$$\therefore \frac{n_{\text{H}_2}}{n_{\text{D}_2}} \sqrt{\frac{M_{\text{D}_2}}{M_{\text{H}_2}}} = \frac{\dot{n}_{\text{H}_2}}{\dot{n}_{\text{D}_2}} \Rightarrow \frac{4}{2} \sqrt{\frac{4}{2}} = \frac{\dot{n}_{\text{H}_2}}{1}$$

$$\Rightarrow \dot{n}_{\text{H}_2} = 2\sqrt{2} \Rightarrow \dot{n}_{\text{D}_2} = 1$$

$$(\% \text{ mole of } \text{H}_2)_{\text{II}} = \frac{(2\sqrt{2})}{(2\sqrt{2} + 1)} \times 100 = \frac{2 \times 1.44}{[(2 \times 1.44) + 1]} \times 100 = \frac{(2.8)}{(2.8 + 1)} \times 100 = \frac{(2.8)}{(2.8 + 1)} \times 100$$

$$= 73.87\%$$

Ex.26 According to Graham's law, at a given temperature the ratio of the rates of diffusion $\frac{r_A}{r_B}$ of gases A and B is given by : [JEE 1998]

(A) $\frac{P_A}{P_B} \left(\frac{M_A}{M_B} \right)^{1/2}$ (B) $\left(\frac{M_A}{M_B} \right) \left(\frac{P_A}{P_B} \right)^{1/2}$ (C) $\frac{P_A}{P_B} \left(\frac{M_B}{M_A} \right)^{1/2}$ (D) $\frac{M_A}{M_B} \left(\frac{P_B}{P_A} \right)^{1/2}$

Ans. (C)

Sol. $r \propto \frac{P}{\sqrt{M}}$

$$\therefore \frac{r_A}{r_B} = \frac{P_A}{P_B} \cdot \sqrt{\frac{M_B}{M_A}}$$

Ex.27 The density of the gaseous mixture in a vessel (CH_4 and He) at 2 atmosphere pressure and 300 K is 0.9756 g/lit. If a small pin-hole is made on the wall of the vessel, through which gases effuse, then which of the followings is the correct composition (by volume) of the gases CH_4 and He effusing out initially ?

(A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 1 : 4

Ans. (A)

Sol. $PM = dRT$

$$\therefore M_{\text{avg}} = \frac{dRT}{P} = \frac{0.9756 \times 0.082 \times 300}{2} \approx 12$$

Let the mixture contains x moles CH_4 and y moles He. Then

$$\therefore M_{\text{avg}} = 12 = \frac{x \times 16 + y \times 4}{x + y} \Rightarrow \frac{x}{y} = \frac{2}{1}$$

$$\text{Now, } \frac{r_{\text{CH}_4}}{r_{\text{He}}} = \frac{P_{\text{CH}_4}}{P_{\text{He}}} \cdot \sqrt{\frac{M_{\text{He}}}{M_{\text{CH}_4}}} = \frac{2}{1} \times \sqrt{\frac{4}{16}} = \frac{1}{1}$$

Ex.28 The number of effusion steps required to convert a mixture of H_2 and O_2 from 240 : 1600 (by mass) to 3072 : 20 (by mass) is

(A) 2 (B) 4 (C) 5 (D) 6

Ans. (C)

Sol. $\left(\frac{n_{\text{H}_2}}{n_{\text{O}_2}} \right)_{\text{final}} = \left(\frac{n_{\text{H}_2}}{n_{\text{O}_2}} \right)_{\text{initial}} \times \left(\sqrt{\frac{M_{\text{O}_2}}{M_{\text{H}_2}}} \right)^n$

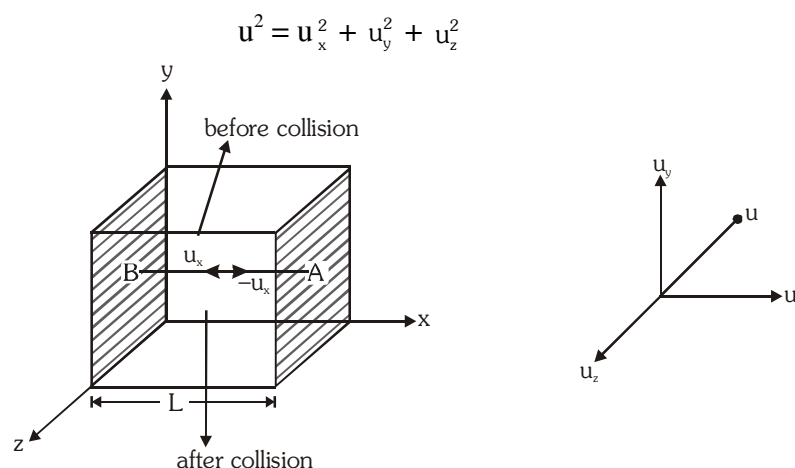
$$\text{or } \frac{3072}{20} = \frac{240}{1600} \times \left(\sqrt{\frac{32}{2}} \right)^n \Rightarrow n = 5$$

9. KINETIC THEORY OF GASES

This is a theoretical model for ideal gas which can correlate the experimental facts (like Boyle's law, Charle's law & Avogadro's law etc.). It was presented by **Bernoulli in 1738** and developed in 1860 by **Clausius, Maxwell, Kroning and Boltzmann**. Postulates of kinetic theory of gases are :

- (i) All the gases consist of very small molecules or atoms whose volume is negligible compared to volume of container.
- (ii) There are no attractive or repulsive forces between the molecules.
- (iii) The gaseous molecules are under a continuous state of motion which is unaffected by gravity (the random straight line motion is known as brownian motion)
- (iv) Due to the continuous motion, collision between gaseous molecules and with the wall of container occurs. The collision with the wall of container are responsible for pressure exerted by the gas on the wall of container.
- (v) The molecule moves with different speed, however the speed of each molecule keep on changing as the collision occur.
- (vi) Collision among gas particles molecules is perfectly elastic, i.e., there is no loss in kinetic energy and moment during such collision.
- (vii) The average kinetic energy of gas particles will depends on absolute temperature only.

9.1 Kinetic gas equation : Let us consider a cube of side length L , that has N molecules each of mass ' m ' moving with velocity $\vec{u}$ in all direction and thus colliding with one each other and against sides of the container. Velocity $\vec{u}$ can be resolved into three components u_x , u_y and u_z along there axis such that



For a simplest case we consider motion of a molecule along x-axis only in which it moves towards face B with velocity u_x . After collision against face B it moves towards face A with velocity $(-u_x)$, collision being elastic (which results in change in direction but not speed)

Hence, Momentum before collision on face B = mu_x

Momentum after collision on face B = $-mu_x$

Change in momentum due to one collision on face B = $mu_x - (-mu_x) = 2mu_x$

To strike face B again distance travelled = $2L$

Time taken to strike face B again = $\frac{2L}{u_x}$ seconds

∴ Number of collisions per second on face B along x-axis = $\frac{u_x}{2L}$

∴ Rate of change in momentum due to $\frac{u_x}{2L}$ collisions per second on face B along x-axis.

$$= 2 m u_x \cdot \frac{u_x}{2L} = \frac{m u_x^2}{L}$$

Similarly for y-axis change in momentum per second = $\frac{m u_y^2}{L}$ and for z-axis = $\frac{m u_z^2}{L}$

Net force by N molecules on a wall, $F_x = \frac{m u_{x1}^2}{L} + \frac{m u_{x2}^2}{L} + \dots + \frac{m u_{xN}^2}{L} = \frac{M}{L} \cdot \Sigma u_x^2$

Now pressure = $\frac{\text{Force}}{\text{Area of six faces}} = \frac{\frac{m}{L} \cdot \Sigma u_x^2}{L^2} = \frac{m \cdot \Sigma u_x^2}{L^3} = \frac{m \cdot \Sigma u^2}{V}$ [$L^3 = \text{volume } V$]

$$\therefore PV = m \cdot \Sigma u_x^2$$

As $\Sigma u_x^2 = \Sigma u_y^2 = \Sigma u_z^2$ and $\Sigma u_x^2 + \Sigma u_y^2 + \Sigma u_z^2 = \Sigma u^2$

$$\therefore PV = m \cdot \Sigma u_x^2 = \frac{1}{3} m \cdot \Sigma u^2$$

$$PV = \frac{1}{3} m \left(\frac{u_1^2 + u_1^2 + \dots + u_N^2}{N} \right) \cdot N$$

$$\therefore \boxed{PV = \frac{1}{3} m N u_{\text{rms}}^2}$$

This equation is called kinetic gas equation.

9.2 Kinetic energy of gas molecules :

Total translational K.E. of molecules

$$= \frac{1}{2} m u_1^2 + \frac{1}{2} m u_2^2 + \dots + \frac{1}{2} m u_N^2 = \frac{1}{2} m \cdot \Sigma u^2 = \frac{1}{2} m N \cdot u_{\text{rms}}^2 = \frac{3}{2} PV = \frac{3}{2} nRT$$

$$\therefore \text{Average translational K.E. per mole} = \frac{3}{2} RT$$

$$\text{and (K.E.)}_{\text{per molecule}} = \frac{3}{2} \left(\frac{R}{N_A} \right) T = \frac{3}{2} kT$$

$$\text{Where } k = \text{Boltzman constant} = \frac{R}{N_A} = \frac{8.314 \text{ J/molK}}{6.02 \times 10^{23}} = 1.3806 \times 10^{-23} \text{ J K}^{-1}$$

Note : (K.E.)_{per molecule} and (K.E.)_{per mol} is only depend on absolute temperature. It is does not depend on the nature of gas. This conclusion is known as "*Maxwell's Generalisation*".

9.3 Root Mean Square Velocity (u_{rms}) by kinetic gas equation :

$$PV = \frac{1}{3} m N u_{\text{rms}}^2$$

$$\therefore u_{\text{rms}} = \sqrt{\frac{3PV}{mN}} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3RT}{M}}$$

9.4 Different kind of speed of molecules :

$$(i) \text{ Average or mean speed, } u_{av} = \frac{u_1 + u_2 + \dots + u_N}{N} = \sqrt{\frac{8RT}{\pi M}}$$

$$(ii) \text{ Root mean square speed, } u_{rms} = \sqrt{\frac{u_1^2 + u_2^2 + \dots + u_N^2}{N}} = \sqrt{\frac{3RT}{M}}$$

$$(iii) \text{ Most probable speed, } u_{mp} = \sqrt{\frac{2RT}{M}}$$

It is the speed at which maximum fraction of molecules are travelling

9.5 Ratio of speeds :

$$\begin{aligned} U_{rms} : U_{avg} : U_{mps} &= \sqrt{\frac{3RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{2RT}{M}} \\ &= \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} = 1.225 : 1.128 : 1.000 = 1.00 : 0.92 : 0.816 \end{aligned}$$

Ex.29 Four particles have speed 2, 3, 4 and 5 cm/s respectively. Find their average & rms speed.

Sol. $U_{avg.} = \frac{U_1 + U_2 + U_3 + \dots + U_N}{N} = \frac{2 + 3 + 4 + 5}{4} = 3.5 \text{ cm/s}$

$$U_{r.m.s.} = \sqrt{\frac{U_1^2 + U_2^2 + U_3^2 + \dots}{N}} = \sqrt{\frac{2^2 + 3^2 + 4^2 + 5^2}{4}} = 3.67 \text{ cm/s.}$$

Ex.30 The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of the gas : [JEE 2000]

(A) $T(H_2) = T(N_2)$

(B) $T(H_2) > T(N_2)$

(C) $T(H_2) < T(N_2)$

(D) $T(H_2) = \sqrt{7} T(N_2)$

Ans. (C)

Sol. $(u_{rms})_{H_2} = \sqrt{7} \times (u_{rms})_{N_2}$

$$\sqrt{\frac{3RT_1}{2}} = \sqrt{7} \times \sqrt{\frac{3RT_2}{28}}$$

$$\frac{3RT_1}{2} = \frac{7 \times 3RT_2}{28}$$

$$2T_1 = T_2 \quad \Rightarrow \quad T_{N_2} > T_{H_2}$$

Ex.31 At what temperature do the average speed of CH_4 molecule equal the average speed of O_2 molecule at 300 K ?

Sol. $(U_{\text{avg}})_{\text{CH}_4} = (U_{\text{avg}})_{\text{O}_2}$

$$\sqrt{\frac{8RT}{\pi \times 16}} = \sqrt{\frac{8 \times R \times 300}{\pi \times 32}}$$

$$T = 150 \text{ K}$$

Ex.32 At 27°C, find the ratio of root mean square speeds of ozone to oxygen.

Sol.
$$\frac{U_{\text{rms}}(\text{O}_3)}{U_{\text{rms}}(\text{O}_2)} = \sqrt{\frac{\frac{3RT}{M.W_{\text{O}_3}}}{\frac{3RT}{M.W_{\text{O}_2}}}} = \sqrt{\frac{M.W_{\text{O}_2}}{M.W_{\text{O}_3}}} = \sqrt{\frac{32}{48}} = \sqrt{\frac{2}{3}}$$

Ex.33 The temperature at which U_{rms} of He becomes equal to U_{mp} of CH_4 at 500 K.

Sol. $(U_{\text{rms}})_{\text{He}} = (U_{\text{mp}})_{\text{CH}_4}$

$$\sqrt{\frac{3RT_{\text{He}}}{M_{\text{He}}}} = \sqrt{\frac{2RT_{\text{CH}_4}}{M_{\text{CH}_4}}}$$

$$\frac{3T}{4} = \frac{2 \times 500}{16}$$

$$T = \frac{250}{3} \text{ K}$$

Ex.34 Calculate the root mean square speed of H_2 molecules under following condition.

(a) 2 mole of H_2 at 27°C.

(b) 3 mole of H_2 in a 5 lit container at 10^5 Pa .

(c) 4 mole of H_2 at the density of 1 gm/ml at 10^5 Pa .

Sol. (a) $U_{\text{r.m.s.}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \text{ J/molK} \times 300 \text{ K}}{2 \times 10^{-3} \text{ kg}}} = 1934.24 \text{ m/sec.}$

(b) $U_{\text{r.m.s.}} = \sqrt{\frac{3PV}{nM}} = \sqrt{\frac{3 \times 10^5 \text{ Pa} \times 5 \times 10^{-3} \text{ m}^3}{3 \text{ mol} \times 2 \times 10^{-3} \text{ kg}}} = 500 \text{ m/sec.}$

(c) $U_{\text{r.m.s.}} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3 \times 10^5 \text{ Pa}}{10^3 \text{ kg/m}^3}} = 17.32 \text{ m/sec.}$

Ex.35 Calculate the total translational kinetic energy of 8 gram methane (CH_4) at 27°C temperature.

Sol. $n = \frac{8}{16} = \frac{1}{2}$, $T = (27^\circ + 273) = 300 \text{ K}$, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

$$(\text{K.E.})_{\text{n mol}} = n \times \frac{3}{2} RT = \frac{1}{2} \times \frac{3}{2} \times 8.314 \times 300 = 1870.65 \text{ J}$$

Ex.36 Calculate the pressure exerted by 10^{23} gas molecules, each of mass 10^{-25} kg, in a container of volume $1 \times 10^{-3} \text{ m}^3$ and having root mean square velocity of 10^3 ms^{-1} . Also calculate total kinetic energy and Temperature of the gas.

Sol. By kinetic theory

$$P = \frac{1}{3} \frac{mNu^2}{V} = \frac{1 \times 10^{-25} \times 10^{23} \times (10^3)^2}{3 \times 10^{-3}} = 3.33 \times 10^6 \text{ N m}^{-2}$$

$$\text{Total KE} = \left(\frac{1}{2} mu_{\text{rms}}^2 \right) \times N = \frac{1}{2} \times 10^{-25} \times (10^3)^2 \times 10^{23} = \frac{1}{2} \times 10^4 = 0.5 \times 10^4 \text{ J}$$

$$\text{Also total KE} = \frac{3}{2} nRT, \text{ where } n (\text{mole}) = \frac{10^{23}}{N_A} = \frac{10^{23}}{6.023 \times 10^{23}}$$

$$0.5 \times 10^4 = \frac{3}{2} \times \frac{10^{23}}{6.023 \times 10^{23}} \times 8.314 \times T$$

$$\therefore T = \frac{0.5 \times 10^4 \times 2 \times 6.023}{3 \times 8.314} = 2415 \text{ K}$$

Ex.37 At what temperature will be total kinetic energy (KE) of 0.30 mole of He be the same as the total KE of 0.40 mole of Ar at 400 K :

- (A) 400 K (B) 373 K (C) 533 K (D) 300 K

Ans. (C)

Sol. Total K.E. = $\frac{3}{2} nRT$

$$\text{From question, } \frac{3}{2} \times 0.3 \times R \times T = \frac{3}{2} \times 0.4 \times R \times 400$$

$$\therefore T = 533.33 \text{ K.}$$

10. MAXWELL'S DISTRIBUTION OF SPEEDS

It has already been pointed out that a gas is a collection of tiny particles separated from one another by large empty spaces and moving rapidly at random in all directions. In the course of their motion, they collide with one another and also with the walls of the container. Due to frequent collisions, the speeds and directions of motion of the molecules keep on changing. Thus, all the molecules in a sample of gas do not have same speed. Although it is not possible to find out the speeds of individual molecule, yet from probability considerations it has become possible to work out the distribution of molecules in different speed intervals. This distribution is referred to as the Maxwell-Boltzmann distribution in honour of the scientists who developed it. It may be noted that the distribution of speeds remains constant at a particular temperature although individual speeds of molecules may change.

$$dN = 4\pi N \left[\frac{M}{2\pi RT} \right]^{\frac{3}{2}} e^{-\frac{Mu^2}{2RT}} u^2 du = 4\pi N \left[\frac{m}{2\pi kT} \right]^{\frac{3}{2}} e^{-\frac{mu^2}{2kT}} u^2 du$$

Here, dN = Number of molecules having speeds between u and $u + du$.

N = Total number of molecules.

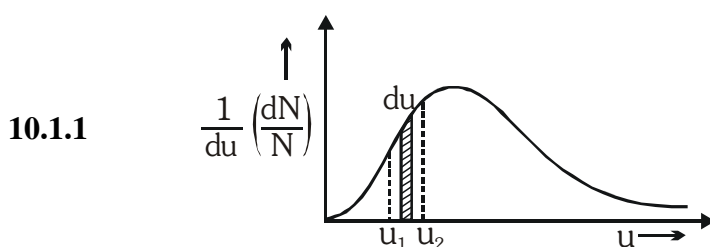
M = Molar mass of gas

dN/N = fraction of molecules having speeds between u and $u + du$.

and $\frac{1}{N} \left(\frac{dN}{du} \right) =$ fraction of molecules having speed between u to $u + du$ per unit interval of speed.
 = Maxwell distribution function = $f(u)$

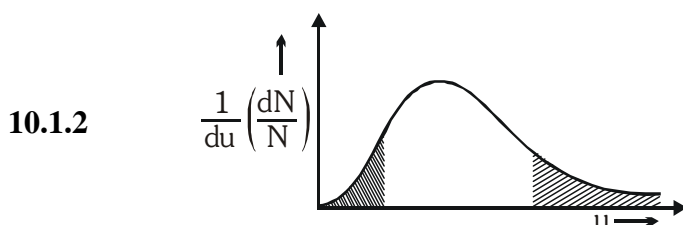
According to this expression, the fraction $\left(\frac{dN}{N} \right)$ of molecules depends only on temperature having speeds between u and $u + du$ for a gas of molar mass M . Thus for a given temperature, this fraction has a constant value.

10.1 Properties of Maxwell's graph :

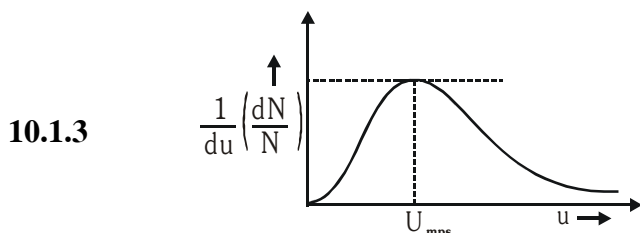


$$\text{Area between } u_1 \text{ and } u_2 = \int_{u_1}^{u_2} \frac{1}{du} \left(\frac{dN}{N} \right) du = \int_{u_1}^{u_2} \left(\frac{dN}{N} \right)$$

Hence, total fraction of particles with speed between u_1 and u_2
 = Area under the curve represents fraction of molecules.

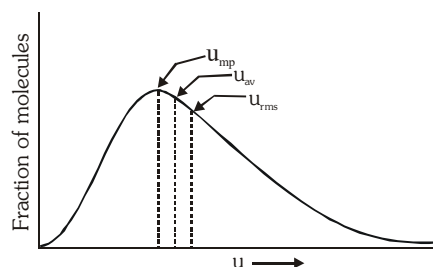


It can be seen from the above figure, that the fraction of molecules having either very low speeds or very high speeds are small in numbers.

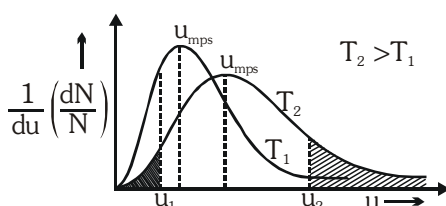


The curve at any temperature is parabolic near the origin, since the factor u^2 is dominant in this region, the exponential function being approximately equal to unity. At high values of u , however, the exponential factor dominates the behaviour of the function, causing it to decrease rapidly in value. As a consequence of the contrasting behaviour of two factors, the product function passes through a maximum at a speed known as the most probable speed (u_{mps}). Thus, the most probable speed is the speed possessed by the maximum fraction of the molecules.

10.1.4 Graph between fraction of molecules vs molecular speeds :



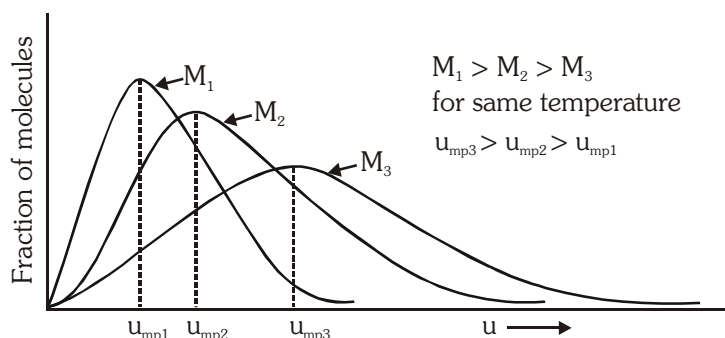
10.1.5



Total area under the curve will be constant and will be unity at all temperatures. The above figure illustrates the distribution of speeds at two temperatures T_1 and T_2 . Since the total no. of molecules is same at both temperatures, increase in the K.E. of the molecules results decrease in fraction of molecules having lower speed range and increase in fraction of molecules having higher speed range on increasing the temperature.

On increasing temperature the value of u_{mps} (most probable speed) will increase. Also the curve at the higher temperature T_2 has its u_{mps} shifted to a higher value compared with that for T_1 , whereas corresponding fraction of molecules has decreased. But at the same time, the curve near u_{mps} has become broader at the higher temperature indicating the more molecules possess speeds near to most probable speed.

10.1.6



At a given temperature u_{mps} will be more for lighter gas (M_3) but fraction of molecules moving with u_{mps} will be more for heavier gas (M_1).

Note : Effect of M and T are opposite.

11.4 Mean free path :

The mean free path is the average distance travelled by a molecule between two successive collisions. We can express it as follows :

$$\lambda = \frac{\text{distance travelled per unit time}}{\text{No. of collisions made by single molecule per unit time}}$$

$$= \frac{U_{\text{avg}}}{Z_1} = \frac{U_{\text{avg}}}{\sqrt{2}\pi\sigma^2 U_{\text{avg}} N^*} = \frac{1}{\sqrt{2}\pi\sigma^2 N^*}$$

11.5 Wall collision : It represents the total number of molecules colliding at the wall per unit area per unit time.

$$Z_w = \frac{1}{4} \cdot N^* \cdot u_{\text{av}} = \frac{P \cdot N_A}{\sqrt{2}\pi MRT}$$

11.6 Rate of effusion :

If the cross-section area of orifice in the vessel is 'A', then the number of molecules effusing out per unit time is

$$r_{\text{eff.}} = Z_w \cdot A = \frac{P \cdot N_A \cdot A}{\sqrt{2}\pi MRT}$$

Ex.38 Calculate λ , Z_1 and Z_{11} for oxygen at 298 K and 10^{-3} mm Hg. Given $\sigma = 3.61 \times 10^{-8}$ cm.

Sol. $N^* = \frac{P}{kT} = \frac{10^{-3} \times 101325}{760 \times 1.38 \times 10^{-23} \times 298} = 0.324 \times 10^{20}$

$$U_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8}{3.14} \times \frac{8.314 \times 298}{32 \times 10^{-3}}} = 444.138 \text{ m/sec}$$

$$Z_1 = \sqrt{2} \pi \sigma^2 U_{\text{avg}} N^* = \sqrt{2} \times 3.14 \times 3.61 \times 10^{-10} \text{ m} \times 444.138 \text{ m} \times 0.324 \times 10^{20}$$

$$= 8.326 \times 10^3 \text{ sec}^{-1}$$

$$Z_{11} = \frac{1}{2} Z_1 N^* = \frac{1}{2} \times 8.326 \times 10^3 \text{ sec}^{-1} \times 0.324 \times 10^{20} = 13.488 \times 10^{22} \text{ m}^{-3} \text{ sec}^{-1};$$

$$\lambda = \frac{U_{\text{avg.}}}{Z_1} = \frac{444.138 \text{ m/sec}}{8.326 \times 10^3 \text{ sec}^{-1}} = 5.334 \times 10^{-2} \text{ m}$$

Ex.39 Two flask A and B have equal volume. A is maintained at 300 K and B at 600 K while A contains H_2 gas, B has an equal mass of CH_4 gas. Assuming ideal behaviour for both the gases, find the following.

- Flask containing greater number of moles
- Flask in which pressure is greater
- Flask in which U_{avg} of the molecules are greater
- Flask with greater mean free path of molecules
(Collision diameters of H_2 & CH_4 may be taken same)
- Flask with greater molar kinetic energy.
- Flask in which the total kinetic energy is greater
- Flask in which Z_1 and Z_{11} are greater

Sol.

A H ₂	B CH ₄
300K	600K

- (a) $N_{H_2} = \frac{m}{2} N_A$; $N_{CH_4} = \frac{m}{16} N_A$
 $\therefore$ molecules of H₂ in flask A > molecules of CH₄ in flask B
- (b) $P_A V = n_A R T_A$; $P_B V = n_B R T_B$
 $\Rightarrow \frac{P_A}{P_B} = \frac{n_A T_A}{n_B T_B} = \frac{m/2}{m/16} \times \frac{300}{600} = 4$
 $\therefore$ pressure of H₂ in flask A > pressure of CH₄ in flask B
- (c) $(U_{avg})_A = \sqrt{\frac{8 R T_A}{\pi M_A}}$; $(U_{avg})_B = \sqrt{\frac{8 R T_B}{\pi M_B}}$
 $\frac{(U_{avg})_A}{(U_{avg})_B} = \sqrt{\frac{T_A}{T_B} \times \frac{M_B}{M_A}} = \sqrt{\frac{300}{600} \times \frac{16}{2}} = 2$
 $\therefore U_{avg}$ of H₂ in flask A > U_{avg} of CH₄ in flask B
- (d) $\lambda = \frac{1}{\sqrt{2\pi\sigma^2 N^*}} = \frac{1}{\sqrt{2\pi\sigma^2 \times P}} \times kT$ [where $N^* = P/kT$]
 $\frac{\lambda_A}{\lambda_B} = \frac{T_A}{T_B} \times \frac{P_B}{P_A} = \frac{300}{600} \times \frac{1}{4} = \frac{1}{8}$
 $\therefore \lambda_B > \lambda_A$
- (e) molar K.E. = $\frac{3}{2} RT$
 $\therefore T_B > T_A$
 $\therefore$ KE of CH₄ in flask B > KE of H₂ in flask A
- (f) $(KE)_{total} = \frac{3}{2} RT \times n$
 $\frac{(KE)_{T,A}}{(KE)_{T,B}} = \frac{(300 \times \frac{m}{2})}{(600 \times \frac{m}{16})} = \frac{1}{2} \times 8 = 4$
 $\therefore (KE)_{T,A} > (KE)_{T,B}$
- (g) (i) $Z_1 = \sqrt{2\pi\sigma^2} U_{avg} N^*$
 $\frac{(Z_1)_A}{(Z_1)_B} = \frac{(U_{avg})_A}{(U_{avg})_B} \times \frac{N_A^*}{N_B^*} = 2 \times \frac{N_A}{N_B} = 2 \times 8 = 16$
 $\therefore (Z_1)_A > (Z_1)_B$
- (ii) $Z_{11} = \frac{1}{\sqrt{2}} \pi \sigma^2 U_{avg} N^{*2}$
 $\frac{(Z_{11})_A}{(Z_{11})_B} = \frac{(U_{avg})_A}{(U_{avg})_B} \times \frac{(N_A^*)^2}{(N_B^*)^2} = 2 \times \frac{N_A}{N_B} = 2 \times 8^2 = 128$
 $\therefore (Z_{11})_A > (Z_{11})_B$

SELF STUDY

12. PRESSURE MEASURING DEVICES :

Generally, the instruments used for the measurement of pressure of a gas are **barometer** and **manometer**, which measures the pressure in terms of height of mercury column.

$$\begin{aligned} \text{Pressure}(P) &= \text{Force}(F) / \text{Area}(A) \\ &= \text{Mass}(m) \times \text{Acceleration}(g) / \text{Area}(a) \\ &= \text{Volume} \times \text{density} \times \text{Acceleration}(g) / \text{Area}(a) \\ &= \text{Area}(a) \times \text{height}(h) \times \text{density}(\rho) \times \text{Acceleration}(g) / \text{Area}(a) \end{aligned}$$

$$\text{Pressure}(P) = h\rho g$$

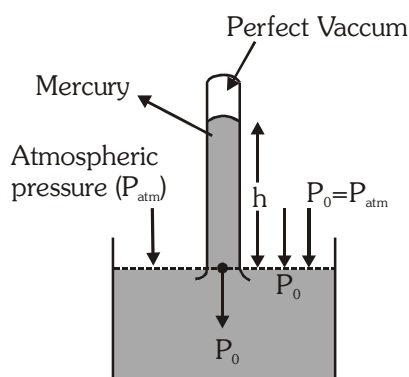
where h = Height of mercury column in the barometer.

ρ = Density of mercury.

g = Acceleration due to gravity.

Pressure does not depend on the cross section of tube, but only on the vertical height of the Hg. If area is doubled, volume also gets doubled and mass will also get doubled. Now it will rest on twice area but pressure exerted remains same.

- (i) **Barometer** : A barometer is an instrument that is used for the measurement of atmospheric pressure. The construction of the barometer is as follows -



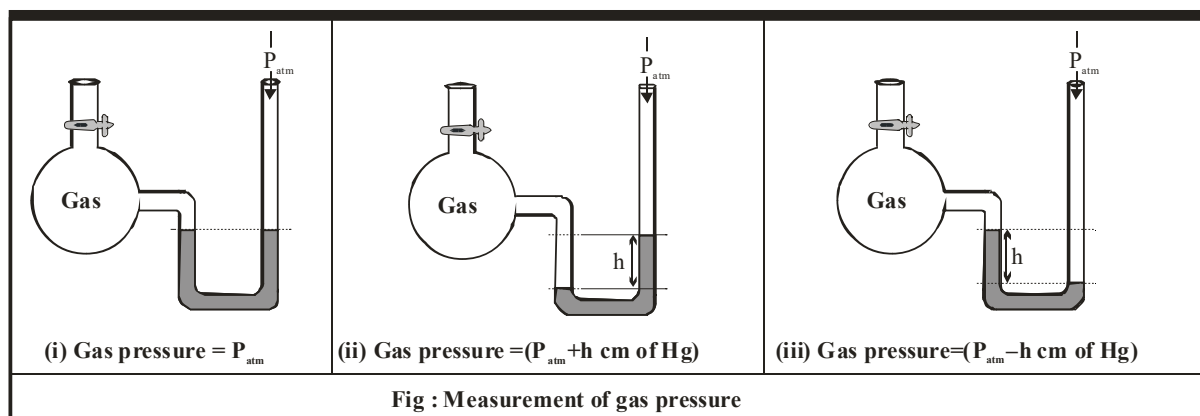
A thin narrow calibrated capillary tube is filled up to the brim, with a liquid such as mercury, and is inverted into a trough filled with the same fluid. Now depending on the external atmospheric pressure, the level of the mercury inside the tube will adjust itself, the reading of which can be monitored. When the mercury column inside the capillary comes to rest, then the net forces on the column should be balanced.

$$\Rightarrow P_0 \times A = \rho A \times gh$$

$$\Rightarrow \boxed{P_0 = \rho gh} \quad ; \text{ where } \rho \text{ is the density of the fluid.}$$

(ii) Manometer :**(a) Open end manometer :**

It consists of a U-shaped tube partially filled with mercury. One limb of the tube is shorter than the other. The shorter limb is connected to the vessel containing the gas whereas the longer limb is open as shown in fig. The mercury in the longer tube is subjected to the atmospheric pressure while mercury in the shorter tube is subjected to the pressure of the gas.



Where $P_{\text{atm}} = 76 \text{ cm of Hg}$ and $h = \text{Height in cm of Hg}$

There are three possibilities as described below :

(i) If the level of Hg in the two limbs is same, then gas pressure = atmospheric pressure (P_{atm}).

(ii) If the level of Hg in the longer limbs is higher, gas pressure

$$= P_{\text{atm}} + (\text{difference between the two levels})$$

$$= P_{\text{atm}} + h \text{ cm of Hg}$$

(iii) If the level of Hg in the shorter limb is higher, then gas pressure

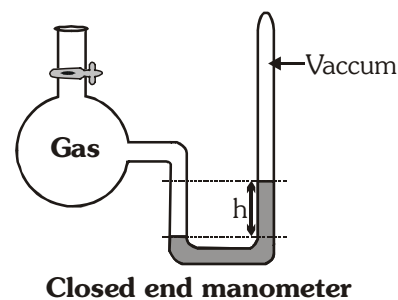
$$= P_{\text{atm}} - (\text{difference between the two levels})$$

$$= P_{\text{atm}} - h \text{ cm of Hg}$$

(b) Closed end manometer :

This is generally used to measure low gas pressure.

It also consists of U-tube with one limb shorter than the other and partially filled with mercury as shown in fig. The space above mercury on the closed end is completely evacuated. The shorter limb is connected to the vessel containing gas. The gas exerts pressure on the mercury in the shorter limb and forces its level down.



$$\text{Gas pressure} = [\text{Difference in the Hg level in two limbs}]$$

Ex.40 Why mercury is used in the barometer tube?

Sol. Mercury, a liquid with very high density, is normally used in the barometer because it does not stick to the surface of the glass tube. Mercury is also non-volatile at room temperature. Therefore, there are hardly any vapours of mercury above the liquid column and their pressure, if any, can be neglected. Due to high density of mercury, height of mercury column will be small and can be easily measured.

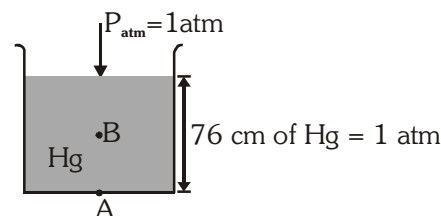
Ex.41 An open tank is filled with Hg upto a height of 76cm.

Find the pressure at the

(a) Bottom (A) of the tank

(b) Middle (B) of the tank.

(If atmospheric pressure is 1 atm)



Sol. (a) At bottom,

$$P_A = P_{\text{atm}} + P_{\text{Hg}} \\ = 1 + 1 = 2 \text{ atm}$$

(b) At middle,

$$P_B = P_{\text{atm}} + P_{\text{Hg}} \\ = 1 + \frac{1}{2} = 1.5 \text{ atm}$$

Ex.42 Find the height of water upto which water must be filled to create the same pressure at the bottom, as in above problem.

Given that $d_w = 1 \text{ gm/cm}^3$, $d_{\text{Hg}} = 13.6 \text{ gm/cm}^3$, $h_{\text{Hg}} = 76 \text{ cm}$

Sol. $P_{\text{water}} = P_{\text{Hg}}$

$$h_w d_w g = h_{\text{Hg}} d_{\text{Hg}} g$$

$$h_w d_w = h_{\text{Hg}} d_{\text{Hg}}$$

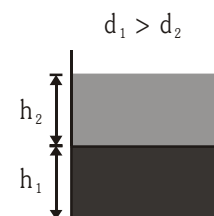
$$h_w \times 1 \text{ g/cm}^3 = 76 \text{ cm} \times 13.6 \text{ g/cm}^3$$

$$h_w = 1033.6 \text{ cm} = 10.336 \text{ m.}$$

Ex.43 What will be the pressure if two immiscible fluid is filled according to given diagram.

(a) Find the pressure at the bottom of tank.

(b) Find the pressure at the middle point of bottom layer.

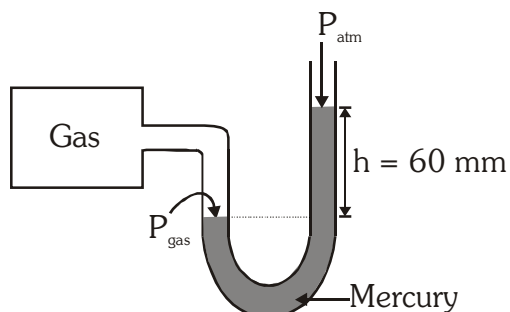


Sol. (a) $P_{\text{atm}} + h_2 d_2 g + h_1 d_1 g$;

(b) $P_{\text{atm}} + h_2 d_2 g + \frac{h_1}{2} d_1 g$

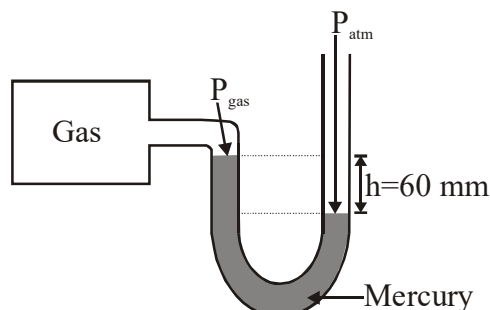
Ex.44 Find the pressure of the gas inside a container if the open manometer attached to the container shows a difference of 60 mm.

Sol. Case-I : $P_{\text{atm}} < P_{\text{gas}}$



$$\begin{aligned} P_{\text{gas}} &= P_{\text{atm}} + 60 \text{ mm} \\ &= 760 \text{ mm} + 60 \text{ mm} \\ &= 820 \text{ mm of Hg} \end{aligned}$$

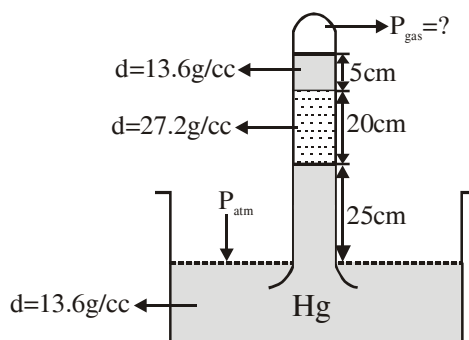
Case-II : $P_{\text{atm}} > P_{\text{gas}}$



$$\begin{aligned} P_{\text{atm}} &= P_{\text{gas}} + 60 \text{ mm} \\ 760 \text{ mm} &= P_{\text{gas}} + 60 \\ P_{\text{gas}} &= 700 \text{ mm} \end{aligned}$$

Ex.45 In the following arrangement find the pressure of gas (in cm of Hg).

(Assume that atmospheric pressure $P_{\text{atm}} = 75 \text{ cm of Hg}$)



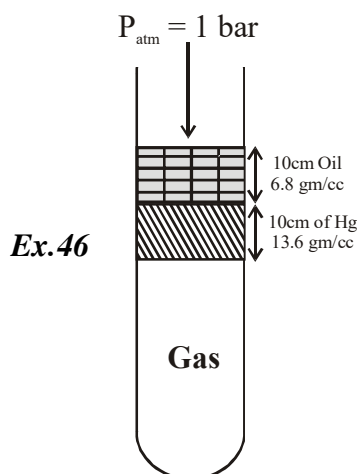
Ans.(5)

Sol. $P_{\text{atm}} = P_{\text{gas}} + P_1 + P_2 + P_3$

or $75 \text{ cm Hg} = P_{\text{gas}} + 5 \text{ cm of liquid (1)} + 20 \text{ cm of liquid (2)} + 25 \text{ cm of Hg}$

$$= P_{\text{gas}} + \frac{5 \times 13.6}{13.6} \text{ cm Hg} + \frac{20 \times 27.2}{13.6} \text{ cm Hg} + 25 \text{ cm Hg}$$

$$\therefore P_{\text{gas}} = 5 \text{ cm Hg}$$



Find pressure of gas (in cm of Hg) enclosed in tube.

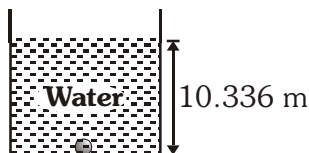
Ans. (90)

Sol. $P_{\text{gas}} = 1 \text{ bar} + 10 \text{ cm oil} + 10 \text{ cm Hg}$

$$= 75 \text{ cm Hg} + \frac{10 \times 6.8}{13.6} \text{ cm Hg} + 10 \text{ cm Hg}$$

$$= 90 \text{ cm Hg.}$$

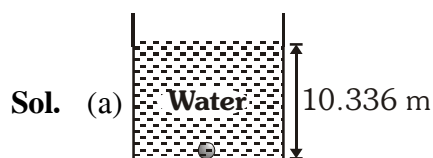
Ex.47.(a) Radius of a bubble at the bottom of the tank shown below was found to be 1 cm, then find the radius (in cm) of the bubble at the surface of water considering the temperature at the surface & bottom being same. $[(1.26)^3 = 2.0, (1.14)^3 = 1.5]$



(b) If absolute temperature at the surface is 4 times that at the bottom, then find radius (in cm) of bubble at the surface.

IG0009

Ans. a. (1.26) b. (2.00)



Pressure of 10.336 m water = 1 atm

R_B = Radius of bubble at bottom = 1 cm

R_S = Radius of bubble at surface

Apply $PV = nRT$ at bottom

$$P_B V_B = nRT$$

$$2 \times \frac{4}{3} \pi (R_B)^3 = nRT \quad \dots\dots\dots (1)$$

Apply $PV = nRT$ at surface $P_S \cdot V_S = nRT$

$$1 \times \frac{4}{3} \pi (r_s)^3 = nRT \quad \dots\dots\dots (2)$$

Diving (1) / (2)

$$\Rightarrow \frac{2}{7} \times \left(\frac{R_B}{R_S} \right)^3 = 1$$

$$\Rightarrow 2 = \left(\frac{R_S}{R_B} \right)^3$$

$$\therefore R_B = 1 \text{ cm}$$

$$R_S = 2^{1/3} \text{ cm}$$

$$(b) T_S = 4T_B$$

T_S = temperature at surface

T_B = temperature at bottom

Apply $PV = nRT$ at bottom

$$P_B \times V_B = nRT_B \quad \dots\dots\dots (1)$$

$$2 \times \frac{4}{3} \pi (r_B)^3 = nRT_B$$

Apply $PV = nRT$ at surface

$$1 \times \frac{4}{3} \pi (r_S)^3 = nRT_S \quad \dots\dots\dots (2)$$

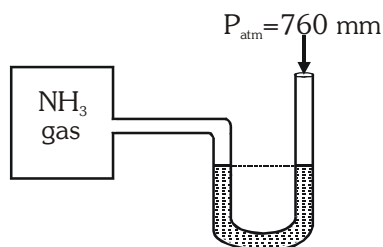
On dividing equation 2 by 1

$$\frac{2 \times \frac{4}{3} \pi (r_B)^3}{1 \times \frac{4}{3} \pi (r_S)^3} = \frac{nRT_B}{nR(4T_B)}$$

$$2 \times \left(\frac{r_B}{r_S} \right)^3 = \frac{1}{4}$$

$$r_S = 2r_B = 2 \text{ cm}$$

Ex.48. A manometer attached to a flask contains NH_3 gas have no difference in mercury level initially as shown in diagram. After the sparking into the flask, it have difference of 19 cm in mercury level in two columns. Calculate % dissociation of ammonia.



IG0026

Ans. (25.00)



$$t = 0 \quad 76 \text{ cm} \quad 0 \quad 0$$

$$t = t_{\text{eq}} \quad 76-x \quad \frac{x}{2} \quad \frac{3x}{2}$$

$$\text{Total pressure of system} = (76 - x) + \frac{x}{2} + \frac{3x}{2} = (76 + x) \text{ cm of Hg}$$

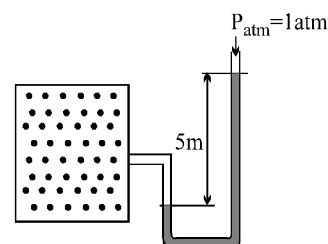
$$76 + x = 95 \quad [P_{\text{total}} = 76 + 19 = 95]$$

$$x = 19 \text{ cm of Hg}$$

$$\% \text{ dissociation of } \text{NH}_3 = \frac{19}{76} \times 100 = 25\%$$

Ex.49 Calculate the number of moles of gas present in the container of volume 10 lit at 300 K. If the manometer containing glycerin shows 5m difference in level as shown in diagram.

Given: $d_{\text{glycerin}} = 2.72 \text{ gm/ml}$, $d_{\text{mercury}} = 13.6 \text{ gm/ml}$.



IG0046

Sol. Height of glycerin column = 5m = 500 cm

For glycerin

$$(\text{h}\rho\text{g})_{\text{glycerin}} = (\text{h}\rho\text{g})_{\text{Hg}}$$

$$500 \times 2.72 \times g = h \times 13.6 \times g$$

$$h_{Hg} = 100 \text{ cm}$$

Hence $P_{\text{gas}} = P_{\text{atm}} + P_{\text{glycerin}}$

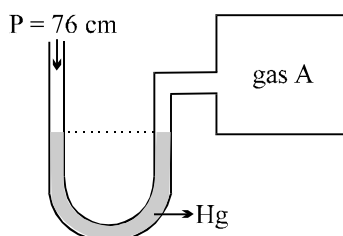
$$(\text{Initial}) = 76 + 100 = 176 \text{ cm of Hg}$$

$$PV = nRT$$

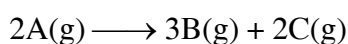
$$\left(\frac{176}{76}\right) \times 10 = n \times 0.082 \times 300$$

Moles of gas (n) = 0.94 mole

Ex.50 An open ended mercury manometer is used to measure the pressure exerted by a trapped gas as shown in the figure. Initially manometer shows no difference in mercury level in both columns as shown in diagram.



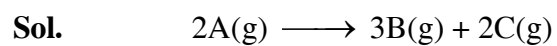
After sparking 'A' dissociates according to following reaction



If pressure of Gas "A" decreases to 0.8 atm. Then (Assume temperature to be constant and is 300 K)

- (A) total pressure increased by 1.3 atm
(B) total pressure increased by 0.3 atm
(C) total pressure increased by 22.3 cm of Hg
(D) difference in mercury level is 228 mm.

Ans. (B,D)



$$t = 0 \quad 1 \text{ atm} \quad 0 \quad 0$$

$$t \quad 0.8 \text{ atm} \quad 0.3 \text{ atm} \quad 0.2 \text{ atm}$$

$$P_{\text{total}} = P_A + P_B + P_C$$

$$0.8 + 0.3 + 0.2 = 1.3 \text{ atm.}$$

* Total pressure is increased by 0.3 atm or 22.8 cm of Hg or 228 mm of Hg.

SOLVED EXAMPLES

1. An evacuated glass vessel weighs 50.0 g when empty, 148.0 gm when filled with a liquid of density 0.98 g/mL and 50.5 g when filled with an ideal gas at 760 mm Hg at 300 K. Determine the molecular weight of the gas. [JEE 1998]

Ans. 123

Sol. Weight of liquid = 148 – 50 = 98 g

$$\text{Volume of liquid} = \frac{98}{0.98} = 100 \text{ mL} = \text{volume of vessel}$$

Thus, vessel of 100 ml contains ideal gas at 760 mm of Hg at 300K.

$$\text{Weight of gas} = 50.5 - 50 = 0.5 \text{ g}$$

Using $PV = nRT$

$$\frac{760}{760} \times \frac{100}{1000} \times \frac{0.5}{m} = 0.0821 \times 300$$

$$\therefore \text{Molecular weight of gas (m)} = 123$$

2. The pressure exerted by 12 g of an ideal gas at temperature $t^\circ\text{C}$ in a vessel of volume V is one atmp. When the temperature is increased by 10 degrees at the same volume, the pressure increases by 10 %. Calculate the temperature ' t ' and volume ' V '. [Molecular weight of gas = 120] [JEE 1999]

Ans. -173°C , 0.82 L

Sol. $P_1 = 1 = 1 \text{ atm}$, $T_1 = t + 273$

$$P_2 = 1 + \frac{10}{100} = 1.1 \text{ atm}, T_2 = t + 283$$

$$\frac{1 \times V}{1.1 \times V} = \frac{\frac{12}{M} \times R(t + 273)}{\frac{12}{M} \times R(t + 283)}$$

$$t = -173^\circ\text{C} = 100\text{K}$$

$$1 \times V = \frac{12}{120} \times 0.082 \times 100$$

$$V = 0.82 \text{ L}$$

3. One mole of N_2 gas at 0.8 atm takes 38 sec to diffuse through a pin hole, whereas one mole of an unknown compound of Xenon with F at 1.6 atm takes 57 sec to diffuse through the same hole. Calculate the molecular formula of the compound. (At. wt. Xe = 138, F = 19) [JEE 1999]

Ans. XeF_6

Sol. $\frac{r_1}{r_2} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$

$$\frac{n_1}{t_1} \times \frac{t_2}{n_2} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$$

$$\frac{1}{38} \times \frac{57}{1} = \frac{0.8}{1.6} \sqrt{\frac{M}{28}}$$

$$M = 252$$

$$138 + 19 \times n = 252 \Rightarrow n = 6 \Rightarrow \text{XeF}_6$$

4. The root mean square velocity of an ideal gas at constant pressure varies with density as

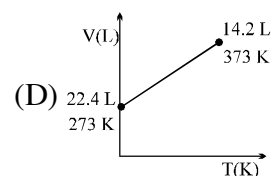
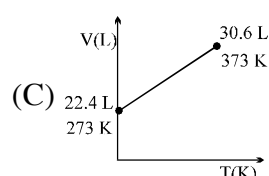
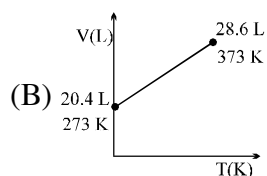
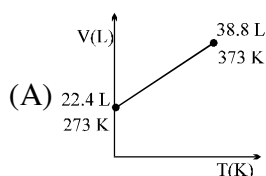
- (A) d^2 (B) d (C) $d^{1/2}$ (D) $1/d^{1/2}$ [JEE 2001]

Ans. (D)

Sol. $u_{\text{rms}} = \sqrt{\frac{3P}{d}}$

$$u_{\text{rms}} \propto \frac{1}{d^{1/2}}$$

5. Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atmp? [JEE 2002]



Ans. (C)

Sol. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$$\frac{22.4}{273} = \frac{30.6}{373}$$

6. The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature. [JEE 2003]

Ans. 434.17 m/sec

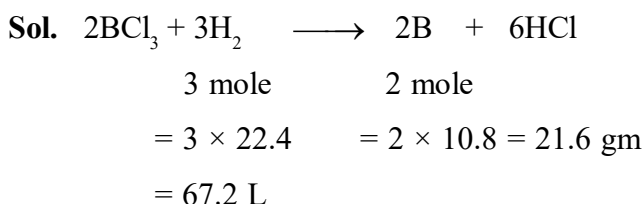
Sol. $u_{\text{rms}} = \sqrt{\frac{3\pi}{8}} \times u_{\text{avg}}$

$$= \sqrt{\frac{3 \times 3.14}{8}} \times 400 = 434 \text{ m/s}$$

7. What volume of hydrogen gas, at 273K and 1 atm pressure will be consumed in obtaining 21.6g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ? [AIEEE-2003]

(A) 44.8 L (B) 22.4 L (C) 89.6 L (D) 67.2 L

Ans. (D)



8. The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be [JEE 2005]
- (A) 4 (B) 2 (C) 1 (D) 0.5

Ans. (B)

Sol. $\frac{r_{\text{He}}}{r_{\text{CH}_4}} = \sqrt{\frac{16}{4}} = 2$

9. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is [JEE 2009]

Ans. (4)

Sol. $(u_{\text{rms}})_x = (u_{\text{mp}})_y$

$$\frac{\sqrt{3RT_x}}{M_x} = \frac{\sqrt{2RT_y}}{M_y}$$

$$\frac{3 \times 400}{40} = \frac{2 \times 60}{M_y}$$

$$M_y = \frac{2 \times 60 \times 40}{3 \times 400} = 4$$

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

EXERCISE # S-I

BOYLE'S LAW

1. Certain mass of a gas occupy 400 ml at 1.5 atm and 27°C. Calculate the volume (in ml) occupied by same mass of the gas at 1.2 atm and 27°C.

IG0170

2. 1.80 dm³ of a gas is expanded upto 9.0 dm³ at constant temperature. If the final pressure of gas becomes 20 kPa, the initial pressure of gas (in bar) was

IG0171

3. The pressure of certain mass of a gas is increased by 60% at constant temperature. The percentage decrease in the volume of gas is

IG0172

4. Two glass bulbs A and B are connected by a very small tube having a stop cock. Bulb A has a volume of 100 ml and contained some ideal gas, while bulb B was empty. On opening the stop cock, the pressure fell down to 40% at constant temperature. Find out the volume of bulb B (in mL).

IG0004

CHARLES LAW

5. Calculate the volume (in L) which 4 litre of gas at 0°C will occupied at 100°C and the same pressure.

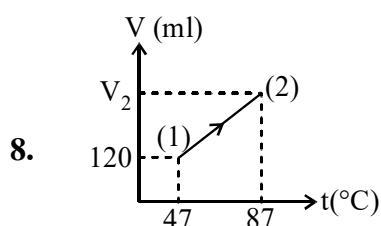
IG0006

6. A certain amount of a gas at 27°C and 1 bar pressure occupies a volume of 25 m³. If the pressure is kept constant and the temperature is raised to 87°C, what will be the volume (in m³) of the gas ?

IG0008

7. A balloon is filled upto $\frac{3}{4}$ th of its maximum stretching capacity at 27°C. The maximum temperature (in °C) upto which the balloon may be heated safely at constant pressure, is.

IG0173



The value of V_2 is.

IG0174

PRESSURE LAW

9. In a rigid closed vessel, the pressure exerted by gas is 57 cm Hg at 40°C. If it is heated to 80°C, the final pressure of gas (in cm Hg) becomes.
10. The temperature (in °C) of certain mass of a gas is increased by 10% at constant volume. If the initial temperature was 100°C, the percentage increase in the pressure of gas is %.

IG0175

IG0176

AVOGADRO'S LAW

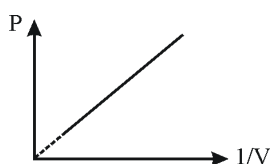
11. 10 gm of a gas occupy 123 cm³. Calculate the volume (in cm³) occupied by 2 gm of the same gas at the same temperature and pressure.
12. 1.20 moles of CH₄ gas occupy 38.94 dm³ at certain pressure and temperature. How many moles of SO₃ gas will occupy 19.47 dm³ at the same pressure and temperature.

IG0177

IG0178

IDEAL GAS EQUATION

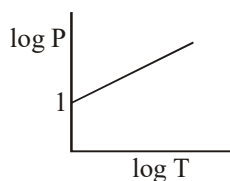
13. Consider the following graph



Graph is plotted for 1 mol of gas at 400K, find slope of curve (in L-atm). [Take : $R = 0.08 \frac{\text{L-atm}}{\text{mol-K}}$]

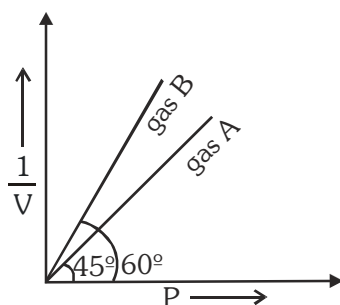
IG0003

14. In constant volume, container of 0.82 litre, log P vs log T is plotted as shown in graph. Calculate number of moles of ideal gas present in container :



IG0007

15. At constant temperature of 273 K, $\left(\frac{1}{V}\right)$ v/s P are plotted for two ideal gases A and B as shown.
(V in litre & P in atm)

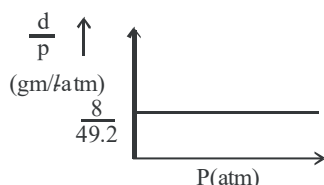


Find the ratio of number of moles of A and B gases.

IG0011

16. For an ideal gas, the following graph is obtained at constant temperature of 300 K.

The molar mass of gas (in gm/mol) is -



IG0012

17. The density of phosphorus vapours at 327°C and 1 atm is 2.52 gm/lit. If molecular formula of phosphorus is P_x , then calculate 'x'. (Atomic weight of : $P = 31$)

IG0013

18. Density of ideal gas at 2.46 atm and 300 K is 0.8 gm/L. Hence molar mass of gas (in gm mol⁻¹) is.

IG0014

19. In a hospital respiratory unit, the pressure gauge reads 4500 mm Hg for 10 L tank containing compressed O_2 . How many litres of O_2 can be delivered from this tank at atmospheric pressure of 750 mm Hg.

(Take temperature to be constant at TK)

IG0015

20. A human adult breathes in approximately 0.50 dm³ of air at 1.00 bar with each breath. If an air tank holds 100 dm³ of air at 200 bar, how many breathes (in order of 10⁴) the tank will supply ?

IG0017

21. A gas at a pressure of 5.0 bar is heated from 0°C to 546°C and simultaneously compressed to one third of its original volume. What will be the final pressure (in bar) ?

IG0018

22. 3.6 g of an ideal gas was injected into a bulb of internal volume of 8 L at pressure P atm and temp T K. The bulb was then placed in a thermostat maintained at $(T + 15)K$, 0.6 g of the gas was let off to keep the original pressure. The value of ' P ' is (Molar mass of gas = 41 gm mol⁻¹)

G0019

23. While resting, the average human male use 0.2 dm^3 of O_2 per hour at 1 atm & 300 K for each kg of body mass. Assume that all this O_2 is used to produce energy by oxidising glucose in the body . (i) What is the mass (in gm) of glucose required per hour by a resting male having mass 60 kg . (ii) What volume (in dm^3), at 1 atm & 300 K of CO_2 would be produced ($R = 0.08 \frac{\text{L-atm}}{\text{mol-K}}$)

IG0016

DALTON'S LAW OF PARTIAL PRESSURE

24. The partial pressures of N_2 , O_2 and CO_2 in a vessel are 38 cm of Hg, 190 torr and 0.5 atm, respectively . The total pressure of the mixture (in atm) at the same temperature is.

IG0020

25. Equal masses of ethane and hydrogen are mixed in an empty container at 25°C . The percentage of the total pressure exerted by hydrogen is

IG0021

26. A mixture of hydrogen and oxygen at one bar pressure contains 20% , by weight of hydrogen. The partial pressure (in bar) of hydrogen will be

IG0022

27. A closed vessel contains equal number of nitrogen and oxygen molecules at pressure of 38.96 mm. If nitrogen is removed from the system, then the pressure (in mm) will be

IG0023

28. If 2 lit. of gas A at 1.5 atm and 3 lit. of gas B at 2 atm are mixed in a 5 lit. container then find the final pressure (in atm), considering all are at same temperature.

IG0024

29. One mole of N_2 and 3 moles of H_2 are taken in a container of capacity 8.2 lit. at 300 K to produce NH_3 . Find the partial pressures (in atm) of N_2 , if partial pressure of NH_3 after sufficient time was found to be 3 atm.

IG0025

30. A gaseous mixture at 760 mm in a cylinder has 65% N_2 , 15% O_2 and 20 % CO_2 by volume. Calculate the partial pressure (in atm) of O_2 gas.

IG0027

31. The density of a mixture of O_2 and N_2 gases at 1 atm and 273K is 0.0013 gm/ml. the partial pressure (in atm) of O_2 in the mixture is

IG0029

32. A containers contains air above liquid water. Total pressure was 800 torr. What will be the final pressure (in torr) if volume is doubled. (Aqueous tension = 40 torr)

IG0030

33. One litre flask contains air, water vapour and a small amount of liquid water at a pressure of 200 mmHg. If this is connected to another one litre evacuated flask, what will be the final pressure (in mm Hg) of the gas mixture at equilibrium? Assume that temperature to be 50°C. Aqueous tension at 50°C = 93 mmHg.

IG0031

GRAHAM'S LAW OF DIFFUSION OR EFFUSION

34. If helium and methane are allowed to diffuse out of the container under the similar conditions of temperature and pressure, then the ratio of rate of diffusion of helium to methane is $x : 1.00$. The value of 'x' is

IG0032

35. 100 ml SO_2 gas diffuses in 5 minutes. Under identical conditions, 200 ml of an unknown gas takes 7.5 minutes in diffusion. The molar mass of unknown gas (in gm mol^{-1}) is.

IG0179

36. 4 gm HF gas effuses through a pin hole in a certain time. What mass (in gm) SO_3 gas will effuse through the same pin hole, under identical conditions ? ($F = 19$)

IG0180

37. A sample of pure Ne gas ($M = 20$) diffuses $\sqrt{2}$ times faster than a mixture of CH_4 and SO_2 gases, under identical conditions. The mass percent of CH_4 in the mixture of CH_4 and SO_2 gases is.

IG0181

38. Under identical conditions, gas 'X' diffuses 1.2 times faster than gas 'Y'. If the density of gas 'Y' is 7.2 gm/litre, the density of gas 'X' in the same conditions (in gm/litre) is.

IG0182

39. At a particular fixed temperature the gas A_n is 52% dissociated according to the following reaction

$$A_n(g) \rightleftharpoons nA(g)$$

The equilibrium mixture effuses 1.25 times slower than pure oxygen gas under identical condition. If atomic wt. of A is 32, then find "n" ?

IG0033

40. A mixture of H_2 and O_2 in 2 : 1 mole ratio is allowed to diffuse through a orifice. If the mole ratio of gases coming out initially is $r_{\text{H}_2} : r_{\text{O}_2} = x : 1$, the value of 'x' is

IG0035

41. H_2 and O_2 are kept in mass ratio 1 : 8 respectively at 6 atm. If small orifice is made, then moles of H_2 with respect to O_2 coming out initially, is

IG0034

42. A mixture of N_2 and H_2 has initially mass ratio of 196 : 1 then find after how many steps we can obtain a mixture containing 1 : 14 mole ratio of N_2 and H_2 .

IG0037

43. The ratio of moles of SO_2 to CH_4 after fifth diffusion steps is $x : 1.00$, if their initially mole ratio is $8 : 1$. The value of 'x' is. IG0038

KINETIC THEORY OF GASES

44. Calculate the average speed (in m/s) of molecules of an ideal gas -
(Molar mass = $\frac{166.28}{\pi}$ gm mol⁻¹) at 127°C. IG0183
45. Root mean square speed of an unknown gas at 727°C is 10⁵ cm/second. Calculate molar mass of unknown gas (in gram/mole) [Take $R = \frac{25}{3}$ J/mole-K]. IG0040
46. How many times is the rms speed of molecules in 8 gm O_2 gas at 1200 K and 10 bar, relative to rms speed of molecules in 16 gm O_2 gas at 300K and 20 bar ? IG0041
47. The most probable speed of molecules of an ideal gas is 300 m/s at 27°C. The molar mass of gas (in gm mol⁻¹) is ($R = \frac{25}{3}$ J/K.mol) IG0184
48. Calculate the most probable speed (in 10⁵ m/s) of electrons at 127°C, assuming them as isolated gaseous particles. (Mass of each electron = 9.1×10^{-31} kg, $N_A = 6 \times 10^{23}$) IG0185
49. Calculate the total translational kinetic energy (in J) of CH_4 gas molecules, present in its 0.8 gm sample, at 27°C) IG0186
50. Average translational kinetic energy of an ideal gas molecule at 27°C is 3.88×10^{-x} eV. Hence x is (1 eV = 1.602×10^{-19} J) IG0042
51. When the temperature of an ideal gas is increased from 27°C to 927°C, the kinetic energy will be increased by x times. Find the value of x ? IG0043
52. At what temperature (in °C), the graph between $\left(\frac{1}{du} \cdot \frac{dN}{N}\right)$ (Y-axis) and u (X-axis) will be maximum at 200 ms⁻¹ for a sample of ideal gas (molar mass = 41.57 gm mol⁻¹) ? IG0187
53. A sample of gas contains only 100 molecules in which 40 molecules are travelling at 10 m/s, 35 molecules at 20 m/s and rest, at 30 m/s. The ratio of u_{rms} to u_{av} of molecules is $x : 1$. The value of 'x' is
[$\sqrt{405} = 20.12$, $\sqrt{410} = 20.25$, $\sqrt{415} = 20.37$, $\sqrt{425} = 20.62$]. IG0188

COLLISION PARAMETERS

54. If the mean free path is 10 cm at one bar pressure, then its value (in cm) at 5 bar pressure, if temperature is kept constant is

IG0045

55. Two flask A and B have equal volume at 100 K and 200 K and have pressure 4 atm and 1 atm respectively. The flask A contains H_2 gas and B contains CH_4 gas. The collision diameter of CH_4 is twice that of H_2 . If ratio of mean free path of CH_4 to H_2 is 1 : x, the value of 'x' is

IG0044

56. The temperature of a sample of ideal gas is doubled at constant volume. The collision frequency (Z_{11}) of gas molecules becomes times.

IG0189

57. The temperature of a sample of ideal gas is doubled at constant pressure. The collision frequency (Z_{11}) of gas molecules becomes times.

IG0190

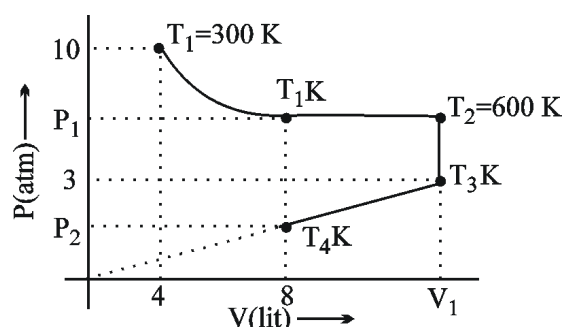
58. The temperature of a sample of ideal gas is increased at constant volume such that the most probable speed of molecules becomes four times. The collision number (Z_1) of gas molecules becomes times.

IG0191

EXERCISE # S-II

- The degree of dissociation is 0.4 at 400K and 1 atm for the gaseous reaction :
$$2\text{HI(g)} \rightarrow \text{H}_2\text{(g)} + \text{I}_2\text{(g)}.$$
Assume ideal behaviour of all the gases, calculate the density (in gm/litre) of mixture at 400K and 1 atm [At. wt. of Iodine = 127 ; Use R = 0.08 atm-litre/K-mole]

IG0047
- Fixed mass of a gas is subjected to the changes as shown in diagram, calculate T_3 , T_4 , P_1 , P_2 and V_1 as shown in diagram. Considering gas obeys $PV = nRT$ equation.



- IG0048**

3. A balloon containing 1 mole air at 1 atm initially is filled further with air till pressure increases to 3 atm. The initial diameter of the balloon is 1 m and the pressure at each state is proportion to diameter of the balloon. Calculate :

 - No. of moles of air added to change the pressure from 1 atm to 3 atm.
 - balloon will burst if either pressure increases to 7 atm or volume increases to $36\pi \text{ m}^3$. Calculate the number of moles of air that must be added after initial condition to burst the balloon.

IG0049

4. One mole of NH_4Cl (s) is kept in an open container & then covered with a lid. The container is now heated to 600 K where all NH_4Cl (s) dissociates into NH_3 & HCl (g). The Volume of the container is 24.6 litres Assume that outside air is at 300 K and 1 atm pressure.

 - What will be the final pressure (in atm) of gases inside the container.
 - Find whether the lid would stay or bounce off if it can with stand a pressure difference of 5.5 atm.

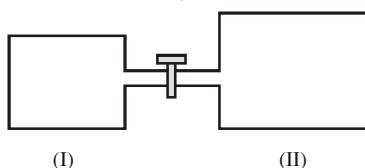
IG0050

5. One mole of an ideal gas is subjected to a process in which $P = \frac{1}{8.2} V$ where P is in atm & V in litre. If the process is operating from 1 atm to finally 3 atm (no higher pressure achieved during the process) then what would be the maximum temperature (in $^\circ\text{C}$) obtained.

6. A compound exists in the gaseous state both as a monomer (A) and dimer (A_2). The molecular weight of the monomer is 48. In an experiment, 96 g of the compound was confined in a vessel of volume 33.6 litres and heated to 273°C . Calculate the pressure (in atm) developed, if the compound exists as a dimer to the extent of 50 per cent by weight, under these conditions.

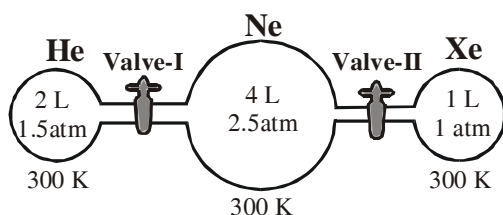
IG0052

7. Two container (I) & (II) with volume 2 litre & 3 litre are separated through a stopcock as shown in the diagram. Container (I) contains three gases A, B, & C with mole ratio 1 : 2 : 3 and total pressure 30 atm. If stopcock is opened then, calculate the pressure (in atm) of gas C in container (II). (Assume temperature remains constant)



IG0053

8. For the following diagram, calculate final pressure (in atm) when both valves are opened. Given that temperature (T) is same for each container.



IG0054

9. At 30°C dry air [75% N₂ + 25% O₂] is placed over H₂O (l) at 800 torr combined pressure of all 3 gases. If pressure is gradually increased isothermally to 1560 torr, then calculate partial pressure of O₂ at high pressure, in torr. [V.P._{H₂O} = 40 torr at 30°C].

IG0055

- 10.** 16 gm of O_2 was filled in a container of capacity 8.2 lit. at 300 K. Calculate
- (i) Pressure exerted by O_2 gas, in atm
 - (ii) Total pressure (in atm) of mixture of O_2 and O_3 , if 50 % of oxygen is converted into ozone at same temperature.
 - (iii) Total pressure (in atm) exerted by gases, if 50% of oxygen is converted into ozone (O_3) at temperature 50 K.

IG0056

11. To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to **[JEE 2011]**

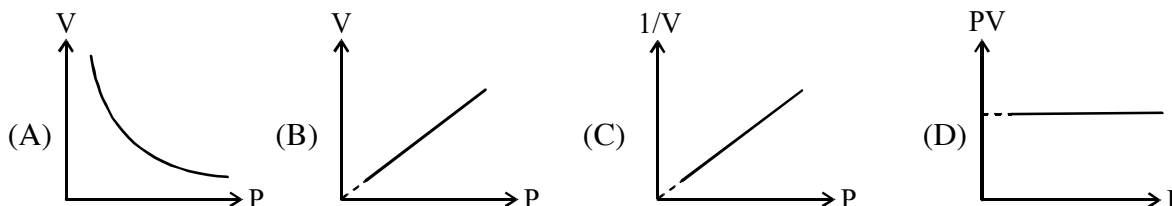
IG0161

EXERCISE # O-I

Single Correct :

BOYLE'S LAW

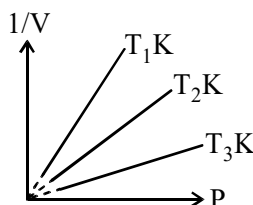
1. Certain mass of a gas occupy 222 cm^3 at 950 torr . The volume occupied by same mass of gas at same temperature is 333 cm^3 . The final pressure of gas is :
(A) 0.833 atm (B) 1425 torr (C) 1.875 atm (D) 633.33 cm Hg
2. Which of the following graph is incorrect according to Boyle's law ?



IG0192

IG0193

3. The following graph is obtained for a fixed mass of a gas, at different temperatures :



The correct relation between T_1 , T_2 and T_3 is :

- (A) $T_1 = T_2 = T_3$ (B) $T_1 < T_2 < T_3$ (C) $T_1 > T_2 > T_3$ (D) $T_1 < T_2 > T_3$

IG0194

4. Which of the following relation is correct for fixed moles of a gas at constant temperatures ?

- (A) $\frac{d(\log P)}{d(\log V)} = +ve$ (B) $\frac{dP}{dV} = \text{zero}$ (C) $\frac{d\left(\frac{1}{V}\right)}{dP} = -ve$ (D) $P_1 \cdot d_2 = P_2 \cdot d_1$
(d = density of gas)

IG0195

5. The volume of fixed mass of a gas is increased by 25%, at constant temperature. For it, the pressure of gas is :
(A) Increased by 25% (B) Decreased by 25%
(C) Increased by 20% (D) Decreased by 20%

IG0196

CHARLES LAW

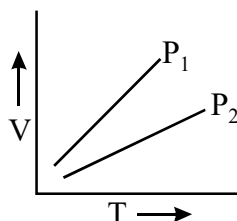
6. Certain mass of a gas occupy 0.4 m^3 at 27°C . At what temperature and same pressure, the gas will occupy 0.8 m^3 ?
(A) 54°C (B) 600°C (C) 327°C (D) -123°C

IG0197

7. 10 g of a gas at 1 atm, 273 K occupies 5 litres. The temp. at which the volume becomes double for the same mass of gas at the same pressure is ?
 (A) 273K (B) -273°C (C) 273°C (D) 546°C

IG0065

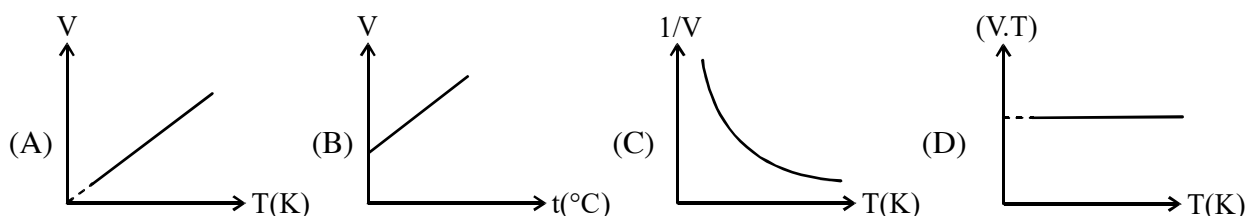
8. V versus T curves at constant pressure P_1 and P_2 for an ideal gas are shown in Fig. Which is correct



- (A) $P_1 > P_2$ (B) $P_1 < P_2$ (C) $P_1 = P_2$ (D) All

IG0067

9. Which of the following graph is incorrect for Charle's law ?



IG0198

10. The absolute temperature of a sample of gas is increased by 20% at constant pressure. The volume of gas will :
 (A) Increase by 20% (B) Decreased by 20%
 (C) Increase by 25% (D) Increase by 16.67%

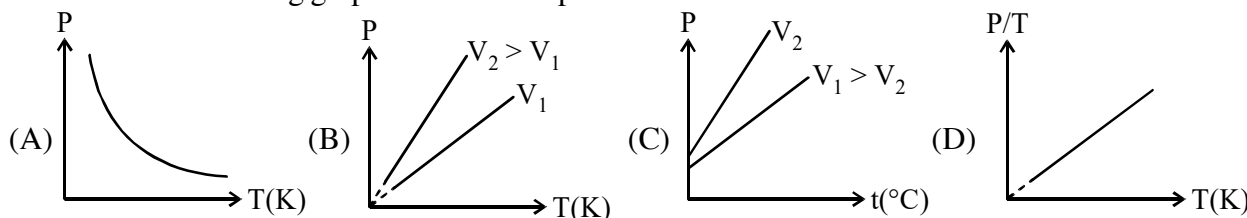
IG0199

PRESSURE LAW

11. The pressure of a gas in a cylinder is 3.0 atm at 87°C . At what temperature, the pressure of gas will drop to 912 mm Hg ?
 (A) 144°C (B) -129°C (C) 417°C (D) 1167°C

IG0200

12. Which of the following graph is correct for pressure law ?



IG0201

13. On increasing the temperature (in $^{\circ}\text{C}$) of certain mass of a gas by 20%, at constant volume, the pressure of gas increased by 5%. The initial temperature of gas is :
 (A) 91°C (B) 91 K (C) 364°C (D) -182°C

IG0202

AVOGADRO'S LAW

14. 5.0 gm of a gas occupy 5.0 dm³. What mass of this gas will occupy 250 ml at the same temperature and pressure ?

(A) 2.5 gm (B) 250 mg (C) 5.0 gm (D) 100 gm

IG0203

15. A glass bulb contains 10 gm O₂ gas. Now, the bulb is completely evacuated and re-filled with CH₄ gas at the same temperature and pressure. The mass of CH₄ gas used in refilling is :

(A) 20 gm (B) 10 gm (C) 5 gm (D) 2.5 gm

IG0204

IDEAL GAS

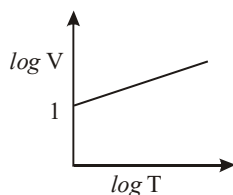
16. A vessel contains mono atomic 'He' at 1 bar and 300 K, determine its number density -

(A) $2.4 \times 10^{25} \text{ m}^{-3}$ (B) $6.8 \times 10^{23} \text{ m}^{-3}$ (C) $4.8 \times 10^{26} \text{ m}^{-3}$ (D) $9.2 \times 10^{27} \text{ m}^{-3}$

IG0069

17. At constant pressure of 0.82 atm ; log V vs log T is plotted as shown in-

Then number of moles present in experiment -



(A) 1 (B) 10 (C) 100 (D) 0.1

IG0062

18. A gas is found to have a formula [CO]_x. If its vapour density is 70 the value of x is

(A) 2.5 (B) 3.0 (C) 5.0 (D) 6.0

IG0066

19. A container when is empty weighs 50 gm. After certain liquid of density 25 gm/dm³ is filled its mass becomes equal to 100 gm. The volume of the container will be :

(A) 0.25 dm³ (B) 0.5 dm³ (C) 1 dm³ (D) 2 dm³

IG0068

20. A gas at a pressure of 5.0 atm is heated from 0°C to 546°C and simultaneously compressed to one-third of its original volume. Hence final pressure is

(A) 10 atm (B) 30 atm (C) 45 atm (D) 5 atm

IG0063

21. A flask containing air (open to the atmosphere) is heated from 300 K to 500 K. Then % of air escaped to the atmosphere is -

(A) 20.0 (B) 40 (C) 60 (D) 80

IG0064

22. A rigid container containing 10 gm gas at some pressure and temperature. The gas has been allowed to escape from the container due to which pressure of the gas becomes half of its initial pressure and temperature become $(2/3)^{\text{rd}}$ of its initial. The mass of gas (in gms) escaped is

(A) 7.5 (B) 1.5 (C) 2.5 (D) 3.5

IG0070

- 23.** The density of gas A is twice that of B at the same temperature the molecular weight of gas B is thrice that of A. The ratio of pressure of gas A and gas B will be
(A) 1 : 6 (B) 7 : 8 (C) 6 : 1 (D) 1 : 4
- IG0071**
- 24.** In a rigid container NH_3 is kept at certain temperature, if on doubling the temperature it is completely dissociated into N_2 and H_2 . Find final pressure to initial pressure ratio :
(A) 4 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
- IG0072**
- 25.** 4.0 g of argon has pressure P and temperature T K in a vessel. On keeping the vessel at 50° higher temperature, 0.8 g of argon was given out to maintain the pressure P. The original temperature was :
(A) 73 K (B) 100 K (C) 200 K (D) 510 K

DALTON'S LAW OF PARTIAL PRESSURE

26. The total pressure exerted by a number nonreacting gases is equal to the sum of partial pressure of the gases under the same conditions is known as :
- (A) Boyle's law (B) Charle's law (C) Avogadro's law (D) Dalton's law
- IG0076**
27. Dalton's law cannot be applied for which gaseous mixture at normal temperatures:
- (A) O_2 and N_2 (B) NH_3 and HCl (C) He and N_2 (D) CO_2 and O_2
- IG0077**
28. Equal masses of methane and oxygen are mixed in an empty container at $25^\circ C$. The fraction of the total pressure exerted by oxygen is -
- [AIEEE-2008]**
- (A) $2/3$ (B) $\frac{1}{3} \times \frac{273}{298}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$
- IG0078**
29. A closed vessel contains helium and ozone at a pressure of P atm. The ratio of He and oxygen atoms is $1 : 1$. If helium is removed from the vessel, the pressure of the system will reduce to :
- (A) $0.5 P$ atm (B) $0.75 P$ atm (C) $0.25 P$ atm (D) $0.33 P$ atm
- IG0079**
30. At constant temperature 200 cm^3 of N_2 at 720 mm and 400 cm^3 of O_2 at 750 mm pressure are put together in a one litre flask. The final pressure of mixture is
- (A) 111 mm (B) 222 mm (C) 333 mm (D) 444 mm
- IG0080**
31. If saturated vapours are compressed slowly (temperature remaining constant) to half the initial volume, the vapour pressure will :
- (A) Become four times (B) become doubled
- (C) Remain unchanged (D) Become half

- 32.** A box of 1L capacity is divided into two equal compartments by a thin partiion which are filled with 2g H₂ and 16g CH₄ respectively. The pressure in each compartment is recorded as P atm. The total pressure when partition is removed will be :
- (A) P (B) 2P (C) P/2 (D) P/4

IG0082

- 33.** A vessel has N₂ gas and water vapours at a total pressure of 1 atm. The vapour pressure of water is 0.3 atm. The contents of this vessel are transferred to another vessel having one third of the capacity of original volume, completely at the same temperature the total pressure of this system in the new vessel is -
- (A) 3.0 atm (B) 1 atm (C) 3.33 atm (D) 2.4 atm

IG0083

GRAHAM'S LAW OF DIFFUSION OR EFFUSION

- 34.** Which gas effuses fastest under identical conditions -
(A) N_2 (B) O_2 (C) Cl_2 (D) CH_4
- IG0084**
- 35.** The rate of diffusion of methane at a given temperature is twice that of a gas X. The molecular weight of X is :
(A) 64 (B) 32 (C) 4.0 (D) 8.0

IG0085

- 36.** The rate of diffusion of two gases A and B is in the ratio of 1 : 4 and that of B and C in the ratio of 1 : 3 the rate of diffusion of C with respect to A is -
- (A) $\frac{1}{12}$ (B) 12 (C) 6 (D) 4

IG0086

37. A gas X diffuses three times faster than another gas Y the ratio of their densities i.e., $D_x : D_y$ is
(A) 1/3 (B) 1/9 (C) 1/6 (D) 1/12

IG0087

- 38.** Rate of diffusion of a gas is :
- (A) directly proportional to its density
 - (B) directly proportional to its molecular weight
 - (C) directly proportional to the square of its molecular weight
 - (D) inversely proportional to the square root of its molecular weight

IG0088

- 39.** Since the atomic weights of carbon nitrogen and oxygen are 12, 14 and 16 respectively, among the following pairs of gases, the pair that will diffuse at the same rate is :
- (A) CO_2 and N_2O (B) CO_2 and N_2O_3
(C) CO_2 and CO (D) CO_2 and NO

IG0089

- 40.** If the four tubes of a car are filled to the same pressure with N_2 , O_2 , H_2 and He separately then which one will be filled first :
- (A) N_2 (B) O_2 (C) H_2 (D) He

IG0090

41. The increasing order of effusion among the gases, H_2 , O_2 , NH_3 and CO_2 is –
 (A) H_2 , CO_2 , NH_3 , O_2 (B) H_2 , NH_3 , O_2 , CO_2
 (C) H_2 , O_2 , NH_3 , CO_2 (D) CO_2 , O_2 , NH_3 , H_2
IG0091
42. The rate of diffusion of hydrogen is about–
 (A) $\frac{1}{2}$ that of Helium (B) 1.4 times that of Helium
 (C) twice that of H (D) Four times that of Helium
IG0092
43. Under identical experiment conditions which of the following pairs of gases will be most easy to separate by using effusion process –
 (A) H_2 and T_2 (B) SO_2 and SO_3 (C) NH_3 and CH_4 (D) $U^{235}O_2$ and $U^{238}O_2$
IG0095
44. Certain volume of He gas takes 10 sec for its diffusion, how much time will be taken by CH_4 gas to diffuse its same volume under identical conditions –
 (A) 5 sec (B) 10 sec (C) 20 sec (D) 40 sec
IG0096
45. A football bladder contains equimolar proportions of H_2 and O_2 . The composition by mass of the mixture effusing out of punctured football is in the ratio ($H_2 : O_2$)
 (A) 1 : 4 (B) $2\sqrt{2} : 1$ (C) $1 : 2\sqrt{2}$ (D) 4 : 1
IG0097
46. Consider the following pairs of gases A and B.
- | | A | B |
|-----|--------------|--------------|
| (a) | CO | N_2 |
| (b) | O_2 | O_3 |
| (c) | $^{235}UF_6$ | $^{238}UF_6$ |
- Relative rates of effusion of gases A to B under similar condition is in the order:
 (A) $a < b < c$ (B) $a < c < b$ (C) $a > b > c$ (D) $a > c > b$
IG0098
47. A 4 : 1 molar mixture of He & CH_4 kept in a vessel at 20 bar pressure. Due to a hole in the vessel, gas mixture leaks out. What is the composition of mixture effusing out initially –
 (A) 8 : 1 (B) 4 : 1 (C) 1 : 4 (D) 4 : 3
IG0100
48. Calculate the ratio of rate of effusion of O_2 and H_2 from a container containing 16gm O_2 and 2gm H_2
 (A) 1 : 8 (B) 8 : 1 (C) 1 : 4 (D) 4 : 1
IG0101

KINETIC THEORY OF GASES

49. At STP, the order of root mean square speed of molecules H_2 , N_2 , O_2 and HBr is :
 (A) $H_2 > N_2 > O_2 > HBr$ (B) $HBr > O_2 > N_2 > H_2$
 (C) $HBr > H_2 > O_2 > N_2$ (D) $N_2 > O_2 > H_2 > HBr$
IG0104
50. Four particles have speed 2, 3, 4 and 5 cm/s respectively. Their rms speed is :
 (A) 3.5 cm/s (B) $\left(\frac{27}{2}\right)$ cm/s (C) $\sqrt{54}$ cm/s (D) $\left(\frac{\sqrt{54}}{2}\right)$ cm/s
IG0105
51. A flask has 10 molecules out of which four molecules are moving at 7ms^{-1} & the remaining are moving at same speed of $X\text{ms}^{-1}$. If U_{rms} of the gas is 5ms^{-1} . The value of 'X' will be
 (A) 5 (B) 3 (C) 9 (D) 16
IG0106
52. If the average velocity of N_2 molecules is 0.3 m/sec. at 27°C , then the velocity of 0.6 m/sec will take place at :
 (A) 273 K (B) 927 K (C) 1000 K (D) 1200 K
IG0107
53. Which of the following expression does not give root mean square velocity –
 (A) $\left(\frac{3RT}{M_w}\right)^{\frac{1}{2}}$ (B) $\left(\frac{3P}{DM_w}\right)^{\frac{1}{2}}$ (C) $\left(\frac{3P}{D}\right)^{\frac{1}{2}}$ (D) $\left(\frac{3PV}{nM_w}\right)^{\frac{1}{2}}$
IG0108
54. Which one of the following gases would have the highest R.M.S. velocity at 25°C ?
 (A) Oxygen (B) Carbon dioxide (C) Sulphur dioxide (D) Carbon monoxide
IG0109
55. At what temperature would the rms speed of a gas molecule have twice its value at 100°C ?
 (A) 4192 K (B) 1492 K (C) 9142 K (D) 2491 K
IG0110
56. Temperature at which most probable speed of O_2 becomes equal to root mean square speed of N_2 is [Given : N_2 at 427°C]
 (A) 732 K (B) 1200 K (C) 927 K (D) 800 K
IG0112
57. The density ratio of O_2 and H_2 is 16 : 1. The ratio of their U_{rms} is :-
 (A) 4 : 1 (B) 16 : 1 (C) 1 : 4 (D) 1 : 16
IG0113
58. The av. K.E./mole of an ideal monoatomic gas at 27°C is
 (A) 900 cal (B) 1800 cal (C) 300 cal (D) None
IG0115

59. The average kinetic energy of an ideal gas per molecule in SI units at 25°C will be :
 (A) 6.17×10^{-21} kJ (B) 6.17×10^{-21} J (C) 6.17×10^{-20} J (D) 7.16×10^{-20} J
IG0116
60. Average K.E. of CO_2 at 27°C is E. The average kinetic energy of N_2 at the same temperature will be
 (A) E (B) 22E (C) E/22 (D) $E/\sqrt{2}$
IG0118
61. If a gas expands at constant temperature then :
 (A) No. of gaseous molecule decreases (B) kinetic energy of molecule decreases
 (C) K.E. remain same (D) K.E. increases
IG0119
62. Total translational kinetic energy possessed by 8 gm methane at 273°C
 (R = 2 Cal/mol-K)
 (A) 819 calorie (B) 409.5 calorie (C) 1638 calorie (D) None of these
IG0120
63. Ideal gas equation in terms of K.E. per unit volume E, is-
 (A) $P = \frac{3}{2}RT$ (B) $P = \frac{2}{3}E$ (C) $P = \frac{2}{3}RT$ (D) $P = \frac{3}{2}RT$
IG0121
64. According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels -
[AIEEE-2003]
 (A) In a straight line path (B) with an accelerated velocity
 (C) In a circular path (D) In a wavy path
IG0122
65. As the temperature is raised from 20°C to 40°C, the average kinetic energy of neon atoms changes by factor of which of the following ?
[AIEEE-2004]
 (A) 1/2 (B) $\sqrt{(313/293)}$ (C) 313/293 (D) 2
IG0124
66. Three gases A, B and C are at same temperature. If their r.m.s. speed are in the ratio $1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$ then their molar masses will be in the ratio :
 (A) 1 : 2 : 3 (B) 3 : 2 : 1 (C) $1 : \sqrt{2} : \sqrt{3}$ (D) $\sqrt{3} : \sqrt{2} : 1$
IG0125
67. The molecular velocity of any gas is :-
[AIEEE-2011]
 (A) inversely proportional to the square root of temperature
 (B) inversely proportional to absolute temperature
 (C) directly proportional to square of temperature
 (D) directly proportional to square root of temperature
IG0142

68. α , v and u represent most probable velocity, average velocity and root mean square velocity respectively of a gas at a particular temperature. The correct order among the following is -[JEE(Main)-2012]
 (A) $\alpha > u > v$ (B) $v > u > \alpha$ (C) $u > v > \alpha$ (D) $u > \alpha > v$

IG0143

69. The relationship among most probable velocity, average velocity and root mean square velocity is respectively :- [JEE(Main-online)-2012]
 (A) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (B) $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$ (C) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$ (D) $\sqrt{8/\pi} : \sqrt{3} : \sqrt{2}$

IG0146

COLLISION PARAMETERS

70. The following gases are present under similar condition of T , P & V . The longest mean free path stands for
 (A) H_2 (B) N_2 (C) O_2 (D) Cl_2
71. At constant volume, Z_{11} is directly proportional to -
 (A) $\sqrt{P}$ (B) P (C) T^2 (D) T

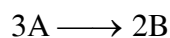
IG0126

IG0127

EXERCISE # O-II

Single Correct :

1. A gaseous reaction,



is carried out in a 0.082 litre closed container initially containing 1 mole of gas A. After sufficient time a curve of P (atm) vs T (K) is plotted and the angle with x-axis was found to be 42.95° . The degree of association of gas A is [Given : $\tan 42.95 = 0.8$]

- (A) 0.4 (B) 0.6 (C) 0.5 (D) 0.8

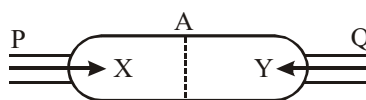
IG0074

2. Under identical conditions of pressure and temperature, 4 L of gaseous mixture (H_2 and CH_4) effuses through a hole in 5 min whereas 4 L of a gas X of molecular mass 36 takes to 10 min to effuse through the same hole. The mole ratio of $H_2 : CH_4$ in the mixture is -

- (A) 1 : 2 (B) 2 : 1 (C) 2 : 3 (D) 1 : 1

IG0093

3. 3 mole of gas "X" and 2 moles of gas "Y" enters from end "P" and "Q" of the cylinder respectively. The cylinder has the area of cross-section A, shown as under -

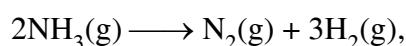


The length of the cylinder is 150 cm. The gas "X" intermixes with gas "Y" at the point A. If the molecular weight of the gases X and Y is 20 and 80 respectively, then what will be the distance of point A from Q?

- (A) 75cm (B) 50cm (C) 37.5 cm (D) 90 cm

IG0094

4. For the reaction



what is the % of NH_3 converted if the mixture diffuses twice as fast as that of SO_2 under similar conditions.

- (A) 3.125 % (B) 31.25 % (C) 6.25 % (D) 62.5 %

IG0099

5. Two flasks X and Y have capacity 1L and 2L respectively and each of them contains 1 mole of a gas. The temperature of the flask are so adjusted that average speed of molecules in X is twice as those in Y. The pressure in flask X would be

- (A) same as that in Y (B) half of that in Y (C) twice of that in Y (D) 8 times of that in Y

IG0111

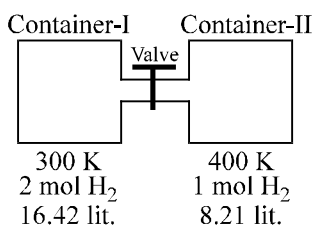
6. Which of the gas have highest fraction of molecules at $27^\circ C$ in most probable speed region -

- (A) H_2 (B) N_2 (C) O_2 (D) CO_2

IG0114

One or more may be correct :

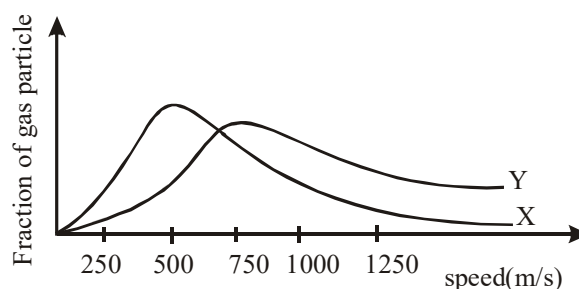
7. If both the containers are maintained at constant temperature ($R = 0.0821 \text{ L-atm/K-mol}$)
Select the correct option(s) :



- (A) Pressure in container-I is 3 atm before opening the valve.
(B) Pressure after opening the valve is 3.57 atm.
(C) Moles in each container are same after opening the valve.
(D) Pressure in each container are same after opening the valve.

IG0129

8. The graph below shows the distribution of molecular speed of two ideal gases X and Y at 200K. on the basis of the below graph identify the correct statements -



- (A) If gas X is methane, then gas Y can be CO_2
- (B) Fraction of molecules of X must be greater than Y in a particular range of speed at 200K
- (C) Under identical conditions rate of effusion of Y is greater than that of X
- (D) The molar kinetic energy of gas X at 200K is equal to the molar kinetic energy of Y at 200K

IG0130

9. Identify the correct statements when a fixed amount of ideal gas is heated in a container fitted with a movable piston always operating at constant pressure.
- (A) Average distance travelled between successive collisions will decrease.
 - (B) Collision frequency increases since speed of the molecules increases with increase in temperature.
 - (C) Average relative speed of approach remains unaffected.
 - (D) Average angle of approach remains unaffected.

IG0131

10. Choose the correct statement(s) among the following
- (A) Average molecular speed of gases increases with decrease in fraction of molecules moving slowly
 - (B) Rate of effusion of gases increases with increase in collision frequency at constant volume.
 - (C) Rate of effusion is inversely proportional to molecular weight of gas
 - (D) Mean free path does not change with change in temperature at constant pressure.

IG0132

11. Which of the following quantities is the same for all ideal gases at the same temperature :
(A) The kinetic energy of 1 mol
(B) The kinetic energy of 1 g
(C) The number of molecules in 1 mol
(D) The number of molecules in 1 g
- IG0133
12. Choose the correct statement(s) among the following -
(A) The mean free path (λ) of gaseous molecules is directly proportional to temperature of gas at constant volume
(B) The mean free path (λ) of gaseous molecules is inversely proportional to pressure of gas at constant volume
(C) The mean free path (λ) of gaseous molecules is directly proportional to volume of gas at constant T
(D) The mean free path (λ) of gaseous molecules is directly proportional to volume of gas at constant P
- IG0134
13. Which statement is/are correct for postulates of kinetic theory of gases -
(A) Gases are composed of molecules whose size is negligible compared with the average distance between them
(B) Molecules moves randomly in straight lines in all directions and at various speeds.
(C) When molecules collide with one another the collisions are elastic. In an elastic collision the loss of kinetic energy takes place
(D) The average kinetic energy of a molecule is proportional to the absolute temperature.

IG0135

Paragraph for Question 14 to 16

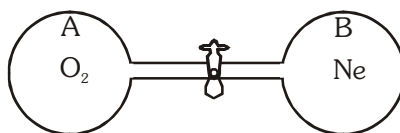
The constant motion and high velocities of gas particles lead to some important practical consequences. One such consequence is that as minimum rapidly when they come in contact. The mixing of different gases by random molecular motion and with frequent collisions is called diffusion. A similar process in which gas molecules escape through a tiny hole into a vacuum is called effusion.

14. Helium gas at 1 atm and SO_2 at 2 atm pressure, temperature being the same, are released separately at the same moment into 1 m long evacuated tubes of equal diameters. If helium reaches the other end of the tube in t sec, what distance SO_2 would traverse in the same time interval in the other tube ?
(A) 25 cm (B) 50 cm (C) 60 cm (D) 75 cm
- IG0136
15. 4 g of H_2 effused through a pinhole in 10 sec at constant temperature and pressure. The amount of oxygen effused in the same time interval and at the same conditions of temperature and pressure would be :
(A) 4 g (B) 8 g (C) 16 g (D) 32 g
- IG0137
16. For 10 min. each at 27°C , from two identical bulbs helium and an unknown gas X at equal pressures are leaked into a common vessel of 3 L capacity. The resulting pressure is 4.1 atm and the mixture contains 0.4 mol of helium. The molar mass of gas X is :
(A) 16 (B) 32 (C) 64 (D) None of these

IG0138

Paragraph for Question 17 to 18

Initially, flask A contained oxygen gas at 27°C and 950 mm of Hg, and flask B contained neon gas at 27°C and 900 mm. Finally, the two flasks were joined by means of a narrow tube of negligible volume equipped with a stopcock and gases were allowed to mix up freely. The final pressure in the combined system was found to be 910 mm of Hg.



17. What is the correct relationship between volumes of the two flasks ?

(A) $V_B = 3V_A$ (B) $V_B = 4V_A$ (C) $V_B = 5V_A$ (D) $V_B = 4.5V_A$

IG0139

18. How many moles of gas are present in flask A in the final condition, if volume of flask B is 304 litre ? ($R = 0.08 \text{ atm L mol}^{-1} \text{ K}^{-1}$)

(A) 7.58 (B) 3.79 (C) 15.16 (D) None of these

IG0139

TABLE TYPE QUESTION :

Column-I (Gases at different conditions)	Column-II (Value of speed (m/s))	Column-III
(A) CH_4 at 27°C	(P) $U_{\text{rms}} = 342.5$	(I) Molar K.E. = 3750 Joule
(B) SO_2 at 27°C	(Q) $U_{\text{mp}} = 1100$	(II) Molar K.E. = 5000 Joule
(C) 4 gm He at 1 atm and 24.6 litre	(R) $U_{\text{rms}} = 685$	(III) Average K.E. per gram = 937.5 Joule
(D) 32 gm O_2 at 127°C	(S) $U_{\text{rms}} = 550$	(IV) Average K.E. per gram = 234 Joule.

(Given : $R = \frac{25}{3} \frac{\text{J}}{\text{mol} \times \text{K}}$, $\sqrt{30} = 5.48$, $\sqrt{\frac{2}{3}} = 0.8$, $\sqrt{5} = 2.2$)

19. Which of the following is correct

(A) A ; R ; 4 (B) A ; R ; 2 (C) A ; Q ; 4 (D) B ; R ; 1

IG0140

20. Which of the following is correct

(A) B ; Q ; 1 (B) B ; Q ; 2 (C) B ; P ; 1 (D) C ; S ; 3

IG0140

21. Which of the following is correct

(A) C ; Q ; 3 (B) D ; R ; 2 (C) D ; S ; 1 (D) D ; S ; 4

IG0140

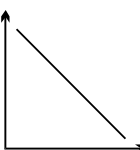
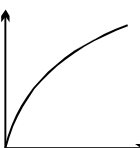
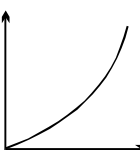
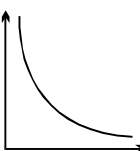
MATCH THE COLUMN:

22. Match the entries in column I with entries in Column II and then pick out correct options.

Column I

- (A) $\frac{1}{V^2}$ vs P for ideal gas at constant T and n.
- (B) V vs $\frac{1}{T}$ for ideal gas at constant P and n
- (C) log P vs log V for ideal gas at constant T and n.
- (D) V vs $\frac{1}{P^2}$ for ideal gas at constant T and n.

Column II

- (P) 
- (Q) 
- (R) 
- (S) 

IG0141

EXERCISE # JEE-MAINS

1. Which one of the following is the wrong assumption of kinetic theory of gases ?

[JEE(Main-online)-2013]

- (1) All the molecules move in straight line between collision and with same velocity.
- (2) Molecules are separated by great distances compared to their sizes.
- (3) Pressure is the result of elastic collision of molecules with the container's wall.
- (4) Momentum and energy always remain conserved.

IG0147

2. By how many folds the temperature of a gas would increase when the root mean square velocity of the gas molecules in a container of fixed volume is increased from 5×10^4 cm/s to 10×10^4 cm/s ?

[JEE(Main-online)-2013]

- (1) Four (2) three (3) Two (4) Six

IG0148

3. A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is :

[JEE(Main-online)-2014]

- (1) NH_2 (2) NH_3 (3) N_3H (4) N_2H_4

IG0150

4. The initial volume of a gas cylinder is 750.0 mL. If the pressure of gas inside the cylinder changes from 840.0 mm Hg to 360.0 mm Hg, the final volume the gas will be [JEE(Main-online)-2014]

- (1) 1.750 L (2) 7.50 L (3) 3.60 L (4) 4.032 L

IG0151

5. The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at 300 K is : (Atomic masses : He = 4 u, O = 16 u) [JEE(Main-online)-2014]

- (1) 1200 K (2) 600 K (3) 300 K (4) 2400 K

IG0152

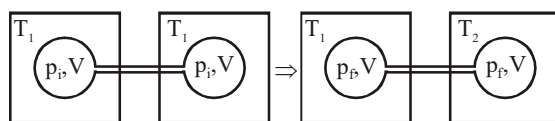
6. Which of the following is not an assumption of the kinetic theory of gases ?

[JEE-Mains (online)-2015]

- (1) Gas particles have negligible volume.
- (2) A gas consists of many identical particles which are in continual motion.
- (3) At high pressure, gas particles are difficult to compress.
- (4) Collisions of gas particles are perfectly elastic.

IG0153

7. Two closed bulbs of equal volume (V) containing an ideal gas initially at pressure p_i and temperature T_1 are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then raised to T_2 . The final pressure p_f is :- [JEE-Mains-2016]



- (1) $2p_i \left(\frac{T_1 T_2}{T_1 + T_2} \right)$ (2) $p_i \left(\frac{T_1 T_2}{T_1 + T_2} \right)$ (3) $2p_i \left(\frac{T_1}{T_1 + T_2} \right)$ (4) $2p_i \left(\frac{T_2}{T_1 + T_2} \right)$

IG0154

8. At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen (N_2) at 4 bar. The molar mass of gaseous molecule is:- [JEE-Mains-2017(ONLINE)]

- (1) 28 g mol^{-1} (2) 56 g mol^{-1} (3) 224 g mol^{-1} (4) 112 g mol^{-1}

IG0155

9. Assuming ideal gas behaviour, the ratio of density of ammonia to that of hydrogen chloride at same temperature and pressure is : (Atomic wt. of Cl = 35.5 u) [JEE-Mains-2018(ONLINE)]

- (1) 0.64 (2) 1.64 (3) 1.46 (4) 0.46

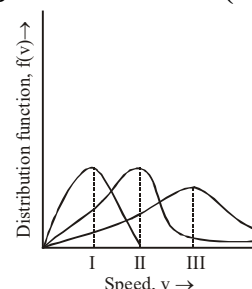
IG0156

10. Points I, II and III in the following plot respectively correspond to.

(V_{mp} : most probable velocity)

- (1) V_{mp} of N_2 (300K); V_{mp} of H_2 (300K); V_{mp} of O_2 (400K)
 (2) V_{mp} of H_2 (300K); V_{mp} of N_2 (300K); V_{mp} of O_2 (400K)
 (3) V_{mp} of O_2 (400K); V_{mp} of N_2 (300K); V_{mp} of H_2 (300K)
 (4) V_{mp} of N_2 (300K); V_{mp} of O_2 (400K); V_{mp} of H_2 (300K)

[JEE-Mains-2019(ONLINE)]



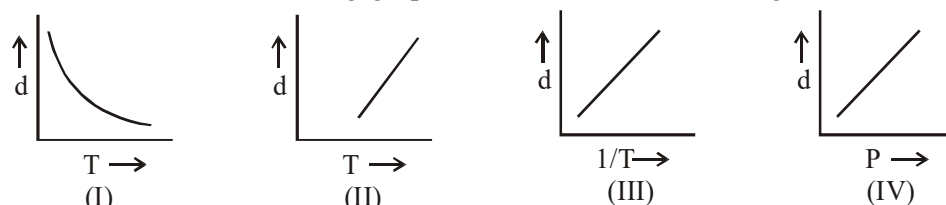
IG0157

11. An open vessel at 27°C is heated until two fifth of the air (assumed as an ideal gas) in it has escaped from the vessel. Assuming that the volume of the vessel remains constant, the temperature at which the vessel has been heated is : [JEE-Mains-2019(ONLINE)]

- (1) 750°C (2) 500°C (3) 750 K (4) 500 K

IG0159

12. Which one of the following graphs is **not correct** for ideal gas ?



d = Density, P = Pressure, T = Temperature

- (1) II (2) III (3) I (4) IV

[JEE(Main-online)-2020]

IG0205

13. A mixture of one mole each of H_2 , He and O_2 each are enclosed in a cylinder of volume V at temperature T . If the partial pressure of H_2 is 2 atm, the total pressure of the gases in the cylinder is :

(1) 14 atm (2) 22 atm (3) 6 atm (4) 38 atm

[JEE(Main-online)-2020]

IG0206

14. A spherical balloon of radius 3 cm containing helium gas has a pressure of 48×10^{-3} bar. At the same temperature, the pressure, of a spherical balloon of radius 12 cm containing the same amount of gas will be $\times 10^{-6}$ bar.

[JEE(Main-online)-2020]

IG0207

15. The pressure exerted by a non-reactive gaseous mixture of 6.4 g of methane and 8.8 g of carbon dioxide in a 10 L vessel at 27°C is _____ kPa.

(Round off to the Nearest Integer).

[Assume gases are ideal, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]

[JEE(Main-online)-2021]

IG0208

16. An LPG cylinder contains gas at a pressure of 300 kPa at 27°C . The cylinder can withstand the pressure of 1.2×10^6 Pa. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is _____ $^\circ\text{C}$. (Nearest integer)

[JEE(Main-online)-2021]

IG0209

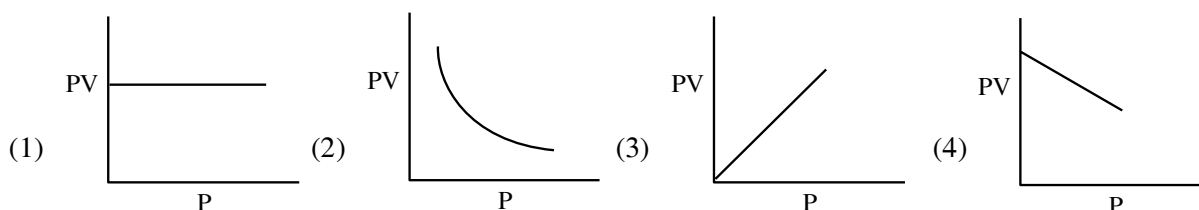
17. A home owner uses $4.00 \times 10^3 \text{ m}^3$ of methane (CH_4) gas, (assume CH_4 is an ideal gas) in a year to heat his home. Under the pressure of 1.0 atm and 300 K, mass of gas used is $x \times 10^5$ g. The value of x is _____

(Nearest integer) (Given $R = 0.083 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

[JEE(Main-online)-2021]

IG0210

18. Which one of the following is the correct PV vs P plot at constant temperature for an ideal gas ? (P and V stand for pressure and volume of the gas respectively)



[JEE(Main-online)-2021]

IG0212

19. An empty LPG cylinder weighs 14.8 kg. When full, it weighs 29.0 kg and shows a pressure of 3.47 atm. In the course of use at ambient temperature, the mass of the cylinder is reduced to 23.0 kg. The final pressure inside of the cylinder is _____ atm. (Nearest integer)

(Assume LPG to be an ideal gas)

[JEE(Main-online)-2021]

IG0213

20. At 300 K, a sample of 3.0 g of gas A occupies the same volume as 0.2 g of hydrogen at 200 K at the same pressure. The molar mass of gas A is _____ g mol⁻¹ (nearest integer) Assume that the behaviour of gases is ideal. (Given: The molar mass of hydrogen (H₂) gas is 2.0 g mol⁻¹)

[JEE(Main-online)-2022]

IG0214

21. A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is 27 °C. At 03:00 pm, when the temperature is 45°C, the pressure in the tank will be _____ atm. [nearest integer]

[JEE(Main-online)-2022]

IG0215

22. An evacuated glass vessel weighs 40.0 g when empty, 135.0 g when filled with a liquid of density 0.95 g mL⁻¹ and 40.5 g when filled with an ideal gas at 0.82 atm at 250 K. The molar mass of the gas in g mol⁻¹ is : (Given : R = 0.082 L atm K⁻¹ mol⁻¹)

[JEE(Main-online)-2022]

(A) 35

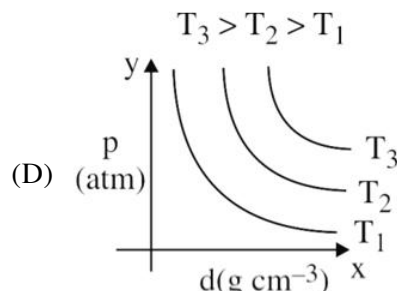
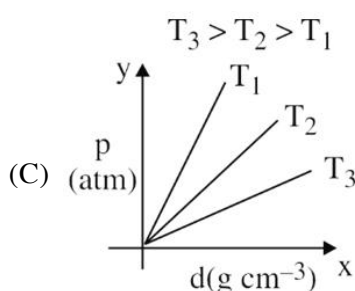
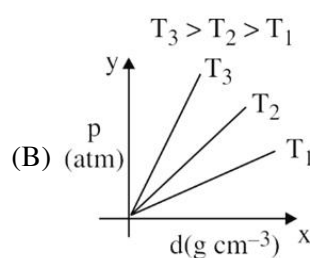
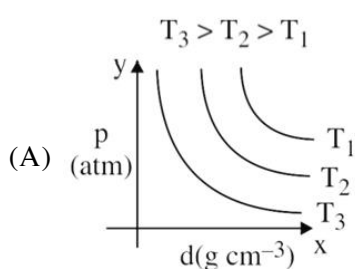
(B) 50

(C) 75

(D) 125

IG0216

23. Which amongst the given plots is the correct plot for pressure (p) vs density (d) for an ideal gas ?



[JEE(Main-online)-2022]

IG0217

24. CNG is an important transportation fuel. When 100 g CNG is mixed with 208 oxygen in vehicles, it leads to the formation of CO₂ and H₂O and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is _____. [nearest integer]

[Assume CNG to be methane]

[JEE(Main-online)-2022]

IG0218

25. A box contains 0.90 g of liquid water in equilibrium with water vapour at 27°C. The equilibrium vapour pressure of water at 27°C is 32.0 Torr. When the volume of the box is increased, some of the liquid water evaporates to maintain the equilibrium pressure. If all the liquid water evaporates, then the volume of the box must be _____ litre. [nearest integer]

(Given: $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

[JEE(Main-online)-2022]

(Ignore the volume of the liquid water and assume water vapours behave as an ideal gas.)

IG0219

26. Geraniol, a volatile organic compound, is a component of rose oil. The density of the vapour is 0.46 gL⁻¹ at 257°C and 100 mm Hg. The molar mass of geraniol is _____ (Nearest Integer)

[Given $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

[JEE(Main-online)-2022]

IG0220

27. A sealed flask with a capacity of 2 dm³ contains 11 g of propane gas. The flask is so weak that it will burst if the pressure becomes 2 MPa. The minimum temperature at which the flask will burst is _____ °C. [Nearest integer]

(Given: $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$. Atomic masses of C and H are 12u and 1u respectively.) (Assume that propane behaves as an ideal gas.)

[JEE(Main-online)-2022]

IG0221

28. A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is _____ bar. (Nearest Integer)

[JEE(Main-online)-2022]

IG0222

29. 'x' g of molecular oxygen (O₂) is mixed with 200 g of neon (Ne). The total pressure of the non-reactive mixture of O₂ and Ne in the cylinder is 25 bar. The partial pressure of Ne is 20 bar at the same temperature and volume. The value of 'x' is _____.

[Given: Molar mass of O₂ = 32 g mol⁻¹.

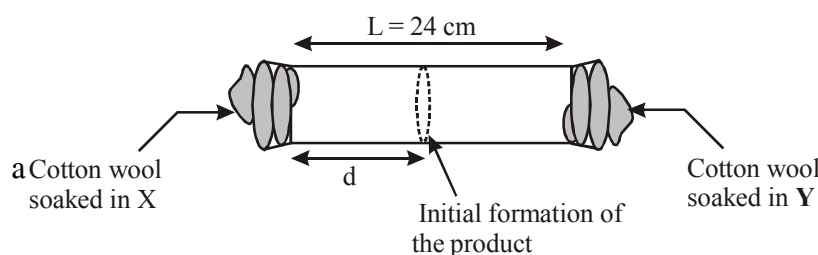
Molar mass of Ne = 20 g mol⁻¹]

[JEE(Main-online)-2022]

IG0223

EXERCISE # JEE-ADVANCED**Paragraph for Question 1 & 2**

X and Y are two volatile liquids with molar weights of 10 g mol^{-1} and 40 g mol^{-1} respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length $L = 24 \text{ cm}$, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300 K. Vapours of X and Y react to form a product which is first observed at a distance $d \text{ cm}$ from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours. [JEE 2014]



- The value of d in cm (shown in the figure), as estimated from Graham's law, is -
(A) 8 (B) 12 (C) 16 (D) 20
IG0162
- The experimental value of d is found to be smaller than the estimate obtained using Graham's law. This is due to -
(A) Larger mean free path for X as compared to that of Y
(B) Larger mean free path for Y as compared to that of X
(C) Increased collision frequency of Y with the inert gas as compared to that of X with the inert gas
(D) Increased collision frequency of X with the inert gas as compared to that of Y with the inert gas
IG0163
- The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. The absolute temperature of an ideal gas is increased 4 times and its pressure is increased 2 times. As a result, the diffusion coefficient of this gas increases x times. The value of x is [JEE 2016]

IG0224

4. A closed tank has two compartments A and B, both filled with oxygen (assumed to be ideal gas). The partition separating the two compartments is fixed and is a perfect heat insulator (Figure 1). If the old partition is replaced by a new partition which can slide and conduct heat but does NOT allow the gas to leak across (Figure 2), the volume (in m^3) of the compartment A after the system attains equilibrium is _____. [JEE 2018]

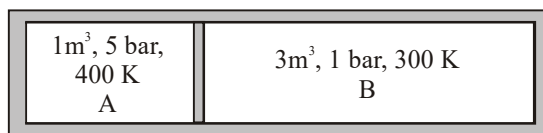


Figure 1

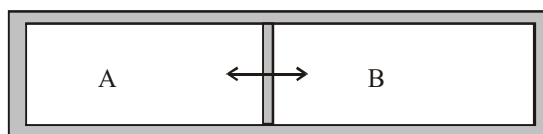


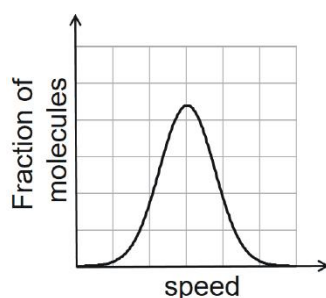
Figure 2

IG0164

5. Which of the following statement(s) is (are) correct regarding the root mean square speed (U_{rms}) and average translational kinetic energy (ϵ_{av}) of a molecule in a gas at equilibrium ?
- (1) U_{rms} is doubled when its temperature is increased four times [JEE 2019]
 - (2) ϵ_{av} at a given temperature does not depend on its molecular mass
 - (3) U_{rms} is inversely proportional to the square root of its molecular mass
 - (4) ϵ_{av} is doubled when its temperature is increased four times

IG0165

6. If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average and the roots mean square speeds, respectively, is [JEE 2020]



- (A) 1 : 1 : 1 (B) 1 : 1 : 1.224 (C) 1 : 1.128 : 1.224 (D) 1 : 1.128 : 1

IG0225

7. Aluminium reacts with sulfuric acid to form aluminium sulfate and hydrogen. What is the volume of hydrogen gas in liters (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulfuric acid are combined for the reaction ?
(Use molar mass of aluminium as 27.0 g mol^{-1} , $R = 0.082 \text{ atm L mol}^{-1} \text{ K}^{-1}$) [JEE 2020]

IG0226

ANSWER KEY**EXERCISE # S-I**

- | | | | |
|------------------------------------|--------------------|----------------------------|-------------------------|
| 1. (500.00) | 2. (1.00) | 3. (37.50) | 4. (150.00) |
| 5. (5.46 or 5.47) | 6. (30.00) | 7. (127.00) | 8. (135.00) |
| 9. (64.28) | 10. (2.68) | 11. (24.60) | 12. (0.60) |
| 13. (32.00) | 14. (100.00) | 15. (1.73) | 16. (4.00) |
| 17. (4.00) | 18. (8.00) | 19. (50.00) | 20. (3.98) |
| 21. (45.00) | 22. (0.06 or 0.07) | 23. i. (15.00) ii. (12.00) | 24. (1.25) |
| 25. (93.75) | 26. (0.80) | 27. (19.48) | 28. (1.80) |
| 29. (1.50) | 30. (0.15) | 31. (0.28) | 32. (420.00) |
| 33. (146.50) | 34. (2.00) | 35. (36.00) | 36. (8.00) |
| 37. (20.00) | 38. (5.00) | 39. (4.00) | 40. (8.00) |
| 41. (8.00) | 42. (4.00) | 43. (0.25) | 44. (400.00) |
| 45. (25.00) | 46. (2.00) | 47. (55.55 or 55.56) | 48. (1.10) |
| 49. (187.06) | 50. (2.00) | 51. (4.00) | 52. (-173.00) |
| 53. (1.08 or 1.09) | 54. (2.00) | 55. (0.50) | 56. $(1.41 = \sqrt{2})$ |
| 57. $(0.35 = \frac{1}{2\sqrt{2}})$ | 58. (4.00) | | |

EXERCISE # S-II

- | | | | |
|---|------------------------|-------------|-------------|
| 1. (4.00) | | | |
| 2. ($T_3 = 360.00$, $T_4 = 90.00$, $P_1 = 5.00$, $P_2 = 1.50$, $V_1 = 16.00$) | | | |
| 3. a. (80.00), b. (1295.00) | 4. i. (6.00), ii. (No) | 5. (627.00) | |
| 6. (2.00) | 7. (6.00) | 8. (2.00) | 9. (380.00) |
| 10. i. (1.50), ii. (1.25), iii. (0.20 or 0.21) | 11. (7.00) | 12. (4.00) | |
| 13. (8.00) | | | |
| 14. i. (0.012 atm), ii. (2.50×10^{22}) , iii. (3.60 J), iv. (300.00 J), v. $(1 : \sqrt{2})$, vi. $(1 : 2500)$ | | | |

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	D	D	C	C	B	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	A	B	C	A	C	C	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	C	A	C	D	B	C	C	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	D	D	A	B	B	D	A	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	B	A	C	A	B	A	A	A	D
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	D	B	D	B	B	C	A	B	A
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	A	B	A	C	A	D	C	A	A
Que.	71									
Ans.	A									

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	C	C	D	D	A,D	C,D	D	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	C,D	A,B,D	B	C	C	B	B	A	C
Que.	21	22								
Ans.	A	A-R	B-S	C-P	D-Q					

EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	4	1	4	3	4	4	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	3	750.00	150	927	26	1	2	45
Que.	21	22	23	24	25	26	27	28	29	
Ans.	32	D	B	143	29	152	1655	2	80	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7			
Ans.	C	D	4	2.22	1,2,3	B	6.15			

Important Notes

REAL GAS

1. Introduction :

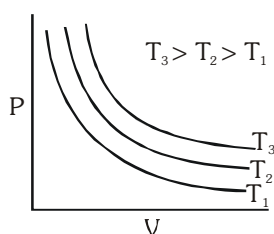
An ideal gas is a hypothetical gas whose pressure, volume and temperature behaviour is completely described by the ideal gas equation. Actually no gas is ideal or perfect in nature. All gases are real gases.

Real gases do not obey the ideal gas laws exactly under all conditions of temperature and pressure.

1.1 COMPARISON BETWEEN REAL AND IDEAL GAS :

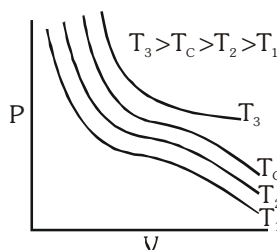
IDEAL GAS

- (i) $PV = nRT$
- (ii) No liquifaction is possible.

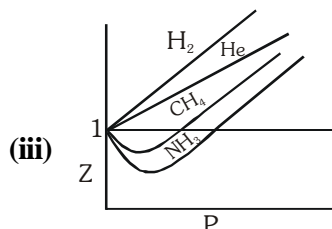
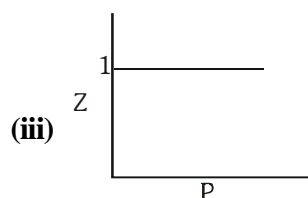


REAL GAS

- (i) $PV \neq nRT$
- (ii) Liquifaction is possible below a certain temperature.



⇒ Follow critical phenomena and can not liquefy above T_c .



- (iv) No interaction force is present between gas particles.
- (iv) Interaction force exist between gas particles which vary, depending upon conditions.
- (v) Volume of gas particles is negligible w.r.t. volume of container.
- (v) Volume of gas particles has significant value and can not be neglected normally w.r.t. volume of container.

1.2 REASON OF DEVIATION IN GASES FROM IDEAL BEHAVIOUR :

Real gases deviates from ideal behaviour because of mainly two assumptions of "Kinetic theory of gases".

- (i) The volume of gas particle is negligible compared to volume of container (while the real gas particle may have some significant volume).
- (ii) There is no interaction between gaseous particles (while intermolecular forces exist between real gas particles).

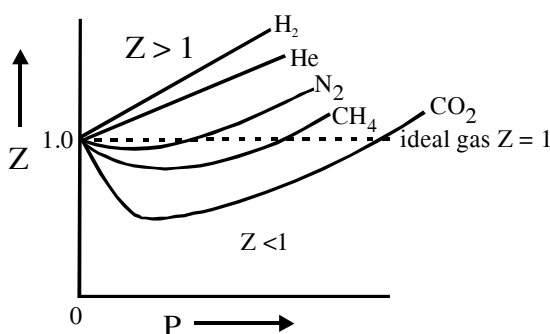
1.3 COMPRESSIBILITY FACTOR (Z) :

The extent to which a real gas departs from the ideal behaviour may be expressed in terms of compressibility factor or compression factor (Z),

$$Z = \frac{(PV)_{\text{real}}}{(PV)_{\text{ideal}}} = \left(\frac{V_{m, \text{real}}}{V_{m, \text{ideal}}} \right)_p = \frac{PV_m}{RT} \quad [V_m = \text{molar volume}]$$

1.3.1 Compressibility factor vs pressure :

- (i) For an ideal gas, $Z = 1$ and is independent of temperature and pressure.
- (ii) Exceptional behaviour of H_2 and He :
For these gases $Z > 1$ at 0°C .
- (iii) Effect of pressure :
 - (a) At very low P , $PV_m \approx RT$ i.e. $Z \approx 1$
 - (b) At low P , $PV_m < RT$ i.e. $Z < 1$
 $\Rightarrow$ attractive forces dominant
 $\Rightarrow$ Negative deviation from ideal behaviour.
 $\Rightarrow$ Gas is more compressible than ideal behaviour.
 - (c) At high P , $PV_m > RT$ i.e. $Z > 1$
 $\Rightarrow$ repulsive forces dominant
 $\Rightarrow$ Positive deviation from ideal behaviour.
 $\Rightarrow$ Gas is less compressible than ideal behaviour.
- (iv) For the gases which are easily liquefied (e.g. CO_2) Z dips sharply below the ideal line in the low pressure region.
- (v) Effect of temperature : An increase in temperature shows a decrease in deviation from ideal behaviours, Z approaches unity with increase in temperature.



1.4 VAN DER WAAL EQUATION OF REAL GASES :

The ideal gas equation does not consider the effect of attractive forces and molecular volume.

Van der Waal corrected the ideal gas equation by taking the effect of

- (a) Molecular volume
- (b) Molecular attraction

(A) **Volume correction :**

In the ideal gas equation, $P_i V_i = nRT$, V_i represents the ideal volume where the molecules can move freely. In real gases, a part of the total volume is occupied by the gas molecules. Hence the free volume V_i is the total volume V minus the volume occupied by the gas molecules.

Real volume of gas = Actual volume of container – volume occupied by molecules in motion.

$$V_i = V - nb \quad \text{for } n \text{ mole of gas}$$

Where b is termed the 'excluded volume' or 'co-volume' per mole.

It is constant and characteristic for each gas.

$$b = 4 \times \text{volumes of one molecules} \times N_A$$

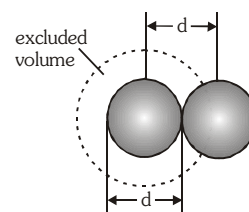
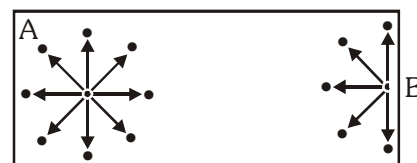


Illustration of excluded volume

(B) **Pressure correction :**

In order to take account the effect of intermolecular forces of attraction, let us consider a molecule A in the midst of the vessel.

This molecule is surrounded by other molecules in a symmetrical manner and is being attracted uniformly on all sides by the neighbouring molecules with the result that this molecule on the whole experiences no net force of attraction.



Now, consider a molecule B near the side of the vessel, which is about to strike one of its sides, thus contributing towards the total pressure of the gas. There are molecules only in one side of the vessel, i.e. towards its centre, with the result of that, this molecule experiences a net force of attraction towards the centre of the vessel. This results in decreasing the velocity of the molecule, and hence its momentum. Thus, the molecule does not contribute as much force as it would have, had there been no force of attraction. Thus, the pressure of a real gas would be smaller than the corresponding pressure of an ideal gas.

Van der Waals noted that the total force of attraction on any molecule about to hit a wall is proportional to the concentration of neighbouring molecules, n/V . However, the number of molecules about to hit the wall per unit wall area is also proportional to the concentration n/V . Therefore, the force per unit wall area, or pressure, is reduced from that assumed in the ideal gas wall by a factor proportional to n^2/V^2 . Letting a be the proportionality constant, we can write

$$P(\text{actual}) = P(\text{ideal}) - \frac{an^2}{V^2}$$

$$\text{or } P(\text{ideal}) = P(\text{actual}) + \frac{an^2}{V^2}$$

$$\text{or } P_i = P + \frac{an^2}{V^2}$$

'a' is a constant which depends upon the nature of the gas, Greater the intermolecular attraction, greater the volume of 'a'.

Combining the two corrections,

$$\Rightarrow P_i \cdot V_i = nRT$$

$$\text{or, } \left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT$$

For 1 mole of gas

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

1.4.1 The constants 'a' and 'b' :

Van der Waals constant for attraction 'a' and volume 'b' are characteristic constants for a given gas, independent from pressure and temperature conditions.

- The 'a' values for a given gas are measure of intermolecular forces of attraction. More are the intermolecular forces of attraction, more will be the value of a.
- The gas having higher value of 'a' can be liquefied easily.
- Unit of 'a' is $\text{atm L}^2 \text{mole}^{-2}$ or $\text{dyne cm}^4 \text{mole}^{-2}$ or $\text{Nm}^4 \text{mol}^{-2}$
- Unit of 'b' is L mole^{-1} or $\text{cm}^3 \text{mole}^{-1}$ or $\text{m}^3 \text{mol}^{-1}$

1.4.2 The van der waal's constants for some common gases

Gas	a (atmL ² mol ⁻²)	b (L mol ⁻¹)
Ammonia	4.17	0.0371
Argon	1.337	0.0322
Carbon dioxide	3.59	0.0427
Carbon monoxide	1.49	0.0399
Chlorine	6.49	0.0562
Ethane	5.49	0.0638
Ethanol	2.56	0.087
Ethylene	4.552	0.0582
Helium	0.034	0.0237
Hydrogen	0.024	0.0266
Hydrogen chloride	3.67	0.0408
Hydrogen bromide	4.45	0.0433
Methane	2.25	0.0428
Neon	0.21	0.0171
Nitric oxide	1.34	0.0279
Nitrogen	1.39	0.0319
Oxygen	1.36	0.0318
Sulphur dioxide	3.71	0.0564
Water	5.44	0.0305

(iii) **At low pressure and very high temperature.**

V_m will be very large, hence 'b' can be neglected and $\frac{a}{V_m^2}$ can also be neglected as V_m is very large.

$$PV_m = RT \text{ (ideal gas condition)}$$

(iv) For H_2 or He $a \approx 0$ because molecules are smaller in size or vander Waal's forces will be very weak.

$$P(V_m - b) = RT$$

So $Z = 1 + \frac{Pb}{RT}$

This explains type I plot.

1.5 LIMITATIONS OF VANDER WAAL'S GAS EQUATION :

- Vander waal considered only attractive forces in gas molecules, but in reality, the molecules experience attractive as well as repulsive forces. The positive deviation ($Z > 1$) was explained due to finite size of particles ('b' is not negligible relative to V_m)
- The constants 'a' and 'b' are considered as characteristic constants for a gas, independent from temperature and pressure. But in reality, they should change with conditions.
- At critical point (discussed later in this sheet, 3.2), Z becomes 3/8 for all gases obeying Vander waal's gas equation. But in reality, different gases have different values of Z at critical point and none of them have the value 3/8.

1.6 SOME IMPORTANT INFORMATIONS :

- There is no any general equation relating P, V and T of gases, obeyed by all the gas, in all conditions.
- Gases obeying $PV = nRT$ in all the conditions are called **ideal** or **perfect** gases and none of the gas in nature is ideal gas.
- Gases obeying $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ in all the conditions are called **Vander waal** gases and none of the gas in nature is Vander waal gas.
- Virial form of gas equation :**

$$PV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Where B = second virial coefficient

C = third virial coefficient

Ex.1. Calculate the pressure exerted by 5 mole of CO_2 in one litre vessel at $47^\circ C$ using van der waal's equation. Also report the pressure of gas if it behaves ideal in nature.

Given that $a = 3.592 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 0.0427 \text{ L/mol}$. Also, if the volume occupied by CO_2 molecules is negligible, then calculate the pressure exerted by one mole of CO_2 gas at 273 K .

Sol. Vander waal's equation

$$\left[p + \frac{n^2 a}{V^2} \right] [V - nb] = nRT$$

$$n_{\text{CO}_2} = 5, V = 1 \text{ litre}, T = 320 \text{ K}, a = 3.592, b = 0.0427$$

$$\therefore \left[P + 25 \times \frac{3.592}{1} \right] [1 - 5 \times 0.0427] = 5 \times 0.0821 \times 320$$

$$\therefore P = 77.218 \text{ atm}$$

For ideal behaviour of gas, $PV = nRT$

$$\therefore P \times 1 = 5 \times 0.0821 \times 320$$

$$\therefore P = 131.36 \text{ atm}$$

$$\text{For one mole } \left[P + \frac{a}{V^2} \right] [V - b] = RT$$

$$\therefore P = \frac{RT}{V} - \frac{a}{V^2}$$

$$\therefore P = \frac{0.0821 \times 273}{22.4} - \frac{3.592}{(22.4)^2}$$

$$\therefore P = 0.9922 \text{ atm}$$

The volume occupied by 1 mole at 273 K is 22.4 litre if b is negligible.

Ex.2 One mole of CCl_4 vapours at 77°C occupies a volume of 35.0 L. If vander waal's constants are $a = 20.39 \text{ L}^2 \text{ atm mol}^{-2}$ and $b = 0.1383 \text{ L mol}^{-1}$, calculate compressibility factor Z under, (a) low pressure region. (b) high pressure region.

Sol. (a) Under low pressure region, V is high

$$\therefore (V - b) \approx V$$

$$\left(P + \frac{a}{V^2} \right) V = RT$$

$$PV + \frac{a}{V} = RT$$

$$\frac{PV}{RT} + \frac{a}{RTV} = 1$$

$$Z = \frac{PV}{RT} = \left(1 - \frac{a}{RTV} \right) = 1 - \frac{20.39}{0.0821 \times 350 \times 35} = 0.98$$

(b) Under high pressure region, P is high,

$$\left(P + \frac{a}{V^2} \right) \approx P$$

$$\therefore P(V - b) = RT$$

$$PV - Pb = RT$$

$$\frac{PV}{RT} - \frac{Pb}{RT} = 1$$

$$\therefore Z = \frac{PV}{RT} = 1 + \frac{Pb}{RT} \quad \left(\because \frac{PV}{RT} = 1, \text{ or } \frac{P}{RT} = \frac{1}{V} \right)$$

$$Z = 1 + \frac{b}{V} = 1 + \frac{0.1383}{35} = 1 + 0.004 = 1.004$$

Ex.3 One way of writing the equation of state for real gases is $PV = RT \left[1 + \frac{B}{V} + \dots \right]$ where B is a constant and V is the molar volume of gas. Derive an expression for B in terms of van der Waal's constants a and b .

Sol. According to van der Waal's equation

$$\left[P + \frac{a}{V^2} \right] [V - b] = RT \text{ or } P = \frac{RT}{(V-b)} - \frac{a}{V^2}$$

Multiply by V , then

$$PV = \frac{RTV}{(V-b)} - \frac{a}{V} \quad \text{or} \quad PV = RT \left[\frac{V}{V-b} - \frac{a}{VRT} \right]$$

$$\text{or } PV = RT \left[\left(1 - \frac{b}{V} \right)^{-1} - \frac{a}{VRT} \right]$$

$$\therefore \left[1 - \frac{b}{V} \right]^{-1} = 1 + \frac{b}{V} + \left(\frac{b}{V} \right)^2 + \dots$$

$$\therefore PV = RT \left[1 + \frac{b}{V} + \dots - \frac{a}{VRT} \right]$$

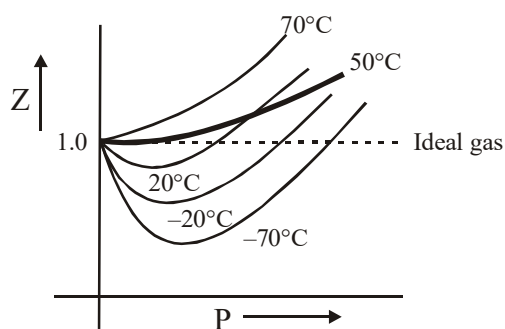
$$PV = RT \left[1 + \left(b - \frac{a}{RT} \right) \cdot \frac{1}{V} + \dots \right]$$

Comparing with the given equation :

$$\therefore B = b - \frac{a}{RT}$$

2. BOYLE TEMPERATURE :

- (i) It is the temperature at which a real gas behave ideally in a wide range of pressure, in low pressure region.
- (ii) (a) Temperature < Boyle temperature
 $Z < 1$, low pressure range
 $Z > 1$, high pressure range
- (b) Temperature = Boyle temperature
 $Z = 1$, low pressure range
 $Z > 1$, high pressure range
- (c) At temperature > Boyle temperature
 $Z > 1$, at all pressure
- (d) $T \rightarrow \infty$, $Z \rightarrow 1$



On increasing the temperature, the thermal energy increases and simultaneously the attractive forces decreases. Hence a temperature comes at which the thermal energy become too high that it balances the effect of attraction and gas molecules becomes independent.

If at Boyle temperature, pressure is increased, molecules come more closer. Due to repulsive force, Z becomes greater than 1.

2.1 CALCULATION OF T_B :

$$(i) \quad PV_m = RT = \left(1 + \frac{B}{V_m} + \underbrace{\frac{C}{V_m^2} + \dots}_{\text{negligible}} \right) \text{ at low pressure.}$$

At $T = T_B$, the second virial coefficient should be 0

$$B = 0$$

$$\text{or} \quad b - \frac{a}{RT_B} = 0$$

$$\therefore \quad \boxed{T_B = \frac{a}{Rb}}$$

(ii) **Calculus method :**

$$\text{At Boyle temperature, } \left(\frac{\partial Z}{\partial P} \right)_T = 0 \text{ at low pressure.}$$

Ex.4 Derive the expression for compressibility factor of a vanderwaal gas at Boyle temperature.

Solution : $Z = \frac{V_m}{V_m - b} - \frac{a}{V_m - RT}$

At Boyle temperature,

$$Z = \frac{V_m}{V_m - b} - \frac{a}{V_m R \times \frac{a}{Rb}}$$

$$Z = \frac{V_m}{V_m - b} - \frac{b}{V_m}$$

$$Z = \frac{V_m^2 - V_m b + b^2}{V_m (V_m - b)}$$

$$Z = 1 + \frac{b^2}{V_m (V_m - b)}$$

Ex.5 Calculate the volume occupied by 2 moles of a vanderwall gas at 5 atm 800 K.

Given : $a = 4.0 \text{ atm } \ell^2 \text{ mol}^{-2}$, $b = 0.0625 \text{ } \ell \text{ mol}^{-1}$, $R = 0.08 \text{ } \ell \text{-atm/K-mol}$

Solution : $T_B = \frac{a}{Rb} = 800\text{K}$

Gas behave ideally at given condition.

$$PV = nRT$$

$$5 \times V = 2 \times 0.08 \times 800$$

$$V = 25.6 \text{ litre}$$

3. LIQUEFACTION OF GASES AND CRITICAL CONSTANTS :

The phenomenon of converting a gas into liquid is known as liquefaction. The liquefaction of a gas takes place when the intermolecular forces of attraction become so high that they exist in liquid state. A gas can be liquefied by :

- Increasing pressure** : An increase in pressure results decrease in intermolecular distance.
- Decreasing temperature** : A decrease in temperature results decrease in kinetic energy of molecules.

Note : Due to absence of intermolecular forces, ideal gases can never be liquified.

3.1 Andrews Isotherms :

The essential conditions for liquefaction of gases were discovered by Andrews (1869) as a result of his study of P–V–T relationship for CO_2 . The types of isotherms are shown in figure.

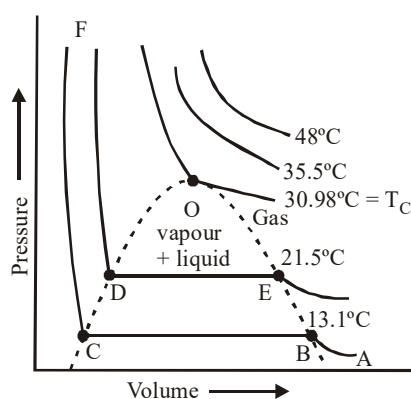


Fig : Isotherms for carbon dioxide

Observations from figure :

(a) At low temperatures :

For the curve ABCF, as the pressure increases, volume of the gas decreases (curve A to B). At point B, at constant pressure, liquefaction commences and the volume decreases rapidly (because gas is converted to liquid with higher density). At point C, liquefaction is complete. The line CF represents the variation of V with P of the liquid state. The steepness of the line CF indicates that the liquid cannot be easily compressed. Thus AB, represent gaseous state, BC represent liquid and vapour in equilibrium and CF represent liquid state.

The pressure corresponding to the line BC is vapour pressure of the liquid at that temperature.

(b) At lower temperatures :

Similar type of curve as in case (A) is obtained but the width of the horizontal portion is reduced. The pressure corresponding to this portion is higher than at lower temperatures.

(c) At high temperatures :

(say 48°C), the isotherms are like those of ideal gas. Gas does not liquify, even at very high pressure.

(d) At temperature (30.98°C) : The horizontal portion is reduced to a point.

The isotherm at T_C is called **critical isotherm**.

3.2 Critical parameters or critical constants :

(i) **Critical temperature (T_C) :**

The temperature above which a system can never be liquefied by the application of pressure alone i.e. the temperature above which the liquid cannot exist is called the critical temperature T_C .

(ii) **Critical pressure (P_C) :**

The minimum pressure required to liquefy the gas at critical temperature (T_C) is called the critical pressure, P_C .

(iii) **Critical volume (V_C) :**

The volume occupied by one mole of the system at critical temperature, (T_C) and critical pressure, (P_C) is called the critical volume (V_C) of the gas.

T_C , P_C and V_C are collectively called **critical parameters** or **critical constants of the gas**.

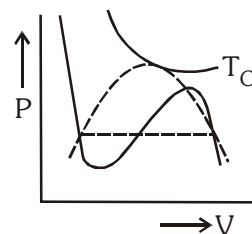
3.3 Determination of value of P_C , V_C and T_C :

The Vander waal's equation is

$$\left(P + \frac{a}{V_m^2}\right) (V_m - b) = RT$$

or
$$V_m^3 - \left(b + \frac{RT}{P}\right) V_m^2 + \frac{a}{P} V_m - \frac{ab}{P} = 0 \quad \dots (1)$$

This equation has three roots in V_m for given values of a , b , P and T . It is found that either all the three roots are real or one is real and the other two are imaginary.



At temperature lower than T_C , the isotherm exhibits a maximum and a minimum for certain values of pressures, the equation gives three real roots of volume e.g., V_1 , V_2 and V_3 at pressure P_1 . On increasing the temperature, the three roots become closer to each other and ultimately at critical temperature, they become identical. Thus, the cubic equation V_m can be written as

$$(V_m - V') (V_m - V'') (V_m - V''') = 0$$

At the critical point $V' = V'' = V''' = V_C$, and hence the equation becomes,

$$(V_m - V_C)^3 = 0$$

or
$$V_m^3 - V_C^3 - 3V_C V_m^2 + 3V_C^2 V_m = 0 \quad \dots (2)$$

By comparing the coefficients in eq.(1) and eq(2)

$$3V_C = b + \frac{RT_C}{P_C}, \quad 3V_C^2 = \frac{a}{P_C}, \quad V_C^3 = \frac{ab}{P_C}$$

By solving, $V_C = 3b$, $P_C = \frac{a}{27b^2}$ and $T_C = \frac{8a}{27Rb}$

3.4 The value of critical compressibility factor in terms of vander waal's constants is given by

$$Z = \frac{P_c V_c}{RT_c} = \frac{\frac{a}{27b^2} \times 3b}{R \times \frac{8b}{27Rb}} = \frac{3}{8} = 0.375$$

If we compare the value of $\frac{P_c V_c}{RT_c} = 0.375$, with the experimental values, it has been found that the agreement is very poor.

Critical constants of gases

Substance	T_c / K	P_c / bar	$V_c / \text{dm}^3 \text{mol}^{-1}$
H_2	33.2	12.97	0.0650
He	5.3	2.29	0.0577
N_2	126.0	33.9	0.0900
O_2	154.3	50.4	0.0744
CO_2	304.10	73.9	0.0956
H_2O	647.1	220.6	0.0450
NH_3	405.5	113.0	0.0723

Ex.6 The critical temperature and pressure of CO_2 gas are 304.2 K and 72.9 atm respectively. What is the radius of CO_2 molecule assuming it to behave as vander Waal's gas ?

Sol. $T_c = 304.2 \text{ K}$ $P_c = 72.9 \text{ atm}$

$$T_c = \frac{8a}{27Rb} \quad P_c = \frac{a}{27b^2}$$

$$\therefore \frac{T_c}{P_c} = \frac{\frac{8a}{27Rb}}{\frac{a}{27b^2}} = \frac{8a}{27Rb} \times \frac{27b^2}{a} = \frac{8b}{R} \quad \text{or} \quad b = \frac{RT_c}{8P_c} = \frac{1}{8} \times \frac{0.082 \times 304.2}{72.9} = 0.04277 \text{ lit mol}^{-1}$$

$$b = 4 N_A \times \frac{4}{3} \pi r^3 = 42.77 \text{ cm}^3$$

$$\text{or } r = (4.24)^{1/3} \times 10^{-8} \text{ cm} = 1.62 \times 10^{-8} \text{ cm}$$

$$\therefore \text{radius of } CO_2 \text{ molecule} = 1.62 \text{ \AA}$$

SELF STUDY

4. THE LIQUID STATE

Liquid state is intermediate between gaseous and solid states. The liquids possess fluidity like gases but incompressibility like solids.

The behaviour of liquids explained above gives some characteristic properties to the liquids such as definite shape, incompressibility, diffusion, fluidity (or viscosity), evaporation (or vapour pressure), surface tension, etc.

The following general characteristics are exhibited by liquids :

(i) **Shape :**

Liquids have no shape of their own but assume the shape of the container in which they are kept. No doubt, liquids are mobile but they do not expand like gases as to fill up all the space offered to them but remain confined to the lower part of the container.

(ii) **Volume :**

Liquids have definite volume as the molecules of a liquid are closely packed and the cohesive forces are strong.

(iii) **Density :**

As the molecules in liquids are closely packed, the densities of liquids are much higher than in gaseous state. For example, density of water at 100° C and 1 atmospheric pressure is 0.958 g mL⁻¹ while that of water vapour under similar conditions as calculated from ideal gas law $\left(d = \frac{MP}{RT}\right)$ is 0.000588 g mL⁻¹.

(iv) **Compressibility :**

The molecules in a liquid are held in such close contact by their mutual attractive forces (cohesive forces) that the volume of any liquid decreases very little on increasing pressure. Thus, liquids are relatively incompressible compared to gases.

(v) **Diffusion :**

When two miscible liquids are put together, there is slow mixing as the molecules of one liquid move into the other liquid. As the space available for movement of molecules in liquids is much less and their velocities are small. Liquids diffuse slowly in comparison to gases.

(vi) **Evaporation :**

The process of changes of liquid into vapour state on standing is termed **evaporation**. Evaporation may be explained in terms of motion of molecules. At any given temperature all the motion of molecules do not possess the same kinetic energy (average kinetic energy is, however same). Some molecules move slowly, some at intermediate rates and some move very fast. A rapidly moving molecule near the surface of the liquid may possess sufficient kinetic energy to overcome the attraction of its neighbours and escape. Evaporation depends on the following factor :

(a) **Nature of the liquid :** The evaporation depends on the strength of intermolecular forces (cohesive forces). The liquids having low intermolecular forces evaporate faster in comparison to the liquids having high intermolecular forces. For example, ether evaporates more quickly than alcohol, and alcohol evaporates more quickly than water, as the intermolecular forces in these liquids are in the order :



(b) **Surface area :** Evaporation is a surface phenomenon. Larger the surface area, greater is the opportunity of the molecules to escape. Thus, rate of evaporation increases with increase of surface area.

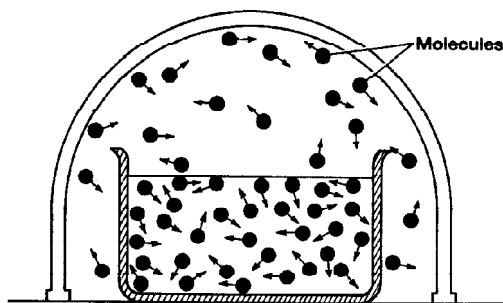
(c) **Temperature :** Rate of evaporation increases with the increase of temperature as the kinetic energy of the molecules increases with the rise of temperature.

(d) **Flow of air current over the surface :** Flow of air helps the molecules to go away from the surface of liquid and, therefore, increases the evaporation of liquid in open vessel.

(vii) **Latent heat of vaporisation :**

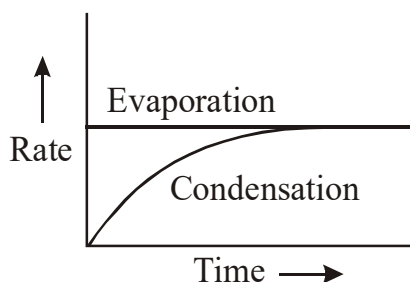
The quantity of heat required to vaporise unit mass of a given liquid at constant temperature is known as latent heat of vaporisation. The latent heat of vaporisation depends on the strength of the intermolecular forces within the liquid. The value of latent heat of vaporisation decreases with increase in temperature and it becomes zero at the critical temperature. When the vapour is condensed into a liquid, heat is evolved. This is called **heat of condensation**. It is numerically equal to the heat of vaporisation at the same temperature.

(viii) **Vapour pressure :**



When the space above the liquids is closed, the molecules cannot escape into open but strike the walls of the container, rebound and may strike the surface of the liquid, where they may be trapped. The return of the molecules from the vapour state to the liquid state is known as **condensation**. As evaporation proceeds, the number of molecules in the vapour state increases and, in turn, the rate of condensation increases.

The rate of condensation soon becomes equal to the rate of the evaporation, i.e., the vapour in the closed container is in equilibrium with the liquid.



At equilibrium the concentration of molecules in the vapour phase remains unchanged. The pressure exerted by the vapour in equilibrium with liquid, at a given temperature, is called the **vapour pressure**.

Since the rate of evaporation increases and rate of condensation decreases with increasing temperature, vapour pressure of liquids always increases as temperature increases. At any given temperature, the vapour pressures of different liquids are different because their cohesive forces are different. Easily vapourisable liquids are called **volatile liquids** and they have relatively high vapour pressure. Vapour pressure values (in mm of Hg) for water, alcohol and ether at different temperatures are given in the following table :

Substance	Temperatures				
	0° C	20° C	40° C	80° C	100° C
Water	4.6	17.5	55.0	355.5	760.3
Ethyl alcohol	12.2	43.9	812.6	1693.3	
Diethyl ether	185.3	442.2	921.1	2993.6	4859.4

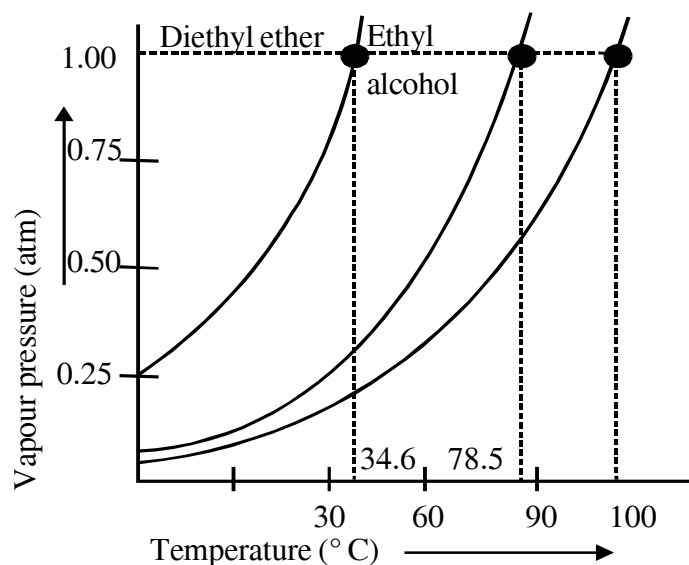
The vapour pressure of a given liquid at two different temperatures can be compared with the help of **Clausius-Clapeyron equation**.

$$\ln \frac{P_2}{P_1} = \frac{\Delta H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Where ΔH is the latent heat of vaporisation and R is the molar gas constant.

(ix) Boiling point :

The temperature at which the vapour pressure of the liquid becomes equal to the atmospheric pressure is called the **boiling point** of the liquid. When a liquid is heated under a given applied pressure, bubbles of vapour begin to form below the surface of the liquid. They rise to the surface and burst releasing the vapour into the air. This process is called **boiling**. The normal boiling point is the temperature at which the vapour pressure of a liquid is equal to one atmospheric pressure (760 mm of Hg). Figure shows that normal boiling points of di-ethyl ether, ethyl alcohol and water are 34.6° C, 78.5° C and 100° C respectively.



The temperature of the liquid remains constant until all the liquid has been vaporised. Heat must be added to the boiling liquid to maintain the temperature because in the boiling process, the high energy molecules are lost by the liquid.

The boiling point of a liquid changes with the change in external pressure. A liquid may boil at temperatures higher than normal under external pressures greater than one atmosphere. Conversely, the boiling point of a liquid may be lowered than normal below one atmosphere. Thus, at high altitudes where the atmospheric pressure is less than 760 mm, water boils at temperatures below its normal boiling water.

Boiling and evaporation are similar processes (conversion of liquid into vapour) but differ in following respects :

- Evaporation takes place spontaneously at all temperatures but boiling occurs at a particular temperature at which the vapour pressure is equal to the atmospheric pressure.
- Evaporation is surface phenomenon. It occurs only at the surface of the liquid whereas boiling involves formation of bubbles below the surface of the liquid.

Note. : Boiling does not occur when liquid is heated in a closed vessel.

(x) Freezing point :

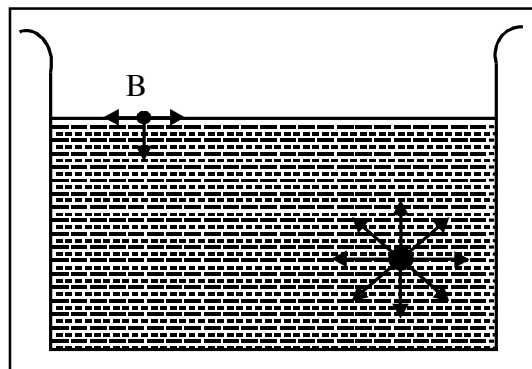
When a liquid is cooled, the kinetic energy of the molecules goes on decreasing. A stage comes when the intermolecular forces become stronger and the rotational motion is siezed. At this stage, the formation of solid begins and the liquid is seen to freeze out. The point (temperature) at which the vapour pressure of solid and liquid forms of a substance become equal is termed as **freezing point**.

Normal freezing point of a liquid is the temperature at which is liquid and solid forms are in equilibrium with each other under a pressure of one atmosphere. The freezing point of a liquid is the same as the melting point of its solid. The amount of heat that must be removed to freeze a unit mass of the liquid at the freezing point, which is called the **latent heat of fusion**.

The freezing point of a liquid is affected by the change of external pressure. With increased external pressure, the freezing point of some liquids rises while of others falls. But the effect of pressures is very small because solid as well as liquid are almost incompressible.

(xi) Surface tension :

It is the property of liquids caused by the intermolecular attractive forces. A molecule within the bulk of the liquid is attracted equally in all the directions by the neighbouring molecules. The resultant force on any one molecule in the centre of the liquid is, therefore, zero. However, the molecules on the surface of the liquid are attracted only inward and sideways. This unbalanced molecular attraction pulls some of the molecules into the bulk of the liquid, i.e., are pulled inward and the surface area is minimized.



Surface tension is a measure of this inward force on the surface of the liquid. It acts downwards perpendicular to the plane of the surface. The unit of surface tension is dyne cm^{-1} . Surface tension is, defined as **the force acting on the surface at right angles to any line of unit length.**

As the intermolecular forces of attraction decreases with the rise of temperature, the surface tension of a liquid, thus, decreases with increase in temperature. Similarly, addition of chemicals to a liquid may reduce its surface tension. For example, addition of chemicals like soaps, detergents, alcohol, camphor, etc., lowers the surface tension of water.

Many common phenomenon can be explained with the help of surface tension. Some are described here :

(a) Small droplets are spherical in shape :

The surface tension acting on the surface of the liquid tries to minimise the surface area of a given mass of a liquid. It is known that for a given volume, a sphere has the minimum surface area. On account of this, drops of liquids acquire a spherical shape.

(b) Insects can walk on the surface of water :

Many insects can walk on the surface of water without drowning. This is due to the existence of surface tension. The surface tension makes the water surface to behave like an elastic membrane and prevents the insects from drowning.

(c) Cleaing action of soap and detergents :

Soap and detergent solutions due to their lower surface tensions penetrate into the fibre and surround the greasy substances and wash them away.

(d) Capillary action :

The tendency of a liquid to rise into narrow pores and tiny openings is called capillary action. The liquids rise in the capillary tubes due to the surface tension.

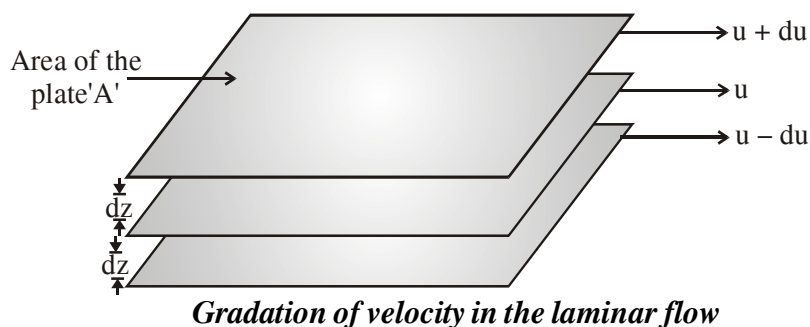
(e) Surface Energy :

The work required to be done to increase or extend surface area by unit area is called surface energy. The units of surface energy is joule per sq. metre, i.e. J m^{-2} in S.I. system.

(xii) Viscosity :

Viscosity is a measure of resistance to flow which arises due to the internal friction between layers of fluid as they slip past one another while liquid flows. Strong intermolecular forces between molecules hold them together and resist movement of layers past one another.

When a liquid flows over a fixed surface, the layer of molecules in the immediate contact of surface is stationary. The velocity of upper layers increase as the distance of layers from the fixed layer increases. This type of flow in which there is a regular gradation of velocity in passing from one layer to the next is called **laminar flow**. If we choose any layer in the flowing liquid, the layer above it accelerates its flow and the layer below this retards its flow.



If the velocity of the layer at a distance dz is changed by a value du then velocity gradient is given by the amount $\frac{du}{dz}$. A force is required to maintain the flow of layers. This force is proportional to the area of contact of layers and velocity gradient i.e.

$F \propto A$ (A is the area of contact)

$F \propto A \frac{du}{dz}$ (where, $\frac{du}{dz}$ is velocity gradient; the change in velocity with distance)

$$F \propto A \cdot \frac{du}{dz} \Rightarrow F = \eta A \frac{du}{dz}$$

' η ' is proportionality constant and is called **coefficient of viscosity**. Viscosity coefficient is the force when velocity gradient is unity and the area of contact is unit area. SI unit of viscosity coefficient is 1 newton second per square metre (N s m^{-2}) = pascal second ($\text{Pa s} = 1 \text{ kg m}^{-1} \text{ s}^{-1}$).

Greater the viscosity, the more slowly the liquid flows. Hydrogen bonding and van der Waals forces are strong enough to cause high viscosity. Glass is an extremely viscous liquid. It is so viscous that many of its properties resemble solids. However, property of flow of glass can be experienced by measuring the thickness of windowpanes of old buildings. These become thicker at the bottom than at the top.

Viscosity of liquids decreases as the temperature rises because at high temperature molecules have high kinetic energy and can overcome the intermolecular forces to slip past one another between the layers.

3. Calculate vander waal's constant a and b if critical temperature and critical pressure are 30°C and 72 atm respectively.

Sol. $T_c = \frac{8a}{27Rb}$ and $P_c = \frac{a}{27b^2}$

$$\therefore \frac{T_c}{P_c} = \frac{8b}{R} \text{ or } b = \frac{RT_c}{8P_c}$$

$$\therefore b = \frac{0.0821 \times 303}{8 \times 72} = 0.043 \text{ litre mol}^{-1}$$

$$\alpha = 27P_c \times b^2 = 27 \times 72 \times (0.043)^2 \\ = 3.59 \text{ litre}^2\text{atm mol}^{-2}$$

4. Using vander waal's equation, calculate the constant, ' a ' when two mole of a gas confined in a four litre temperature of 300K . The value of ' b ' is $0.05 \text{ litre mol}^{-1}$.

Sol. Vander waal's equation for n mole of gas is

$$\left[P + \frac{n^2 a}{V^2} \right] [V - nb] = nRT$$

Given $V = 4 \text{ litre}$; $P = 11.0 \text{ atm}$, $T = 300 \text{ K}$

$b = 0.05 \text{ litre mol}^{-1}$; $n = 2$

$$\text{Thus, } \left[11 + \frac{2^2 a}{4^2} \right] [4 - 2 \times 0.05] = 2 \times 0.082 \times 300.$$

$$\therefore a = 6.46 \text{ atm litre}^2 \text{ mol}^2$$

5. One way of writing the equation of state for real gases is,

[JEE 1997]

$$P \bar{V} = RT \left[1 + \frac{B}{\bar{V}} + \dots \right] \text{ where } B \text{ is a constant.}$$

Derive an approximate expression for ' B ' in terms of van der Waals' constants ' a ' & ' b '.

Ans. $B = \left(b - \frac{a}{RT} \right)$

Sol. $\left[P + \frac{a}{V_m^2} \right] [V_m - b]$

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$\frac{PV_m}{RT} = \frac{V_m}{V_m - b} - \frac{a}{V_m RT}$$

$$Z = \left(1 - \frac{b}{V_m} \right)^{-1} - \frac{a}{V_m RT}$$

$$= 1 + \frac{b}{V_m} + \frac{b^2}{V_m^2} + \dots - \frac{a}{V_m RT}$$

$$= 1 + \left[b - \frac{a}{RT} \right] \frac{1}{V_m} + \frac{b^2}{V_m^2} + \dots \quad \dots (1)$$

$$\text{Also, } p\bar{V} = RT \left[1 + \frac{B}{V} + \dots \right]$$

$$\Rightarrow \frac{p\bar{V}}{RT} = \left[1 + \frac{B}{V} + \dots \right] \quad \dots (2)$$

$\Rightarrow$ Comparing equation (1) & (2), we get

$$B = b - \frac{a}{RT}$$

6. Using Vander Waals equation, calculate the constant "a" when 2 moles of a gas confined in a 4 litre flask exerts a pressure of 11.0 atm at a temperature of 300 K. The value of "b" is 0.05 litre mol⁻¹. [JEE 1998]

Ans. 6.52 atm L² mol⁻²

Sol. $\left(P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$

$$\left[11 + \frac{2^2 \times a}{4 \times 4} \right] [4 - 2 \times 0.05] = 2 \times 0.0821 \times 300$$

$$\Rightarrow a = 6.52 \text{ atm L}^2 \text{ mol}^{-2}$$

7. A gas will approach ideal behaviour at :

[JEE 1999]

- (A) low temperature and low pressure
- (B) low temperature and high pressure
- (C) low pressure and high temperature
- (D) high temperature and high pressure .

Ans. (C)

Sol. A gas approaches ideal behavior at High T & low P

8. The compressibility of a gas is less than unity at STP. Therefore ,

[JEE 2000]

(A) $V_m > 22.4 \text{ L}$

(B) $V_m < 22.4 \text{ L}$

(C) $V_m = 22.4 \text{ L}$

(D) $V_m = 44.8 \text{ L}$

Ans. (B)

Sol. $Z = \frac{(V_m)_{\text{real}}}{(V_m)_{\text{ideal}}} ; Z < 1$

$$\Rightarrow (V_m)_{\text{real}} < (V_m)_{\text{ideal}}$$

$$\therefore (V_m)_{\text{real}} < 22.4 \text{ L}$$

EXERCISE # S-I

1. Calculate the pressure (in atm) exerted by 22 g of carbon dioxide in 0.5 dm³ at 300 K using:
 - (a) the ideal gas law and
 - (b) Van der Waal's equation respectively.

Given : [a = 3.6 atm litre²mol⁻² , b = 0.04 litre mol⁻¹, R = 0.08 L-atm/K-mol]

RG0001

2. Calculate from the Van der Waals equation, the temperature (in Kelvin) at which 192 g of SO_2 would occupy a volume of 6 dm^3 at 15 atm pressure.

$$[a = 5.68 \text{ atm L}^2 \text{ mol}^{-2}, b = 0.06 \text{ L mol}^{-1}, R = 0.0821 \text{ L-atm/K-mol}]$$

RG0002

3. The density of water vapour at 328.0 atm and 800 K is 135.0 g/dm³. Determine (i) the molar volume (in L mol⁻¹) and (ii) the compression factor of water vapour. (R = 0.082 L-atm/K-mol)

RG0003

4. At 300 K and under a pressure of 10.1325 MPa, the compressibility factor of O_2 is 0.9. Calculate the mass (in kg) of O_2 necessary to fill a gas cylinder of 45 dm^3 capacity under the given conditions.

[R = 0.08L-atm/K-mol]

RG0004

5. 1 mole of CCl_4 vapours at 27°C occupies a volume of 40 L. If Van der Waals constants are $24.6 \text{ L}^2 \text{ atm mol}^{-2}$ and 0.5 L mol^{-1} , then, calculate compressibility factor in

(a) Low pressure region (b) High Pressure region [R = 0.082 L -atm/K-mol]

RG0005

6. If at 200 K & 500 atm, density of CH_4 is 0.25 gm/ml then its compressibility factor (Z) is (R = 0.08 L-atm/K-mol)

RG0006

7. Certain mass of a gas occupy 500 ml at 2 atm and 27°C. Calculate the volume (in L) occupied by same mass of the gas at 0.3 atm and 227°C. The compressibility factors of gas at the given condition are 0.8 and 0.9, respectively. **RG0007**

RG0007

BOYLE TEMPERATURE

8. The vander waal's constant for a gas are $a = 1.92 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 0.06 \text{ L mol}^{-1}$. If $R = 0.08 \text{ L atm K}^{-1} \text{ mol}^{-1}$, what is the Boyle's temperatrue (in K) of this gas.

RG0008

9. The Van der Waals constant for O_2 are $a = 1.64 \text{ atm L}^2 \text{ mol}^{-2}$ and $b = 0.04 \text{ L mol}^{-1}$. Calculate the temperature (in $^{\circ}\text{C}$) at which O_2 gas behaves ideally for long range of pressure, in low pressure region.

(R = 0.082 L-atm/K-mol)

RG0009

LIQUIFICATION OF GASES, CRITICAL PHENOMENON

10. The Van der Waals constants for gases A, B and C are as follows

Gas	a [atm L ² mol ⁻²]	b [L mol ⁻¹]
A	8.21	0.050
B	4.105	0.030
C	1.682	0.040

Which gas has (i) the highest critical temperature, (ii) the largest molecular volume, and (iii) most ideal behaviour around 500 K?

RG0010

11. For a real gas, if at critical conditions molar volume of gas is 8.2 litre at 3 atm, then critical temperature (in K) will be :

RG0011

12. An unknown gas behaves ideally at 540K in low pressure region, then calculate the temperature (in K) below which it can be liquified by applying pressure.

RG0012

EXERCISE # S-II

1. The density of mercury is 13.6 g/cm^3 . Estimate the value of 'b' (in cm^3/mole). ($Hg = 200$)
RG0013
2. The molarity of O_2 gas at 72 atm and 300K is 6M. Calculate the value of Z for O_2 .
(Use : $R = 0.08 \text{ atm-litre/K-mole}$).
RG0014
3. Calculate the amount of He (in gm) present in the 10 litre container at 240 atm and 300K. Given value of "b" for He is $0.08 \text{ dm}^3 \text{ mol}^{-1}$. ($R = 0.08 \text{ atm lit mol}^{-1} \text{ K}^{-1}$)
RG0015
4. For a real gas (mol. mass = 30) if density at critical point is 0.40 g/cm^3 and its $T_c = \frac{10^4}{41} \text{ K}$, then calculate Van der Waals constant a (in $\text{atm L}^2\text{mol}^{-2}$).
RG0016
5. Calculate the volume (in L) occupied by 0.2 mole of a Vander waal gas at 27°C and 0.082 atm.
 $[a = 4.10 \text{ L}^2 \text{ atm mol}^{-2}, b = \frac{1}{6} \text{ Lmol}^{-1}]$
RG0017
6. At what pressure (in atm) and 127°C , the density of O_2 gas becomes 1.6 g/L ?
 $[a = 4.0 \text{ atm L}^2 \text{ mol}^{-2}, b = 0.4 \text{ Lmol}^{-1}, R = 0.08 \text{ L - atm/ K-mol}]$
RG0018

EXERCISE # O-I

1. The correct expression for the Van der Waals equation of state is :

(A) $\left(P + \frac{a}{n^2 V^2}\right)(V - nb) = nRT$ (B) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = \Delta nRT$
 (C) $\left(P + \frac{an^2}{V^2}\right)(V - b) = nRT$ (D) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

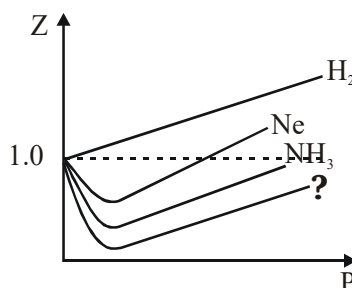
RG0019

2. At relatively high pressure, Van der Waals equation reduces to :

(A) $PV_m = RT$ (B) $PV_m = RT + \frac{a}{V_m}$ (C) $PV_m = RT + Pb$ (D) $PV_m = RT - \frac{a}{V_m^2}$

RG0020

3. Observe the following Z vs P graph.



The missing gas in the above graph can be :

- (A) He (B) Ar (C) C_5H_{12} (D) All are correct

RG0021

4. Correct option regarding a container containing 1 mol of a gas in 22.4 litre container at 273 K is

- (A) If compressibility factor (Z) > 1 then 'P' will be less than 1 atm.
 (B) If compressibility factor (Z) > 1 then 'P' will be greater than 1 atm.
 (C) If 'b' dominates, pressure will be less than 1 atm.
 (D) If 'a' dominates, pressure will be greater than 1 atm.

RG0022

5. If 'V' is actual volume of 1 molecule of gas then, excluded volume (b) of 1 mole of gaseous molecule is -

- (A) $4 N_A V$ (B) $N_A V$ (C) V/N_A (D) V

RG0023

6. Consider the equation $Z = \frac{PV}{RT}$, Which of the following statements is correct :

- (A) When $Z > 1$ real gases are easier to compress
(B) When $Z = 1$ real gases are easier to compress
(C) When $Z > 1$ real gases are difficult to compress
(D) When $Z < 1$ real gases are difficult to compress

RG0024

7. Compressibility factor of ideal gas is :-

- (A) $z > 1$ (B) $z > 1$ (C) $z = 1$ (D) $z = \infty$

RG0025

8. The density of a gaseous substance at 1 atm pressure and 750 K is 0.30 g/lit. If the molecular weight of the substance is 27, the dominant forces existing among gas molecules is -

- (A) Attractive (B) Repulsive (C) Both (A) and (B) (D) None of these

RG0026

9. The third virial coefficient of a He gas is $4 \times 10^{-2} \text{ (lit/mol)}^2$. What will be volume of 2 mole He gas at 1 atm and 273K ($273\text{K} > T_r$)

- (A) 22.0 L (B) 44.0 L (C) 44.8 L (D) 45.2 L

RG0027

10. At low pressure the vander waals equation is reduced to -

- (A) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RTV_m}$ (B) $Z = \frac{pV_m}{RT} = 1 + \frac{a}{RT}p$

(C) $pV_m = RT$ (D) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RT}$

RG0028

11. The values of Van der Waals constant 'a' for the gases O₂, N₂, NH₃ and CH₄ are 1.360, 1.390, 4.170 and 2.253 L atm mol⁻² respectively. The gas which can most easily be liquefied is :

- (A) O_2 (B) N_2 (C) NH_3 (D) CH_4

RG0029

12. The values of critical temperatures of few gases are given gases :

Gases :	H ₂	He	O ₂	N ₂
T_c(K)	33.2	5.2	154.3	126

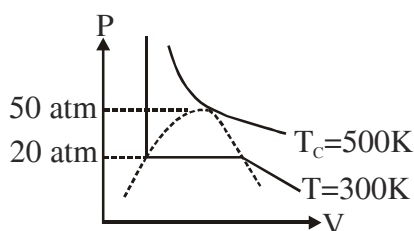
From the above data arrange the given gases in the increasing order of ease of their liquification.

- (A) O_2 , N_2 , H_2 , He
(B) He, N_2 , O_2 , H_2
(C) He, H_2 , N_2 , O_2
(D) H_2 , N_2 , O_2 , He

RG0030

EXERCISE # O-II

1. For real gas the P–V curve was experimentally plotted and it had the following appearance. With respect to liquification, choose the incorrect statement :



- (A) At $T = 500\text{ K}$, $P = 40\text{ atm}$, the state will be liquid
 (B) At $T = 300\text{ K}$, $P = 50\text{ atm}$, the state will be gas
 (C) At $T < 300\text{ K}$, $P = 20\text{ atm}$, the state will be gas
 (D) At $300\text{ K} < T < 500\text{ K}$, $P > 50\text{ atm}$, the state will be liquid

RG0031

2. Select the incorrect statement (s)

- (A) The critical constant for a Vander Waal's gas is $V_c = 3b$, $P_c = \frac{a}{27b^2}$ and $T_c = \frac{a}{27Rb}$
 (B) At 56 K a gas may be liquified if its critical temperature is -156°C .
 (C) U_{avg} of gas in a rigid container can be doubled when the pressure is quadrupled by pumping in more gas at constant temperature
 (D) At extremely low pressure, all real gases behave ideally.

RG0032

3. A 1 litre vessel contains 2 moles of a vanderwaal's gas.

Given data : $a = 2.5\text{ atm-L}^2\text{ mole}^{-2}$ $T = 240\text{ K}$
 $b = 0.4\text{ L-mole}^{-1}$ $RT = 20\text{ L-atm mole}^{-1}$

Identify the correct options about the gas sample :

- (A) Pressure of gas = 190 atm
 (B) Compressibility factor = 4.75
 (C) Attraction forces are dominant in the gaseous sample
 (D) T_B (Boyle temperature) = 75 K

RG0033

4. Choose the correct statement(s) among the following -

- (A) A gas having higher value T_c is easy to liquify
 (B) The radius of molecules of gas having same value if T_c/P_c is same
 (C) Hydrogen gas can be liquified at its boyle temperature by application of pressure.
 (D) Real gas show negative deviation from ideal behaviour at low pressure condition.

RG0034

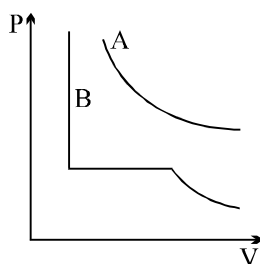
5. Select the **INCORRECT** statement(s):

- (A) At Boyle's temperature, a real gas behaves like an ideal gas irrespective of pressure.
(B) At critical condition, a real gas behaves like an ideal gas.
(C) At low pressure, Van der Waal's constant 'a' dominates over 'b'.
(D) At high pressure, Van der Waal's constant 'b' dominates over 'a'.

RG0050

Question No. 6 & 7 (2 questions)

For two gases A and B, P v/s V isotherms are shown at same temperature, T_K . T_A & T_B are critical temperatures of A & B respectively



6. Which of the following is true?

- (A) $T_A < T < T_B$
 (B) $T_A > T > T_B$
 (C) $T_A > T_R > T$
 (D) none of above

RG0036

7. The correct statement(s) is/are

- (I) Pressure correction term will be more negligible for gas B at T K.
 (II) The curve for gas 'B' will be of same shape as for gas A if $T > T_B$
 (III) Gas 'A' will show same P v/s V curve as of gas 'B' if $T > T_A$
- (A) III only (B) II and III (C) II only (D) All

RG0037

Match the column:

8. Match the column :

Column-I

- (A) Boyle's temperature
(B) Compressibility factor
(C) Real gas with very large molar volume
(D) Critical temperature

Column-II

- (P) Depends on 'a' and 'b'
- (Q) Depends on identity of real gas
- (R) The temperature at which $\frac{dZ}{dP} = 0$
at low pressure region.
- (S) $PV = nRT$
- (T) $\frac{8a}{27R.b.}$

RG0038

EXERCISE # JEE-MAINS

1. 'a' and 'b' are Van der Waals constants for gases. Chlorine is more easily liquefied than ethane because :- [AIEEE-2011]

- (1) a for $\text{Cl}_2 < a$ for C_2H_6 but b for $\text{Cl}_2 > b$ for C_2H_6
 (2) a for $\text{Cl}_2 > a$ for C_2H_6 but b for $\text{Cl}_2 < b$ for C_2H_6
 (3) a and b for $\text{Cl}_2 > a$ and b for C_2H_6
 (4) a and b for $\text{Cl}_2 < a$ and b for C_2H_6

RG0039

2. If Z is the compressibility factor, Van der Waals equation at low pressure can be written as :

[JEE-MAINS-2014]

- (1) $Z = 1 - \frac{Pb}{RT}$ (2) $Z = 1 + \frac{Pb}{RT}$ (3) $Z = 1 + \frac{RT}{Pb}$ (4) $Z = 1 - \frac{a}{V_m RT}$

RG0041

3. When does a gas deviate the most from its ideal behaviour ? [JEE-MAINS(ONLINE)-2015]

- (1) At high pressure and low temperature
 (2) At high pressure and high temperature
 (3) At low pressure and low temperature
 (4) At low pressure and high temperature

RG0040

4. Among the following, the incorrect statement is :

[JEE-Mains-2017(ONLINE)]

- (1) At low pressure, real gases show ideal behaviour
 (2) At very large volume, real gases show ideal behaviour
 (3) At Boyle's temperature, real gases show ideal behaviour
 (4) At very low temperature, real gases show ideal behaviour

RG0042

5. The volume of gas A is twice that of gas B. The compressibility factor of gas A is thrice that of gas B at same temperature. The pressures of the gases for equal number of moles are :

[JEE-MAINS-2019]

- (1) $2P_A = 3P_B$ (2) $P_A = 3P_B$ (3) $P_A = 2P_B$ (4) $3P_A = 2P_B$

RG0043

6. At a given temperature T , gases Ne, Ar, Xe and Kr are found to deviate from ideal gas behaviour.

Their equation of state is given as $p = \frac{RT}{V-b}$ at T .

[JEE-MAINS-2019]

Here, b is the van der Waals constant. Which gas will exhibit steepest increase in the plot of Z (compression factor) vs p ?

- (1) Ne (2) Ar (3) Xe (4) Kr

RG0044

7. Consider the van der Waals constants, a and b , for the following gases.

Gas	Ar	Ne	Kr	Xe
$a/(\text{atm dm}^6 \text{ mol}^{-2})$	1.3	0.2	5.1	4.1
$b/(\text{10}^{-2} \text{ dm}^3 \text{ mol}^{-1})$	3.2	1.7	1.0	5.0

[JEE-MAINS-2019]

Which gas is expected to have the highest critical temperature?

- (1) Kr (2) Ne (3) Ar (4) Xe

RG0045

8. Consider the following table :

[JEE-MAINS-2019]

Gas	$a/(\text{k Pa dm}^6 \text{ mol}^{-1})$	$b/(\text{dm}^3 \text{ mol}^{-1})$
A	642.32	0.05196
B	155.21	0.04136
C	431.91	0.05196
D	155.21	0.4382

a and b are vander waals constant. The correct statement about the gases is :

- (1) Gas C will occupy lesser volume than gas A; gas B will be lesser compressible than gas D
 (2) Gas C will occupy more volume than gas A; gas B will be lesser compressible than gas D
 (3) Gas C will occupy more volume than gas A; gas B will be more compressible than gas D
 (4) Gas C will occupy lesser volume than gas A; gas B will be more compressible than gas D

RG0046

9. A certain gas obeys $P(V_m - b) = RT$. The value of $\left(\frac{\partial Z}{\partial P}\right)_T$ is $\frac{xb}{RT}$. The value of x is_____.

(Integer answer) (Z : compressibility factor)

[JEE-MAINS-2021]

RG0051

10. The unit of the van der Waals gas equation parameter ' a ' in $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ is :

- (1) kg m s^{-2} (2) $\text{dm}^3 \text{ mol}^{-1}$
 (3) kg m s^{-1} (4) $\text{atm dm}^6 \text{ mol}^{-2}$

[JEE-MAINS-2021]

RG0052

RG0052

11. For a real gas at 25°C temperature and high pressure (99 bar) the value of compressibility factor is 2, so the value of Vander Waal's constant ' b ' should be _____ $\times 10^{-2} \text{ L mol}^{-1}$ (Nearest integer)
 (Given $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)

[JEE-MAINS-2022]

RG0053

EXERCISE # JEE-ADVANCED

1. The compression factor (compressibility factor) for one mole of a Van der Waals' gas at 0° C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the Van der Waals' constant 'a'. [JEE 2001]

RG0054

2. The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 Kg m⁻³. The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition.

Determine

[JEE 2002]

- (i) mol. wt. ;
 (ii) molar volume ;
 (iii) compression factor z of the vapour and
 (iv) which forces among the gas molecules are dominating, the attractive or the repulsive

RG0055

3. Positive deviation from ideal behaviour takes place because of

[JEE 2003]

(A) molecular attractions between atoms and $\frac{PV}{nRT} > 1$

(B) molecular attractions between atoms and $\frac{PV}{nRT} < 1$

(C) finite size of atoms and $\frac{PV}{nRT} > 1$

(D) finite size of atoms and $\frac{PV}{nRT} < 1$

RG0056

4. For a real gas obeying van der Waals' equation a graph is plotted between PV_m (y-axis) and P (x-axis) where V_m is molar volume. Find y-intercept of the graph. [JEE 2004]

RG0057

5. The given graph represents the variation of Z (compressibility factor = $\frac{PV}{nRT}$) versus P , for three real gases A, B and C. Identify the only **INCORRECT** statement.

(A) for the gas A, $a = 0$ and its dependence on P is linear at all pressure

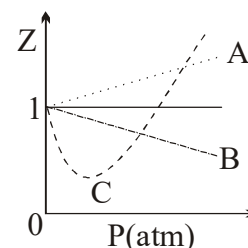
(B) for the gas B, $b = 0$ and its dependence on P is linear at all pressure

(C) for the gas C, which is typical real gas for which neither a nor $b = 0$. By knowing the minima and the point of intersection, with $Z = 1$, a and b can be calculated.

(D) At high pressure, the slope is positive for all real gases A, B and C.

[JEE 2006]

RG0058



- 6.** Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I

- (A) Hydrogen gas ($P = 200 \text{ atm}$, $T = 273 \text{ K}$)
(B) Hydrogen gas ($P \sim 0$, $T = 273 \text{ K}$)
(C) CO_2 ($P = 1 \text{ atm}$, $T = 273 \text{ K}$)
(D) Real gas with very large molar volume

Column II

- (P) Compressibility factor $\neq 1$
 (Q) Attractive forces are dominant
 (R) $PV = nRT$
 (S) $P(V - nb) = nRT$ **[JEE]**

[JEE 2007]

RG0059

7. A gas described by van der Waals' equation

[JEE 2008]

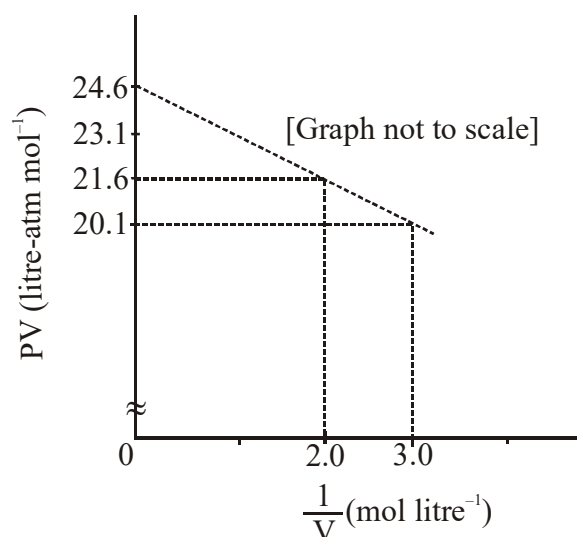
- (A) behaves similar to an ideal gas in the limit of large molar volumes
- (B) behaves similar to an ideal gas in the limit of large pressures
- (C) is characterised by van der Waals' coefficients that are dependent on the identity of the gas but are independent of the temperature
- (D) has the pressure that is lower than the pressure exerted by the same gas behaving ideally

RG0060

- 8.** For one mole of a Van der Waals gas when $b = 0$ and $T = 300$ K, the PV vs. $1/V$ plot is shown below.

The value of the Van der Waals constant a (atm. litre² mol⁻²) is

[JEE 2012]



- (A) 1.0 (B) 4.5 (C) 1.5 (D) 3.0

RG0047

ANSWER KEY**EXERCISE # S-I**

- | | | |
|----------------------------|-----------------------|--------------------------------|
| 1. (a) 24.00, (b) 21.40 | 2. 388.00 | 3. (i) 0.13, (ii) 0.66 or 0.67 |
| 4. 6.66 or 6.67 | 5. (a) 0.98, (b) 1.01 | 6. 2.00 |
| 7. 6.25 | 8. 400.00 | 9. 227.00 |
| 10. (i) A, (ii) A, (iii) C | 11. 800.00 | 12. 160.00 |

EXERCISE # S-II

Que.	1	2	3	4	5	6	
Ans.	58.82	0.50	222.22	1.68 or 1.69	60.00	1.62	

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	C	B	A	C	C	B	D	A
Que.	11	12								
Ans.	C	C								

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	
Ans.	A,B,C	A,C	A,B,D	A,B,D	A,B	A	C	
Que.	11	12	13	14				
Ans.	A-P,Q,R,S	B-P,Q	C-S	D-P,Q,T				

EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	4	1	3	1	3	1	4
Que.	11									
Ans.	25									

EXERCISE # JEE-ADVANCED

- | | | | |
|--|-------------------------------|--------------|----------------|
| 1. 1.256 atm L ² mol ⁻² | | | |
| 2. (i) 18 g/mol, | (ii) 50 L mol ⁻¹ , | (iii) 1.218, | (iv) repulsive |
| 3. (C) | 4. RT | 5. (D) | |
| 6. (A) - P, S ; (B) - R ; (C) - P, Q ; (D) - R | 7. (A,C,D) | 8. (C) | |

ATOMIC STRUCTURE

1. INTRODUCTION :

The existence of atoms has been proposed since the time of early Indian and Greek philosophers (400 B.C.) who were of the view that atoms are the fundamental building blocks of matter. According to them, the continued subdivisions of matter would ultimately yield atoms which would not be further divisible. The word 'atom' has been derived from the Greek word 'a-tomio' which means 'uncuttable' or 'non-divisible'. These earlier ideas were mere speculations and there was no way to test them experimentally. These ideas remained dormant for a very long time and were revived again by scientists in the nineteenth century. The atomic theory of matter was first proposed on a firm scientific basis by **John Dalton**, a British school teacher in 1808. His theory, called **Dalton's atomic theory**, regarded the atom as the ultimate particle of matter.

2. DALTON ATOMIC THEORY :

- (i) Atom is the smallest particle of any substance which cannot be divided further.
- (ii) Atoms can never be created nor be destroyed. Only the rearrangement of atoms occurs in chemical reactions.
- (iii) All the atoms are hard and dense.
- (iv) All the atoms of an element are identical but the atoms of different elements will be different.
- (v) A compound is formed by the combinations of atoms of different elements in a fixed ratio by mass.

We start with the experimental observations made by scientists towards the end of nineteenth and beginning of twentieth century. These established that atoms can be further divided into subatomic particles, i.e., electrons, protons and neutrons, a concept very different from that of Dalton. The major problems before the scientists at that time were:

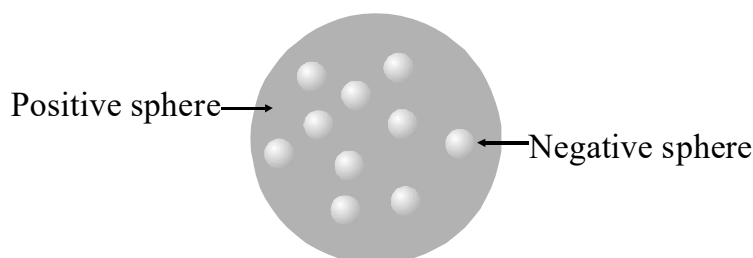
- to account for the stability of atom after the discovery of sub-atomic particles,
- to compare the behaviour of one element from other in terms of both physical and chemical properties,
- to explain the formation of different kinds of molecules by the combination of different atoms and,
- to understand the origin and nature of the characteristics of electromagnetic radiation absorbed or emitted by atoms.

3. ATOMIC MODELS :

Observations obtained from the experiments mentioned in the previous sections have suggested that Dalton's indivisible atom is composed of sub-atomic particles carrying positive and negative charges. Different atomic models were proposed to explain the distributions of these charged particles in an atom. Although some of these models were not able to explain the stability of atoms, two of these models, proposed by J. J. Thomson and Ernest Rutherford are discussed below.

3.1 Thomson Model of Atom :

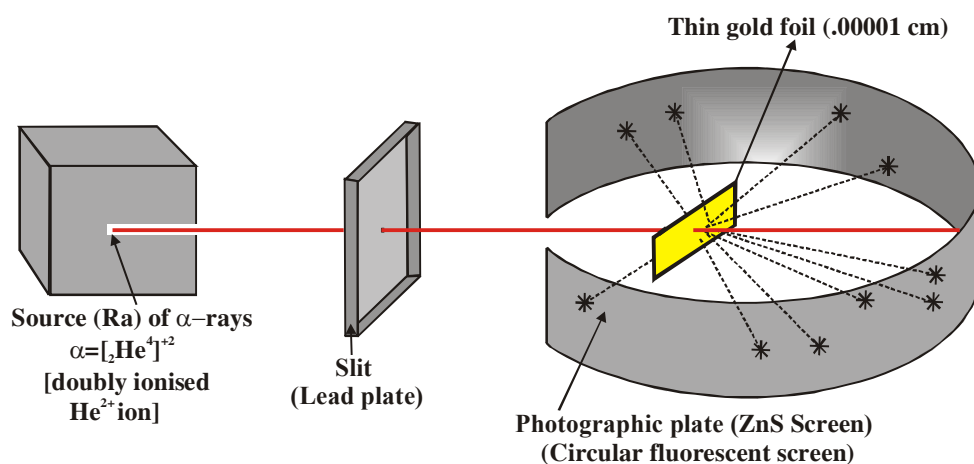
J. J. Thomson, in 1898, proposed that an atom possesses a spherical shape (radius approximately 10^{-10} m) in which the positive charge is uniformly distributed. The electrons are embedded into it in such a manner as to give the most stable electrostatic arrangement. Many different names are given to this model, for example, **plum pudding, raisin pudding or watermelon**. This model can be visualised as a pudding or watermelon of positive charge with plums or seeds (electrons) embedded into it. An important feature of this model is that the mass of the atom is assumed to be uniformly distributed over the atom. Although this model was able to explain the overall neutrality of the atom, but was not consistent with the results of later experiments. Thomson was awarded Nobel Prize for physics in 1906, for his theoretical and experimental investigations on the conduction of electricity by gases.

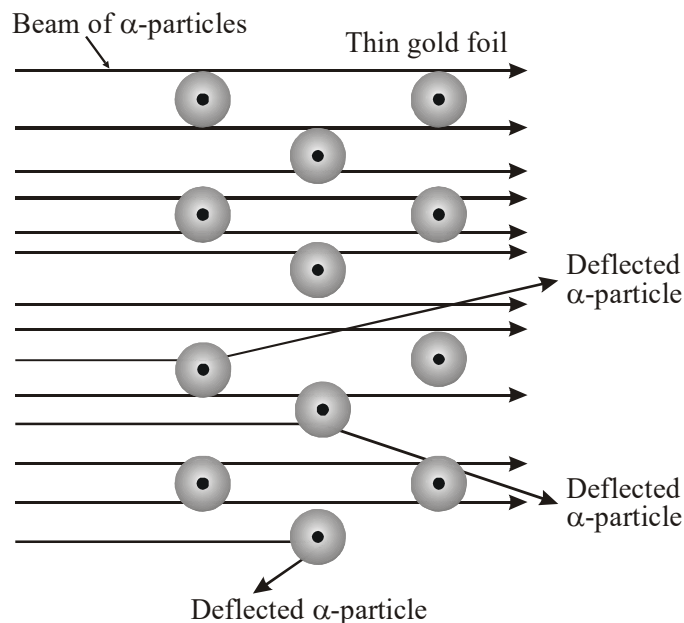


Thomson model of atom

3.2 Rutherford's model :

Rutherford's α -scattering experiment





B. Schematic molecular view of the gold foil

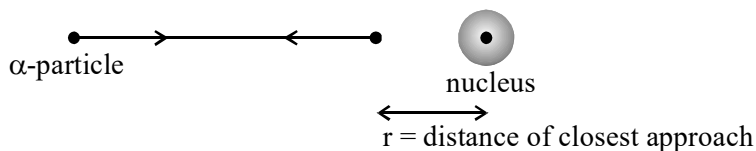
❖ **Observations and conclusions :**

- (i) Most of the α -particles passed through the gold foil undeflected. Hence, most of the space in the atom is empty.
- (ii) A small fraction of α -particles was deflected by small angles. A very few α -particles (~ 1 in 20,000) bounced back, that is, were deflected by nearly 180° .

The deflection must be due to enormous repulsive force showing that the positive charge of the atom is not spread throughout the atom as Thomson had presumed. The positive charge has to be concentrated in a very small volume that repelled and deflected the positively charged α -particles. Rutherford given the name **nucleus** to this positively charged center of atom.

- (iii) The volume occupied by the nucleus is negligibly small as compared to the total volume of atom. The radius of the atom is about 10^{-10} m, while that of nucleus is 10^{-15} m.

Rutherford estimated the size of nucleus by calculating the **distance of closest approach**.



The initial kinetic energy of α -particle must be equal to potential energy at distance of closest approach.

$$\text{K.E.} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 \cdot q_2}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{(2e) \cdot (Ze)}{r}$$

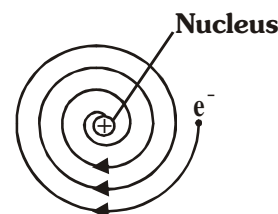
$$r = \frac{2Ze^2}{(4\pi\epsilon_0)(\text{K.E.})}$$

where Z = atomic number of element used in foil.

- (iv) Almost all mass of the atom is densely concentrated in extremely small region (nucleus).
- (v) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called **orbits**. Electrons and the nucleus are held together by electrostatic forces of attraction.

3.2.1 Drawbacks of Rutherford model :

- (i) According to the electromagnetic theory of Maxwell, charged particles when accelerated should emit electromagnetic radiation. Therefore, an electron in an orbit will emit radiation, the energy carried by radiation comes from electronic motion. The orbit will thus continue to shrink.



Calculations show that it should take an electron only 10^{-8} s to spiral into the nucleus. But this does not happen. Thus, the Rutherford model cannot explain the stability of an atom.

- (ii) It does not explain the arrangement of electrons revolving round nucleus.
- (iii) It does not explain the stability of nucleus against strong repulsive forces.
- (iv) It does not explain atomic spectrum.

Ex.1 An α -particles of kinetic energy of 5.4 MeV is projected towards gold nucleus. Calculate the distance of closet approach. (Atomic number of gold = 79, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$)

Sol :
$$\text{K.E.} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}$$

or
$$5.4 \times 10^6 \times 1.6 \times 10^{-19} = 9 \times 10^9 \times \frac{(2 \times 1.6 \times 10^{-19}) \times (79 \times 1.6 \times 10^{-19})}{r}$$

$$\therefore r = 4.21 \times 10^{-14} \text{ m}$$

Ex.2 An α -particles is projected towards the following nucleus with same kinetic energy in different experiment the distance of closet approach is maximum for

- | | |
|-----------------|-----------------|
| (A) Na (Z = 11) | (B) Ca (Z = 20) |
| (C) Ag (Z = 47) | (D) Au (Z = 79) |

Answer.(A)

Sol. $r \propto Z$

Ex.3 An α -particle, a proton, a deuteron and a neutron are projected towards the same nucleus with the same kinetic energy in different experiment. The distance of closet approach is minimum for

- | | | | |
|--------------|-------|-------|-------|
| (A) α | (B) p | (C) d | (D) n |
|--------------|-------|-------|-------|

Answer.(D)

Sol. Neutron is chargeless and hence it will not experience repulsion.

Ex.4 An α -particle having K.E. = 7.7 MeV is scattered by gold (Z = 79) nucleus through 180° . Find distance of closest approach.

$$\text{K.E.} = 7.7 \text{ M eV}$$

$$= 7.7 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 1.23 \times 10^{-12} \text{ J}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

Using we get : $\frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{1.23 \times 10^{-12}}$

$$r_0 = 3 \times 10^{-14} \text{ m}$$

4. WAVE THEORY :

A wave is defined as a periodic disturbance in space or in a medium that involves elastic displacement of material particles or a periodic change in some physical quantities such as T, P, V etc. Thus, wave motion represents propagation of a periodic disturbance carrying energy. The wave travels at right angles to the vibratory motion of the object.

When an object moves up and down or vibrates continuously, energy in the form of waves is transmitted by a vibrating object to a distant place. For example, when a stone is thrown in a still water of a pond, a disturbance is produced at a place where the stone strikes the water.

This disturbance advances outwards in the same form and reaches the edges of the pond. Such a disturbance is called a ***mechanical wave***. The mechanical waves transmit only in a material medium. Besides the mechanical waves, there are waves which do not require any medium for their transmission. These waves are called ***electromagnetic waves*** or ***electromagnetic radiations***.

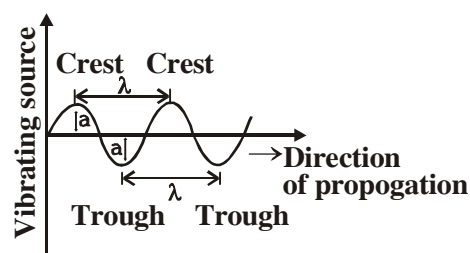
4.1 WAVE CHARACTERISTICS :

4.1.1 Wavelength (λ) :

It is defined as the distance between two nearest crests or nearest troughs.

It is measured in terms of a Å (Angstrom), pm (Picometre), nm (nanometer), cm (centimetre), m (metre)

$$1 \text{ \AA} = 10^{-10} \text{ m}, \quad 1 \text{ pm} = 10^{-12} \text{ m}, \quad 1 \text{ nm} = 10^{-9} \text{ m}, \quad 1 \text{ cm} = 10^{-2} \text{ m}$$



4.1.2 Frequency (ν) :

Frequency of a wave is defined as the number of waves which pass through a point in 1 sec.

- It is measured in terms of Hertz (Hz), sec^{-1} , or cycle per second (cps)

$$1 \text{ Hertz} = 1 \text{ sec}^{-1}$$

4.1.3 Time period (T) : Time taken by a wave to pass through one point.

$$T = \frac{1}{\nu} \text{ sec.}$$

4.1.4 Speed (c) :

Speed of a wave is defined as distance covered by a wave in 1 sec.

$$c = \lambda / T = \lambda \nu$$

$$\nu = c/\lambda$$

$$c = \nu (\text{sec}^{-1}) \times \lambda (\text{m})$$

$$c = \nu \lambda (\text{m sec}^{-1})$$

Since c is constant for em-waves, frequency is inversely proportional to λ

4.1.5 Wave number ($\bar{\nu}$) :

It is defined as number of waves per unit length. It is denoted by $\bar{\nu}$ & is expressed in cm^{-1} .

$$\bar{\nu} = \frac{1}{\lambda}$$

$$(1 \text{ cm}^{-1} = 100 \text{ m}^{-1})$$

- It is measured in terms of cm^{-1} , m^{-1} etc.

4.1.6 Amplitude (a) :

It is the height of the crest or depth of the trough of a wave and is denoted by 'a'. It is half the vertical distance from the top of the wave to the bottom of the wave. It determines the intensity or brightness of the beam of light.

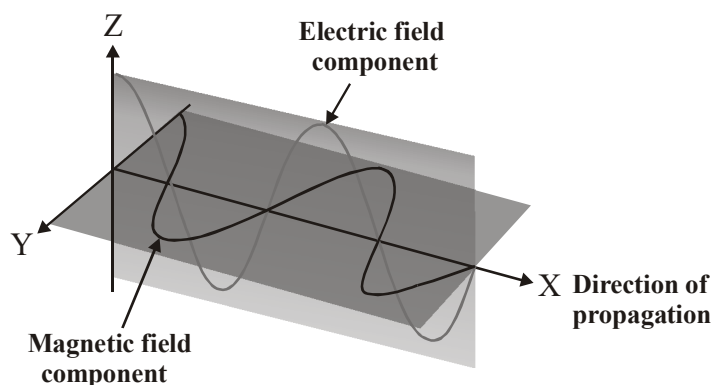
The amplitude of a wave in stretched string is the maximum displacement of the string from its normal position that of water waves is the maximum height of the water surface relative to its normal level, That of a sound wave is the maximum change in pressure relative to the normal pressure.

4.2 ELECTRO MAGNETIC (EM) WAVES :

James Maxwell (1870) suggested that when electrically charged particle moves under acceleration, alternating electrical and magnetic fields are produced and transmitted by it. These fields are transmitted in the forms of waves called **electromagnetic waves** or **electromagnetic radiations**.

4.2.1 Main assumptions of Maxwell EM theory :

- The oscillating electric and magnetic fields produced by oscillating charged particles are perpendicular to each other and both are perpendicular to the direction of propagation of the wave.



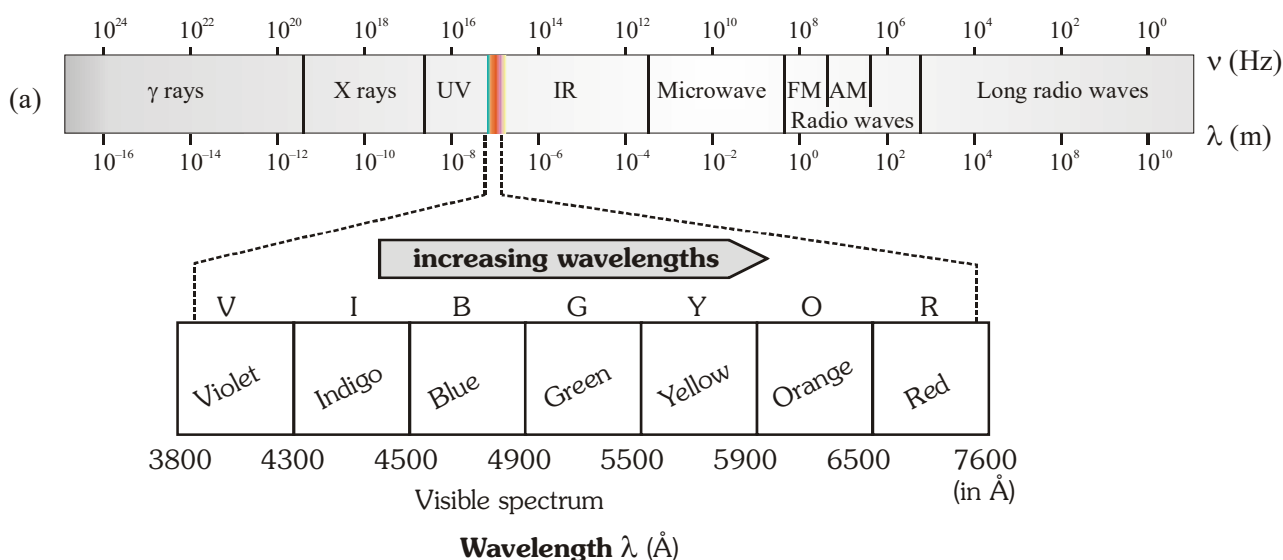
(The electric and magnetic field components of an electromagnetic wave. These components have the same wavelength, frequency, speed and amplitude, but they vibrate in two mutually perpendicular planes.)

- (ii) Unlike sound waves or water waves, electromagnetic waves do not require medium and can move in vacuum. However speed changes in different mediums
- (iii) The EM waves travel with the speed of light (3×10^8 m/sec) in vacuum.
- (iv) Energy of EM waves depends on its amplitude not on its frequency or wavelength.
(This assumptions was later proved to be wrong in some situations).

4.3 ELECTRO MAGNETIC SPECTRUM :

The arrangement of various types of electromagnetic radiations in the order of their increasing or decreasing wavelength or frequencies is known as electromagnetic spectrum. The wavelength decreases in the following order.

Radio waves > microwaves > Infrared > Television waves > Visible > Ultraviolet > X-rays > γ -rays > cosmic rays



(a) The spectrum of electromagnetic radiation. (b) Visible spectrum.
The visible region is only a small part of the entire spectrum.

Ex.5 Calculate the frequency of EMR (Electromagnetic radiation) of wave number 10^4 cm^{-1} .

Sol : $\nu = \frac{c}{\lambda} = c\bar{\nu} = (3 \times 10^8 \text{ m/s})(10^4 \text{ cm}^{-1}) = (3 \times 10^{10} \text{ cm/s})(10^4 \text{ cm}^{-1}) = 3 \times 10^{14} \text{ Hz}.$

Ex.6 A radio station radiate the radiowaves of frequency 20kHz. What is meter band of that radio station?

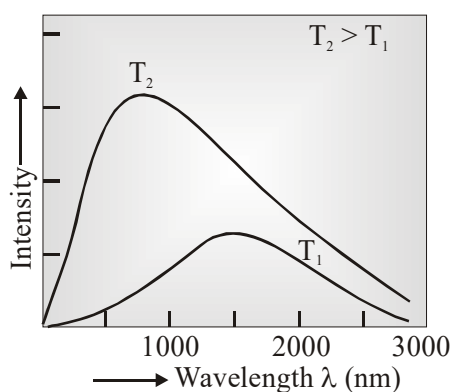
Sol : $20\text{kHz} = \frac{3 \times 10^8}{\lambda} \Rightarrow \lambda = 1.5 \times 10^4 \text{ m}$

5. PARTICLE NATURE OF ELECTROMAGNETIC RADIATIONS

(PLANCK'S QUANTUM THEORY OF RADIATION)

Hot objects emit electromagnetic radiations over a wide range of wavelengths. At high temperatures, an appreciable proportion of radiation is in the visible region of the spectrum. As the temperature is raised, a higher proportion of short wavelength (blue light) is generated. For example, when an iron rod is heated in a furnace, it first turns to dull red and then progressively becomes more and more red as the temperature increases. As this is heated further, the radiation emitted becomes white and then becomes blue as the temperature becomes very high. This means that red radiation is most intense at a particular temperature and the blue radiation is more intense at another temperature. This means intensities of radiations of different wavelengths emitted by hot body depend upon its temperature.

By late 1850's it was known that objects made of different material and kept at different temperatures emit different amount of radiation. Also, when the surface of an object is irradiated with light (electromagnetic radiation), a part of radiant energy is generally reflected as such, a part is absorbed and a part of it is transmitted. The reason for incomplete absorption is that ordinary objects are as a rule imperfect absorbers of radiation. **An ideal body, which emits and absorbs radiations of all frequencies uniformly, is called a black body and the radiation emitted by such a body is called black body radiation.** In practice, no such body exists. Carbon black approximates fairly closely to black body. A good physical approximation to a black body is a cavity with a tiny hole, which has no other opening. Any ray entering the hole will be reflected by the cavity walls and will be eventually absorbed by the walls. A black body is also a perfect radiator of radiant energy. Furthermore, a black body is in thermal equilibrium with its surroundings. It radiates same amount of energy per unit area as it absorbs from its surrounding in any given time. The amount of light emitted (intensity of radiation) from a black body and its spectral distribution depends only on its temperature. At a given temperature, intensity of radiation emitted increases with the increase of wavelength, reaches a maximum value at a given wavelength and then starts decreasing with further increase of wavelength. Also, as the temperature increases, maxima of the curve shifts to short wavelength. Several attempts were made to predict the intensity of radiation as a function of wavelength.



Wavelength intensity relationship

But the results of the above experiment could not be explained satisfactorily on the basis of the wave theory of light. Max Planck arrived at a satisfactory relationship by making an assumption that absorption and emission of radiation arises from oscillator i.e., atoms in the wall of black body. Their frequency of oscillation is changed by interaction with oscillators of electromagnetic radiation. Planck assumed that radiation could be sub-divided into discrete chunks of energy.

Max Planck in 1900, put forward a theory known as "**Planck's Quantum Theory**". It regards electromagnetic radiations made up of particles. The main points of the theory are :

- (i) The radiant energy is emitted or absorbed by atoms or molecules discontinuously in the form of small energy packets called **quantum**. In case of light, these energy packets are known as **photons**.
- (ii) The energy of each quantum is directly proportional to the frequency of the radiation i.e.

$$E \propto \nu \quad \text{or} \quad E = h\nu = \frac{hc}{\lambda}$$

Here h is a constant known as **Planck's constant**. Its value is 6.626×10^{-34} Joules sec.

- (iii) The total amount of energy emitted or absorbed by a body is some whole number multiple of quantum i.e.,

$$E = nh\nu \quad (\text{Here } n \text{ is a positive integer : } 1, 2, 3, 4 \dots \text{etc.})$$

Thus, Planck for the first time has given a relationship between the frequency (or wavelength) of the radiations and the energy associated with them. Cosmic rays, gamma rays and X-rays are high energy radiations since they have very high frequency. At the same time, microwaves and radiowaves with small frequency are regarded as low energy radiations.

❖ Important features of Photon :

- Source of energy (light) emits radiation in the form photons, which travel with speed of light.
- Energy of single photon, $E = h\nu = \frac{hc}{\lambda}$.
- All the photons in vacuum travel with speed 3×10^8 m/s but their speed is changed in different medium, however frequency remains same. Speed does not depend on energy.
- Photons travel as waves but are absorbed or emitted as particles.

Ex.7 Calculate the energy per quanta of an EMR of frequency 400 MHz.

Sol : $E = 6.626 \times 10^{-34} \times 4 \times 10^6 \text{ J/quanta}$

Ex.8 Calculate the energy per quanta of an EMR of wavelength 662.6 nm.

Sol : $E = n \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{662.6 \times 10^{-9} \text{ m}} = 3 \times 10^{-19} \text{ J/quanta}$

Ex.9 Calculate the wavelength (in Å) of an EMR of energy 3.1 eV/quanta.

$$\text{Sol: } E = h\nu = h \frac{c}{\lambda} \Rightarrow E(\text{eV}) \times 1.602 \times 10^{-19} = \frac{1 \times 6.62 \times 10^{-34} \times 3 \times 10^8}{\lambda(\text{Å})}$$

$$\boxed{E(\text{eV}) \approx \frac{12400}{\lambda(\text{Å})} = \frac{1240}{\lambda(\text{nm})}}$$

Ex.10 In order to see an object, 10^{-19} J must be received by our eyes. How many photons of green light must be received by our eyes for its visibility. ($\lambda = 550 \text{ nm}$, $h = 6.6 \times 10^{-34} \text{ J}$)

$$\text{Sol: } E = n \frac{hc}{\lambda}$$

$$10^{-19} = n \times \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5500 \times 10^{-9}}$$

$$n = \frac{5}{18} \approx 1$$

Ex.11 A bulb is rated as 110 watt. If it emits 25% of absorbed energy as red light ($\lambda = 6626 \text{ Å}$), how many photons are emitted out by the bulb per second.

$$\text{Sol: } 110 \times \frac{25}{100} = n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{6626 \times 10^{-10}}$$

Ex.12 The wavelength of microwave radiation is 0.08 m. How many moles of photons is needed to increase the temperature of 400 gm water from 25° to 45°C , assuming 25% efficiency.

Specific heat capacity = 4.2 J/K-gm

$$\text{Sol: } \left(\frac{nhc}{\lambda} \right) \times \frac{25}{100} = ms\Delta t$$

$$n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{0.08} \times \frac{25}{100} = 400 \times 4.2 \times 20$$

$$\therefore \text{Number of moles of photon} = \frac{n}{N_A}$$

Ex.13 A dye absorbs the radiation of $4000 \text{ \AA}$ and fluoresces the radiation of $5000 \text{ \AA}$. If only 40% of the absorbed energy is emitted out, calculate the ratio of number of quanta emitted out and the number of quanta absorbed.

Sol: $E_a \times \frac{40}{100} = E_e$

$$n_a \times \frac{hc}{4000 \text{ \AA}} \times \frac{40}{100} = n_e \times \frac{hc}{5000 \text{ \AA}}$$

$$\frac{n_e}{n_a} = \frac{40}{100} \times \frac{5000}{4000} = \frac{1}{2}$$

Ex.14 The bond dissociation energy of Cl–Cl bond in chlorine gas is 240 kJ/mol . Calculate the longest wavelength of EMR needed to dissociate bond. Assume one photon may dissociate only one bond.

Sol: $\frac{240 \times 10^3}{6 \times 10^{23}} = \frac{1 \times 6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda}$

Ex.15 A near ultra violet photon of wavelength 300 nm is absorbed by a gas and then emitted as two photons. One photon is of red light with wavelength 760 nm . What would be the wave length of the second photon ?

Sol. It may noted that energy of photon which adsorbed is emitted as sum of the energy of two photons.

$$\text{Energy absorbed } h\nu = \frac{hc}{\lambda}$$

According to available information,

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}; \frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}; \frac{1}{\lambda_2} = \left[\frac{1}{\lambda} - \frac{1}{\lambda_1} \right]$$

Now, $\lambda = 300 \text{ nm}$; $\lambda_1 = 760 \text{ nm}$; λ_2 can be calculated as :

$$\frac{1}{\lambda_2} = \left[\frac{1}{300} - \frac{1}{760} \right] = \frac{760 - 300}{300 \times 760} (\text{nm}^{-1})$$

$$\frac{1}{\lambda_2} = \frac{460}{760 \times 300} (\text{nm}^{-1})$$

or $\lambda_2 = 496 \text{ nm}$

6. PHOTOELECTRIC EFFECT :

Hertz in 1887 observed that when a light of certain frequency strikes the surface of a metal, electrons are ejected from the metal. This phenomenon is known as **photoelectric effect** and the ejected **electrons** are called **photoelectrons**.

A few metals, which are having low ionisation energy like Cesium, show this effect under the action of visible light but many more show it under the action of more energetic ultraviolet light.

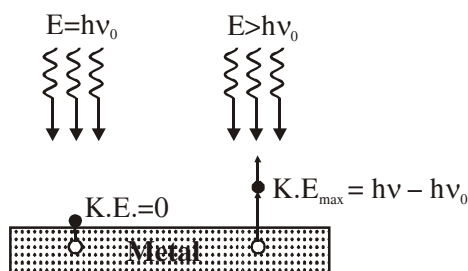
6.1 The experimental findings are summarized as below :

- (i) Electrons come out as soon as the light (of sufficient energy) strikes the metal surface. There is no time lag between the two events.
- (ii) The light of any frequency will not be able to cause ejection of electrons from a metal surface. For each metal, there is a minimum frequency, called the **threshold (or critical) frequency**, which can just cause the ejection. This frequency varies with the nature of the metal. The higher the frequency of the light, the more energy the photoelectrons have. Blue light results in faster electrons than red light.
- (iii) Photoelectric current is increased with increase in intensity of light of same frequency, if emission is permitted, i.e. a bright light yields more photoelectrons than a dim one of the same frequency, but the electron energies remain the same.
- (iv) The number of electrons ejected is proportional to the intensity or brightness of light.

All the above results could not be explained on the basis of laws of classical physics. According to latter, the energy content of the beam of light depends upon the brightness of the light. In other words, number of electrons ejected and kinetic energy associated with them should depend on the brightness of light. It has been observed that though the number of electrons ejected does depend upon the brightness of light, the kinetic energy of the ejected electrons does not. For example, red light [$\nu = (4.3 \text{ to } 4.6) \times 10^{14} \text{ Hz}$] of any brightness (intensity) may shine on a piece of potassium metal for hours but no photoelectrons are ejected. But, as soon as even a very weak yellow light ($\nu = 5.1\text{--}5.2 \times 10^{14} \text{ Hz}$) shines on the potassium metal, the photoelectric effect is observed. The threshold frequency (ν_0) for potassium metal is $5.0 \times 10^{14} \text{ Hz}$. Einstein (1905) was able to explain the photoelectric effect using Planck's quantum theory of electromagnetic radiation as a starting point.

6.2 Einstein's explanation :

Light must have stream of energy particles or quanta of energy ($h\nu$). Suppose, the threshold frequency of light required for ejecting electrons from a metal is ν_0 , when a photon of light of this frequency strikes a metal it imparts its entire energy ($h\nu_0$) to the electron.



This energy enables the electron to break away from the surface by overcoming the attractive influence of the nucleus. Thus each photon can eject one electron. If the frequency of light is less than ν_0 , there is no ejection of electron. If the frequency of light is higher than ν_0 (let it be ν), the photon of this light having higher energy ($h\nu$), will impart some energy to the electron that

is needed to remove it away from the atom. Einstein proposed that light consisted of quanta, which we call photons with a frequency over a certain threshold would have sufficient energy to eject a single electron, producing the photoelectric effect.

6.3 Einstein's Equation for the Photoelectric Effect :

Einstein's interpretation of the photoelectric effect results in equation :

Energy of photon = Energy needed to remove an electron + Max. Kinetic energy of the emitted electron

The excess energy would give a certain velocity (i.e. kinetic energy) to the electron.

$$h\nu = h\nu_0 + K.E._{max.}$$

$$h\nu = h\nu_0 + \frac{1}{2}mv_{max}^2$$

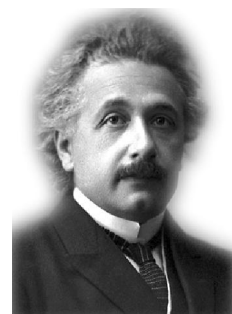
$$\frac{1}{2}mv_{max}^2 = h\nu - h\nu_0$$

where ν = frequency of the incident light, ν_0 = threshold frequency

v_{max} = max. speed of photoelectron.

$h\nu_0$ is the **threshold energy** (or) the **work function** denoted by $\phi = h\nu_0$ (minimum energy of the photon to liberate electron). It is constant for a particular metal.

The maximum kinetic energy of the photoelectrons increases linearly with the frequency of incident light. Thus, if the energy of the ejected electrons is plotted as a function of frequency, it results in a straight line whose slope is equal to Planck's constant 'h' and whose intercept is $h\nu_0$.



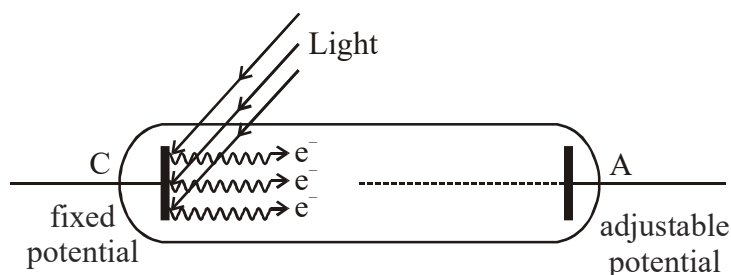
Albert Einstein (1879 - 1955)

Albert Einstein, a German born American physicist, is regarded by many as one of the two great physicists the world has known (the other is Isaac Newton). His three research papers (on special relativity, Brownian motion and the photoelectric effect) which he published in 1905, **Albert Einstein (1879 - 1955)** while he was employed as a technical assistant in a Swiss patent office in Berne have profoundly influenced the development of physics. He received the Nobel Prize in Physics in 1921 for his explanation of the photoelectric effect.

6.4 Important conclusions from photoelectric effect :

- Photoelectric effect demonstrates particle nature of radiation.
- A photon is quanta of energy. Its rest mass is zero. This is why photon can give up its all energy to the particle it strikes.
- There is no effect of frequency of incident light on the number of the emitted photoelectrons.
- There is no effect of intensity of incident light on the K.E. of the emitted photoelectrons.

6.5 SATURATION CURRENT & STOPPING POTENTIAL



Case-I : $V_C = V_A$

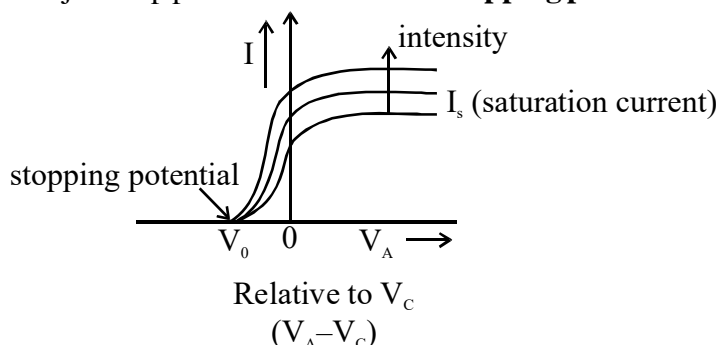
Some of the ejected electrons reach at electrode A resulting **photocurrent**.

Case-II : $V_C < V_A$

As electrode (A) is at high potential, it attracts the electron & even a slower electron will reach at electrode A. It will result in increase in photocurrent. Further increase in the potential difference, a situation may result when the slowest photocurrent electron reach at electrode. It results maximum photocurrent called **saturation current**. Further increase in potential will not increase photocurrent.

Case-III : $V_C > V_A$

As electrode (A) is at low potential, it will repel electron resulting decrease in photocurrent. Further decrease in potential at electrode (A) may result a situation when the fastest electron just fails to reach at (A) and the photocurrent drops to zero. The potential of (A) relative to C to just stop photocurrent is called **stopping potential**.



On increasing the intensity of light the stopping potential does not change because the maximum K.E. of photoelectron does not change. But the photocurrent increases, because the number of photon falling on the surface increase.

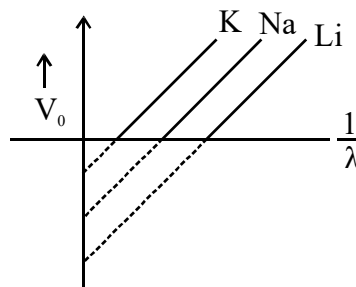
If the frequency of light is changed, the stopping potential will change.

$$(KE)_{\max} = eV_0$$

$$(KE)_{\max} = h\nu - h\nu_0$$

$$eV_0 = \frac{hc}{\lambda} - \phi$$

$$V_0 = \frac{hc}{e} \frac{1}{\lambda} - \frac{\phi}{e}$$



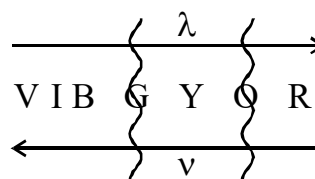
Ex.16 From a metal surface, photoelectron never comes out by orange light but comes from green light. Predict about the injection of photoelectron from the same metal by

(i) Red (ii) Blue (iii) Yellow light

Sol: (i) Red light = No

(ii) Blue light = Yes

(iii) Yellow light = Can't say



Ex.17 The work function of a metal is 3 eV. If EMR of 200 nm fall on the metal surface, calculate the maximum speed of photoelectron ejected.

Sol: $E = \frac{1240}{200} = 6.2 \text{ eV}$

$$(KE)_{\max} = h\nu - \phi$$

$$\frac{1}{2}mv^2 = (6.2 - 3) \text{ eV}$$

$$\frac{1}{2} \times 9.1 \times 10^{-31} v_{\max}^2 = 3.2 \times 1.6 \times 10^{-19}$$

$$\therefore v_{\max} = 1.06 \times 10^6 \text{ ms}^{-1}$$

Ex.18 When EMR of frequency $5 \times 10^{15} \text{ Hz}$ fall on a metal surface, the maximum kinetic energy of photoelectron is double than the photoelectron which emits when EMR of frequency $3 \times 10^{15} \text{ Hz}$ fall on the same metal. The threshold frequency for the metal is

Sol. $(K.E.)_1 = h \times 5 \times 10^{15} - h\nu_0$

$$(K.E.)_2 = h \times 3 \times 10^{15} - h\nu_0$$

$$2E_1 = E_2$$

$$\nu_0 = 1 \times 10^{15} \text{ Hz.}$$

Ex.19 A photon of wavelength $3000 \text{ \AA}$ strikes a metal surface, the work function of the metal being 2.20 eV. Calculate (i) the energy of the photon in eV (ii) the kinetic energy of the emitted photoelectron and (iii) the velocity of the photoelectron.

Sol. (i) Energy of the photon

$$E = h\nu = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ ms}^{-1})}{3 \times 10^{-7} \text{ m}} = 6.6 \times 10^{-19} \text{ J}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$\text{Therefore } E = \frac{6.6 \times 10^{-19} \text{ J}}{1.6 \times 10^{-19} \text{ J/eV}} = 4.125 \text{ eV}$$

(ii) Kinetic energy of the emitted photo electron

$$\text{Work function} = 2.20 \text{ eV}$$

$$\text{Therefore, KE} = 2.475 - 2.20$$

$$= 1.925 \text{ eV} = 3.08 \times 10^{-19} \text{ J}$$

(iii) Velocity of the photo electron

$$\text{KE} = \frac{1}{2} mv^2 = 3.08 \times 10^{-19} \text{ J}$$

$$\text{Therefore, velocity (v)} = \sqrt{\frac{2 \times 3.08 \times 10^{-19}}{9.1 \times 10^{-31}}} = 8.22 \times 10^5 \text{ ms}^{-1}$$

7. SPECTRUM

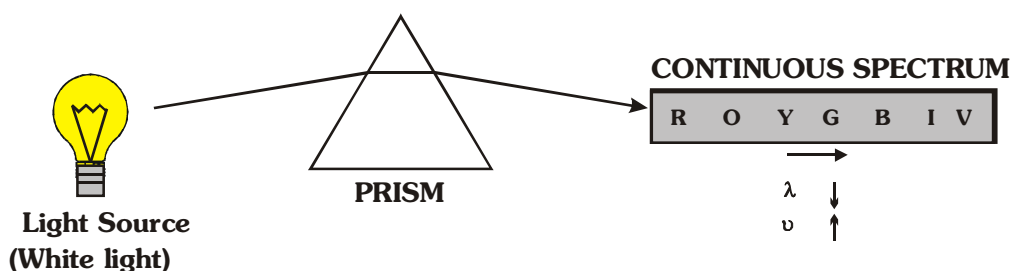
It is the impressions produced on any screen when a light falls on it after passing through prism or prism like material.

Classification of spectrum :

(i) **Continuous and discontinuous spectrum :**

In continuous spectrum, the impression produced on screen overlap each other, but in discontinuous spectrum, some gap exist between the impression.

The spectrum of white light that we can see ranges from violet at $7.50 \times 10^{14} \text{ Hz}$ to red at $4 \times 10^{14} \text{ Hz}$. This spectrum is called *continuous spectrum* because violet merges into blue, blue into green, and so on. A similar spectrum is produced when a rainbow forms in the sky. In a continuous spectrum, radiations corresponding to all the wavelengths are present.



(ii) **Emission and Absorption Spectra :** The spectrum of radiation emitted by a substance that has absorbed energy is called an **emission spectrum**. Atoms, molecules or ions that have absorbed radiation are said to be '**excited**'. To produce an emission spectrum, energy is supplied

to a sample by heating it or irradiating it and the wavelength (or frequency) of the radiation emitted, as the sample gives up the absorbed energy, is recorded.

An absorption spectrum is like the photographic negative of an emission spectrum. A continuum of radiation is passed through a sample which absorbs radiation of certain wavelengths. The missing wavelength which corresponds to the radiation absorbed by the matter, leave dark spaces in the bright continuous spectrum.

The study of emission or absorption spectra is referred to as **spectroscopy**. The spectrum of the visible light, as discussed above, was continuous as all wavelengths (red to violet) of the visible light are represented in the spectra. The emission spectra of atoms in the gas phase, on the other hand, do not show a continuous spread of wavelength from red to violet, rather they emit light only at specific wavelengths with dark spaces between them. Such spectra are called **line spectra** or **atomic spectra** because the emitted radiation is identified by the appearance of bright lines in the spectra.

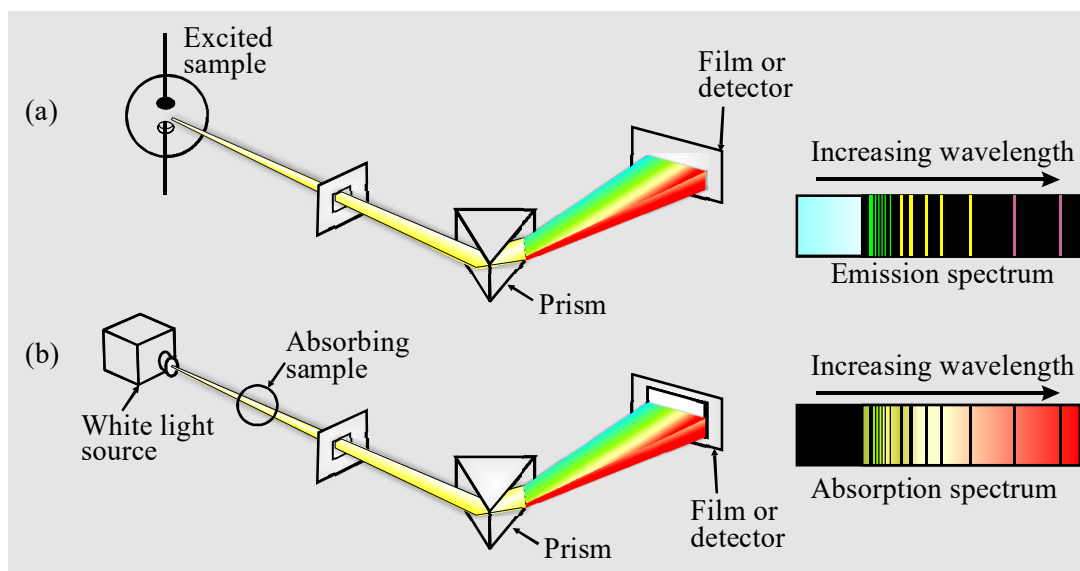


Fig. (a) Atomic emission. The light emitted by a sample of excited hydrogen atoms (or any other element) can be passed through a prism and separated into certain discrete wavelengths. Thus an emission spectrum, which is a photographic recording of the separated wavelengths is called as line spectrum. Any sample of reasonable size contains an enormous number of atoms. Although a single atom can be in only one excited state at a time, the collection of atoms contains all possible excited states. The light emitted as these atoms fall to lower energy states is responsible for the spectrum. **(b) Atomic absorption.** When white light is passed through unexcited atomic hydrogen and then through a slit and prism, the transmitted light is lacking in intensity at the same wavelengths as are emitted in (a). The recorded absorption spectrum is also a line spectrum and the photographic negative of the emission spectrum.

Line emission spectra are of great interest in the study of electronic structure. Each element has a unique line emission spectrum. The characteristic lines in atomic spectra can be used in chemical analysis to identify unknown atoms in the same way as finger prints are used to identify people. The exact matching of lines of the emission spectrum of the atoms of a known element with the lines from an unknown sample quickly establishes the identity of the latter, German

chemist, Robert Bunsen (1811-1899) was one of the first investigators to use line spectra to identify elements.

Elements like rubidium (Rb), caesium (Cs), thallium (Tl), indium (In), gallium (Ga) and scandium (Sc) were discovered when their minerals were analysed by spectroscopic methods. The element helium (He) was discovered in the sun by spectroscopic method.

8. BOHR'S ATOMIC MODEL

Bohr's model for hydrogen atom is based on the following postulates:

- (i) The electron in the hydrogen atom can move around the nucleus in a circular path of fixed radius and energy. These paths are called **orbits, stationary states** or **allowed energy states**. These orbits are arranged concentrically around the nucleus.
- (ii) The energy of an electron in the orbit does not change with time. However, the electron will move from a lower stationary state to a higher stationary state when required amount of energy is absorbed by the electron or energy is emitted when electron moves from higher stationary state to lower stationary state. The energy change does not take place in a continuous manner.
- (iii) The frequency of radiation absorbed or emitted when transition occurs between two stationary states that differ in energy by ΔE , is given by :

$$\nu = \frac{\Delta E}{h} = \frac{E_2 - E_1}{h}$$

where E_1 and E_2 are the energies of the lower and higher allowed energy states respectively. This expression is commonly known as **Bohr's frequency rule**.

- (iv) The angular momentum of an electron in a given stationary state can be expressed as in equation

$$m_e v r = n \cdot \frac{h}{2\pi} \quad n = 1, 2, 3$$

Thus an electron can move only in those orbits for which its angular momentum is integral multiple of $h/2\pi$. That is why, only certain fixed orbits are allowed.

8.1 APPLICATION OF BOHR'S MODEL

When electron revolves in fixed circular orbit then electrostatic force of attraction and centripetal force are equal.

$$\text{Electrostatic force} = \frac{Kq_1q_2}{r^2} = \frac{K \cdot Ze \cdot e}{r^2} = \frac{KZe^2}{r^2}$$

Where, constant $K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$ (MKS) = 1 (CGS)

$$\text{Centrifugal force} = \frac{mv^2}{r}$$

In balanced condition

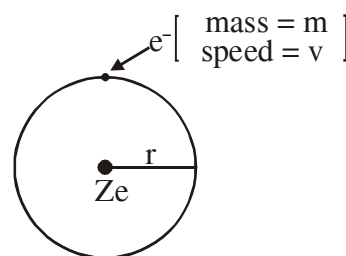
Electrostatic force = Centrifugal force

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r} \text{ or } \frac{KZe^2}{r} = mv^2 \quad \dots\dots(i)$$



Niels Bohr
(1885-1962)

Niels Bohr, was a Danish physicist. After first world war, Bohr worked for peaceful uses of atomic energy. He was awarded the Nobel Prize in physics in 1922.



8.1.1 Radius of various orbits (shell) :

According to Bohr model, $mvr = \frac{nh}{2\pi}$

$$v = \frac{nh}{2\pi mr} \quad \text{.....(ii)}$$

Now putting the value of v from eq.(ii) into eq.(i)

$$\frac{KZe^2}{r} = m \left(\frac{nh}{2\pi mr} \right)^2$$

$$\frac{KZe^2}{r} = \frac{mn^2h^2}{4\pi^2 m^2 r^2}$$

$$\boxed{r = \frac{n^2 h^2}{4\pi^2 m K Z e^2}} \quad \text{.....(iii)}$$

Putting the value of π , h , m , K , & e (Constants) in the above eq.(iii)

$$r = 0.529 \times 10^{-10} \times \frac{n^2}{Z} \text{ m} \quad \{ \text{\AA} = 10^{-10} \text{ m} = 10^{-8} \text{ cm} \}$$

$$\text{or } \boxed{r_n = 0.529 \times \frac{n^2}{Z} \text{\AA}}$$

8.1.2. Speed of electron in Bohr orbit :

According to Bohr postulate

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr} = \frac{nh}{\frac{2\pi m \times n^2 h^2}{4\pi^2 m K Z e^2}}$$

$$\boxed{v = \frac{2\pi K Z e^2}{nh}} \quad \text{.....(iv)}$$

Putting the value of π , h , K , & e (Constants) in the above eq (iv)

$$\boxed{v = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}}$$

8.1.3. Total energy of electron in Bohr orbit :

Total energy of an electron is the sum of kinetic and potential energy.

$$\text{i.e. } T.E. = K.E. + P.E.$$

$$(i) \text{ Potential energy : } P.E. = -\frac{Kq_1q_2}{r} = -\frac{K Z e^2}{r} = -\frac{KZe^2}{r}$$

$$(ii) \text{ Kinetic energy : } K.E. = \frac{1}{2}mv^2$$

$$\text{But } \frac{KZe^2}{r} = mv^2 \quad (\text{By eq. i})$$

$$K.E. = \frac{KZe^2}{2r}$$

(iii) Total energy : T.E. = K.E. + P.E.

$$\text{T.E.} = \frac{KZe^2}{2r} - \frac{KZe^2}{r} = -\frac{KZe^2}{2r}$$

Now putting the value of r from eq. (iii)

$$\text{T.E.} = -\frac{KZe^2 \times 4\pi^2 m KZe^2}{2n^2 h^2}$$

$$\boxed{\text{T.E.} = -\frac{2\pi^2 m \times K^2 Z^2 e^4}{n^2 h^2}}$$

Now putting the value of π , K, e, m, h, we get :

$$\begin{aligned}\text{T.E.} &= -2.18 \times 10^{-18} \times \frac{Z^2}{n^2} \text{ J / atom} = -1312 \times \frac{Z^2}{n^2} \text{ kJ/mol} \\ &= -2.18 \times 10^{-11} \times \frac{Z^2}{n^2} \text{ erg/atom} = -313.6 \times \frac{Z^2}{n^2} \text{ Kcal/mol} \\ &= -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}\end{aligned}$$

8.1.4 Some extra points :

- (i) $\text{K.E.} = \frac{KZe^2}{2r}$ i.e. $\text{K.E.} \propto \frac{1}{r}$ On increasing radius, K.E. decreases.
- (ii) $\text{P.E.} = -\frac{KZe^2}{r}$ i.e. $\text{P.E.} \propto \left(-\frac{1}{r}\right)$ On increasing radius, P.E. increases.
- (iii) $\text{T.E.} = -\frac{KZe^2}{2r}$ i.e. $\text{E.} \propto \left(-\frac{1}{r}\right)$ On increasing radius, total energy increases.
- (iv) Relation between T.E., P.E. and K.E.
 $\text{P.E.} = -2 \text{ KE}$
 $\text{KE} = -\text{T.E.}$
 $\text{P.E.} = 2 \text{ T.E.}$

8.1.5 Important Definitions :

- (i) **Ionization energy :**
Minimum energy required to liberate an electron from the ground state of an isolated atom is called the ionization energy.
- (ii) **Separation energy :**
Minimum energy required to remove an electron from its excited state is called as separation energy.
- (iii) **Excitation energy :**
Amount of energy required to shift an electron from ground state to any excited state.

Note : All these kinds of energy are always positive. (We only consider their magnitude)

Ex.20 Calculate the radius of first four orbits of hydrogen atom

Sol: $r_1 = 0.529 \times \frac{1^2}{1} = 0.529 \text{ \AA}$

$$r_2 = 0.529 \times \frac{2^2}{1} = 2.116 \text{ \AA} = r_1 \times 2^2$$

$$r_3 = 0.529 \times \frac{3^2}{1} = 4.761 \text{ \AA} = r_1 \times 3^2$$

$$r_4 = 0.529 \times \frac{4^2}{1} = 8.464 \text{ \AA}$$

From this, for same Z : $r_n = r_1 \times n^2$

Ex.21 Calculate the ratio of radius of 2nd orbits of Li^{2+} atom & 3rd orbits He^+ ion.

Sol:
$$\frac{r_{2, \text{Li}^{2+}}}{r_{3, \text{He}^+}} = \frac{0.529 \times \frac{4}{3}}{0.529 \times \frac{9}{2}} = \frac{8}{27}$$

Ex.22 Calculate the radius ratio of 3rd & 5th orbit of He^+ .

Sol. For same Z , $r_n = r_1 \cdot n^2$

$$\therefore \frac{r_3}{r_5} = \frac{r_1 \times 3^2}{r_1 \times 5^2} = \frac{9}{25}$$

Ex.23 Calculate the energy of Li^{+2} atom for 2nd excited state.

Sol. $E = -13.6 \times \frac{Z^2}{n^2}$

$\therefore Z = 3$ and e^- exist in 2nd excited state, means e^- present in 3rd shell i.e. $n = 3$

$$\therefore E = -13.6 \times \frac{(3)^2}{(3)^2} = -13.6 \text{ eV/atom}$$

Ex.24 If the P.E. of an electron is -6.8 eV in hydrogen atom then find out K.E. of electron and radius of orbit, where electron exist.

Sol. (i) P.E. = -2K.E.

$$-6.8 = -2\text{K.E.}$$

$$\frac{6.8}{2} = \text{K.E.} \quad \Rightarrow \quad \text{K.E.} = 3.4 \text{ eV}$$

(ii) T.E. = $\frac{\text{P.E.}}{2} = -3.4 \text{ eV}$

$$\therefore E = -13.6 \times \frac{Z^2}{n^2}$$

$$\text{or } -3.4 = -13.6 \times \frac{1^2}{n^2}$$

$$\therefore n = 2$$

$$\begin{aligned} \text{(iii) } r &= 0.529 \times \frac{n^2}{Z} \text{ \AA} \\ r &= 0.529 \times \frac{(2)^2}{1} \text{ \AA} \\ &= 0.529 \times 4 \text{ \AA} = 2.16 \text{ \AA} \end{aligned}$$

Ex.25 Calculate the required energy in eV to excite electron from the ground state to 1st excited state in a hydrogen atom.

Sol. $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$

$$E_1 = -13.6 \times \frac{1^2}{1^2} = -13.6 \text{ eV}$$

$$E_2 = -13.6 \times \frac{1^2}{2^2} = -3.4 \text{ eV}$$

$\therefore$ Required energy = $E_2 - E_1$
 $= (-3.4) - (-13.6) = 10.2 \text{ eV}.$

Ex.26 Calculate the amount of energy absorbed in the transition $n = 1$ to $n = 3$ in Li^{2+} ion.

Sol. $n_1 \text{ orbit} \longleftrightarrow n_2 \text{ orbit}$

$$\Delta E = E_{n_2} - E_{n_1} = \left(-13.6 \frac{Z_1^2}{n_2^2} \right) - \left(-13.6 \frac{Z_2^2}{n_1^2} \right)$$

$$\Delta E = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

$$\Delta E = 13.6 \times 3^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = 108.8 \text{ eV}$$

Ex.27 Calculate the excitation energy of Be^{3+} ion in ground state.

Sol. $\Delta E = 13.6 \times 16 \left(\frac{1}{1} - \frac{1}{4} \right) = 163.2 \text{ eV}$

Ex.28 The ionisation energy of He^+ ion is $x \text{ kJ/mole}$. Calculate ionisation energy of Li^{2+} ion.

Sol. For I.E. $\Rightarrow n = 1 \longrightarrow n = \infty$

$$IE = 13.6 Z^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right)$$

$$\therefore \boxed{\text{I.E.} = 13.6 Z^2 \text{ eV}} \Rightarrow \text{I.E.} \propto Z^2$$

$$\frac{(\text{I.E.})_{\text{Li}^{2+}}}{(\text{I.E.})_{\text{He}^+}} = \frac{3^2}{2^2} \Rightarrow (\text{I.E.})_{\text{Li}^{2+}} = \frac{9}{4} \times \text{kJ/mol}$$

Ex.29 The ionisation energy for a single electron system is 14.4 eV . Calculate the amount of energy released when electron jumps from 3rd orbits to 2nd orbit.

Sol. $\Delta E = (IE) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 14.4 \times \left(\frac{1}{4} - \frac{1}{9} \right) = 2 \text{ eV}$

Ex.30 Calculate the speed of an electron in the 3rd orbit of the Li²⁺ ion. Also calculate the number of revolutions per second that it makes around the nucleus.

Sol. Radius of 2nd orbit = $r_1 \times \frac{(n)^2}{Z} = 0.529 \times \frac{(3)^2}{3} = 1.587 \text{ Å}$

Speed of electron in 2nd orbit, $v = 2.18 \times 10^6 \frac{Z}{n} \text{ m/sec} = 2.18 \times 10^6 \text{ m/sec}$

No. of revolutions/sec = $\frac{1}{2\pi r/v} = \frac{v}{2\pi r} = \frac{2.18 \times 10^6 \text{ m/sec}}{2 \times 3.14 \times 1.587 \times 10^{-10} \text{ m}}$
 $= 2.187 \times 10^{15} \text{ rev/sec}$

8.2 RYDBERG FORMULA

If an electron shows transition from n_2 to n_1 energy level then energy change ΔE will be.

$$\Delta E = E_{n_2} - E_{n_1}$$

$$\Delta E = \left[\frac{-2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2} \right] - \left[\frac{-2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} \right]$$

or, $\frac{hc}{\lambda} = \frac{2\pi^2 m K^2 Z^2 e^4}{h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

$$\frac{1}{\lambda} = \frac{2\pi^2 m K^2 e^4 Z^2}{ch^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where $\frac{2\pi^2 m K^2 e^4}{ch^3}$ is a constant called **Rydberg constant (R)**

So, $\bar{\nu} = \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

* Value of $R = 109677 \text{ cm}^{-1} = 10967700 \text{ m}^{-1} \approx 1.09 \times 10^7 \text{ m}^{-1}$

* $\frac{1}{R} = 912 \text{ Å}$

Ex.31 What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from the energy level with $n = 4$ to the energy level with $n = 1$?

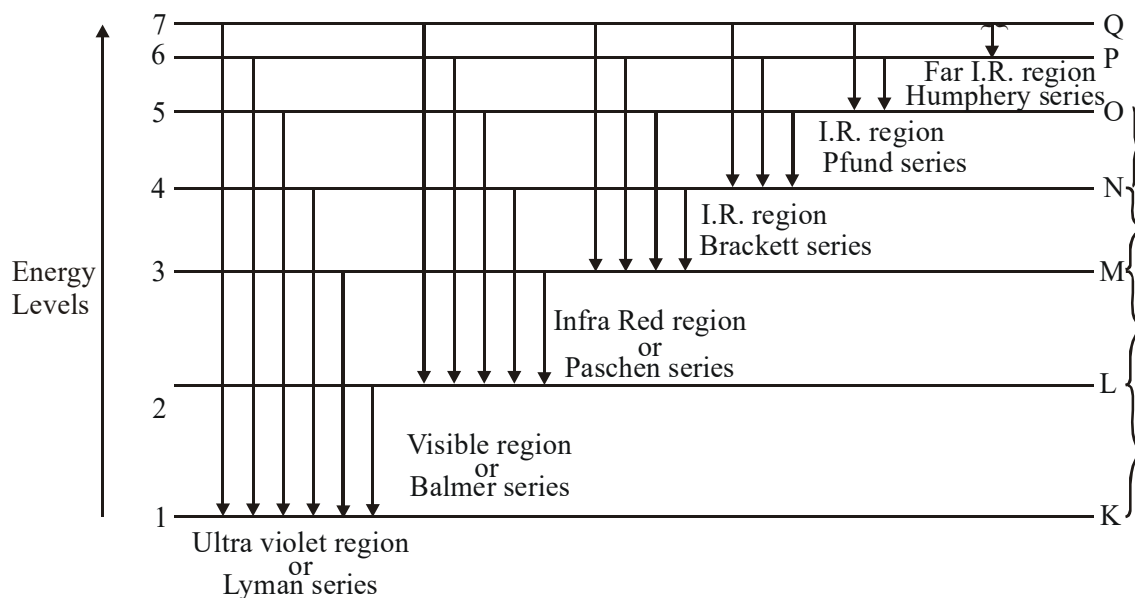
Sol. $\bar{\nu} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

or, $\frac{1}{\lambda} = R \times 1^2 \left(\frac{1}{1^2} - \frac{1}{4^2} \right) = \frac{15R}{16}$

$\therefore \lambda = \frac{16}{15} \times \frac{1}{R} = 9.7859 \times 10^{-8} \text{ m} = 978.59 \text{ Å}$

8.3 HYDROGEN LINE SPECTRUM :

When an electric excitation is applied on atomic hydrogen gas at low pressure, a bluish light is emitted. When a ray of this light is passed through a prism, a spectrum of several isolated sharp lines is obtained. The wavelength of various lines show that spectrum lines lie in visible, Ultraviolet and Infra red region. These lines are grouped into different series.



Series	Discovered by	Regions	n_2	n_1
Lyman	Lyman	U.V. region	$n_2 = 2, 3, 4 \dots$	$n_1 = 1$
Balmer	Balmer	Visible region	$n_2 = 3, 4, 5 \dots$	$n_1 = 2$
Paschen	Paschen	Infra red (I.R.)	$n_2 = 4, 5, 6 \dots$	$n_1 = 3$
Brackett	Brackett	I.R. region	$n_2 = 5, 6, 7 \dots$	$n_1 = 4$
Pfund	Pfund	I.R. region	$n_2 = 6, 7, 8 \dots$	$n_1 = 5$
Humphery	Humphery	Far I.R. region	$n_2 = 7, 8, 9 \dots$	$n_1 = 6$

8.3.1 KEY POINTS :

- First line / Starting line / Initial line ($\lambda_{\max.}$ and $\nu_{\min.}$)
- Last line / limiting line / Series limit ($\lambda_{\min.}$ and $\nu_{\max.}$)
- First line of any series = α line
Second line of any series = β line
Third line of any series = γ line
- Total no. of emission lines between n_2 & $n_1 = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$, ($n_2 > n_1$)
- For transition from any orbit 'n' to $n = 1$, total no. of emission lines = $\frac{n(n-1)}{2}$

Ex.32 In a hydrogen spectrum if electron moves from orbit 6^{th} to 2^{nd} orbit by transition in multi steps, then find out the number of lines in spectrum

Sol. Total number of line = $4 + 3 + 2 + 1 + 0 = 10$

$$\text{Total number of lines} = \frac{(n_2 - n_1)[(n_2 - n_1) + 1]}{2} = \frac{(6-2)(4+1)}{2} = 10$$

Ex.33 In the spectrum of He^+ ion the wavelength of α line of Balmer series is $x \text{ \AA}$. What is the wavelength of β line of Paschen series.

Sol. $\frac{1}{\lambda_1} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$

$$\frac{1}{\lambda_2} = RZ^2 \left(\frac{1}{3^2} - \frac{1}{5^2} \right)$$

$$\frac{\lambda_2}{\lambda_1} = \frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{3^2} - \frac{1}{5^2}} \Rightarrow \frac{\lambda_2}{\text{x}\text{\AA}} = \frac{5}{16} \times \frac{25}{4}, \quad \lambda_2 = 1.95 \text{ x } (\text{\AA})$$

Ex.34 A sample of He^+ ions in ground state absorbs the radiation of $x \text{ \AA}$. subsequently, the sample emit radiation of 6 different wavelength. Calculate the value of x . $\left(\text{Take } \frac{1}{R} = 912 \text{ \AA} \right)$

Sol. $\frac{n(n-1)}{2} = 6 \Rightarrow n = 4$

$$\frac{1}{x} = R\left(\frac{1}{1^2} - \frac{1}{4^2}\right) \times 2^2$$

$$x = \frac{16}{15} \times \frac{912}{4} \text{ \AA} = 243.20$$

Ex.35 Calculate the wavelength of the radiation of maximum frequency in the Lyman series of hydrogen spectrum.

Sol. For maximum v , $n = \infty$ to $n = 1$.

$$\text{Now, } \frac{1}{\lambda} = R \times 1^2 \left(\frac{1}{1^2} - \frac{1}{\infty} \right)$$

$$\therefore \lambda = 912 \text{ \AA}$$

8.4 Limitation of the Bohr's model

- (i) Bohr's theory does not explain the spectrum of multi electron atom.
- (ii) Why the Angular momentum of the revolving electron is equal to $\frac{nh}{2\pi}$, has not been explained by Bohr's theory.
- (iii) Bohr's theory does not explain the fine structure of spectral lines. Fine structure of the spectral line is obtained when spectrum is viewed by a spectroscope of high resolving power.
- (iv) Bohr's theory does not explain the splitting of spectral lines in the presence of magnetic field (**Zeeman effect**) or electric field (**Stark effect**)

9. DUAL BEHAVIOUR OF MATTER & DE BROGLIE WAVELENGTH :

In 1924, a French physicist, **Louis de Broglie** suggested that, like light, matter also has dual character. It exhibits wave as well as particle nature. According to de Broglie, the wavelength λ of any moving particle is inversely proportional to its momentum p .

$$\lambda \propto \frac{1}{p}$$

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Here h = Planck's constant

p = momentum of particle



de Broglie'
(1892-1987)

A French physicist, studied history as an undergraduate in the early 1910s. His interest turned to science as a result of his assignment to radio communications in world war 1. He was awarded the Nobel Prize in physics in 1929.

It is clear from the above equation that the value of λ decreases on increasing either m or v or both. The wavelength of many fast-moving objects like an aeroplane or a cricket ball, is very low because of their high mass. Thus wave nature of macroscopic objects can be neglected but for microscopic particles like electrons, protons, atoms etc. wave nature is significant & cannot be neglected.

9.1 DERIVATION OF BOHR'S ANGULAR MOMENTUM QUANTIZATION RULE :

We know that according to Bohr theory, $mvr = \frac{nh}{2\pi}$

or $2\pi r = \frac{nh}{mv}$

or $2\pi r = \frac{nh}{p} = n\lambda \quad \left(\because \frac{h}{p} = \lambda \text{ de-Broglie equation} \right)$

According to de Broglie, an electron bound to the nucleus behaves like a standing wave.

A standing wave, also known as a stationary wave, is a wave that remains in a constant position.

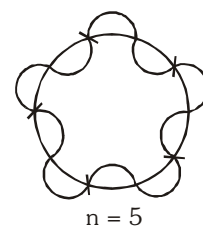
For a circular standing wave to persist, a whole number of wavelength must fit into the circumference of the circle ($2\pi r$).

And if n number of waves of λ wavelength are present in this circle total circumference will be $n\lambda$.

$$2\pi r = n\lambda$$

n = Number of wave made by electron in one complete revolution.

Electron in n^{th} orbit of any unielectron system can create ' n ' waves in one complete revolution.



$\therefore$ **Waves made = 5**

Ex.36 Calculate the de Broglie wavelength of a ball of mass 0.1 kg moving with a speed of 30 ms^{-1} .

Sol. $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{0.1 \times 30} = 2.2 \times 10^{-34} \text{ m}$

This is apparent that this wavelength is too small for ordinary observation.

Although the de Broglie equation is applicable to all material objects but it has significance only in case of microscopic particles.

Ex.37 What should be the mass of the sodium photon if its wavelength is 5894 Å . The speed of light is $3 \times 10^8 \text{ metre/second}$ and the value of h is $6.652 \times 10^{-34} \text{ kg m}^2/\text{sec}.$

(A) 3.746×10^{-26}

(B) 3.746×10^{-30}

(C) 3.746×10^{-34}

(D) 3.746×10^{-36}

Sol. $\lambda = \frac{h}{m \times c} \Rightarrow m = \frac{h}{c\lambda}$
($\because \lambda = 5894 \text{ Å} = 5894 \times 10^{-10} \text{ m}$)

$$m = \frac{6.652 \times 10^{-34}}{3 \times 10^{-8} \times 5894 \times 10^{-10}} \quad \text{or} \quad \frac{6.652}{17682} \times 10^{-32}$$

$$= 0.0003746 \times 10^{-32} = 3.746 \times 10^{-36} \text{ kg}$$

Ex.38 Calculate the de-Broglie wavelength of electron accelerated by the following potential differences.

(i) 750 V

(ii) 300 volt

Sol. (i) $\lambda = \sqrt{\frac{150}{V}} \text{ Å} = \sqrt{\frac{150}{750}} \text{ Å} = \frac{1}{\sqrt{5}} \text{ Å}$ (ii) $\lambda = \sqrt{\frac{150}{300}} \text{ Å} = \frac{1}{\sqrt{2}} \text{ Å}$

Ex.39 Find de-Broglie wavelength of electron with $KE = 9.6 \times 10^{-19} \text{ J}$.

Sol. $KE = \frac{9.6 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 6 \text{ eV}$

$$\lambda = \sqrt{\frac{150}{6}} \text{ Å} = 5 \text{ Å}$$

KE of 6 eV means e^- is accelerated by 6 volt.

Ex.40 Calculate the ratio of de-Broglie wavelength of electron and α -particle.

(i) Moving at same speed

(ii) Moving at same momentum

(iii) Having same K.E.

(iv) Accelerated from rest through the same P.D.

Sol. (i) $\lambda = \frac{h}{mv}$

$$\lambda \propto \frac{1}{m}$$

$$\frac{\lambda_{\text{electron}}}{\lambda_{\alpha}} = \frac{m_{\alpha}}{m_e} = \frac{4 \times 1836}{1}$$

$$(ii) \quad \frac{\lambda_{\text{electron}}}{\lambda_{\alpha}} = \frac{1}{1}$$

$$(iii) \quad \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{\lambda_e}{\lambda_{\alpha}} = \sqrt{\frac{m_{\alpha}}{m_e}} = \sqrt{\frac{1836 \times 4}{1}}$$

$$(iv) \quad \lambda = \frac{h}{\sqrt{2mqV}} = \frac{\lambda_e}{\lambda_{\alpha}} = \sqrt{\frac{(mq)_{\alpha}}{(mq)_e}} = \sqrt{\frac{4 \times 2}{1/1836 \times 1}}$$

Ex.41 In Li^{2+} ion electron jumps from 2^{nd} to 1^{st} orbit. If the emitted radiation is absorbed by H atom.

Calculate the de-Broglie wavelength of the ejected electron.

Sol. $\Delta E = 13.6 \times 9 \left(1 - \frac{1}{4}\right) = 91.8 \text{ eV}$

Excess energy = $91.8 - 13.6 = 78.2 \text{ eV}$

$$\lambda = \sqrt{\frac{150}{78.2}} \text{ \AA} = 1.38 \text{ \AA}$$

Ex.42 Photoelectrons are liberated by ultra violet light of wavelength $2000 \text{ \AA}$ from a metallic surface for which the photoelectric threshold is $4000 \text{ \AA}$. Calculate the de-Broglie wavelength of electrons emitted with maximum kinetic energy.

Solution : K.E. = Quantum Energy – Threshold energy

$$\begin{aligned} &= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2000 \times 10^{-10}} - \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{4000 \times 10^{-10}} \\ &= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{10^{-10}} \left(\frac{1}{2000} - \frac{1}{4000} \right) = 4.969 \times 10^{-19} \text{ Joule.} \end{aligned}$$

$$\frac{1}{2}mv^2 = 4.969 \times 10^{-19} \Rightarrow m^2v^2 = 2 \times 4.969 \times 10^{-19} \times 9.1 \times 10^{-31}$$

$$mv = 9.51 \times 10^{-25} \Rightarrow \lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{9.51 \times 10^{-25}} = 0.696 \times 10^{-9} \text{ m}$$

Ex.43 Calculate the de-Broglie wavelength when proton is accelerated by the 750 V.

Sol. $\lambda = \sqrt{\frac{150}{V \times 1836}} \text{ \AA} = \sqrt{\frac{150}{750 \times 1836}} \text{ \AA} = 0.01 \text{ \AA}.$

9.2 Justification of dual nature of electrons :

I. Particle character :

- If an electron strikes a screen coated with ZnS, it produces a spot of light called scintillation, one electron produces only one scintillation point which means electron are localised not spread out like wave.
- Electron possess definite mass, momentum & KE proving their particle nature.

II. Wave character :

- It was confirmed by phenomenon of diffraction, interference, reflection.
Davisson and Germer showed that when high speed electron strike Ni crystal a diffraction pattern (having number of rings) is obtained like X-rays of electromagnetic spectrum.

10. HEISENBERG UNCERTAINTY PRINCIPLE :

Bohr's theory considers an electron as a material particle. Its position and momentum can be determined with accuracy. But, when an electron is considered in the form of wave as suggested by de-Broglie, it is not possible to ascertain simultaneously the exact position and speed of the electron more precisely at a given instant since the wave is extending throughout a region of space.

In 1927, **Werner Heisenberg** presented a principle known as Heisenberg uncertainty principle which states as : *"It is impossible to measure simultaneously the exact position and exact momentum of a body as small as an electron"*.

The uncertainty of measurement of position, Δx , and the uncertainty of momentum Δp or $m\Delta v$, are related by Heisenberg's relationship as : ($p = mv$, $\Delta p = m\Delta v$)

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta x \cdot m\Delta v \geq \frac{h}{4\pi}$$

$$\text{or} \quad \Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

where h is Planck's constant.

In the case of bigger particles (having considerable mass), the value of uncertainty product is negligible.

Ex.44 A golf ball has a mass of 40 g and a speed of 45 m/s. If the speed can be measured within accuracy of 2 %, calculate the uncertainty in the position.

Sol. Mass of the ball = 40 g = 40×10^{-3} kg

The uncertainty in the speed,

$$\Delta v = 45 \times \frac{2}{100} = 0.9 \text{ ms}^{-1}$$

$$\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-34} \text{ Js}}{4 \times 3.14 \times (40 \times 10^{-3} \text{ kg})(0.9 \text{ ms}^{-1})} = 1.46 \times 10^{-33} \text{ m.}$$

Ex.45 Calculate the uncertainty in the velocity of a cricket ball of mass 150 g, if the uncertainty in its position is of the order of $1 \text{ \AA}$.

11.1 HYDROGEN ATOM AND THE SCHRÖDINGER EQUATION

Erwin Schrodinger, an Austrian physicist received his Ph.D. in theoretical physics from the University of Vienna in 1910. In 1927 Schrodinger succeeded Max Planck at the University of Berlin at Planck's request. In 1933, Schrodinger left Berlin because of his opposition to Hitler and Nazi policies and returned to Austria in 1936. After the invasion of Austria by Germany, Schrodinger fled from his professorship. He moved to Ireland where he remained until his death. Schrodinger shared the Nobel Prize in Physics with P.A.M. Dirac in 1933.



Erwin Schrodinger
(1887-1961)

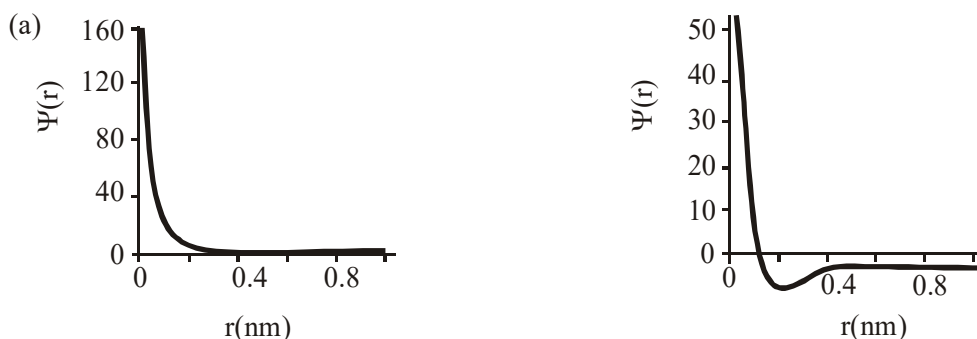
Application of Schrodinger equation to multi-electron atoms presents a difficulty: the Schrodinger equation cannot be solved exactly for a multi-electron atom. This difficulty can be overcome by using approximate methods. Such calculations with the aid of modern computers show that orbitals in atoms other than hydrogen do not differ in any radical way from the hydrogen orbitals discussed above. The principal difference lies in the consequence of increased nuclear charge. Because of this all the orbitals are somewhat contracted. Further, as you shall see later, unlike orbitals of hydrogen or hydrogen like species, whose energies depend only on the quantum number n , the energies of the orbitals in multi-electron atoms depend on quantum numbers n and l .

1. The energy of electrons in atoms is quantized (i.e., can only have certain specific values), for example when electrons are bound to the nucleus in atoms.
2. The existence of quantized electronic energy levels is a direct result of the wave like properties of electrons and are allowed solutions of Schrodinger wave equation.

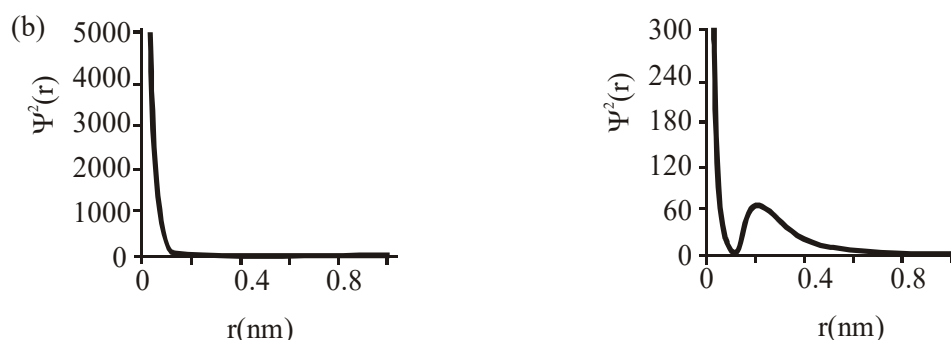
- Both the exact position and exact velocity of an electron in an atom cannot be determined simultaneously (Heisenberg uncertainty principle). The path of an electron in an atom therefore, can never be determined or known accurately. That is why, as you shall see later on, one talks of only probability of finding the electron at different points in an atom.
- An atomic orbital is the wave function ψ for an electron in an atom.** Whenever an electron is described by a wave function, we say that the electron occupies that orbital. Since many such wave functions are possible for an electron, there are many atomic orbitals in an atom. These “one electron orbital wave functions” or orbitals form the basis of the electronic structure of atoms. In each orbital, the electron has a definite energy. An orbital cannot contain more than two electrons. In a multi-electron atom, the electrons are filled in various orbitals in the order of increasing energy. For each electron of a multi-electron atom, there shall, therefore, be an orbital wave function characteristic of the orbital it occupies. All the information about the electron in an atom is stored in its orbital wave function ψ and quantum mechanics makes it possible to extract this information out of ψ .
- The probability of finding an electron at a point within an atom is proportional to the square of the orbital wave function i.e., $|\psi|^2$ at that point. $|\psi|^2$ is known as **probability density** and is always positive. From the value of $|\psi|^2$ at different points within an atom, it is possible to predict the region around the nucleus where electron will most probably be found.

11.3 Shapes of Atomic Orbitals

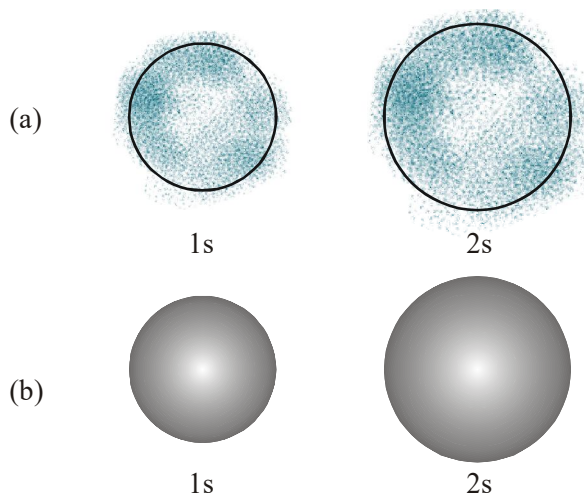
The orbital wave function or ψ for an electron in an atom has no physical meaning. It is simply a mathematical function of the coordinates of the electron. However, for different orbitals the plots of corresponding wave functions as a function of r (the distance from the nucleus) are different. Such plots for 1s ($n = 1, l = 0$) and 2s ($n = 2, l = 0$) orbitals are



According to the German physicist, Max Born, the square of the wave function (i.e., ψ^2) at a point gives the probability density of the electron at that point. The variation of ψ^2 as a function of r for 1s and 2s orbitals is given in fig. Here again, you may note that the curves for 1s and 2s orbitals are different.



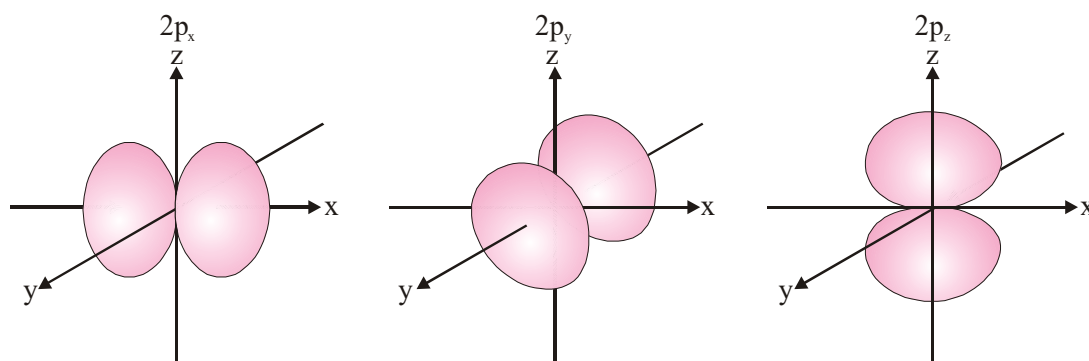
It may be noted that for 1s orbital the probability density is maximum at the nucleus and it decreases sharply as we move away from it. On the other hand, for 2s orbital the probability density first decreases sharply to zero and again starts increasing. After reaching a small maxima it decreases again and approaches zero as the value of r increases further. The region where this probability density function reduces to zero is called **nodal surfaces** or simply **nodes**. In general, it has been found that ns-orbital has $(n - 1)$ nodes, that is, number of nodes increases with increase of principal quantum number n . In other words, number of nodes for 2s orbital is one, two for 3s and so on. These probability density variation can be visualised in terms of charge cloud diagrams. In these diagrams, the density of the dots in a region represents electron probability density in that region.



(a) Probability density plots of 1s and 2s atomic orbitals. The density of the dots represents the probability density of finding the electron in that region. (b) Boundary surface diagram for 1s and 2s orbitals. **Boundary surface diagrams** of constant probability density for different orbitals give a fairly good representation of the shapes of the orbitals. In this representation, a boundary surface or contour surface is drawn in space for an orbital on which the value of probability density $|\psi|^2$ is constant. In principle many such boundary surfaces may be possible. However, for a given orbital, only that boundary surface diagram of constant probability density is taken to be good representation of the shape of the orbital which encloses a region or volume in which the probability of finding the electron is very high, say, 90%. The boundary surface diagram for 1s and 2s orbitals are given in fig. One may ask a question : Why do we not draw a boundary surface diagram, which bounds a region in which the probability of finding the electron is, 100 %? The answer to this question is that the probability density $|\psi|^2$ has always some value, howsoever small it may be, at any finite distance from the nucleus. It is therefore, not possible to draw a boundary surface diagram of a rigid size in which the probability of finding the electron is 100%. Boundary surface diagram for a s orbital is actually a sphere centred on the nucleus. In two dimensions, this sphere looks like a circle. It encloses a region in which probability of finding the electron is about 90%.

Thus we see that 1s and 2s orbitals are spherical in shape. In reality all the s-orbitals are spherically symmetric, that is, **the probability of finding the electron at a given distance is equal in all the directions**. It is also observed that **the size of the s orbital increases with increase in n , that is, $4s > 3s > 2s > 1s$** and the electron is located further away from the nucleus as the principal quantum number increases.

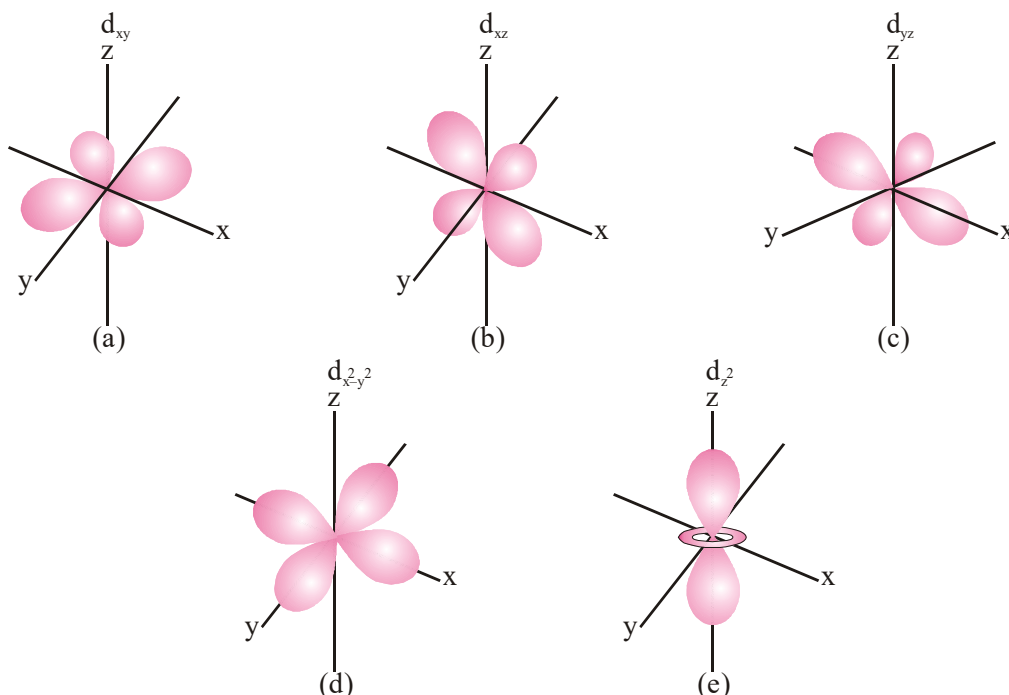
Boundary surface diagrams for three 2p orbitals ($l = 1$) are



Boundary surface diagrams of the three 2p orbitals.

In these diagrams, the nucleus is at the origin. Here, unlike s-orbitals, the boundary surface diagrams are not spherical. Instead each p orbital consists of two sections called lobes that are on either side of the plane that passes through the nucleus. The probability density function is zero on the plane where the two lobes touch each other. The size, shape and energy of the three orbitals are identical. They differ however, in the way the lobes are oriented. Since the lobes may be considered to lie along the x, y or z axis, they are given the designations $2p_x$, $2p_y$, and $2p_z$. It should be understood, however, that there is no simple relation between the values of m_l ($-1, 0$ and $+1$) and the x, y and z directions. For our purpose, it is sufficient to remember that, because there are three possible values of m_l , there are, therefore, three p orbitals whose axes are mutually perpendicular. Like s orbitals, **p orbitals increase in size and energy with increase in the principal quantum number** and hence the order of the energy and size of various p orbitals is **$4p > 3p > 2p$** . Further, like s orbitals, the probability density functions for p-orbital also pass through value zero, besides at zero and infinite distance, as the distance from the nucleus increases. The number of nodes are given by the $n - 2$, that is number of radial node is one for 3p orbital, two for 4p orbital and so on.

For $l = 2$, the orbital is known as d-orbital and the minimum value of principal quantum number (n) has to be 3. as the value of l cannot be greater than $n - 1$. There are five m_l values $-2, -1, 0, +1$ and $+2$ for $l = 2$ and thus there are five d orbitals. The boundary surface diagram of d orbitals are :



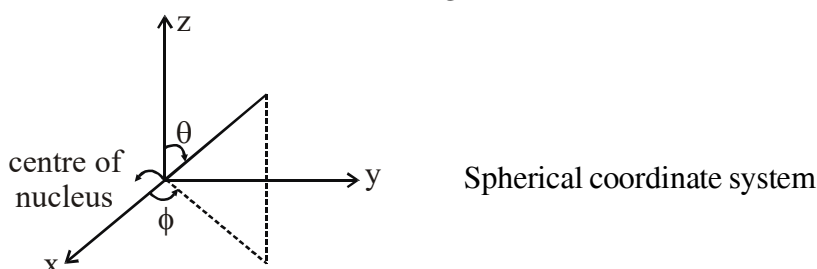
Boundary surface diagrams of the five 3d orbitals.

The five d-orbitals are designated as d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$ and d_{z^2} . The shapes of the first four d-orbitals are similar to each other, whereas that of the fifth one, d_{z^2} , is different from others, but all five 3d orbitals are equivalent in energy. The d orbitals for which n is greater than 3 (4d, 5d...) also have shapes similar to 3d orbital, but differ in energy and size.

Besides the radial nodes (i.e., probability density function is zero), the probability density functions for the np and nd orbitals are zero at the plane(s), passing through the nucleus (origin). For example, in case of p_z orbital, xy-plane is a nodal plane, in case of d_{xy} orbital, there are two nodal planes passing through the origin and bisecting the xy plane containing z-axis. These are called **angular nodes** and number of angular nodes are given by ' l ', i.e., one angular node for p orbitals, two angular nodes for 'd' orbitals and so on.

The total number of nodes are given by $(n-1)$, i.e., sum of l angular nodes and $(n-l-1)$ radial nodes.

11.4 SOLUTION OF SCHRÖDINGER EQUATION :



The solution in spherical coordinates may be represented as :

$$\psi = R(r) \cdot \Theta(\theta) \cdot \Phi(\phi)$$

$R(r)$ = Radial function depends on n and l

$\Theta(\theta) \cdot \Phi(\phi)$ = Angular function depends on l and m = $A(\theta, \phi) = Y(\theta, \phi)$.

11.4.1 Radical part of solution :

$$1s \quad (n = 1, \ell = 0) : \quad R_{1s}(r) = 2 \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot e^{-\sigma/2}$$

$$\text{where } \sigma = \frac{2Zr}{na_0} \quad a_0 = 1^{\text{st}} \text{ Bohr's radius} = 0.529 \text{ \AA}$$

$$2s \quad (n = 2, \ell = 0) : \quad R_{2s}(r) = \frac{1}{2\sqrt{2}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot (2 - \sigma) e^{-\sigma/2}$$

$$2p \quad (n = 2, \ell = 1) : \quad R_{2p}(r) = \frac{1}{2\sqrt{6}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot \sigma \cdot e^{-\sigma/2}$$

$$3s \quad (n = 3, \ell = 0) : \quad R_{3s}(r) = \frac{1}{9\sqrt{3}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot (6 - 6\sigma + \sigma^2) e^{-\sigma/2}$$

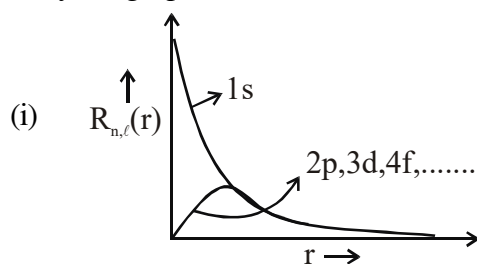
$$3p \quad (n = 3, \ell = 1) : \quad R_{3p}(r) = \frac{1}{9\sqrt{6}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot \sigma(4 - \sigma) e^{-\sigma/2}$$

$$3d \quad (n = 3, \ell = 2) : \quad R_{3d}(r) = \frac{1}{9\sqrt{30}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot \sigma^2 \cdot e^{-\sigma/2}$$

General form : $R_{n\ell}(r) = K \cdot e^{-\sigma/2} \cdot \sigma^\ell$ (Polynomial of order $n - \ell - 1$)

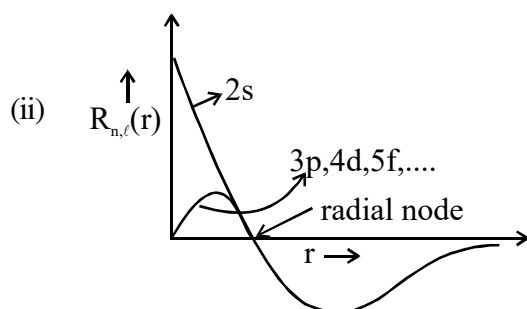
11.4.2 Graph of radial function $[R(r) \text{ or } \Psi(r)]$:

Only the graph of s-orbital does not start from origin.



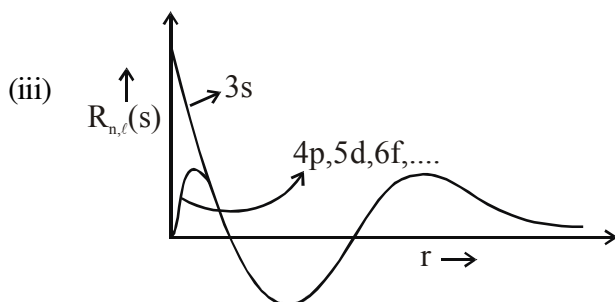
$$n - \ell - 1 = 0$$

1s, 2p, 3d, 4forbitals



$$n - \ell - 1 = 1$$

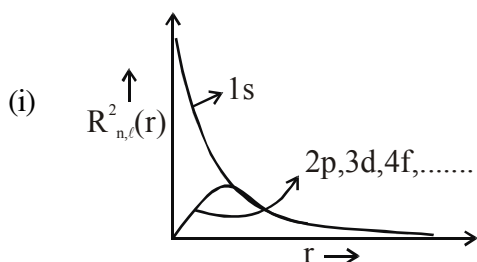
2s, 3p, 4d, 5forbitals



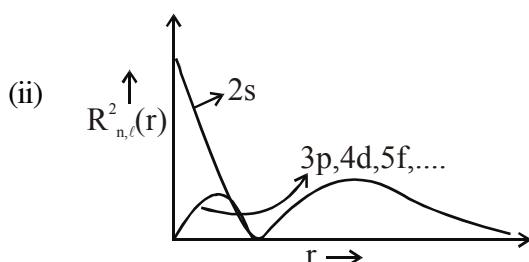
$$n - \ell - 1 = 2$$

3s, 5p, 5d, 6f orbitals

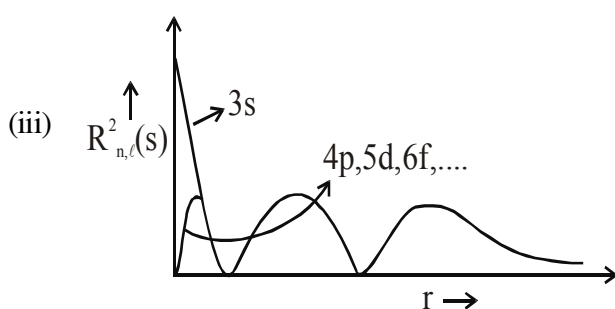
11.4.3 GRAPH OF RADIAL PROBABILITY DENSITY FUNCTION [$R^2(r)$ or $\psi^2(r)$] :



$$n - \ell - 1 = 0, \quad (1s, 2p, 3d, 4f \text{orbitals})$$



$$n - \ell - 1 = 1 \quad (2s, 3p, 4d, 5f \text{orbitals})$$



$$n - \ell - 1 = 2 \quad (3s, 5p, 5d, 6f \text{ orbitals})$$

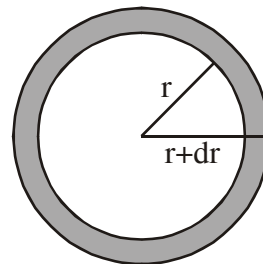
11.4.4 RADIAL PROBABILITY DISTRIBUTION FUNCTION (RPDF), $4\pi r^2 \psi^2(r)$

It is often useful to know the likelihood of finding the electron in an orbital at any given distance away from the nucleus. This enables us to say at what distance from the nucleus the electron is most likely to be found, and also how tightly or loosely the electron is bound in a particular atom. This is expressed by the radial probability distribution function, $4\pi r^2 \psi^2(r)$.

Radial distribution function is the measure of the probability of finding the electron in a spherical shell between thickness r and $(r + dr)$ from the nucleus, irrespective of the direction.

❖ Volume of radial shell :

$$\begin{aligned}
 dV &= \left[\text{Volume of sphere with radius } (r + dr) \right] - \left[\text{Volume of sphere with radius } r \right] \\
 &= \frac{4}{3}\pi(r + dr)^3 - \frac{4}{3}\pi r^3 \\
 &= \frac{4}{3}\pi(r^3 + 3r^2dr + 3rdr^2 + dr^3) - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi[r^3 + 3r^2dr - r^3]
 \end{aligned}$$



(As dr represents an extremely small thickness, the higher powers of dr such as dr^2 and dr^3 may be neglected.)

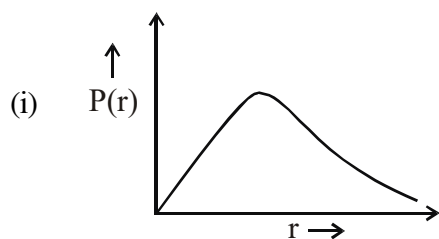
$$\therefore \text{Volume of shell, } dV = \frac{4}{3}(\pi \times 3r^2 dr) = 4\pi r^2 dr$$

$$\text{Now, radial probability density, } R^2(r) = \frac{P}{dV}$$

$\therefore$ Probability of finding electron in the volume element,

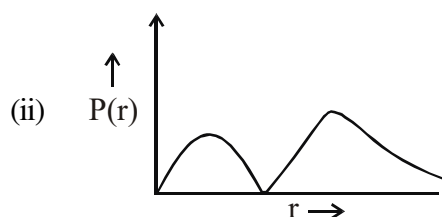
$$P = R^2(r) \cdot dV = R^2(r) \cdot 4\pi r^2 \cdot dr$$

$$\text{Now radial probability distribution function, } P(r) = \frac{P}{dr} = 4\pi r^2 \cdot R^2(r)$$



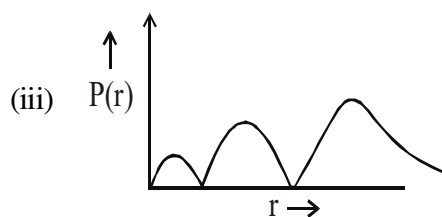
$$n - \ell - 1 = 0$$

1s, 2p, 3d, 4forbitals



$$n - \ell - 1 = 1$$

2s, 3p, 4d, 5forbitals



$$n - \ell - 1 = 2$$

3s, 5p, 5d, 6f orbitals

❖ **Characteristics of radial distribution function :**

- (i) The number of maxima in radial distribution function plot are $(n - \ell)$.
- (ii) The maximum probability of finding the electron, for the ground state hydrogen atom (1s) is found to be at a_0 (first Bohr radius).
- (iii) For 2s, 3s, 3p orbitals, the number of maxima is more than one, indicating that there is maximum probability of finding the electron at the distance corresponding to the highest value of peak. However, there is lesser probability of finding the electron at the other peaks. It shows that in a certain state, the electron spends some portion of its time very close to the nucleus.

11.5 ANGULAR PART OF SOLUTION :

(i) s-orbital :

$$\ell = 0, m = 0 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{1}{4\pi}}$$

As the probability of finding electron is not depending on angle (direction) then it must be same. In all direction and hence the shape of s-orbital is sphere.

(ii) p-orbital :

$$p_x \text{-orbital : } \ell = 1, m = \pm 1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \sin \theta \cos \phi$$

$$p_y \text{-orbital : } \ell = 1, m = \pm 1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \sin \theta \sin \phi$$

$$p_z \text{-orbital : } \ell = 1, m = 0 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \cos \theta$$

(iii) d-orbital :

$$d_{z^2} \text{-orbital : } \ell = 2, m = 0 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$$

$$d_{x^2-y^2} \text{-orbital : } \ell = 2, m = \pm 2 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin^2 \theta \cos 2\phi$$

$$d_{xy} \text{-orbital : } \ell = 2, m = \pm 2 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin^2 \theta \sin 2\phi$$

$$d_{xz} \text{-orbital : } \ell = 2, m = \pm 1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin \theta \cos \theta \cos \phi$$

$$d_{yz} \text{-orbital : } \ell = 2, m = \pm 1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin \theta \cos \theta \cdot \sin \phi$$

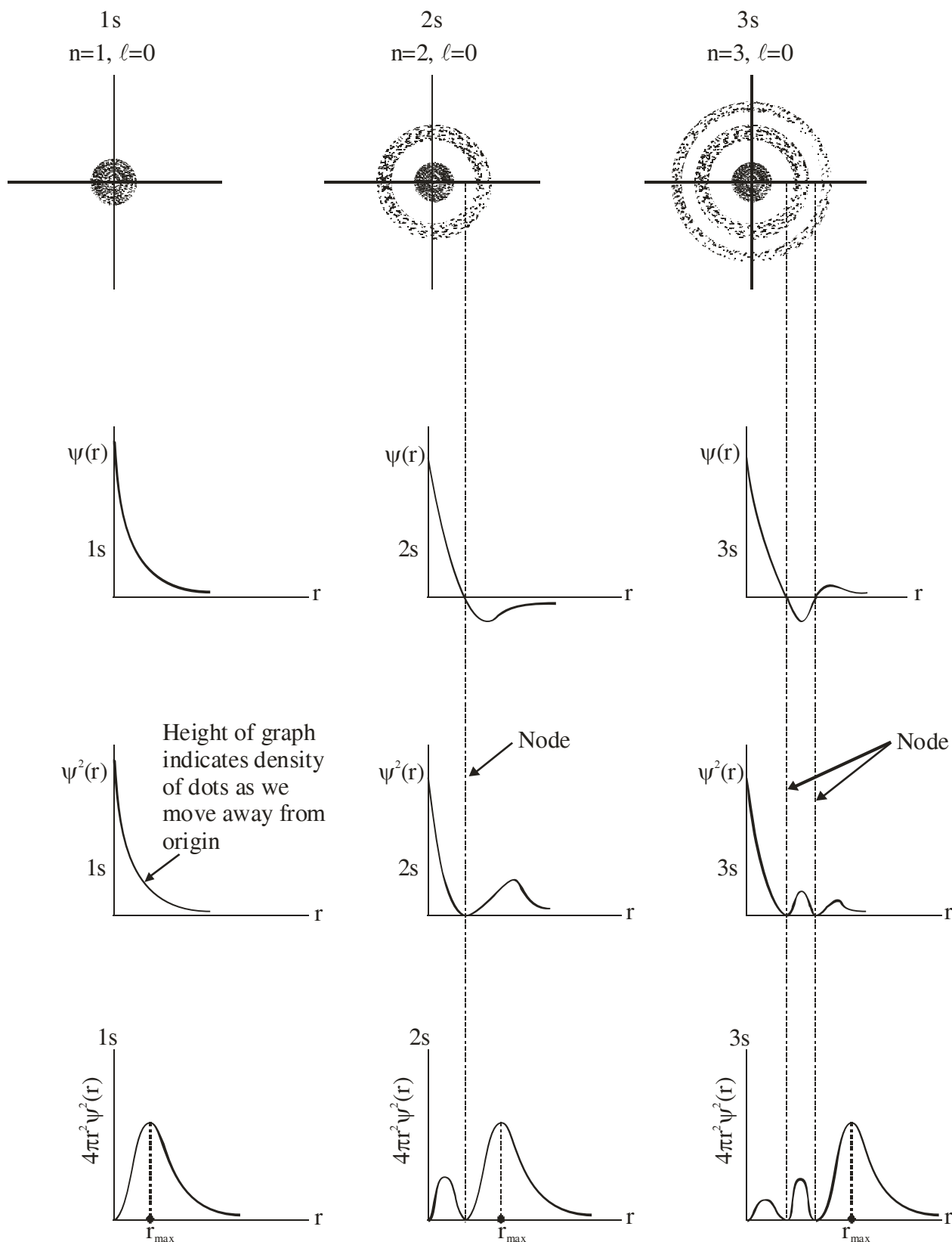
Note : (i) Number of radial nodes = $n - \ell - 1$

Number of angular nodes = ℓ

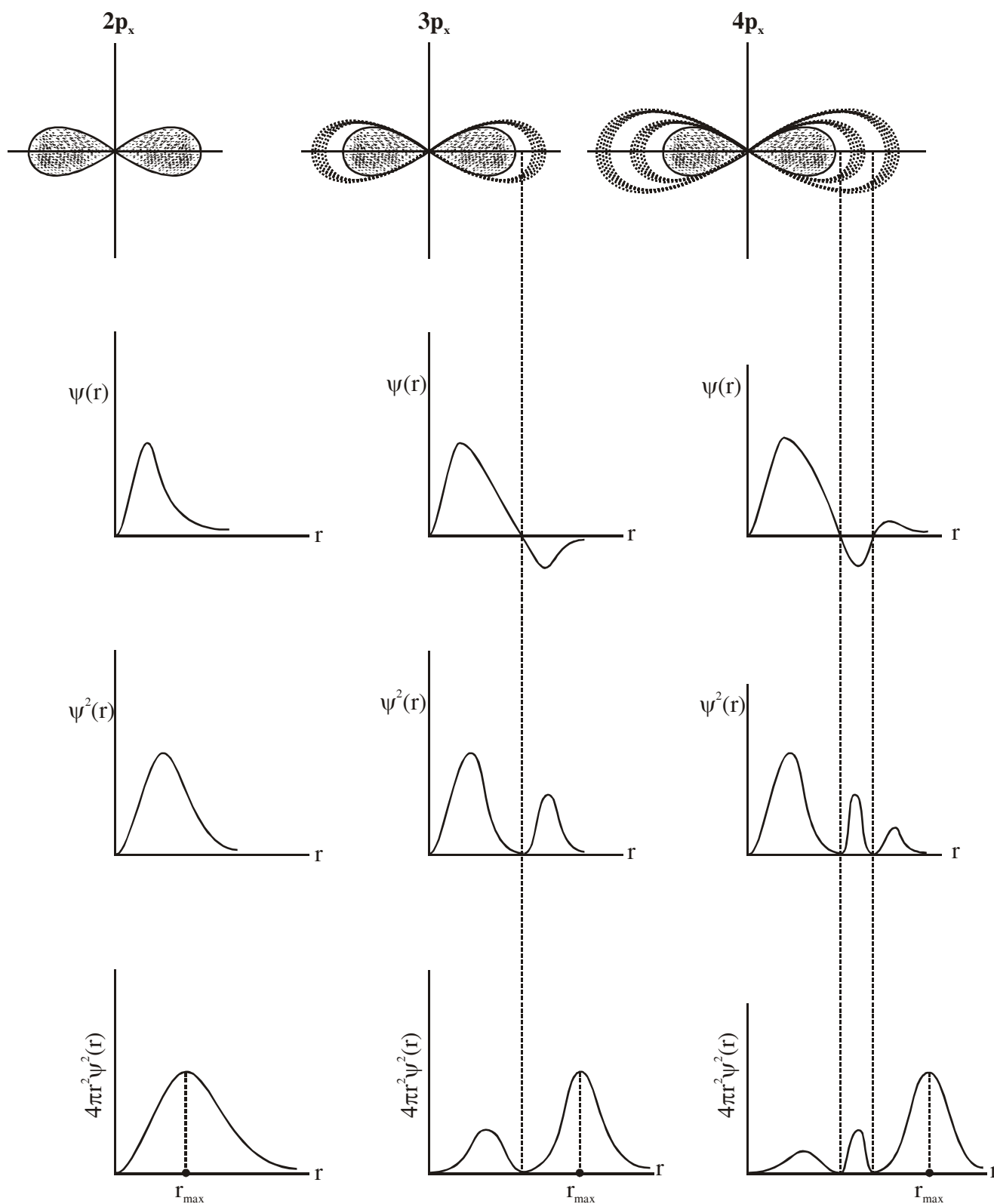
Total of number nodes = $n - 1$

- (ii) On standard axis consideration, the orbital with $m = 0$ have angular dependency only on θ and they represent orbitals with z-component like p_z, d_{z^2} , etc.**

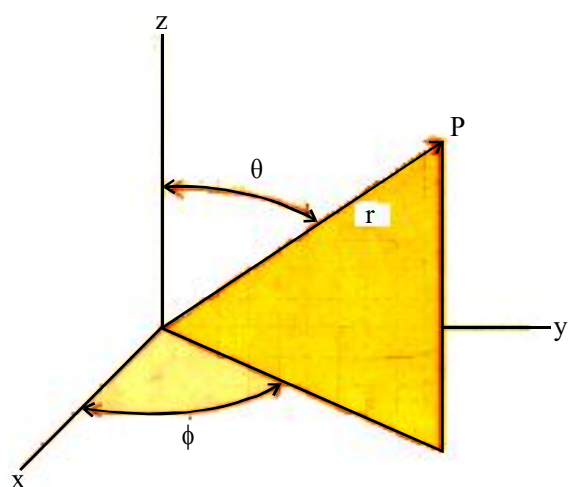
11.6 Electron-density distribution in 1s, 2s and 3s-orbitals. The lower part of the fig. shows how the electron density, represented by ψ^2 varies as a function of distance from the nucleus. In the 2s and 3s - orbitals, the electron-density function drops to zero at certain distances from the nucleus. The spherical surfaces around the nucleus at which ψ^2 is zero are called nodes.



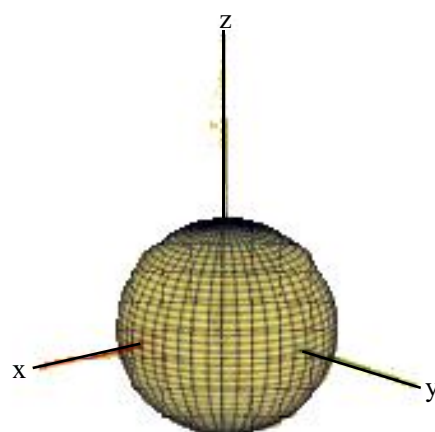
11.7 Electron-density distribution in $2p_x$, $3p_x$ and $4p_x$ -orbitals :



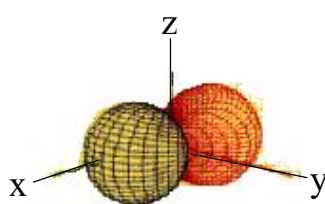
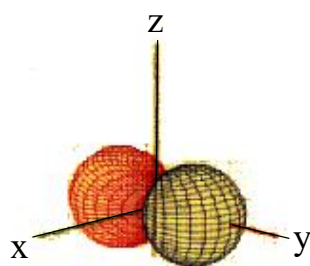
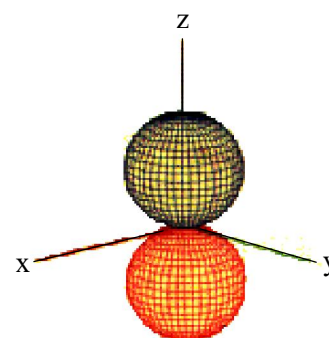
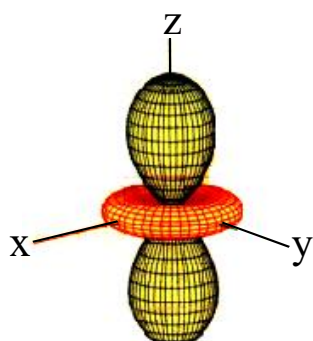
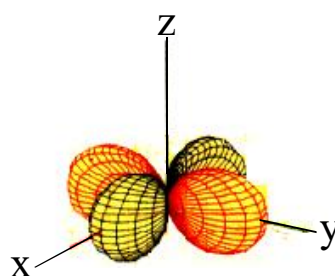
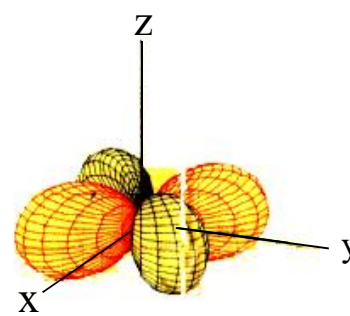
11.8 SHAPES OF ATOMIC ORBITALS

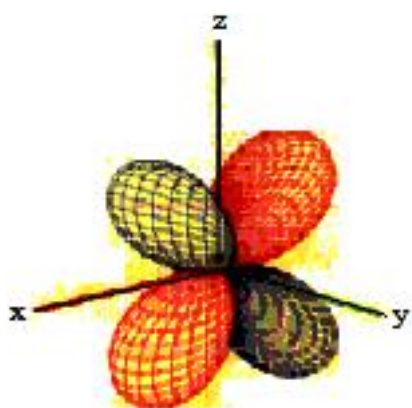


The spherical Polar Coordinates

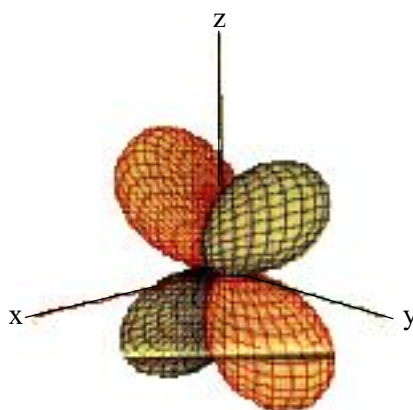


S

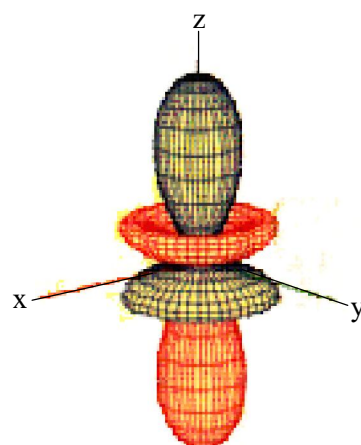
 p_x  p_y  p_z  d_{z^2}  $d_{x^2-y^2}$  d_{xy}



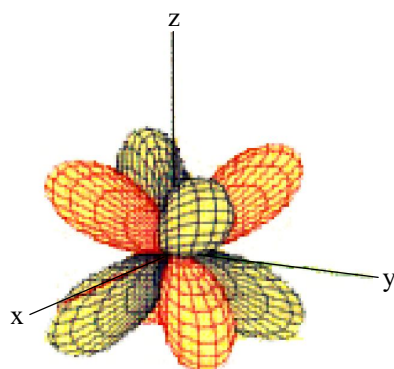
d_{xz}



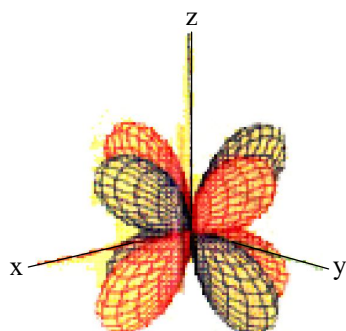
d_{yz}



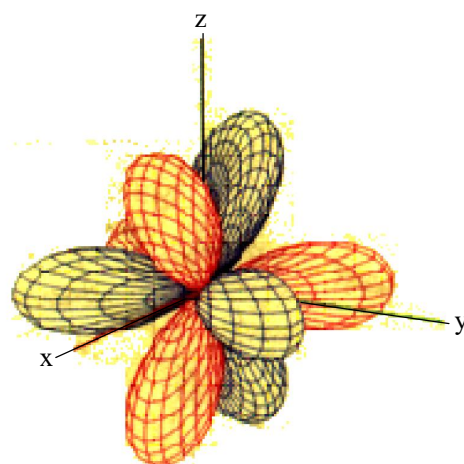
f_{z^3}



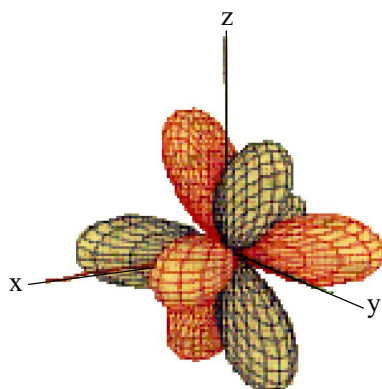
f_{xyz}



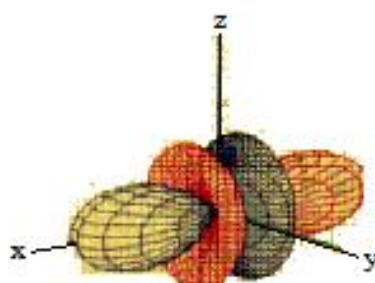
$f_{z(x^2-y^2)}$



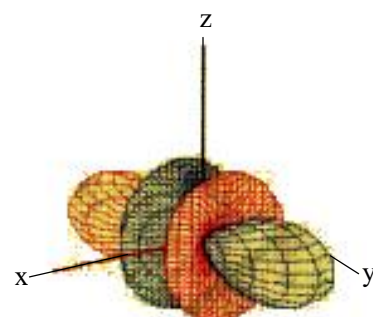
$f_{x(y^2-z^2)}$



$f_{y(z^2-x^2)}$



f_{x^3}



f_{y^3}

Ex.46 Find the distance at which probability of finding electron is maximum for 1s orbital in a hydrogen atom. The wave function of orbital is given as.

$$\psi_{1s} = \frac{2}{a_0^{3/2}} \cdot e^{-\frac{r}{a_0}}$$

Sol. Probability distribution function is $P(r) = \psi^2 \cdot 4\pi r^2 = \left(\frac{4}{a_0^3}\right) e^{\left(-\frac{2r}{a_0}\right)} \cdot 4\pi r^2$

$$\text{or, } P(r) = k \cdot r^2 \cdot e^{-\frac{2r}{a_0}}$$

$$\text{Now, } \frac{dP(r)}{dr} = \frac{16\pi}{a_0^3} \left[2r \left(e^{-\frac{2r}{a_0}} \right) - r^2 e^{-\frac{2r}{a_0}} \times \left(-\frac{2}{a_0} \right) \right] = 0$$

$$\text{or, } \frac{16\pi}{a_0} \cdot 2r \cdot e^{-\frac{2r}{a_0}} \left(1 - \frac{r}{a_0} \right) = 0$$

$$\text{or } r = a_0$$

$\Rightarrow$ probability of finding electron is maximum at distance a_0 from nucleus.

Ex.47 Consider ψ (wave function) of 2s atomic orbital of H-atom is -

$$\psi_{2s} = \frac{1}{4\sqrt{2}\pi a_0^{3/2}} \left[2 - \frac{r}{a_0} \right] e^{-\frac{r}{2a_0}}$$

Find distance of radial node from nucleus in terms of a_0 .

Sol. $R(r) = 0$

$$\left[2 - \frac{r}{a_0} \right] e^{-\frac{r}{2a_0}} = 0 \Rightarrow 2 - \frac{r}{a_0} = 0 \Rightarrow r = 2a_0$$

SELF STUDY

12. SUB-ATOMIC PARTICLES

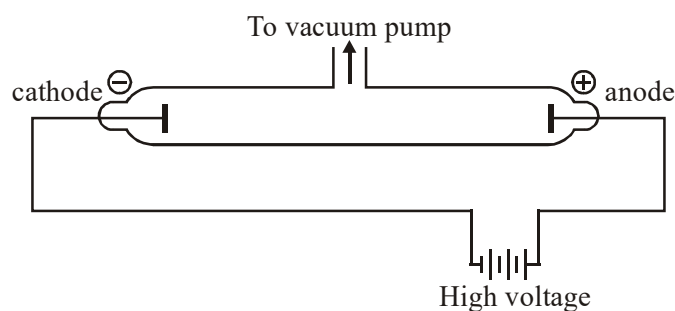
Dalton's atomic theory was able to explain the law of conservation of mass, law of constant composition and law of multiple proportion very successfully. However, it failed to explain the results of many experiments, for example, it was known that substances like glass or ebonite when rubbed with silk or fur generate electricity. Many different kinds of sub-atomic particles were discovered in the twentieth century.

12.1 Discovery of Electron :

In 1830, Michael Faraday showed that if electricity is passed through a solution of an electrolyte, chemical reactions occurred at the electrodes, which resulted in the liberation and deposition of matter at the electrodes. These results suggested the particulate nature of electricity.

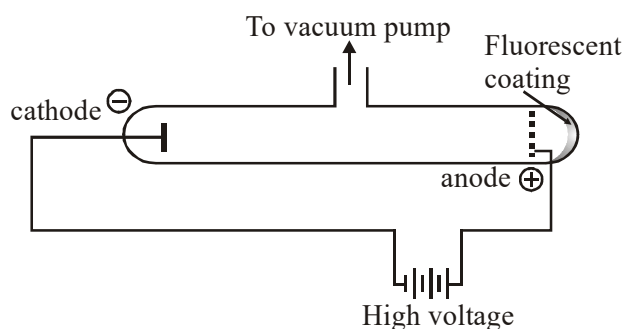
An insight into the structure of atom was obtained from the experiments on electrical discharge through gases. Before we discuss these results we need to keep in mind a basic rule regarding the behaviour of charged particles: "Like charges repel each other and unlike charges attract each other".

In mid 1850s many scientists mainly Faraday began to study electrical discharge in partially evacuated tubes, known as **cathode ray discharge tubes**.



A cathode ray discharge tube

A cathode ray tube is made of glass containing two thin pieces of metal, called electrodes, sealed in it. The electrical discharge through the gases could be observed only at very low pressures and at very high voltages. The pressure of different gases could be adjusted by evacuation. When sufficiently high voltage is applied across the electrodes, current starts flowing through a stream of particles moving in the tube from the negative electrode (cathode) to the positive electrode (anode). These were called **cathode rays or cathode ray particles**. The flow of current from cathode to anode was further checked by making a hole in the anode and coating the tube behind anode with phosphorescent material zinc sulphide. When these rays, after passing through anode, strike the zinc sulphide coating, a bright spot on the coating is developed (same thing happens in a television set).



A cathode ray discharge tube with perforated anode

The results of these experiments are summarised below.

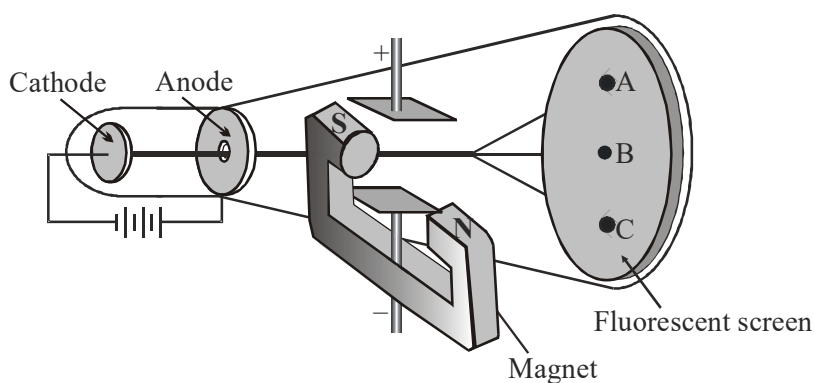
- (i) The cathode rays start from cathode and move towards the anode.
- (ii) These rays themselves are not visible but their behaviour can be observed with the help of certain kind of materials (fluorescent or phosphorescent) which glow when hit by them. Television picture tubes are cathode ray tubes and television pictures result due to fluorescence on the television screen coated with certain fluorescent or phosphorescent materials.
- (iii) In the absence of electrical or magnetic field, these rays travel in straight lines.
- (iv) In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that expected from negatively charged particles, suggesting that the cathode rays consist of negatively charged particles, called **electrons**.
- (v) The characteristics of cathode rays (electrons) do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.

Thus, we can conclude that electrons are basic constituent of all the atoms.

12.1.1 Charge to Mass Ratio of Electron :

In 1897, British physicist J.J. Thomson measured the ratio of electrical charge (e) to the mass of electron (m_e) by using cathode ray tube and applying electrical and magnetic field perpendicular to each other as well as to the path of electrons. Thomson argued that the amount of deviation of the particles from their path in the presence of electrical or magnetic field depends upon :

- (i) the magnitude of the negative charge on the particle, greater the magnitude of the charge on the particle, greater is the interaction with the electric or magnetic field and thus greater is the deflection.
- (ii) the mass of the particle lighter the particle, greater the deflection.
- (iii) the strength of the electrical or magnetic field . the deflection of electrons from its original path increases with the increase in the voltage across the electrodes, or the strength of the magnetic field.



The apparatus to determine the charge to the mass ratio of electron

When only electric field is applied, the electrons deviate from their path and hit the cathode ray tube at point A. Similarly when only magnetic field is applied, electron strikes the cathode ray tube at point C. By carefully balancing the electrical and magnetic field strength, it is possible to bring back the electron to the path followed as in the absence of electric or magnetic field and they hit the screen at point B. By carrying out accurate measurements on the amount of deflections observed by the electrons on the electric field strength or magnetic field strength, Thomson was able to determine the value of e/m_e as:

$$\frac{e}{m_e} = 1.758820 \times 10^{11} \text{ C kg}^{-1}$$

where m_e is the mass of the electron in kg and e is the magnitude of the charge on the electron in coulomb (C). Since electrons are negatively charged, the charge on electron is $-e$.

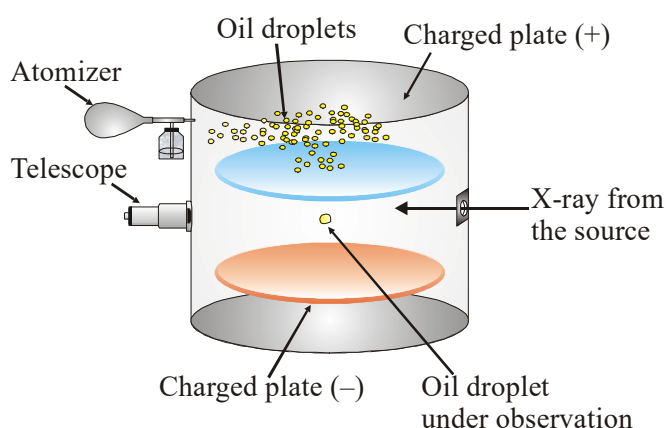
12.1.2 Charge on the Electron :

R.A. Millikan (1868-1953) devised a method known as oil drop experiment (1906-14), to determine the charge on the electrons. He found that the charge on the electron to be $-1.6 \times 10^{-19} \text{ C}$. The present accepted value of electrical charge is $-1.6022 \times 10^{-19} \text{ C}$. The mass of the electron (m_e) was determined by combining these results with Thomson's value of e/m ratio.

$$m_e = \frac{e}{e/m_e} = \frac{1.6022 \times 10^{-19} \text{ C}}{1.758820 \times 10^{11} \text{ C kg}^{-1}} = 9.1094 \times 10^{-31} \text{ kg}$$

Millikan's Oil Drop Method

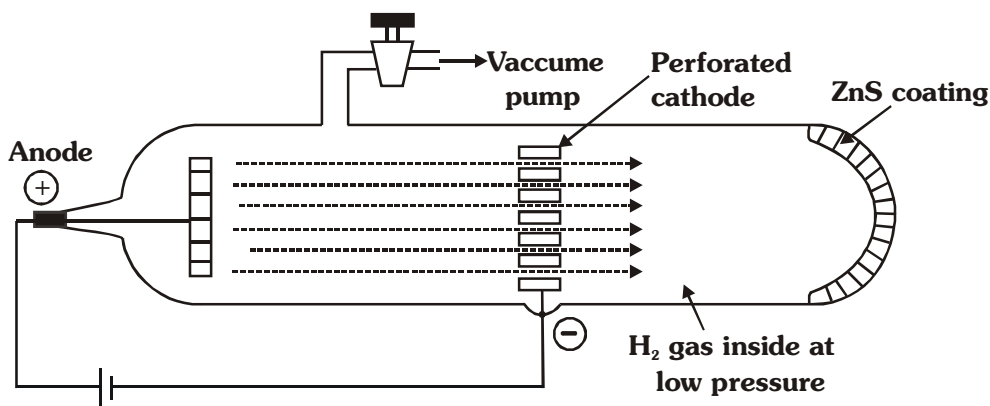
In this method, oil droplets in the form of mist, produced by the atomiser, were allowed to enter through a tiny hole in the upper plate of electrical condenser. The downward motion of these droplets was viewed through the telescope, equipped with a micrometer eye piece. By measuring the rate of fall of these droplets, Millikan was able to measure the mass of oil droplets. The air inside the chamber was ionized by passing a beam of X-rays through it. The electrical charge on these oil droplets was acquired by collisions with gaseous ions. The fall of these charged oil droplets can be retarded, accelerated or made stationary depending upon the charge on the droplets and the polarity and strength of the voltage applied to the plate. By carefully measuring the effects of electrical field strength on the motion of oil droplets, Millikan concluded that the magnitude of electrical charge, q , on the droplets is always an integral multiple of the electrical charge, e , that is, $q = n e$, where $n = 1, 2, 3 \dots$. The highest common factor (HCF) of charges on oil droplet was taken as charge on electron.



The Millikan oil drop apparatus for measuring charge 'e'. In chamber, the forces acting on oil drop are: gravitational, electrostatic due to electrical field and a viscous drag force when the oil drop is moving.

12.2 CANAL RAYS (OR ANODE RAYS) – DISCOVERY OF PROTON :

Atoms are electrically neutral. Hence after the discovery of the negatively charged constituent (electron) of an atom, attempts were made to discover the positively charged counterpart of electrons. By using a discharge tube containing a perforated cathode, **Goldstein (1886)** found that some rays passed through these holes in a direction opposite to that of the cathode rays.



Positive Rays or Canal Rays

Electrical discharge carried out in the modified cathode ray tube led to the discovery of particles carrying positive charge, also known as **canal rays**. The characteristics of these positively charged particles are listed below.

- (i) unlike cathode rays, the positively charged particles depend upon the nature of gas present in the cathode ray tube. These are simply the positively charged gaseous ions.
- (ii) The charge to mass ratio of the particles is found to depend on the gas from which these originate.
- (iii) Some of the positively charged particles carry a multiple of the fundamental unit of electrical charge.
- (iv) The behaviour of these particles in the magnetic or electrical field is opposite to that observed for electron or cathode rays.

The smallest and lightest positive ion was obtained from hydrogen and was called proton. This positively charged particle was characterised in 1919.

The lightest positively charged particle is called a proton (P or P⁺). Positive rays are atomic or molecular cations from which some electrons have been removed. The removed electrons constitute the cathode rays and the positive cations form the positive or canal rays.

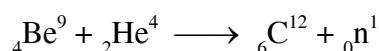
12.3 DISCOVERY OF NEUTRON :

After the discovery of electron and proton **Rutherford (1920)** predicted the existence of a neutral fundamental particle. In **1932, Chadwick** bombarded the element Beryllium with α -particles and noticed the emission of a radiation having the following characteristics.

- (i) The radiation was highly penetrating.
- (ii) The radiation was unaffected by magnetic and electric fields which shows that it is electrically neutral.

(iii) It was found to have approximately the same mass as the protons.

The name 'neutron' was given to this sub-atomic particle. It is denoted by n or ${}_0n^1$. Bombardment of beryllium by α -particles results in the formation of carbon and neutrons are emitted.



Mass of a neutron is 1.00867 amu (1.67493×10^{-24} g or 1.67493×10^{-27} kg)

12.4 PROPERTIES OF FUNDAMENTAL PARTICLES :

Atoms are made up-essentially, of three fundamental particles, which differ in mass and electric charge as follows :

	Electron	Proton	Neutron
Symbol	e or e^-	p	n
Approximate relative mass	1/1836	1	1
Approximate relative charge	-1	+1	0
Mass in kg	9.10982×10^{-31}	$1.6726216 \times 10^{-27}$	1.674927×10^{-27}
Mass in amu	0.00054	1.00727	1.00867
Actual charge/C	$-1.602176 \times 10^{-19}$	$+1.602176 \times 10^{-19}$	0

Ex.48 Arrange the particle in their increasing order of specific charge ratio.

(a) e^- , P, n, α -particle

(b) Na^+ , Li^+ , F^- , Mg^{2+} , Al^{3+}

Sol. (a) $\left(\frac{e}{m}\right)_e = \frac{1e}{\left(\frac{1}{1836}\right)\text{amu}}$

$$\left(\frac{e}{m}\right)_p = \frac{1}{1} = 1 = \frac{1e}{1\text{amu}}$$

$$\left(\frac{e}{m}\right)_n = \frac{0}{1} = 0 = \frac{0e}{1\text{amu}}$$

$$\left(\frac{e}{m}\right)_\alpha = \frac{2}{4} = \frac{1}{2} = \frac{2e}{4\text{amu}}$$

Ans. $n < \alpha < p < e^-$

(b) $\text{Na}^+ = \frac{1}{23}$; $\text{Li}^+ = \frac{1}{7}$; $\text{F}^- = \frac{1}{19}$; $\text{Mg}^{2+} = \frac{2}{24} = \frac{1}{12}$; $\text{Al}^{3+} = \frac{3}{27} = \frac{1}{9}$

$$\text{Na}^+ < \text{F}^- < \text{Mg}^{2+} < \text{Al}^{3+} < \text{Li}^+$$

Ex.49 Which of the following pairs have same specific charge $\left(\frac{e}{m}\right)$?

(a) electron & proton

(b) electron & positron

(c) proton & positron

(d) proton & deuteron

(e) α -particle & deuteron

Answer. b, e

Sol. Positron is antiparticle of electron (same charge and mass, but positively charged).

Ex.50 Through what potential difference an α -particle should be accelerated to have speed 5×10^6 m/s.

Sol. $qV = \frac{1}{2}mv^2$

$$(2 \times 1.6 \times 10^{-19} \times V) = \frac{1}{2} \times (4 \times 1.66 \times 10^{-27}) \times (5 \times 10^6)^2$$

SOME PREVIOUS YEARS QUESTION

1. With what velocity should an α -particle travel towards the nucleus of a Cu atom so as to arrive at a distance 10^{-13} m. [JEE 1997]

Ans. 6.3×10^6 m/s

Sol. $\frac{1}{2}mv^2 = \frac{kq_1q_2}{r}$

$$v^2 = \frac{2kq_1q_2}{mr}$$

$$= \frac{2 \times 9 \times 10^9 \times 2e \times 29e}{(10^{-13}) (1.67 \times 10^{-27})}$$

$$= \frac{18 \times 2 \times 29 \times 10^9 \times (1.6)^2 \times 10^{-38}}{1.67 \times 10^{-23} \times 10^{-27}}$$

$$= \frac{36 \times 29 \times 2.56 \times 10^{-29+40}}{1.67}$$

$$= 1600.3 \times 10^{11}$$

2. A compound of Vanadium has magnetic moment of 1.73 BM work out electronic configuration of Vanadium Ion in the compound. [JEE 1997]

Ans. [Ar] 3d¹

Sol. $r = \sqrt{n(n+2)} \quad n = 1$

$$= \sqrt{3}$$

$$= 1.73 \quad [\text{Ar}] 3d^2$$

3. The energy of an electron in the first Bohr orbit of H atom is -13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is/are :

(A) -3.4 eV (B) -4.2 eV (C) -6.8 eV (D) +6.8 eV [JEE 1998]

Ans. (A)

Sol. $E = -13.6$

$$E = \frac{-13.6}{4} = -3.4 \text{ eV}$$

4. The number of nodal planes in a p_x orbital is: [JEE 2000]

(A) one (B) two (C) three (D) zero

Ans. (A)

Sol. One

5. Calculate the energy required to excite one litre of hydrogen gas at 1 atmp and 298K to the first excited state of atomic hydrogen. The energy for the dissociation of H - H is 436 KJ mol⁻¹.

Ans. 97.819 KJ

Sol. $E = 436$

$T = 298 \text{ K}$

$$p = 1 \text{ atm} \quad \text{Energy req.} = 436 \times \frac{1}{298 \times R}$$

$$v = 1\ell \quad E = \left(10.2 \times 96.47 \times \frac{2}{R \times 298} \right) + \left(\frac{436}{R \times 298} \right)$$

$$n = \frac{1}{298R} = \frac{2 \times 04 \times 96.47 \times 436}{298 \times R}$$

6. **The quantum numbers $+1/2$ and $-1/2$ for the electron spin represent:** [JEE 2001]

- (A) rotation of the electron in clockwise and anticlockwise direction respectively.
 (B) rotation of the electron in anticlockwise and clockwise direction respectively.
 (C) magnetic moment of the electron pointing up and down respectively.
 (D) two quantum mechanical spin states which have no classical analogue.

Ans. (D)

Sol. Two quantum mechanical spin states which have no classical analogue.

7. **Rutherfords experiment, which established the nuclear model of atom, used a beam of :-**

- (A) β - particles, which impinged on a metal foil and get absorbed. [JEE 2002]
 (B) γ - rays, which impinged on a metal foil and ejected electron.
 (C) Helium atoms, which impinged on a metal foil and got scattered.
 (D) Helium nuclei, which impinged on a metal foil and got scattered.

Ans. (D)

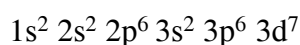
Sol. Helium nuclei, which impinged on a metal foil and got scattered.

8. **The magnetic moment of cobalt of the compound $Hg[Co(SCN)_4]$ is [Given : Co^{+2}]**

- (A) $\sqrt{3}$ (B) $\sqrt{8}$ (C) $\sqrt{15}$ (D) $\sqrt{24}$ [JEE 2004]

Ans. (C)

Sol. Co^{+2}



$$n = 3$$

$$r = \sqrt{n(n+2)}$$

$$= \sqrt{3 \times 5} = \sqrt{15}$$

9. **The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom?**

- (A) He^+ ($n = 2$) (B) Li^{2+} ($n = 2$) (C) Li^{2+} ($n = 3$) (D) Be^{3+} ($n = 2$) [JEE 2004]

Ans. (D)

Sol. $r = 0.529 \times \frac{n^2}{Z} \quad r_H = 0.529$

$$r_{He^+} = 0.529 \times \frac{4}{2} \quad r_{Li^{2+}} = 0.529 \times \frac{9}{3}$$

$$r_{Li^{2+}} = 0.529 \times \frac{4}{3} \quad r_{Be^{3+}} = 0.529 \times \frac{4}{4} = r_H$$

10. (a) The Schrodinger wave equation for hydrogen atom is

[IIT-2004]

$$\Psi_{2s} = \frac{1}{4(2\pi)^{1/2}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{r_0}{a_0} \right) e^{-r/a}$$

Where a_0 is Bohr's radius. Let the radial node in 2s be at r_0 . Then find r_0 in terms of a_0 .
(b) A base ball having mass 100 g moves with velocity 100 m/s. Find out the value of wavelength of base ball.

Ans. (a) $r_0 = 2a_0$, (b) $6.626 \times 10^{-25} \text{ Å}$

Sol. $\Psi_{2s} = \frac{1}{4(2\pi)^{1/2}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{r_0}{a_0} \right) e^{-r/a}$

(b) $\lambda = \frac{h}{mv}$ (a) $2 - \frac{r_0}{a_0} = 0$

$= \frac{6.626 \times 10^{-34}}{(0.1)(100)}$ $\frac{r_0}{a_0} = 2$

$= 6.626 \times 10^{-35} \text{ m}$ $r_0 = 2a_0$

$= 6.626 \times 10^{-35} \times 10^{10}$

$= 6.626 \times 10^{-25} \text{ Å}$

11. (a) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Given $r = a_0$) [IIT-2005]

(b) Find de-Broglie wavelength of the electron in first Bohr orbit.

(c) Find the orbital angular momentum of 2p-orbital in terms of $h/2\pi$ units.

Ans. (a) $2.197 \times 10^6 \text{ m/s}$ (b) 3.31 Å (c) $\sqrt{2} \cdot \frac{h}{2\pi}$

Sol. (a) $v = 2.188 \times 10^6 \times \frac{1}{1} \text{ m/s}$ (b) $\lambda = \frac{h}{mv}$

$= \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 2.188 \times 10^6}$
 $= 0.332 \times 10^{-9} \text{ m}$
 $= 3.31 \times 10^{-10} \text{ m}$
 $= 3.31 \text{ Å}$

(c) $\sqrt{\ell(\ell+1)} \frac{h}{2\pi}$

$= \sqrt{1(1+1)} \frac{h}{2\pi} = \frac{\sqrt{2}h}{2\pi}$

12. Given in hydrogenic atom r_n , V_n , E , K_n stand for radius, potential energy, total energy and kinetic energy in n^{th} orbit. Find the value of U, v, x, y . [JEE 2006]

(A) $U = \frac{V_n}{K_n}$ (P) 1

(B) $\frac{1}{r_n} \propto E^x$ (Q) -2

(C) $r_n \propto Z^y$ (R) -1
(Z = Atomic number)

(D) $v = (\text{Orbital angular momentum of electron in its lowest energy})$ (S) 0

Ans. (A) - Q ; (B) - P ; (C) - R ; (D) - S

Sol. (A) $U = \frac{V_n}{K_n}$
 $= \frac{-27.2}{13.6} = -2$

(B) $\frac{1}{r_n} \propto E^x$

(C) $r_n \propto Z^y$
 $0.529 \frac{n^2}{Z} = (Z)^y$

13. Match the entries in Column I with the correctly related quantum number(s) in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2008]

Column I	Column II
(A) Orbital angular momentum of the electron in a hydrogen-like atomic orbital	(P) Principal quantum number
(B) A hydrogen-like one-electron wave function obeying Pauli principle	(Q) Azimuthal quantum number
(C) Shape, size and orientation of hydrogen like atomic orbitals	(R) Magnetic quantum number
(D) Probability density of electron at the nucleus in hydrogen-like atom	(S) Electron spin quantum number

Ans. (A) - Q,R ; (B) - P,Q,R,S ; (C) - P,Q,R ; (D) - P,Q

Sol. The direction of orbital angular momentum is given by magnetic quantum number.

Important Notes

174



WAVE CHARACTERISTICS

11. Calculate the wave length (in nm) of an electromagnetic radiation of frequency 2×10^{15} Hz.
AS0011
12. A radiostation emits the radiations of wavelength 2000 m (meter band of station). What is the frequency band of that station, in kHz (Frequency of emitted radio waves) ?
AS0012
13. The wave numbers of two electromagnetic radiations are $4 \times 10^6 \text{ m}^{-1}$ and $2 \times 10^5 \text{ cm}^{-1}$. The ratio of their frequencies is $x : 1.00$. the value of 'x' is.
AS0013
14. An electromagnetic radiation makes 5000 waves in 20 cm. The frequency of radiation (in 10^{12} Hz) is.
AS0014

PLANCK'S QUANTUM THEORY

15. Calculate the energy (in 10^{-19} Joule) of a photon of wave length 4000 Å.
AS0015
16. The eyes of certain member of the reptile family pass a single visual signal to the brain when the visual receptors are struck by photons of wavelength 662 nm . If a total energy of 3.0×10^{-18} J is required to trip the signal, what is the minimum number of photons that must strike the receptor.
($h = 6.62 \times 10^{-34} \text{ J} \times \text{s}$)
AS0016
17. Find the number of photons (in the order of 10^{22}) of radiation of frequency $5 \times 10^{13} \text{ s}^{-1}$ that must be absorbed in order to melt one gm ice when the latent heat of fusion of ice is 330 J/g.
($h = 6.6 \times 10^{-34} \text{ J} \times \text{sec.}$)
AS0017
18. A certain dye absorbs 4000 Å and fluoresces at 5000 Å these being wavelengths of maximum absorption that under given conditions 40% of the absorbed energy is emitted. If the ratio of the number of quanta emitted to the number absorbed is $x : 1.00$, the value of 'x' is.
AS0018
19. The reaction between H_2 and Br_2 to form HBr in presence of light is initiated by the photo decomposition of Br_2 into free Br atoms (free radicals) by absorption of light. The bond dissociation energy of Br_2 is 192 KJ/mole. What is the longest wavelength (in nm) of the photon that would initiate the reaction.
 $\left(1 \frac{\text{eV}}{\text{atom}} = 96 \text{ kJ/mol} ; hc = 1240 \text{ eV} \times \text{nm} \right)$
AS0019
20. The quantum yield for decomposition of HI is 0.2. In an experiment 0.01 moles of HI are decomposed. Find the number of photons (in the order of 10^{22}) absorbed. ($N_A = 6 \times 10^{23}$)
AS0020

PHOTOELECTRIC EFFECT

21. The minimum energy required for the emission of a metal electron is 1.324×10^{-18} J. Calculate the threshold wavelength (in nm) required to eject the electron ?

AS0021

22. A light source of wavelength λ illuminates a metal and ejects photo electron with $(KE)_{\max} = 1$ eV. Another light source of wave length $\frac{\lambda}{3}$, ejects photoelectrons from same metal with $(KE)_{\max} = 5$ eV. Find the value of work function (eV) of metal.

AS0022

23. The dissociation energy of H_2 is 482.5 KJ/mol. If H_2 is exposed to radiant energy of wavelength 124 nm, what % of radiant energy will be converted into K.E.

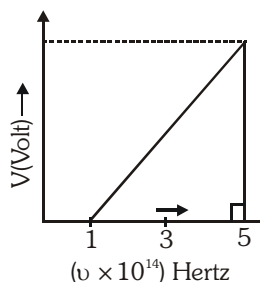
$$(1 \frac{eV}{atom} = 96.5 \text{ kJ/mol}, hc = 1240 \text{ eV} \times \text{nm})$$

AS0023

24. The K.E. of an electron emitted from tungstan surface is 3.06 eV. What voltage (in volt) would be required to bring the electron to rest.

AS0024

25. Stopping potential [V volts] is plotted against frequency of light used [ν]. Find work function (eV). ($h = 6.62 \times 10^{-34}$ J $\times$ s)



AS0025

BOHR'S MODEL

26. If the mass of electron is doubled, then find the new radius (in Å) of 1st orbit of H-atom ?

AS0026

27. If an electron in H-atom jumps from one orbit to other its angular momentum doubles. If the new radius is 'x' times the old radius, then find x ?

AS0027

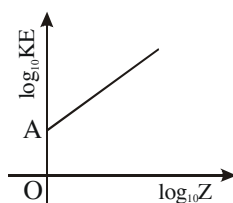
28. The radius of the an orbit of hydrogen atom is 0.85 nm. Calculate the speed (in 10^5 m/sec) of electron in this orbit.

AS0028

29. The velocity of e^- in a certain Bohr orbit of the hydrogen atom bears the ratio 1:275 to the speed of light.
(i) What is the quantum no. "n" of the orbit.
(ii) Calculate the wave number (in 10^6 m^{-1}) of the radiation emitted for the transition from the quantum state (n+1) to the ground state.

AS0029

30. If the average life time of an excited state of H atom is of order 10^{-8} sec, estimate how many revolutions (in the order of 10^6) an e^- makes when it is in the state $n = 2$ and before it suffers a transition to $n = 1$ state.
- AS0030**
31. Calculate the frequency of revolution of electron (in 10^{14} revolutions per sec) in the first Bohr orbit in a H-atom.
- AS0031**
32. Calculate energy (in eV) of electron which is moving in the orbit of a hydrogen atom that has its radius, sixteen times the radius of first Bohr orbit for H-atom.
- AS0032**
33. The energy of electron in an excited H-atom is -3.4 eV. Calculate angular momentum of electron (in 10^{-34} kg m² s⁻¹) in the given orbit.
- AS0033**
34. In a H-like atom for an electron revolving in n^{th} orbit, the variation of $\log_{10} \text{KE}$ versus $\log_{10} Z$ is plotted as following -



where Z is atomic number & KE is kinetic energy of electron (in eV). If $OA = \log_{10} 3.4$ then find the value of principle quantum number of the electron.

- AS0034**
35. The energy of the second stationary state in Li^{+2} ion is -4.95×10^{-18} J. Find ionization energy (in KJ/mole) for He^+ ions in its ground state ($N_A = 6 \times 10^{23}$).
- AS0035**
36. The energy of an electron in the first orbit of He^+ is -8.72×10^{-18} J. The energy of the electron in the first orbit of hydrogen (in 10^{-18} J) would be :
- AS0036**
37. A single electron orbits around a stationary nucleus of charge $+Ze$ where Z is atomic number and ' e ' is the magnitude of the electric charge. The hydrogen like species required 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find
- (i) the value of Z and give the hydrogen like species formed.
- (ii) the kinetic energy (in eV) of the electron in the first Bohr orbit.
- AS0037**
38. A hydrogen like atom with atomic number ' Z ' is in higher excited state of quantum number ' n '. This excited state atom can make a transition to the first excited state by successively emitting two photons of energies 10.2 eV and 17.0 eV respectively. Alternatively, the atom from the same excited state can make a transition to the 2nd excited state by emitting two photons of energies 4.25 eV and 5.95 eV respectively. Calculate the value of ' Z '.
- AS0038**

- AS0039

- AS0040**

- $$\left(\frac{1}{R} = 912 \text{ \AA}\right)$$

AS0041

- (i) The shortest wave length in Lyman series is $\frac{x}{y} \text{ \AA}$. The value of 'y' is.

- (ii) The longest wave length in Paschen series is $\frac{x}{z} \text{ \AA}$. The value of 'z' is.

AS0042

- AS0043

- AS0044

- AS0045

DE-BROGLIE AND HEISENBERG EQUATION

-

AS0046

47. deBroglie wavelength associated with an electron in 4th orbit of hydrogen atom is $a \times (\pi r_0)$ where r_0 is radius of 1st orbit of hydrogen atom, find value of 'a'. AS0047
48. The minimum number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number 3 will be - AS0048
49. Calculate the de-broglie wavelength (in 10^{-66} m) associated with motion of earth (mass 6×10^{24} Kg) orbiting around the sun at a speed of 3×10^6 m/s. AS0049
50. What is de Broglie wavelength (in pm) associated with an e^- accelerated through potential difference = 15 KV. (Answer to the nearest integer) AS0050
51. To what effective potential (in volt) a proton beam be subjected to give its protons a wavelength of 1×10^{-10} m. (Mass of proton = 1.672×10^{-27} kg, $e = 1.6 \times 10^{-19}$ C, $h = 6.626 \times 10^{-34}$ Js) AS0051
52. A proton is accelerated to one-tenth of the speed of light. If its velocity can be measured with a precision $\pm 1\%$. What must be its minimum uncertainty in position (in 10^{-12} m).
(Use : $\frac{h}{4\pi} = 5.28 \times 10^{-35}$) (Mass of proton = 1.6×10^{-24} gm) AS0052
53. The uncertainty in the position (in cm) of an electron (mass 9.1×10^{-28} gm) moving with a speed of 3×10^4 cm sec⁻¹, uncertainty in speed is 0.011% will be : ($h = 6.6 \times 10^{-34}$ J × s) AS0053
54. If uncertainties in measurement of position and momentum of an electron are equal, then the uncertainty in measurement of its speed is $x \times 10^{12}$. The value of x (the closest nearest integer value) is $m_e = 9.1 \times 10^{-31}$ kg, and $h = 6.625 \times 10^{-34}$ Js. AS0054
55. With the help of Heisenberg's uncertainty principle, show that electron can never exist in the nucleus. AS0055

SCHRODINGER EQUATION

56. Calculate the longest distance (in Å) of spherical nodes for '3s' orbital from nucleus in hydrogen atom?

$$R_{3s} = \frac{1}{9\sqrt{3}a_0^{3/2}} (6 - 6\sigma + \sigma^2) e^{-\frac{\sigma}{2}} \quad \text{where } \sigma = \frac{2r}{na_0}$$

AS0056

57. How many non-spherical subshell are possible that have atleast two maxima if a curve is plotted between radial probability distribution function versus radial distance for which principal quantum number : $n \leq 4$?

AS0057

58. The ratio of angular nodes of 3d orbital and radial nodes of 3p orbital is $x : 1$ what is the value of x .

AS0058

59. For an orbital in B^{+4} radial function is :

$$R(r) = \frac{1}{9\sqrt{6}} \left(\frac{Z}{a_0} \right)^{\frac{3}{2}} (4 - \sigma) \sigma e^{-\sigma/2}$$

where $\sigma = \frac{2Zr}{na_0}$ and $a_0 = 0.529 \text{ \AA}$; Z = atomic number, r = radial distance from nucleus.

The radial node of orbital is at distance $\text{\AA}$ from nucleus.

AS0059

60. Find the total number of nodes in 5f-orbital.

AS0060

EXERCISE # S-II

1. ' α particle' of 3.6 MeV are fired towards nucleus A_ZX . At point of closest separation, distance between ' α particle' and 'X' is 1.6×10^{-14} m. Calculate atomic number of 'X'

[Given : $1/4\pi\epsilon_0 = 9 \times 10^9$ in S.I. units]

AS0061

2. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{ke^2}{3r^3}$. Use Bohr's theory to obtain energy of such a hypothetical atom.

AS0062

3. In the Bohr's model, for unielectronic species following symbols are used

$r_{n,z}$ $\longrightarrow$ Radius of n^{th} orbit with atomic number " z "

$U_{n,z}$ $\longrightarrow$ Potential energy of electron in n^{th} orbit with atomic number " z "

$K_{n,z}$ $\longrightarrow$ Kinetic energy of electron in n^{th} orbit with atomic number " z "

$v_{n,z}$ $\longrightarrow$ Velocity of electron in n^{th} orbit with atomic number " z "

$T_{n,z}$ $\longrightarrow$ Time period of revolution of electron in n^{th} orbit with atomic number " z "

Calculate z in all in cases.

(i) $U_{1,2} : K_{1,z} = -8 : 1$

(ii) $r_{1,z} : r_{2,1} = 1 : 8$

(iii) $v_{1,z} : v_{3,1} = 9 : 1$

(iv) $T_{1,2} : T_{2,z} = 9 : 32$

AS0063

4. In a hypothetical H-atom, the mass of electron & its charge is double of what we consider, then calculate the total energy (in eV) of electron in the 1^{st} orbit of such a hypothetical H-atom ?
[Assuming all others concepts and parameter to be same as we considered in Bohr's model.]

AS0064

5. In H-spectrum, longest wavelength of Lyman is 120 nm and shortest wavelength of Balmer is 360 nm. From this data, find longest wavelength (in nm) of photon, that can ionize this H-atom

AS0065

6. A proton and an electron, both at rest initially, combine to form a H-atom in ground state. A single photon is emitted in this process. Find the wavelength (in nm) of this photon

(use : $hc = 1237.6 \text{ eV} \times \text{nm}$)

AS0066

7. A H-like ion is observed to emit six different wavelengths originating from all possible transitions between a group of levels. These levels have energies between -0.85 eV (Min) and -0.544 eV (Max).
(a) Find atomic number (z) of element
(b) Find quantum numbers of levels between which transitions occur.
(c) Calculate the energy (eV) of the longest wavelength emitted in transitions between the levels.

AS0067

8. The angular momentum of an electron in a Bohr's orbit of H-atom is $3.1652 \times 10^{-34} \text{ kg-m}^2/\text{sec}$. Calculate the wavenumber (in 10^6 m^{-1}) of the spectral line emitted when an electron falls from this level to the ground state. [$h = 6.626 \times 10^{-34} \text{ Js}$, $R = 1.09 \times 10^7 \text{ m}^{-1}$]

AS0068

9. The ionisation energy of the hydrogen atom is given to be 13.6 eV. A photon falls on a hydrogen atom which is initially in the ground state and excites it to the ($n = 4$) state. Calculate the wavelength (in Å) of the photon ($hc = 12400 \text{ Å.eV}$).

AS0069

10. The ionisation energy of a H-like Bohr atom is 4 Rydbergs.

(i) What is the wavelength (in Å) of radiation emitted when the electron jumps from the first excited state to the ground state.

(ii) What is the radius (in Å) of first Bohr orbit for this atom.

[1 Rydberg = 2.20×10^{-18} J, $h = 6.60 \times 10^{-34}$ J.s]

AS0070

11. Photon having wavelength 12.4 nm was allowed to strike a metal plate having work function 25 eV. Calculate the

(a) Maximum kinetic energy (in eV) of photoelectrons emitted in eV.

(b) Wavelength (in Å) of electron with maximum kinetic energy in Å.

[Given : $hc = 1240 \text{ nm.eV}$]

AS0071

12. The diameter of a dust particle of mass 10^{-3} gm is $2\text{\AA}$. If uncertainty in speed of this particle is $\frac{3.31}{\pi} \times 10^{-3}$ m/s, then find minimum uncertainty in its position (in 10^{-25} m) (**$h = 6.62 \times 10^{-34}$ J.s.**)

AS0072

13. The vapours of Hg absorb some electrons accelerated by a potential difference of 4.5 volt as a result of which light is emitted. If the full energy of single incident electron is supposed to be converted into light emitted by single Hg atom, find the wave number (in 10^7 m^{-1}) of the light.

AS0073

14. A photon having $\lambda = 960 \text{ \AA}$ causes the ionization of a nitrogen atom. Calculate the I.E. per mole of nitrogen atom (in KJ). (Use : $hc = 1240 \text{ eV} \times \text{nm}$ and $1 \frac{\text{eV}}{\text{atom}} = 96.0 \text{ kJ/mol}$).

AS0074

15. Mr. Santa has to decode a number "ABCDEF" where each alphabet is represented by a single digit. Suppose an orbital whose radial wave function is represented as

$$\Psi_{(r)} = k_1 \cdot e^{-r/k_2} (r^2 - 5k_3r + 6k_3^2)$$

From the following information given about each alphabet then write down the answers in the form of "ABCDEF", for above orbital.

Info A = Value of n where "n" is principal quantum number

Info B = No. of angular nodes

Info C = Azimuthal quantum number of subshell to orbital belongs

Info D = No. of subshells having energy between $(n + 5)s$ to $(n + 5)p$ where n is principal quantum number

Info E = Orbital angular momentum of given orbital.

Info F = Radial distance of the spherical node which is farthest from the nucleus

(Assuming $k_3 = 1$)

AS0075

EXERCISE # O-I

1. Anode rays are made up of :
(A) only protons (B) only nucleus of atom
(C) Positive residue of atoms (D) only electrons
AS0076
2. When Lithium (${}^7_3\text{Li}$) vapours were filled in discharge tube for anode ray experiment, the anode rays were found to contain only Li^+ ions. Thus the anode ray particle contains :
(A) 1 proton only (B) 3 proton and 4 neutron only
(C) 3 proton, 4 neutron and 2 electrons (D) 3 proton, 3 neutrons and 3 electrons
AS0077
3. The specific charge is maximum for
(A) Na^+ (B) Al^{+3} (C) H^+ (D) Mg^{+2}
AS0078
4. Which of the following particle is not deflected in the magnetic field
(A) Electron (B) proton (C) Neutron (D) Deuteron
AS0079
5. e/m ratio of a particle of charge 2 unit and mass 4 amu is
(A) $4.8 \times 10^7 \text{ C/kg}$ (B) 0.5 C/kg (C) $4.8 \times 10^4 \text{ C/kg}$ (D) $8 \times 10^{-20} \text{ C/kg}$
AS0080
6. α -particles are projected towards the nucleus of following metals, with the same kinetic energy. The distance of closest approach will be minimum for which metal?
(A) Cu (B) Ag (C) Au (D) Ca
AS0081
7. Which of the following is not a conclusion of Rutherford's atomic model -
(A) Most of the part inside an atom is empty
(B) Almost all mass of an atom is concentrated in the nucleus.
(C) The size of nucleus is very small in comparison to the size of atom
(D) Electron revolves around the nucleus in definite orbits.
AS0082
8. In the different experiments, α -particles, proton, deuteron and neutron are projected towards gold nucleus with the same kinetic energy. The distance of closest approach will be minimum for
(A) α -particle (B) proton (C) dueteron (D) neutron
AS0083

9. The ratio of the energy of a photon of 2000 Å wavelength radiation to that of 4000 Å radiation is
(A) 1 / 4 (B) 4 (C) 1 / 2 (D) 2
AS0084
10. Small packets of light is called
(A) proton (B) quanta (C) photon (D) spectrum
AS0085
11. Which of the following electromagnetic radiation have greater frequency ?
(A) X - rays (B) Ultraviolet rays (C) Radio waves (D) Visible rays
AS0086
12. Two electromagnetic radiations have wave numbers in the ratio 2 : 3. Their energies per quanta will be in the ratio
(A) 3 : 2 (B) 9 : 4 (C) 4 : 9 (D) 2 : 3
AS0087
13. A radio station is emitting the radiations of frequency 2×10^4 Hz. If its frequency is doubled,
(A) wavelength will be doubled (B) energy per quanta will be doubled
(C) wave number will be halved (D) all of these
AS0088
14. A photon of 400 nm is absorbed by a gas molecule and then the molecule re-emits two photons. One re-emitted photon has wavelength 500 nm. Assuming that there is no change in the energy of molecule, the wavelength of second re-emitted photon is
(A) 100 nm (B) 2000 nm (C) -100 nm (D) 900 nm
AS0089
15. A green bulb and a red bulb are emitting the radiations with equal power. The correct relation between numbers of photons emitted by the bulbs per second is
(A) $n_g = n_r$ (B) $n_g < n_r$ (C) $n_g > n_r$ (D) unpredictable
AS0090
16. The threshold wavelength for ejection of electrons from a metal is 330 nm. The work function for the photoelectric emission from the metal is ($h = 6.6 \times 10^{-34}$ J-s)
(A) 1.2×10^{-18} J (B) 6.0×10^{-19} J (C) 1.2×10^{-20} J (D) 6.0×10^{-12} J
AS0091
17. In the emission of photoelectrons, the number of photoelectrons emitted per unit time depends upon
(A) energy of the incident radiation (B) intensity of the incident radiation
(C) frequency of the incident radiation (D) wavelength of the incident radiation
AS0092
18. Radiations of frequency, ν , are incident on a photosensitive metal. The maximum kinetic energy of photoelectrons is 'E'. When the frequency of the incident radiations is doubled, what is the maximum kinetic energy of the photoelectrons ?
(A) 2E (B) E/2 (C) $E + h\nu$ (D) $E - h\nu$
AS0093

19. Radiation of $\lambda = 155 \text{ nm}$ was irradiated on Li (work function = 5 eV) plate. The stopping potential (in V) is _____.

(A) 3 V (B) 8 V (C) 9 V (D) 5 V

AS0094

20. Electromagnetic radiations having $\lambda = 310 \text{ \AA}$ are subjected to a metal sheet having work function = 12.8 eV. What will be the velocity of photoelectrons with maximum Kinetic Energy..

(A) 0, no emission will occur (B) $2.18 \times 10^6 \text{ m/s}$
(C) $2.18\sqrt{2} \times 10^6 \text{ m/s}$ (D) $8.72 \times 10^6 \text{ m/s}$

AS0095

21. Bohr's model may be applied to

(A) Na^{10+} ion (B) He atom (C) Be^{2+} ion (D) C^{6+} ion

AS0096

22. If the radius of 3rd Bohr's orbit of H is x, then radius of 4th orbit of Li^{2+} ion would be :-

(A) $\frac{27}{16}x$ (B) $\frac{16}{27}x$ (C) $\frac{9}{16}$ (D) None of these

AS0097

23. What would be the approximate quantum number, n, for a circular orbit of hydrogen, $1 \times 10^{-5} \text{ cm}$ in diameter?

(A) 31 (B) 43 (C) 40 (D) 39

AS0098

24. If the mass of electron is doubled, the radius of first orbit of H-atom become about

(A) $0.529 \text{ \AA}$ (B) $0.265 \text{ \AA}$ (C) $1.058 \text{ \AA}$ (D) $0.32 \text{ \AA}$

AS0099

25. The speed of electron revolving in the 4th orbit of a hydrogen like atom or ion is 1094 km/s. The atom or ion is

(A) H (B) He^+ (C) Li^{2+} (D) Be^{3+}

AS0100

26. How much distance an electron revolving in 3rd orbit of He^+ ion will travel in one second

(A) $1.458 \times 10^6 \text{ m}$ (B) $3.28 \times 10^6 \text{ m}$
(C) $4.862 \times 10^5 \text{ m}$ (D) $2.917 \times 10^6 \text{ m}$

AS0101

27. The ratio of time taken by electron in revolutions round the H-nucleus in 2nd and 3rd orbits, are

(A) 2 : 3 (B) 4 : 8 (C) 8 : 27 (D) 27 : 8

AS0102

28. Which of the following is not a permissible value of angular momentum of electron in H-atom?

(A) $1.5 \frac{h}{\pi}$ (B) $0.5 \frac{h}{\pi}$ (C) $1.25 \frac{h}{\pi}$ (D) all of these

AS0103

29. Angular momentum for P-shell electron :-

- (A) $\frac{3h}{\pi}$ (B) Zero (C) $\frac{\sqrt{2}h}{2\pi}$ (D) None

AS0104

30. Angular momentum in 2nd Bohr orbit of H-atom is x. Then find out angular momentum in 1st excited state of Li^{+2} :-

- (A) 3x (B) 9x (C) x/2 (D) x

AS0105

31. The orbit from which when electron will jump in other orbit, energy may be absorbed but not emitted out, will be

- (A) 1st orbit (B) 2nd orbit (C) 7th orbit (D) infinite orbit

AS0106

32. The potential energy of electron revolving in the ground state of H-atom is

- (A) -13.6 eV (B) -6.8 eV (C) -27.2 eV (D) Zero

AS0107

33. If the potential energy (PE) of hydrogen electron is -3.02 eV then in which of the following excited level is electron present :-

- (A) 1st (B) 2nd (C) 3rd (D) 4th

AS0108

34. A single electron is revolving in orbits around a stationary nucleus ($z = 5$). The energy required to excite the electron from third to fourth Bohr orbit will be :-

- (A) 4.5 eV (B) 8.53 eV (C) 25 eV (D) 16.53 eV

AS0109

35. A photon of energy 12.75 eV is completely absorbed by a hydrogen atom initially in ground state. The principle quantum number of the excited state is

- (A) 1 (B) 3 (C) 4 (D) ∞

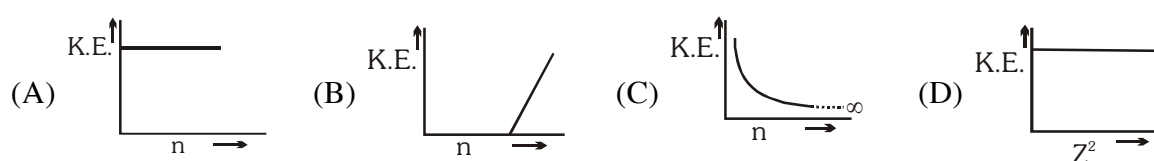
AS0110

36. An hydrogen atom (ionisation energy 13.6 eV) jumps from third excited state to first excited state. The energy of photon emitted in the process is

- (A) 1.89 eV (B) 2.55 eV (C) 12.09 eV (D) 12.75 eV

AS0111

37. Which is correct graph :-



AS0112

38. The energy required to remove an electron from the surface of sodium metal is 3.3 eV. What is the longest wavelength of radiation with which it can show photoelectric effect?

Use $h = 6.6 \times 10^{-34}$ J.s

- (A) 1.75×10^{-7} m (B) 3.75×10^{-5} m (C) 1.75×10^{-5} m (D) 3.75×10^{-7} m

AS0113

39. Which is correct for any H like species :-

- (A) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$ (B) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$
(C) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$ (D) $(E_2 - E_1) = 1/4 (E_3 - E_2) = 1/9 (E_4 - E_3)$

AS0114

40. A single electron orbits a stationary nucleus of charge $+Ze$, where Z is a constant. It requires 47.2 eV to excite electron from second Bohr orbit to third Bohr orbit, find the value of Z :-

- (A) 1 (B) 3 (C) 5 (D) 4

AS0115

41. The energy of H-atom in n^{th} orbit is E_n then energy in n^{th} orbit of singly ionised helium atom will be:

- (A) $4E_n$ (B) $E_n/4$ (C) $2E_n$ (D) $E_n/2$

AS0116

42. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon

- (A) 3s (B) 2p (C) 2s (D) 1s

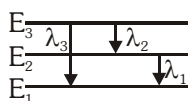
AS0117

43. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen

- (A) $5 \rightarrow 3$ (B) $5 \rightarrow 2$ (C) $4 \rightarrow 3$ (D) $4 \rightarrow 2$

AS0118

44. In the following transition which statement is correct



- (A) $E_{3-1} = E_{3-2} - E_{2-1}$ (B) $\lambda_3 = \lambda_1 + \lambda_2$
(C) $\nu_3 = \nu_2 + \nu_1$ (D) All of these

AS0119

45. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2$ eV. The same energy change is observed in the second Balmer transition of :-

- (A) Li^{2+} (B) Li^+ (C) He^+ (D) Be^{3+}

AS0120

46. In a sample of H-atoms, electron transits from 6^{th} orbit to 2^{nd} orbit in multi step. Then total spectral lines (without Balmer series) will be :-

- (A) 6 (B) 10 (C) 4 (D) 0

AS0121

47. What is the shortest wavelength in the Pfund series of He^+ ion :-

- (A) $\frac{25}{R}$ (B) $\frac{4}{R}$ (C) $\frac{4R}{25}$ (D) $\frac{25}{4R}$

AS0122

48. The shortest wavelength of He^+ ion in Balmer series is x , then longest wavelength in the Paschene series of Li^{+2} is

- (A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$

AS0123

49. The ratio of wave length of photon corresponding to the α -line of Lyman series in H-atom and β -line of Balmer series in He^+ is

- (A) 1 : 1 (B) 1 : 2 (C) 1 : 4 (D) 3 : 16

AS0124

50. Three energy levels P, Q, R of a certain atom are such that $E_P < E_Q < E_R$. If λ_1 , λ_2 and λ_3 are the wave length of radiation corresponding to transition $R \rightarrow Q$; $Q \rightarrow P$ and $R \rightarrow P$ respectively. The correct relationship between λ_1 , λ_2 and λ_3 is

- (A) $\lambda_1 + \lambda_2 = \lambda_3$ (B) $\frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ (C) $\lambda_3 = \sqrt{\lambda_1 \lambda_2}$ (D) $\frac{2}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

AS0125

51. Number of possible spectral lines which may be emitted in bracket series in H atom, if electrons present in 9th excited level returns to ground level, are

- (A) 21 (B) 6 (C) 45 (D) 5

AS0126

52. The value of $(n_2 + n_1)$ and $(n_2^2 - n_1^2)$ for He^+ ion in atomic spectrum are 4 and 8 respectively. The wavelength of emitted photon when electron jump from n_2 to n_1 is

- (A) $\frac{32}{9} R_H$ (B) $\frac{9}{32} R_H$ (C) $\frac{9}{32} R_H$ (D) $\frac{32}{9} R_H$

AS0127

53. An electron, a proton and an alpha particle have kinetic energies of 16E, 4E and E respectively. What is the qualitative order of their de Broglie wavelengths?

- (A) $\lambda_e > \lambda_p = \lambda_\alpha$ (B) $\lambda_p = \lambda_\alpha > \lambda_e$ (C) $\lambda_p > \lambda_e > \lambda_\alpha$ (D) $\lambda_\alpha < \lambda_e \gg \lambda_p$

AS0128

54. The wavelength associated with a golf weighing 200g and moving at a speed of 5m/h is of the order

- (A) 10^{-10}m (B) 10^{-20}m (C) 10^{-30}m (D) 10^{-40}m

AS0129

- 55.** An electron has kinetic energy 2.8×10^{-23} J. de-Broglie wavelength will be nearly :-

$$(m_e = 9.1 \times 10^{-31} \text{ kg})$$

- (A) 9.28×10^{-24} m (B) 9.28×10^{-7} m (C) 9.28×10^{-8} m (D) 9.28×10^{-10} m

AS0130

- 56.** What will be de-Broglie wavelength of an electron moving with a velocity of $1.2 \times 10^5 \text{ ms}^{-1}$:-

- (A) $6.068 \times 10^{-9} \text{ m}$ (B) $3.133 \times 10^{-37} \text{ m}$ (C) $6.626 \times 10^{-9} \text{ m}$ (D) $6.018 \times 10^{-7} \text{ m}$

AS0131

- 57.** For a valid Bohr orbit, its circumference should be:

- (A) $= n\lambda$ (B) $= (n-1)\lambda$ (C) $> n\lambda$ (D) $< n\lambda$

AS0132

- 58.** The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number + 2 :

- (A) 3 (B) 4 (C) 2 (D) 1

AS0133

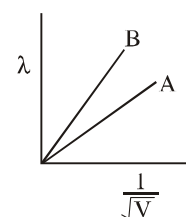
- 59.** How fast is an electron moving if it has a wavelength equal to distance travelled in one second -

- (A) $\sqrt{\frac{m}{h}}$ (B) $\sqrt{\frac{h}{m}}$ (C) $\sqrt{\frac{h}{p}}$ (D) $\sqrt{\frac{h}{2 \times \text{K.E.}}}$

AS0134

- 60.** De-Broglie wavelength of two particles A & B are plotted against $\left(\frac{1}{\sqrt{V}}\right)$; where V is potential on the particles. Which of the following relation is correct about mass of particle (charge is same)

- (A) $M_A = M_B$
 (B) $M_A > M_B$
 (C) $M_A < M_B$
 (D) $M_A \leq M_B$



AS0135

- 61.** An α -particle is accelerated through a potential difference of V volts from rest. The de-Broglie's wavelength associated with it is -

- (A) $\sqrt{\frac{150}{V}} \text{A}^\circ$ (B) $\frac{0.286}{\sqrt{V}} \text{A}^\circ$ (C) $\frac{0.101}{\sqrt{V}} \text{A}^\circ$ (D) $\frac{0.983}{\sqrt{V}} \text{A}^\circ$

AS0136

- 62.** Which quantum number is not related with Schrodinger equation

- (A) Principal (B) Azimuthal (C) Magnetic (D) Spin

AS0137

63. Which is true about ψ :-

- (A) ψ represents the probability of finding an electron around the nucleus
 (B) ψ represent the amplitude of the electron wave
 (C) Both A and B
 (D) None of these

AS0138

64. According to Schrodinger model nature of electron in an atom is as :-

- (A) Particles only (B) Wave only
 (C) Both simultaneously (D) Sometimes waves and sometimes particle

AS0139

65. The orbital angular momentum of an electron in 2s orbital is:

- (A) $+\frac{1}{2} \cdot \frac{h}{2\pi}$ (B) Zero (C) $\frac{h}{2\pi}$ (D) $\sqrt{2} \cdot \frac{h}{2\pi}$

AS0140

66. In an excited state, a calcium atom has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 4d^1$. What is

the orbital angular momentum for d electron :- $\left(\hbar = \frac{h}{2\pi} \right)$

- (A) $\sqrt{4}\hbar$ (B) $\sqrt{16}\hbar$ (C) $\sqrt{6}\hbar$ (D) $\sqrt{10}\hbar$

AS0141

67. Which orbital has two angular nodal planes :-

- (A) s (B) p (C) d (D) f

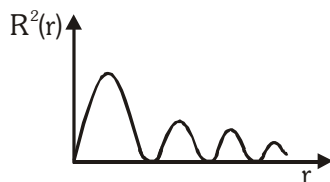
AS0142

68. From the following observations predict the type of orbital :

Observation 1 : x y plane acts as nodal plane

Observation 2 : The angular function of the orbital intersect the three axis at origin only.

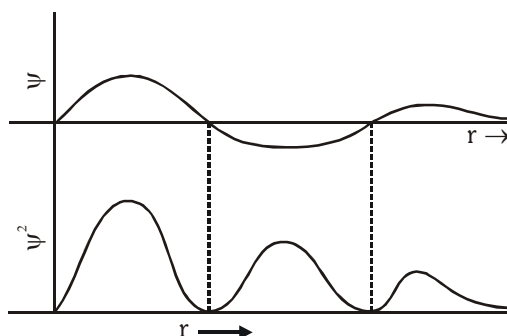
Observation 3 : $R^2(r)$ v/s r curve is obtained for the orbital is



- (A) $5p_z$ (B) $6d_{xy}$ (C) $6d_{x^2-y^2}$ (D) $6d_{yz}$

AS0143

69. The wave function ψ vs radial distance (r) and square of wave function (ψ^2) vs radial distance (r) graph of certain orbital is given. Identify the orbital.



- (A) 3s (B) 4d (C) 3p (D) 4p

AS0144

70. The distance of spherical nodes from nucleus for the given orbital are

$$\psi_{\text{radial}} = \frac{1}{9\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} [\sigma^2 - 4\sigma + 3] \exp(-\sigma/2)$$

where a_0 & Z are the constants and $\sigma = \frac{2Zr}{na_0}$

- (A) Zero, infinity (B) $\frac{a_0}{Z}, \frac{a_0}{2Z}$
(C) $\frac{3a_0}{2Z}, \frac{9a_0}{2Z}$ (D) $\frac{a_0}{Z}, \frac{3a_0}{2Z}$

AS0145

71. The radial function for an orbitals is :

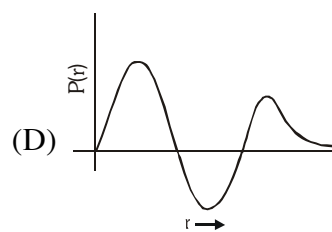
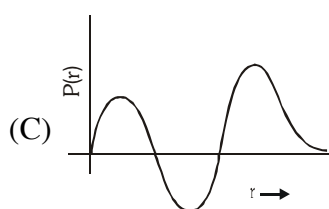
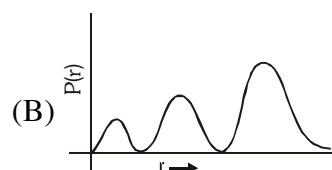
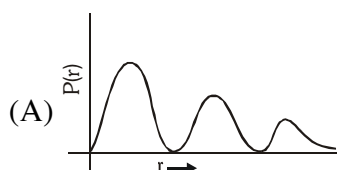
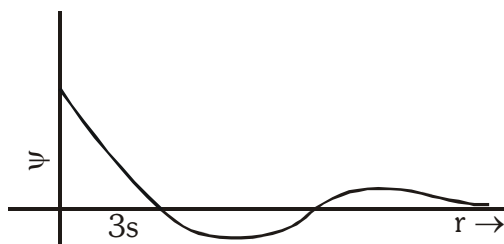
$$R(r) = \frac{1}{9\sqrt{6}} \left(\frac{1}{a_0} \right)^{5/2} \cdot (4 - \sigma) \sigma e^{-\sigma/2}; \sigma = \frac{2r}{na_0}$$

Pick out the incorrect statement :

- (A) orbital must be 3p
(B) orbital must be $3p_x$
(C) the orbital has one radial node
(D) the orbital must have one angular node

AS0146

72. The wave function (ψ) versus radial distance (r) curve for certain orbital is given. Predict the shape of $\psi^2 \cdot 4\pi r^2$ (radial probability distribution function) versus r graph.



AS0147

EXERCISE # O-II

SINGLE CORRECT :

- In a hydrogen like sample electron is in 2nd excited state, the energy of 4th state of this sample is -13.6 eV, then incorrect statement is :
 (A) Atomic number of element is 4.
 (B) 3 different types of spectral line will be observed if electrons make transition upto ground state from the 2nd excited state.
 (C) A 25 eV photon can set free the electron from the 2nd excited state of this sample
 (D) 2nd line of Balmer series of this sample has same energy value as 1st excitation energy of H-atoms.

AS0148

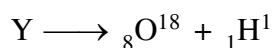
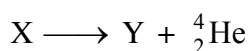
- An electron in a hydrogen atom in its ground state absorbs energy equal to the ionisation energy of Li²⁺. The wavelength of the emitted electron is:
 (A) 3.32×10^{-10} m (B) 1.17 Å (C) 2.32×10^{-9} nm (D) 3.33 pm

AS0149

- Given ΔH for the process $\text{Li(g)} \longrightarrow \text{Li}^{3+}(\text{g}) + 3\text{e}^-$ is 19800 kJ/mole & IE_1 for Li is 520 then IE_2 & IE_3 of Li are respectively (approx, value)
 (A) 7505, 11775 (B) 520, 19280 (C) 11775, 19280 (D) Data insufficient

AS0150

- Consider the following nuclear reactions involving X & Y.



If both neutrons as well as protons in both the sides are conserved in nuclear reaction then moles of neutrons in 4.6 gm of X

- (A) $2.4 N_A$ (B) 2.4 (C) 4.6 (D) $0.2 N_A$

AS0151

ASSERTION AND REASON :

- Statement-1:** Energy emitted when an electron jump from 5 $\rightarrow$ 2 (energy level) is less than when an electron jump from 2 $\rightarrow$ 1 in all 'H' like atom.
Statement-2: The [total energy difference] between 1st & 2nd energy level is greater than that of any two energy level provided level '1' is not part of those two energy levels.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

AS0152

6. **Statement-1** : Emitted radiations will fall in visible range when an electron jump from higher level to $n = 2$ in Li^{+2} ion.

Statement-2 : Balmer series radiations belong to visible range in all H-atoms.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

AS0153

MORE THAN ONE MAY BE CORRECT :

7. Which of the following could be derived from Rutherford's α -particle scattering experiment-

- (A) Most of the space in the atom is empty
(B) The radius of the atom is about 10^{-10} m while that of nucleus is 10^{-15} m
(C) Electrons move in a circular path of fixed energy called orbits
(D) Electrons and the nucleus are held together by electrostatic forces of attraction.

AS0154

8. In a H-like sample electrons make transition from 5th excited state to 2nd excited state

- (A) 10 different spectral lines will be emitted
(B) 6 different spectral lines will be emitted
(C) Number of lines belonging to Balmer series will be 4
(D) Number of lines belonging to paschen series will be 3

AS0155

9. The kinetic energy of photoelectron emitted on irradiating a metal surface with frequency ν is related by $\text{KE} = h\nu - \phi$. The plots of KE vs. incident frequency ν shows :

- (A) A straight line with slope equal to Planck's constant.
(B) A straight line with intercept on x-axis equal to the product of threshold frequency and Planck's constant.
(C) A straight line with extrapolated intercept on y-axis equal to threshold energy.
(D) A straight line with intercept on x-axis equal to threshold frequency.

AS0156

10. Select the correct statement(s):

- (A) All electromagnetic radiation travel with speed of light in vacuum.
(B) Energy of photon of UV light is lower than that of yellow light.
(C) He^+ and H have identical spectrum.
(D) The total energy of an electron in unielectronic specie is greater than zero

AS0157

11. Choose the incorrect statement(s):

(A) Increasing order of wavelength is

Micro waves > Radio waves > IR waves > visible waves > UV waves

(B) The order of Bohr radius is (r_n : where n is orbit number for a given atom) $r_1 < r_2 < r_3 < r_4$

(C) The order of total energy is (E_n : where n is orbit number for a given atom) $E_1 > E_2 > E_3 > E_4$

(D) The order of velocity of electron in H, He^+ , Li^+ , Be^{3+} species in second Bohr orbit is

$\text{Be}^{3+} > \text{Li}^{+2} > \text{He}^+ > \text{H}$

AS0158

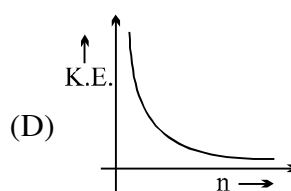
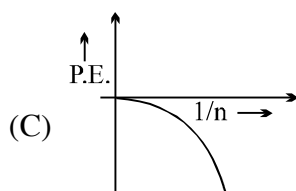
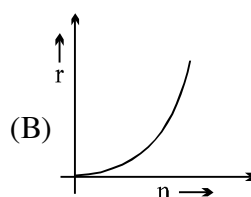
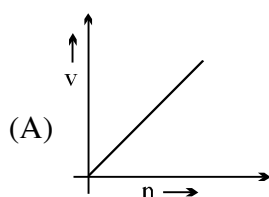
12. Select the correct curve(s):

If v = velocity of electron in Bohr's orbit

r = Radius of electron in Bohr's orbit

P.E. = Potential energy of electron in Bohr's orbit

K.E. = Kinetic energy of electron in Bohr's orbit.



AS0159

13. Which is / are correct statement.

(A) The difference in angular momentum associated with the electron present in consecutive orbits of

H-atom is $(n-1) \frac{h}{2\pi}$.

(B) Energy difference between energy levels will be changed if, P.E. at infinity assigned value other than zero.

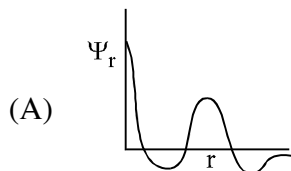
(C) Frequency of spectral line in a H-atom is in the order of $(2 \rightarrow 1) < (3 \rightarrow 1) < (4 \rightarrow 1)$.

(D) On moving away from the nucleus, kinetic energy of electron decreases.

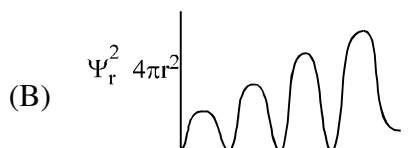
AS0160

MATCH THE COLUMN :

14. Column I & Column II contain data on Schrodinger Wave-Mechanical model, where symbols have their usual meanings. Match the columns.

Column I**Column II (Type of orbital)**

(P) 4s

(Q) 5p_x(C) $\Psi(\theta, \phi) = K$ (independent of θ & ϕ)

(R) 3s

(D) atleast one angular node is present

(S) 6d_{xy}**AS0161****15. Column-I****Column-II**(A) Electron moving in 2nd orbit in He⁺ ion
electron is(P) Radius of orbit in which
moving is 0.529 Å(B) Electron moving in 3rd orbit in H-atom(Q) Total energy of electron is
(-)13.6 × 9 eV(C) Electron moving in 1st orbit in Li⁺² ion(R) Velocity of electron is $\frac{2.188 \times 10^6}{3}$ m/sec(D) Electron moving in 2nd orbit is Be⁺³ ion(S) De-broglie wavelength of
electron is $\sqrt{\frac{150}{13.6}}$ Å**AS0162**

COMPREHENSION :

Question No. 16 to 18 (3 questions)

The French physicist Louis de Broglie in 1924 postulated that matter, like radiation, should exhibit a dual behaviour. He proposed the following relationship between the wavelength λ of a material particle, its linear momentum p and planck constant h .

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

The de Broglie relation implies that the wavelength of a particle should decrease as its velocity increases. It also implies that for a given velocity heavier particles should have shorter wavelength than lighter particles. The waves associated with particles in motion are called matter waves or de Broglie waves. These waves differ from the electromagnetic waves as they

- (i) have lower velocities
- (ii) have no electrical and magnetic fields and
- (iii) are not emitted by the particle under consideration.

The experimental confirmation of the de Broglie relation was obtained when Davisson and Germer, in 1927, observed that a beam of electrons is diffracted by a nickel crystal. As diffraction is a characteristic property of waves, hence the beam of electron behaves as a wave, as proposed by de Broglie.

Werner Heisenberg considered the limits of how precisely we can measure properties of an electron or other microscopic particle like electron. He determined that there is a fundamental limit of how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the "Heisenberg uncertainty principle: It is impossible to determine simultaneously and precisely both the momentum and position of a particle. The product of uncertainty in the position, Δx and the uncertainty in the momentum $\Delta(mv)$ must be greater than

or equal to $\frac{h}{4\pi}$. i.e.

$$\Delta x \Delta(mv) \geq \frac{h}{4\pi}$$

- 16.** The correct order of wavelength of Hydrogen (${}_1\text{H}^1$), Deuterium (${}_1\text{H}^2$) and Tritium (${}_1\text{H}^3$) moving with same kinetic energy is

- (A) $\lambda_{\text{H}} > \lambda_{\text{D}} > \lambda_{\text{T}}$ (B) $\lambda_{\text{H}} = \lambda_{\text{D}} = \lambda_{\text{T}}$ (C) $\lambda_{\text{H}} < \lambda_{\text{D}} < \lambda_{\text{T}}$ (D) $\lambda_{\text{H}} < \lambda_{\text{D}} > \lambda_{\text{T}}$

AS0163

17. The transition, so that the de-Broglie wavelength of electron becomes 3 times of its initial value in He^+ ion will be

- (A) $2 \rightarrow 5$ (B) $3 \rightarrow 2$ (C) $2 \rightarrow 6$ (D) $1 \rightarrow 2$

AS0164

- 18.** If the uncertainty in velocity & position is same, then the uncertainty in momentum will be

- (A) $\sqrt{\frac{hm}{4\pi}}$ (B) $m\sqrt{\frac{h}{4\pi}}$ (C) $\sqrt{\frac{h}{4\pi m}}$ (D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

AS0165

Question No. 19 to 22 (4 questions)

The only electron in the hydrogen atom resides under ordinary conditions on the first orbit. When energy is supplied, the electron moves to higher energy orbit depending on the amount of energy absorbed. When this electron returns to any of the lower orbits, it emits energy. Lyman series is formed when the electron returns to the lowest orbit while Balmer series is formed when the electron returns to second orbit. Similarly, Paschen, Brackett and Pfund series are formed when electron returns to the third, fourth and fifth orbits from higher energy orbits respectively.

Maximum number of lines produced when an electron jumps from n th level to ground level is equal to $\frac{n(n-1)}{2}$. For example, in the case of $n = 4$, number of lines produced is 6. ($4 \rightarrow 3$, $4 \rightarrow 2$, $4 \rightarrow$

1 , $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1$). When an electron returns from n_2 to n_1 state, the number of lines in the spectrum will be equal to

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

If the electron comes back from energy level having energy E_2 to energy level having energy E_1 , then the difference may be expressed in terms of energy of photon as:

$$E_2 - E_1 = \Delta E, \lambda = \frac{hc}{\Delta E}$$

Since h and c are constants, ΔE corresponds to definite energy; thus each transition from one energy level to another will produce a light of definite wavelength. This is actually observed as a line in the spectrum of hydrogen atom.

Wave number of line is given by the formula $\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$.

where R is a Rydberg's constant ($R = 1.1 \times 10^7 \text{ m}^{-1}$)

19. The energy of photon emitted corresponding to transition $n = 3$ to $n = 1$ is [$h = 6 \times 10^{-34} \text{ J-sec.}$]
 (A) $1.76 \times 10^{-18} \text{ J}$ (B) $1.98 \times 10^{-18} \text{ J}$ (C) $1.76 \times 10^{-17} \text{ J}$ (D) None of these

AS0166

20. In a collection of H-atom, electrons make transition from 5th excited state to 2nd excited state then maximum number of different types of photons observed are
 (A) 3 (B) 4 (C) 6 (D) 15

AS0167

21. The difference in the wavelength of the 1st line of Lyman series and 2nd line of Balmer series in a hydrogen atom is
 (A) $\frac{9}{2R}$ (B) $\frac{4}{R}$ (C) $\frac{88}{15R}$ (D) None

AS0168

22. The wave number of electromagnetic radiation emitted during the transition of electron in between two levels of Li^{2+} ion whose principal quantum numbers sum is 4 and difference is 2 is

- (A) 3.5 R (B) 4 R (C) 8 R (D) $\frac{8}{9}$ R

AS0169

1 TABLE (3Q)

Column-I	Column-II	Column-III
(A) Energy of electron in H-like atomic orbital	(P) Principal quantum number(n)	(I) Radial function [R(r)]
(B) Orientation and shape of H-like atomic orbital	(Q) Azimuthal quantum number (l)	(II) Angular function
(C) Magnitude & direction of spin angular momentum	(R) Magnetic quantum number (m)	(III) Wave function [Ψ]
(D) Probability density of electron in s-orbital of H-like atom/ion	(S) Spin quantum number(s)	(IV) Cannot be determined from Schrodinger equation

Assume no external electric or magnetic fields in Q.23 to Q.25

23. Which of the following option is incorrectly matched -

- (A) B – Q – II (B) B – R – II (C) B – P – I (D) B – Q – III

AS0170

24. Which option is correctly matched ?

- (A) C – S – III (B) D – R – III (C) A – P – IV (D) C – S – IV

AS0170

25. If option (A) is considered in presence of external electric or magnetic field then which option is incorrect -

- (A) A – P – III (B) A – Q – III (C) A – R – III (D) A – S – I

AS0170

EXERCISE # J-MAINS

1. The wave number of the first emission line in the Balmer series of H-Spectrum is :
(R = Rydberg constant) : [JEE-Main(online) 2013]
- (1) $\frac{3}{4}R$ (2) $\frac{9}{400}R$ (3) $\frac{5}{36}R$ (4) $\frac{7}{6}R$
- AS0173
2. For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship? [JEE-Main(online) 2014]
- (1) a dust particle (2) an electron (3) a proton (4) an α -particle.
- AS0175
3. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is : [JEE-Main(online) 2014]
- (1) 13.6 eV (2) 30.6 eV (3) 122.4 eV (4) 3.4 eV
- AS0176
4. If λ_0 and λ be the threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is [JEE-Main(online) 2014]
- (1) $\sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right)}$ (2) $\sqrt{\frac{2h}{m} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right)}$ (3) $\sqrt{\frac{2h}{m} (\lambda_0 - \lambda)}$ (4) $\sqrt{\frac{2hc}{m} (\lambda_0 - \lambda)}$
- AS0178
5. At temperature T, the average kinetic energy of any particle is $\frac{3}{2} kT$ - The de Broglie wavelength follows the order : [JEE-Main(online) 2015]
- (1) Visible photon > Thermal electron > Thermal neutron
(2) Thermal proton > Thermal electron > Visible photon
(3) Visible photon > Thermal neutron > Thermal electron
(4) Thermal proton > Visible photon > Thermal electron
- AS0181
6. A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V volts. If e and m are charge and mass of an electron respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by : [JEE-Main(online) 2016]
- (1) $\sqrt{2meV}$ (2) meV (3) 2meV (4) $\sqrt{meV}$
- AS0182
7. If the shortest wavelength in Lyman series of hydrogen atom is A, then the longest wavelength in Paschen series of He^+ is : [JEE-Main(online) 2017]
- (1) $\frac{36A}{5}$ (2) $\frac{9A}{5}$ (3) $\frac{36A}{7}$ (4) $\frac{5A}{9}$
- AS0184

8. The electron in the hydrogen atom undergoes transition from higher orbitals to orbital of radius 211.6 pm. This transition is associated with:- [JEE-Main(online) 2017]

(1) Brackett series (2) Balmer series (3) Lyman series (4) Paschen series

AS0185

9. Ejection of the photoelectron from metal in the photoelectric effect experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function of the metal is :

(1) 5 eV (2) 4 eV (3) 5.5 eV (4) 4.5 eV [JEE-Main(online) 2018]

AS0187

10. Which of the following statements is false ? [JEE-Main(online) 2018]

(1) Photon has momentum as well as wavelength.
 (2) Splitting of spectral lines in electrical field is called Stark effect.
 (3) Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.
 (4) Rydberg constant has unit of energy.

AS0188

11. If p is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength λ , then for $1.5 p$ momentum of the photoelectron, the wavelength of the light should be: [JEE-Main(online) 2019]

(Assume kinetic energy of ejected photoelectron to be very high in comparison to work function)

(1) $\frac{1}{2}\lambda$ (2) $\frac{3}{4}\lambda$ (3) $\frac{2}{3}\lambda$ (4) $\frac{4}{9}\lambda$

AS0189

12. For any given series of spectral lines of atomic hydrogen, let $\Delta\bar{\nu} = \bar{\nu}_{\max} - \bar{\nu}_{\min}$ be the difference in maximum and minimum frequencies in cm^{-1} . The ratio $\Delta\bar{\nu}_{\text{Lyman}} / \Delta\bar{\nu}_{\text{Balmer}}$ is :

(1) 27 : 5 (2) 4 : 1 [JEE-Main(online) 2019]
 (3) 5 : 4 (4) 9 : 4

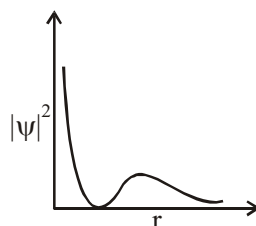
AS0191

13. The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are: [JEE-Main(online) 2019]

(1) Paschen and P fund (2) Lyman and Paschen
 (3) Brackett and Piund (4) Balmer and Brackett

AS0192

14. The graph between $|\psi|^2$ and r (radial distance) is shown below. This represents :-



- (1) 3s orbital
(3) 2p orbital

- (2) 1s orbital
(4) 2s orbital

[JEE-Main(online) 2019]

AS0193

15. Among the following, the energy of 2s orbital is lowest in :

[JEE-Main(online) 2019]

(1) K

(2) Na

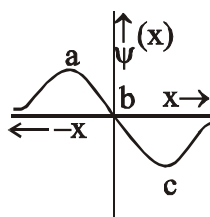
(3) Li

(4) H

AS0194

16. The electrons are more likely to be found :

[JEE-Main(online) 2019]



- (1) in the region a and b
(3) only in the region c

- (2) in the region a and c
(4) only in the region a

AS0195

17. If the de Broglie wavelength of the electron in n^{th} Bohr orbit in a hydrogenic atom is equal to $1.5 \pi a_0$ (a_0 is Bohr radius), then the value of n/z is :

[JEE-Main(online) 2019]

(1) 1.0

(2) 0.75

(3) 0.40

(4) 1.50

AS0197

18. The de Broglie wavelength (λ) associated with a photoelectron varies with the frequency (ν) of the incident radiation as, [ν_0 is threshold frequency] :

[JEE-Main(online) 2019]

(1) $\lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{3}{2}}}$

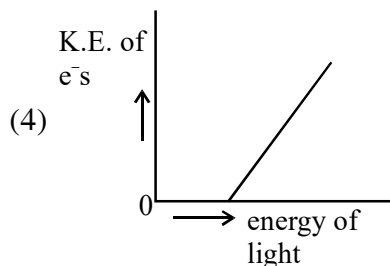
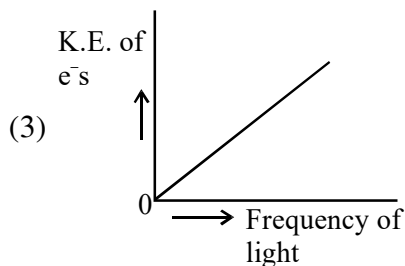
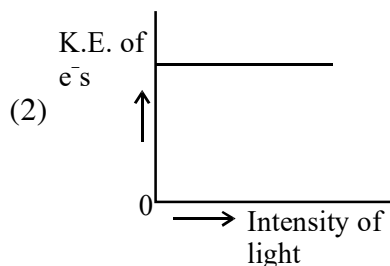
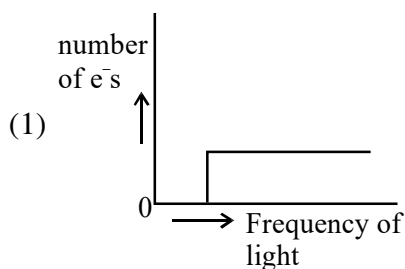
(2) $\lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{1}{2}}}$

(3) $\lambda \propto \frac{1}{(\nu - \nu_0)^{\frac{1}{4}}}$

(4) $\lambda \propto \frac{1}{(\nu - \nu_0)}$

AS0200

19. Which of the graphs shown below does not represent the relationship between incident light and the electron ejected from metal surface ? **[JEE-Main(online) 2019]**



AS0202

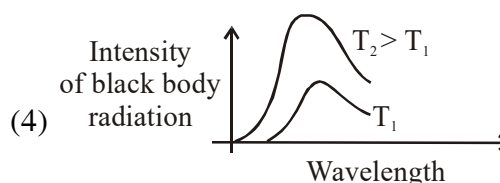
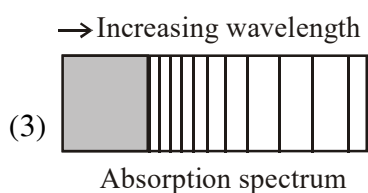
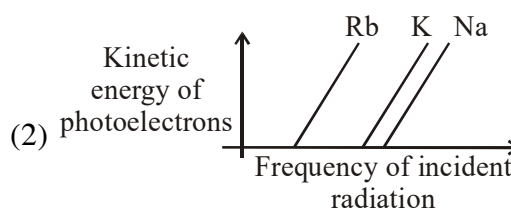
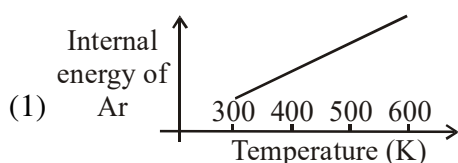
20. Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect?
(The Bohr radius is represented by a_0) **[JEE-Main(online) 2019]**

- (1) The electron can be found at a distance $2a_0$ from the nucleus
- (2) The probability density of finding the electron is maximum at the nucleus.
- (3) The magnitude of potential energy is double that of its kinetic energy on an average.
- (4) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.

AS0232

21. The figure that is not a direct manifestation of the quantum nature of atoms is :

[JEE-Main(online) 2020]



AS0233

22. The correct statement about probability density (except at infinite distance from nucleus) is :

[JEE-Main(online) 2020]

- (1) It can be negative for 2p orbital
(2) It can be zero for 3p orbital
(3) It can be zero for 1s orbital
(4) It can never be zero for 2s orbital

AS0234

23. The difference between the radii of 3rd and 4th orbits of Li^{2+} is ΔR_1 . The difference between the radii of 3rd and 4th orbits of He^+ is ΔR_2 . Ratio $\Delta R_1 : \Delta R_2$ is :

[JEE-Main(online) 2020]

- (1) 8 : 3
(2) 3 : 2
(3) 3 : 8
(4) 2 : 3

AS0235

24. According to Bohr's atomic theory :-

[JEE-Main(online) 2021]

- (A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$.
(B) The product of velocity (v) of electron and principal quantum number (n), 'vn' $\propto Z^2$.
(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.
(D) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$.

Choose the most appropriate answer from the options given below :

- (1) (C) Only
(2) (A) Only
(3) (A), (C) and (D) only
(4) (A) and (D) only

AS0236

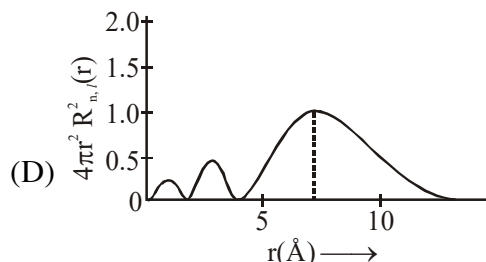
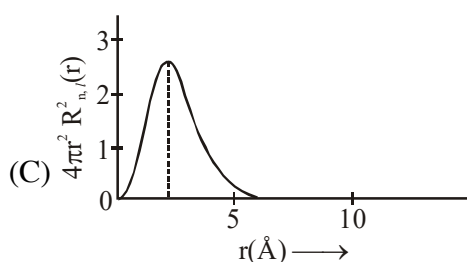
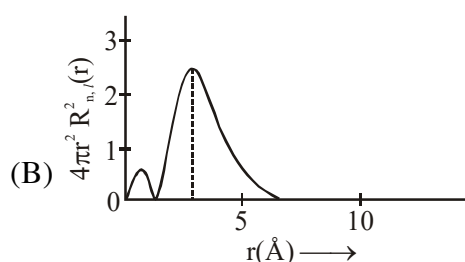
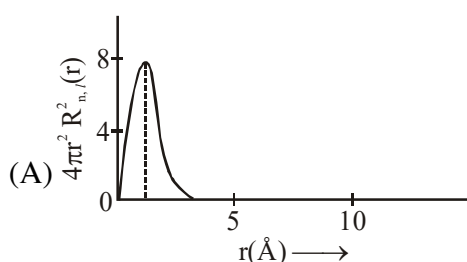
25. A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_{p} denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{\text{Li}}}{\lambda_{\text{p}}}$ is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer) (Mass of $\text{Li}^{3+} = 8.3$ mass of proton)

[JEE-Main(online) 2021]

AS0237

26. The plots of radial distribution functions for various orbitals of hydrogen atom against 'r' are given below:

[JEE-Main(online) 2021]



The correct plot for 3s orbital is:

- (1) (B)
(2) (A)
(3) (D)
(4) (C)

AS0238

27. A certain orbital has $n = 4$ and $m_l = -3$. The number of radial nodes in this orbital is _____.
(Round off to the Nearest Integer). [JEE-Main(online) 2021]
AS0239
28. Given below are two statements : [JEE-Main(online) 2021]
Statement I : Bohr's theory accounts for the stability and line spectrum of Li^+ ion.
Statement II : Bohr's theory was unable to explain the splitting of spectral lines in the presence of a magnetic field.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are true. (2) Statement I is false but statement II is true.
(3) Both statement I and statement II are false. (4) Statement I is true but statement II is false.
AS0240
29. A certain orbital has no angular nodes and two radial nodes. The orbital is :
[JEE-Main(online) 2021]
(1) 2s (2) 3s (3) 3p (4) 2p
AS0241
30. If the Thomson model of the atom was correct, then the result of Rutherford's gold foil experiment would have been : [JEE-Main(online) 2021]
(1) All of the α -particles pass through the gold foil without decrease in speed.
(2) α -Particles are deflected over a wide range of angles.
(3) All α -particles get bounced back by 180°
(4) α -Particles pass through the gold foil deflected by small angles and with reduced speed.
AS0242
31. Given below are two statements : [JEE-Main(online) 2021]
Statement I : Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.
Statement II : Bohr's model of hydrogen atom contradicts Heisenberg's uncertainty principle.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is false but statement II is true. (2) Statement I is true but statement II is false.
(3) Both statement I and statement II are false. (4) Both statement I and statement II are true.
AS0243
32. Given below are two statements. [JEE-Main(online) 2021]
Statement I : According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.
Statement II: According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are false (2) Both Statement I and Statement II are true
(3) Statement I is false but Statement II is true (4) Statement I is true but Statement II is false
AS0244

33. The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1 mW and wavelength of 1000 nm, in 0.1 second is $x \times 10^{13}$. The value of x is _____. (Nearest integer)
($h = 6.63 \times 10^{-34}$ Js, $c = 3.00 \times 10^8$ ms $^{-1}$) [JEE-Main(online) 2021]

AS0245

34. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{xma_0^2}$. The value of $10x$ is _____. (a_0 is radius of Bohr's orbit) (Nearest integer) [Given : $\pi = 3.14$]

AS0246

[JEE-Main(online) 2021]

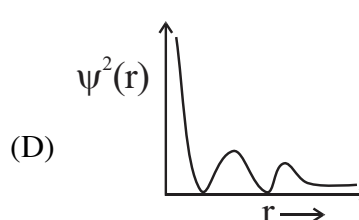
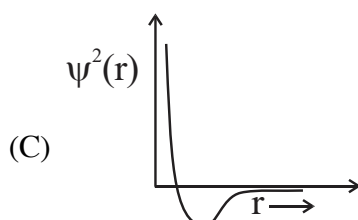
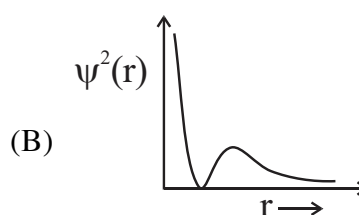
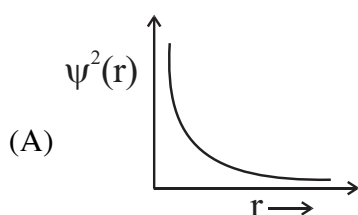
35. The energy of one mole of photons of radiation of wavelength 300 nm is
(Given : $h = 6.63 \times 10^{-34}$ Js, $N_A = 6.02 \times 10^{23}$ mol $^{-1}$, $c = 3 \times 10^8$ ms $^{-1}$) [JEE-Main(online) 2022]
(A) 235 kJ mol $^{-1}$ (B) 325 kJ mol $^{-1}$ (C) 399 kJ mol $^{-1}$ (D) 435 kJ mol $^{-1}$

AS0247

36. The number of radial and angular nodes in 4d orbital are, respectively
[JEE-Main(online) 2022]
(A) 1 and 2 (B) 3 and 2 (C) 1 and 0 (D) 2 and 1

AS0248

37. Which of the following is the correct plot for the probability density $\psi^2(r)$ as a function of distance 'r' of the electron from the nucleus for 2s orbital?
[JEE-Main(online) 2022]



AS0249

38. When the excited electron of a H atom from $n = 5$ drops to the ground state, the maximum number of emission lines observed are _____. [JEE-Main(online) 2022]

AS0250

39. The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is _____. (Nearest Integer)

(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

[JEE-Main(online) 2022]

AS0251

40. Identify the incorrect statement from the following. [JEE-Main(online) 2022]

- (A) A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.
(B) An orbital is the one electron wave function (ψ) in an atom.
(C) The existence of Bohr's orbits is supported by hydrogen spectrum.
(D) Atomic orbital is characterised by the quantum numbers n and l only

AS0252

41. The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_0$ (Where a_0 = Bohr radius 52.9 pm) is _____ km s^{-1} .

(Given : Mass of electron = 9.1×10^{-31} kg, Planck's constant $h = 6.63 \times 10^{-34}$ Js)

[JEE-Main(online) 2022]

AS0253

EXERCISE # J-ADVANCED

Paragraph for questions 1 to 3

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2 . The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom. [JEE 2010]

1. The state S_1 is :-

(A) 1s (B) 2s (C) 2p (D) 3s

AS0209

2. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-

(A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

AS0210

3. The orbital angular momentum quantum number of the state S_2 is :-

(A) 0 (B) 1 (C) 2 (D) 3

AS0211

4. The maximum number of electrons that can have principal quantum number, $n=3$, and spin quantum number, $m_s = -1/2$, is [JEE 2011]

AS0212

5. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is : :- [JEE 2011]

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
$\phi(\text{eV})$	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

AS0213

6. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] [JEE 2012]

(A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$ (C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{64\pi^2 m a_0^2}$

AS0214

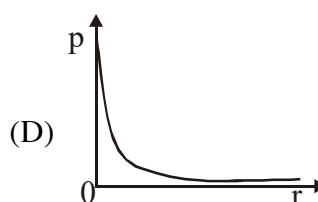
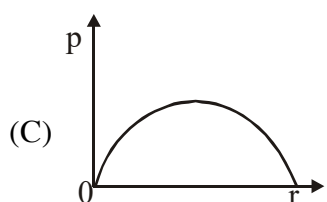
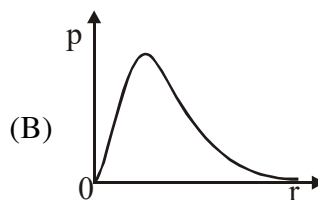
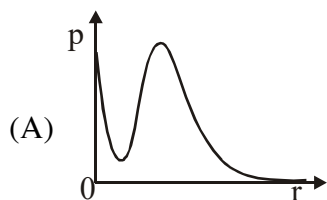
7. The atomic masses of He and Ne are 4 and 20 a.m.u. respectively. The value of the de Broglie wavelength of He gas at -73°C is "M" times that of the de Broglie wavelength of Ne at 727°C . M is. [JEE 2013]

AS0215

8. In an atom, the total number of electrons having quantum numbers $n = 4$, $|m_l| = 1$ and $m_s = -\frac{1}{2}$ is [JEE 2014]

AS0216

9. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr, at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is - [JEE 2016]



AS0217

Answer Q.10, Q.11 and Q.12 by appropriately matching the information given in the three columns of the following table. [JEE 2017]

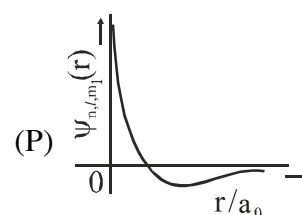
Column-1

Column-2

Column-3

(I) 1s orbital

(i) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$



(II) 2s orbital

(ii) One radial node

(Q) Probability density at nucleus $\propto \frac{1}{a_0^3}$

(III) $2p_z$ orbital

(iii) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{2a_0}\right)} \cos \theta$

(R) Probability density is maximum at nucleus

(IV) $3d_z^2$ orbital

(iv) xy - plane is a nodal plane

(S) Energy needed to excite electron from $n = 2$ state to $n = 4$ state is $\frac{27}{32}$ times the energy needed to excite electron from $n = 2$ state to $n = 6$ state

10. For the given orbital in column 1, the only **CORRECT** combination for any hydrogen - like species is :

(A) (IV) (iv) (R) (B) (II) (ii) (P) (C) (III) (iii) (P) (D) (I) (ii) (S)

AS0218

11. For He^+ ion, the only **INCORRECT** combination is

(A) (II) (ii) (Q) (B) (I) (i) (S) (C) (I) (i) (R) (D) (I) (iii) (R)

AS0219

12. For hydrogen atom, the only **CORRECT** combination is

(A) (I) (iv) (R) (B) (I) (i) (P) (C) (II) (i) (Q) (D) (I) (i) (S)

AS0220

13. The ground state energy of hydrogen atom is -13.6 eV . Consider an electronic state Ψ of He^+ whose energy, azimuthal quantum number and magnetic quantum number are -3.4 eV , 2 and 0 respectively. Which of the following statement(s) is(are) true for the state Ψ ? [JEE 2019]

- (1) It has 2 angular nodes
- (2) It has 3 radial nodes
- (3) It is a 4d state
- (4) The nuclear charge experienced by the electron in this state is less than $2e$, where e is the magnitude of the electronic charge.

AS0221

Question No. 14 to 15 (2 questions)

Answer the following by appropriately matching the lists based on the information given in the paragraph [JEE 2019]

Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following List-I contains some quantities for the n^{th} orbit of the atom and List-II contains options showing how they depend on n .

List-I

- (I) Radius of the n^{th} orbit
- (II) Angular momentum of the electron in the n^{th} orbit
- (III) Kinetic energy of the electron in the n^{th} orbit
- (IV) Potential energy of the electron in the n^{th} orbit

List-II

- (P) $\propto n^{-2}$
- (Q) $\propto n^{-1}$
- (R) $\propto n^0$
- (S) $\propto n^1$
- (T) $\propto n^2$
- (U) $\propto n^{1/2}$

14. Which of the following options has the correct combination considering List-I and List-II?
- (1) (II), (R) (2) (I), (P) (3) (I), (T) (4) (II), (Q)

AS0222

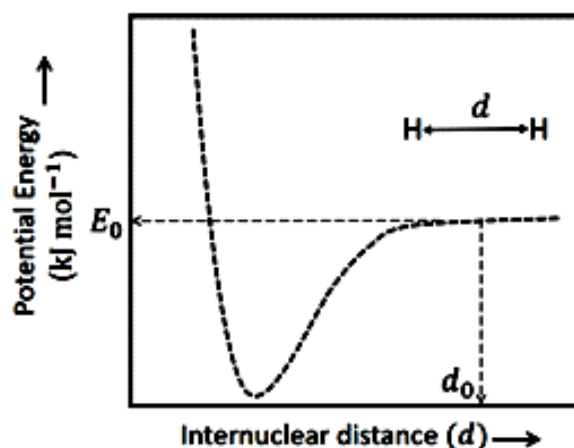
15. Which of the following options has the correct combination considering List-I and List-II?
- (1) (III), (S) (2) (IV), (Q) (3) (IV), (U) (4) (III), (P)

AS0223

- 16.** The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in the figure) in kJ mol^{-1} , for $d = d_0$ at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart.

Use Avogadro constant as $6.023 \times 10^{23} \text{ mol}^{-1}$.

[JEE 2020]



AS0254

17. Consider a helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s⁻¹) of He atom after the photon absorption is ____.

(Assume: Momentum is conserved when photon is absorbed.

Use: Planck constant = 6.6×10^{-34} J s, Avogadro number = 6×10^{23} mol⁻¹,

Molar mass of He = 4 g mol⁻¹)

[JEE 2021]

AS0255

ANSWER KEY**EXERCISE # S-I**

1. 1.00	2. 9.00	3. $n < \alpha < p < e$	4. 6.00
5. 1.60	6. 8.00	7. 2.30	8. 0.50
9. 1.28	10. 1.44	11. 150.00	12. 150.00
13. 0.20	14. 7.50	15. 4.96 or 4.97	16. 10.00
17. 1.00	18. 0.50	19. 620.00	20. 3.00
21. 150.13 or 150.14	22. 1.00	23. 50.00	24. 3.06
25. 0.41	26. 0.26	27. 4.00	28. 5.40 to 5.50
29. (i) 2.00 (ii) 9.60 to 9.80		30. 8.20 to 8.40	31. 65.00 to 67.00
32. 0.85	33. 2.10 or 2.11	34. 2.00	35. 5280.00
36. -2.18	37. (i) 5.00 (ii) 340.00		38. 3.00
39. 6566.40; 1216.00; 1026.00		40. 2.00	41. 114.00
42. (i) 4.00 ; (ii) 0.19	43. 3.00	44. 4.00	45. 3.09
46. 3.00	47. 8.00	48. 4.00	49. 36.81
50. 10.00	51. 0.08	52. 0.11	53. 0.17
54. 8.00	56. 3.75	57. 3.00	58. 2.00
59. 0.63	60. 4.00		

EXERCISE # S-II

1. 20.00	2. $E = \frac{n^6 h^6}{384 m^3 K^2 e^4 \pi^6}$
3. (i) 1, (ii) 2, (iii) 3, (iv) 3	4. - 435.20
5. 90.00	6. 91.00
7. (a) $Z = 3$; (b) $n_1 = 12$, $n_2 = 15$; (c) 0.08	8. 9.68 or 9.69
9. 972.54 or 972.55	10. 300.00, 0.26
11. (a) 75.00; (b) 1.41	12. 0.50
13. 0.36	14. 1240.00
15. 300303	

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	C	C	A	D	D	D	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	B	B	B	B	B	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	A	B	B	A	C	C	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	B	D	C	B	C	D	A	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	D	B	C	C	A	D	B	A	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	A	C	C	A	A	A	B	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	D	B	C	B	C	C	D	D	C
Que.	71	72								
Ans.	B	B								

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	B	A	D	A,B,D	B,D	A,D	A
Que.	11	12	13	14						
Ans.	A,C	B,C,D	C,D	A-P	B-P,Q,S	C-P,R	D-Q,S			
Que.	15				16	17	18	19	20	21
Ans.	A-S	B-R	C-Q	D-P	A	C	A	A	C	B
Que.	22	23	24	25						
Ans.	C	C	D	D						

EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	2	1	1	1	3	2	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	2	4	1	2	2	2	3	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	4	4	2	3	0	2	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	3	50	3155	C	A	B	4	1758	D
Que.	41									
Ans.	548									

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	9	4	C	5	6	B	B
Que.	11	12	13	14	15	16				17
Ans.	D	D	1,3	3	4	-2640.00 to -2620.00		or -5280.00 to -5240.00		30

CHEMICAL EQUILIBRIUM

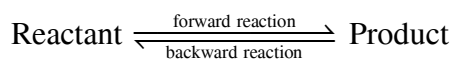
1. TYPES OF CHEMICAL REACTION :

The chemical reactions are classified on the basis of the extent to which they proceed, into the following two classes ;

I. Reversible reactions : Reaction in which entire amount of the reactants is not converted into products is termed as reversible reaction.

(i) Characteristics of reversible reactions :

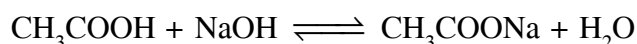
- (a) These reactions can be started from either side.
- (b) These reactions are never complete.
- (c) This sign ($\rightleftharpoons$) represents the reversibility of the reaction.



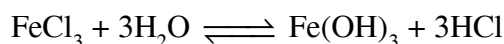
- (d) These reactions attain equilibrium and all the times, reaction mixture contains both reactant and product molecules. When reaction attains equilibrium the concentrations of reactants and product become constant (not same necessarily).

(ii) Examples of reversible reactions :

- (a) Neutralisation between an acid and a base either one or both are weak.



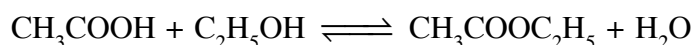
- (b) Salt hydrolysis



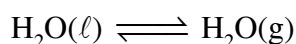
- (c) Thermal decomposition



- (d) Esterification



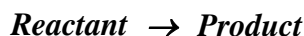
- (e) Evaporation of water in a closed vessel



II. Irreversible reactions : Reaction in which entire amount of the reactants is converted into products is termed as irreversible reaction.

(i) Characteristics of irreversible reactions :

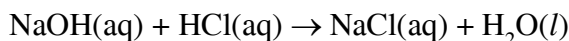
- (a) These reactions proceed only in one direction (forward direction)
- (b) These reactions can proceed to completion and reactant is completely converted into product.
- (c) The arrow ($\rightarrow$) is placed between reactants and products



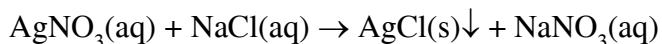
- (d) These reactions never attain equilibrium.

(ii) Examples of irreversible reactions :

- (a) Neutralisation between strong acid and strong base



- (b) Precipitation reactions

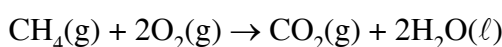


- (c) Thermal decomposition



However, if the above reaction is carried out in closed container, i.e., leaving no scope for gas to escape out, the reaction shows reversible nature.

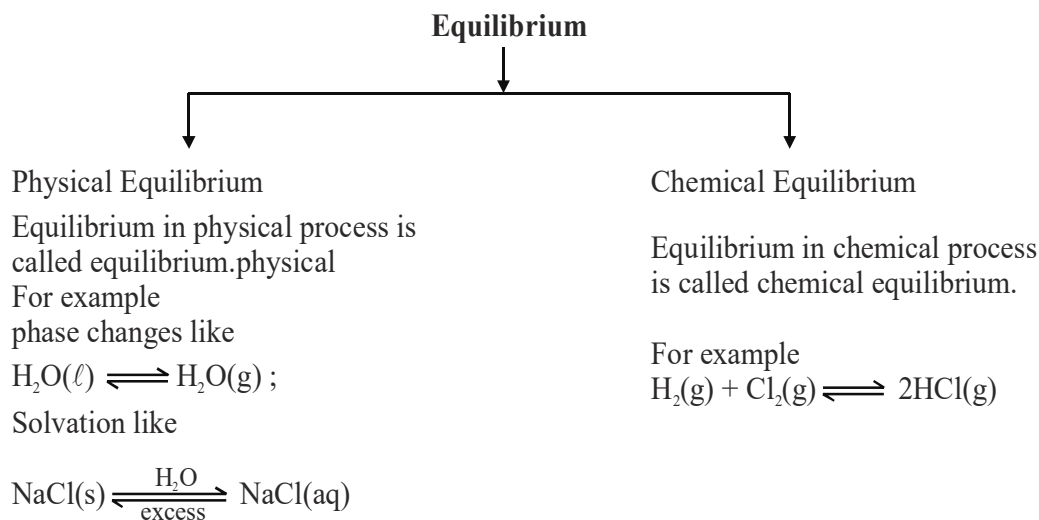
- (d) Combustion reactions

**2. EQUILIBRIUM :**

In the state of equilibrium, system loses its tendency for a change and all the properties associated with system like pressure, temperature, composition, etc become constant and do not vary without external stimulation. On the basis of nature of process in which state of equilibrium is attained, it may be of two types :

(A) Physical equilibrium

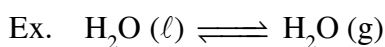
(B) Chemical equilibrium

**2.1 Physical equilibrium :**

If in a system only physical state (phase) is changed and equilibrium is established, (i.e. there is no chemical change), the equilibrium is called **physical equilibrium**. Ex : Fusion of ice, evaporation of water, dissolution of salts, absorption of gases in liquid, etc.

Following are the types of common physical equilibria :

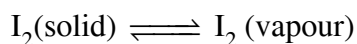
- (i) **Liquid-Vapour equilibria** : In a closed vessel, the vapours above the liquid are in equilibrium at given temperature.



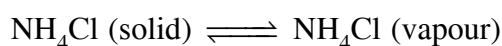
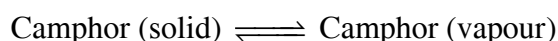
- (ii) **Solid-Liquid equilibria** : This equilibrium can be established only at melting point of solid. At this stage solid and liquid phases exist simultaneously in equilibrium.



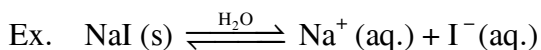
- (iii) **Solid-Vapour equilibria** : Let us now consider the systems where solids sublime to vapour phase. If we place solid iodine in a closed vessel, after sometime the vessel gets filled up with violet vapour and the intensity of colour increases with time. After certain time the intensity of colour becomes constant and at this stage equilibrium is attained. Hence solid iodine sublimes to give iodine vapour and the iodine vapour condenses to give solid iodine. The equilibrium can be represented as,



Other examples showing this kind of equilibrium are,



- (iv) **(Solute-Solvent) Saturated solution equilibria** : If the rate of dissolution of solids in liquid is equal to the rate of crystallization of solid from solution i.e. solution is saturated with respect to solid then saturated solution equilibria established, provided temperature is constant.



- (v) **(Gas + Solvent) Saturated solution equilibria** : In such equilibriums, solvents is saturated with respect to gas i.e. rate of entering of gas molecules in solvent is equal to rate of escaping of gas molecules from solvents. Above phenomenon can be observed in closed container at definite temperature. Ex : Dissolved CO_2 in cold drinks, dissolved O_2 in water, etc.

Note:(i) The solubilities of gases in liquid is a function of pressure of gas over liquid.

- (ii) **Henry's law** can be applied on such system, that states, the mass of gas dissolved in a given mass of solvent at any temperature is proportional to the pressure of the gas above the solvent.

- (iii) One should not compare it with liquid vapour equilibria.

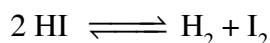
2.2 Chemical equilibrium :

Analogous to the physical systems, chemical reactions also attain a state of equilibrium. These reactions can occur both in forward and backward directions. When the rates of the forward and reverse reactions become equal, the concentrations of the reactants and the products become constant. This is the stage of chemical equilibrium. This equilibrium consists of a forward reaction in which the reactants give product(s) and reverse reaction in which product(s) gives the original reactants.

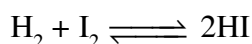
2.2.1 Characteristics of Chemical equilibrium :

- (i) It is attained in reversible chemical reactions only.
- (ii) Equilibrium is possible only in closed system.
- (iii) In this state, all the measurable properties of the system like temperature, concentration, colour, density etc. don't undergo any change with time, i.e. constant.
- (iv) Equilibrium is dynamic in nature i.e., at microscopic level reaction is not stopped. It appears that no change is occurring but both the opposing reactions are proceeding at the same rate. So there is no net change. Thus equilibrium is not static in nature.

- (v) Chemical equilibrium can be approached from both sides



or



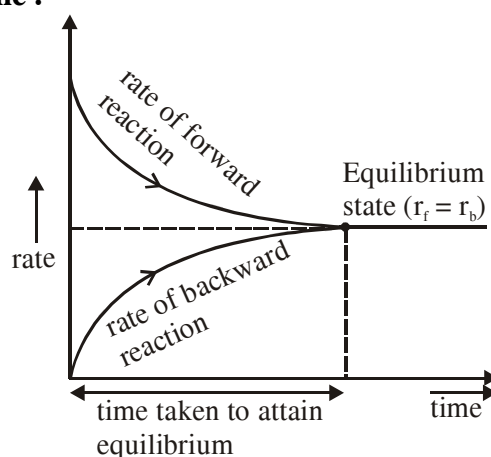
At equilibrium, reactants and products have fixed composition and this is independent of the fact whether the reaction start with the reactant or with the product.

- (vi) Equilibrium is not affected by the presence of catalyst. The catalyst only helps in attaining equilibrium rapidly.
- (vii) At equilibrium, opposing reactions (i.e., forward and backward) proceeds with equal rates.
i.e., rate of forward reaction = rate of backward reaction.

Note: (i) Whenever question doesn't ask about direction, then we take forward direction only.

- (ii) In a reversible reaction if forward reaction is exothermic then the backward reaction will be endothermic and vice versa.

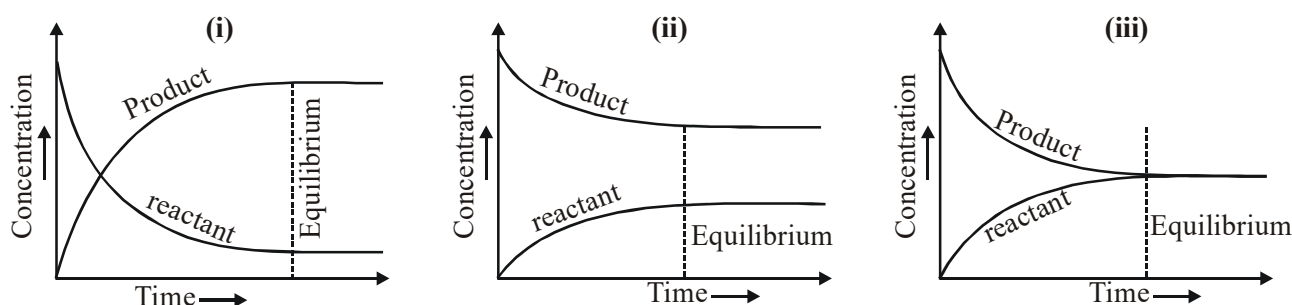
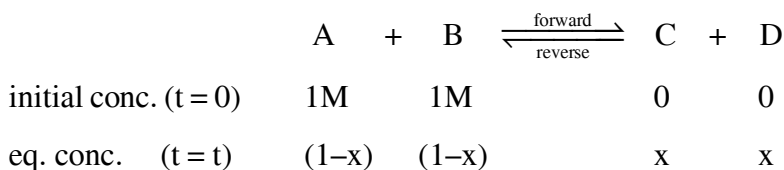
2.2.2 Variation of Rate v/s Time :



From above graph, initially rate of forward reaction decreases and the rate of backward reaction increases. At certain stage, rate of forward reaction becomes equal to rate of backward reaction called *equilibrium state*.

2.2.3 Variation of Concentration v/s Time :

Let us consider a reversible reaction,



In the beginning (at time $t = 0$), the container has only the reacting molecules A and B while the products C and D are nil. With the passage of time, the reactants A and B will be used up or consumed to form the products, C and D. It means that the concentration or molar concentration of the reactants will decrease while those of the products will increase. A stage will be ultimately reached when their concentration becomes constant i.e., there will be no further change in concentration of either of the reactants A and B or of the products C and D. This represents a state of equilibrium.

Note: At equilibrium, the concentration of reactant and product will be constant. It means, it may be different as above graph (i) & (ii) or same as above graph (iii) but the rate of forward reaction and the backward reaction will be always same.

Ex.1 Chemical equilibrium is a condition :

- (A) where all species have same concentration
- (B) where all species have constant concentration with respect to time.
- (C) where all species have unit concentration
- (D) all of above

Sol. (B)

Ex.2 Example of physical equilibria, is :

- (A) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
- (B) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- (C) $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$
- (D) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

Sol. (C)

Physical equilibria does not include any chemical change.

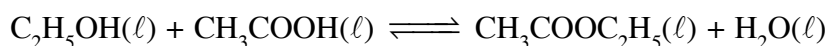
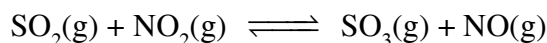
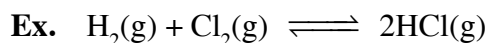
3. TYPES OF CHEMICAL EQUILIBRIUM :

I. Homogeneous equilibrium

II. Heterogeneous equilibrium

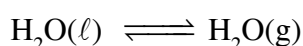
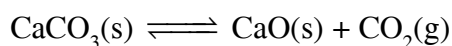
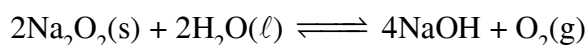
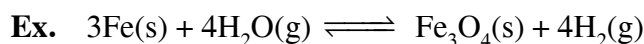
I. Homogeneous equilibrium :

It is the equilibrium when all reactants and products are in same phase.



II. Heterogeneous equilibrium :

It is the equilibrium the reactants and the products are present in different phases. All physical equilibria are heterogeneous.



4. ACTIVE MASS :

The mass of a substance which effect the rate of reaction i.e. mass of substance which take a part actively in a reaction. Active mass depends on state of substance.

(i) **Solution state :** In this state, active mass of a substance is represented by concentration (molarity).

Active mass is usually expressed in concentration by enclosing the symbol of the reactant in square bracket [].

$$\text{Active mass} = \frac{\text{moles}}{\text{Volume in litres}} = \frac{\text{grams(w)}}{\text{mol.wt.}(M_w) \times \text{Volume in litres(V)}} = \frac{w \times 1000}{M_w \times V(\text{mL})}$$

(ii) **Gaseous state :** In this state, active mass of a substance may be represented as concentration (molarity) or partial pressure.

(iii) **Pure solid & pure liquid :** In this state, active mass of solids , pure liquids and solvents in large excess is a constant quantity because there is no change in activity with the change in quantity or volume of system.

$$\text{Molar concentration (M)} = \frac{w}{M \times V} = \frac{d}{M} = \frac{\text{density of the substance}}{\text{molar mass of the substance}} = \text{constant}$$

as density of pure solids and liquids is constant and molar mass is also constant.

5. LAW OF MASS ACTION AND LAW OF EQUILIBRIUM CONSTANT :

The law of mass action is given by **Guldberg** and **Waage**. According to it, "the rate of a chemical reaction at a particular temperature is proportional to the product of active masses of reactants raised to the powers equal to their stoichiometric coefficients".

Consider a reversible reaction : $m_1A + m_2B \rightleftharpoons n_1C + n_2D$

According to law of mass action

rate of forward reaction (r_f) $\propto (a_A)^{m_1} (a_B)^{m_2}$

$$r_f = k_f (a_A)^{m_1} (a_B)^{m_2}$$

rate of backward reaction (r_b) $\propto (a_C)^{n_1} (a_D)^{n_2}$

$$r_b = k_b (a_C)^{n_1} (a_D)^{n_2}$$

At equilibrium $r_f = r_b$

$$k_f (a_A)^{m_1} (a_B)^{m_2} = k_b (a_C)^{n_1} (a_D)^{n_2}$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{(a_C)^{n_1} (a_D)^{n_2}}{(a_A)^{m_1} (a_B)^{m_2}}$$

where, K_{eq} = equilibrium constant

k_f = forward rate (velocity) constant

k_b = backward rate (velocity) constant

a_A, a_B = active mass of reactant A & B

a_C, a_D = active mass of product C & D

m_1, m_2 = stoichiometry coefficient of reactant A & B

n_1, n_2 = stoichiometry coefficient of product C & D

At a given temperature, the product of molar concentrations of the reaction products raised to the respective stoichiometric coefficient in the balanced chemical equation divided by the product of molar concentrations of the reactants raised to their individual stoichiometric coefficients has a constant value. This is known as the **Equilibrium Law or Law of Chemical Equilibrium**.

5.1 Types of equilibrium constant (K_{eq}) :

- (i) **Equilibrium constant in terms of concentration (K_C)** : K_C is defined for reactions in gas phase & solution phase. Molar concentrations are used to express amounts.
- (ii) **Equilibrium constant in terms of partial pressure (K_P)** : K_P is defined for reactions in gas phase. Partial pressures are used to express amounts.

Ex. Consider a reversible reaction in gas phase : $m_1A + m_2B \rightleftharpoons n_1C + n_2D$

$$K_C = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}}$$

$$K_P = \frac{(P_C)^{n_1} (P_D)^{n_2}}{(P_A)^{m_1} (P_B)^{m_2}}$$

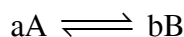
here,

$[A], [B], [C], [D]$ = Molar concentration of A, B, C, D respectively at equilibrium.

$(P_A), (P_B), (P_C), (P_D)$ = partial pressure of A, B, C, D respectively at equilibrium.

5.2 Units of equilibrium constant :

5.2.1 The value of equilibrium constant K_c can be calculated by substituting the concentration terms in mol/L or M and for K_p partial pressure is substituted in Pa, kPa, bar or atm. This results in units of equilibrium constant based on molarity or pressure, unless the exponents of both the numerator and denominator are same. For general reactions,



$$(i) \quad K_c = \frac{[B]^b}{[A]^a}$$

$$\text{unit of } K_c = \left(\frac{\text{mol}}{\text{L}} \right)^{b-a} = [M]^{b-a} = M^{\Delta n_g}$$

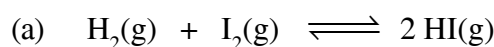
here, $\Delta n_g = b - a = \text{moles of (Product - Reactant)}$

$$(ii) \quad K_p = \frac{(P_B)^b}{(P_A)^a}$$

$$\text{unit of } K_p = (\text{atm})^{b-a} = (\text{atm})^{\Delta n_g} \text{ or } (\text{Pa})^{\Delta n_g} \text{ or } (\text{bar})^{\Delta n_g}$$

here, $\Delta n_g = b - a = \text{moles of (Product - Reactant)}$

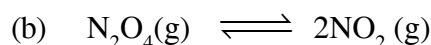
Ex.3 Write down the expression and unit for following reaction :



$$K_c = \frac{[HI]^2}{[H_2][I_2]} \quad \left(\frac{\text{mol}}{\text{L}} \right)^0$$

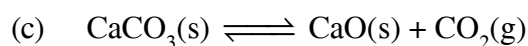
$$K_p = \frac{P_{HI}^2}{P_{H_2} P_{I_2}} \quad (\text{atm})^0$$

$\therefore K_c$ and K_p have no unit.



$$K_c = \frac{[NO_2]^2}{[N_2O_4]} \quad \left(\frac{\text{mol}}{\text{L}} \right)^1$$

$$K_p = \frac{P_{NO_2}^2}{P_{N_2O_4}} \quad (\text{atm})^1$$

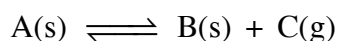


$$K_c = [CO_2] \quad \left(\frac{\text{mol}}{\text{L}} \right)^1$$

$$K_p = P_{CO_2} \quad (\text{atm})^1$$

5.2.2

Equilibrium constants is expressed in standard state that is called standard state equilibrium constant or thermodynamic equilibrium constant, which is dimensionless quantity. It is denoted by K_c^0 and K_p^0 . For a pure gas, the standard state is 1 bar. Therefore a pressure of 4 bar in standard state can be expressed as $4 \text{ bar}/1 \text{ bar} = 4$, which is a dimensionless number and concentration of 3 M solution in standard state can be expressed as $\frac{3\text{M}}{1\text{M}} = 3$, which is a dimensionless number. The numerical value of equilibrium constant depends on the standard state chosen. Thus, in this system both K_c^0 and K_p^0 are dimensionless quantities but have different numerical values due to different standard states. For example,



$$K_c = [C] \quad (\text{mol/L})$$

$$\text{but } K_c^0 \text{ is a dimensionless quantity so } K_c^0 = \frac{[C]}{1\text{M}}$$

$$K_p = P_C \quad (\text{bar})^1$$

$$\text{but } K_p^0 \text{ is a dimensionless quantity so } K_p^0 = \frac{P_C}{1 \text{ bar}}$$

5.3 Relation between K_p & K_c :

For the reaction, $aA + bB \rightleftharpoons cC + dD$

we can write

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \dots\dots\dots(i)$$

Assuming the gaseous components to behave ideally,

$$P_i V_i = n_i RT$$

$$P_i = (n_i/V_i) RT = C_i RT = [i] RT$$

where $[i]$ is the molar concentration of the species 'i'.

$$P_A = [A]RT, P_B = [B]RT, P_C = [C]RT, P_D = [D]RT$$

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b} = \frac{([C]RT)^c ([D]RT)^d}{([A]RT)^a ([B]RT)^b} = \frac{[C]^c [D]^d}{[A]^a [B]^b} \times (RT)^{(c+d)-(a+b)}$$

from eq..(i)

$$K_p = K_c (RT)^{(c+d)-(a+b)}$$

$$\boxed{K_p = K_c (RT)^{\Delta n_g}}$$

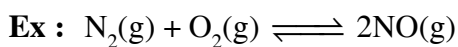
Δn_g = (Number of moles of gaseous products) - (Number of moles of gaseous reactants).

$$\Delta n_g = (c + d) - (a + b)$$

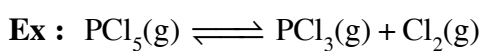
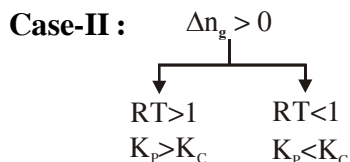
The units of K_p & K_c are not fixed and depend on stoichiometry of the reaction. In case the number of moles of the reactant & that of the product are same K_p & K_c do not have any unit.

5.4 Different cases for $K_p = K_c(RT)^{\Delta n_g}$:

Case-I : If $\Delta n_g = 0$ then $K_p = K_c$



$K_p = \text{unit less} ; K_c = \text{unit less} ;$

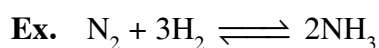
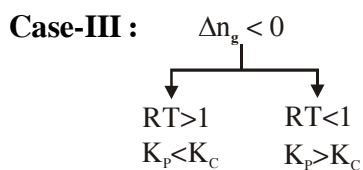


$\Delta n_g = 2 - 1 = 1$ then $K_p = K_c(RT)^1$

if $RT > 1$ then $K_p > K_c$

if $RT < 1$ then $K_p < K_c$

Unit : $K_p = \text{atm}^1 ; K_c = \text{conc.}^1$



$\Delta n_g = 2 - 4 = -2$ then $K_p = K_c(RT)^{-2}$

if $RT > 1$ then $K_p < K_c$

if $RT < 1$ then $K_p > K_c$

$K_p = \text{atm}^{-2} ; K_c = \text{conc.}^{-2}$

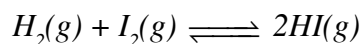
Case-IV : If $T = \frac{1}{R} = \frac{1}{0.0821} \approx 12.2 \text{ K}$

$[K_p = K_c ; \text{for any value of } \Delta n_g]$

Ex.6 In the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ the concentration of H_2 , I_2 and HI at equilibrium are 10.0, 6.0 and 28 moles per litre respectively. What will be the equilibrium constant?

- (A) 30.61 (B) 13.066 (C) 29.40 (D) 20.90

Sol. (B)



Applying law of mass action,

$$K_c = \frac{[HI]^2}{[H_2][I_2]}$$

$$\begin{aligned} \text{Given } [H_2] &= 10 \text{ mol L}^{-1} \\ [I_2] &= 6.0 \text{ mol L}^{-1} \\ [HI] &= 28.0 \text{ mol L}^{-1} \end{aligned}$$

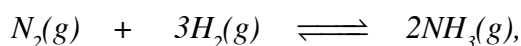
$$\text{So, } K_c = \frac{(28.0)^2}{(10) \times (6.0)} = 13.066$$

Ex.7 For a gas phase reaction at equilibrium,

$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$, the partial pressures of H_2 and N_2 are 0.4 and 0.8 atmosphere respectively. The total pressure of the entire system is 2.4 atmosphere. What will be the value of K_p if all the pressures are given in atmosphere ?

- (A) 32 atm^{-2} (B) 20 atm^{-2} (C) 28.125 atm^{-2} (D) 80 atm^{-2}

Sol. (C)



Partial pressures at equilibrium

$$0.8 \quad 0.4 \quad [2.4 - (0.8 + 0.4) = 1.2]$$

Applying law of mass action,

$$K_p = \frac{[P_{NH_3}]^2}{[P_{N_2}][P_{H_2}]^3} = \frac{1.2 \times 1.2}{0.8 \times 0.4 \times 0.4 \times 0.4} \Rightarrow K_p = 28.125 \text{ atm}^{-2}$$

Ex.8. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; Correct option for reaction is/are :

- (a) $K_p = K_c$
 (b) $K_p > K_c$
 (c) $K_p < K_c$
 (d) any of these, depending on temperature.

Sol. (D)

If (d) option is not given, then the answer is (b) because $\frac{1}{R} \approx 12.2K$ which is very low relative to room temperature.

Ex.9. For which of the following reactions, $K_p > K_c$ at 25°C (298K)

- (a) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 (b) $\text{H}_2(\text{g}) + \text{I}_2(\text{s}) \rightleftharpoons 2\text{HI}(\text{g})$
 (c) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 (d) $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 (e) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

Sol. (b, d, e)

Ex.10. For a gaseous reaction $K_p = 0.4 \text{ atm}^3$ at 27°C . Calculate K_c .

Sol. $K_p = K_c (RT)^{\Delta n}$
 $0.4 = K_c (0.0821 \times 300)^3$
 $K_c = 2.6 \times 10^{-8} \text{ M}^3$

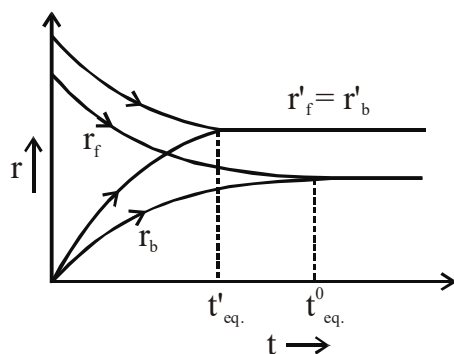
6. Characteristics of equilibrium constant :

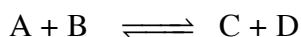
- The expression for equilibrium constant K is applicable only when concentrations of the reactants and products have attained their equilibrium values and do not change with time.
- The value of equilibrium constant is independent of initial concentration of the reactants and product.
- Equilibrium constant has one unique value for a particular reaction represented by a balanced equation at a given temperature.
- Value of equilibrium constant is not affected by catalyst. Catalyst simply helps in attaining equilibrium earlier.

The relative increase in the rate of forward as well as backward reaction remains same on using the catalyst.

$$K_{\text{eq. uncatalyst}} = \frac{K_f}{K_b}$$

$$K_{\text{eq. catalyst}} = \frac{K_f \times x}{K_b \times x} = \frac{K_f}{K_b} = K_{\text{eq. (uncatalyst)}}$$

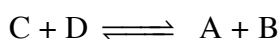


6.1 Factor affecting the equilibrium constant :**(A) Mode of representation of the reaction :**

The equilibrium constant for the reaction

$$K_c = \frac{[C][D]}{[A][B]}$$

If the reaction is reversed



$$\text{then, } K'_c = \frac{[A][B]}{[C][D]}$$

The two equilibrium constant related as ; $K_c = \frac{1}{K'_c}$

If the reaction is reversed, the value of the equilibrium constant is inverted.

(B) Stoichiometry of the reaction :

When a reversible reaction can be written with the help of two or more stoichiometric equation, the value of equilibrium constant will be numerically different.

(i) For reaction, $2NO_2 \rightleftharpoons N_2 + 2O_2$

$$K_c = \frac{[N_2][O_2]^2}{[NO_2]^2}$$

For reaction $NO_2 \rightleftharpoons \frac{1}{2}N_2 + O_2$

$$K'_c = \frac{[N_2]^{\frac{1}{2}}[O_2]}{[NO_2]}$$

The two constants are related as $K'_c = \sqrt{K_c}$

(ii) For reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

$$K_p = \frac{P_{HI}^2}{P_{H_2} \cdot P_{I_2}}$$

For reaction $2H_2(g) + 2I_2(g) \rightleftharpoons 4HI(g)$

$$K'_p = \frac{P_{HI}^4}{P_{H_2}^2 \cdot P_{I_2}^2}$$

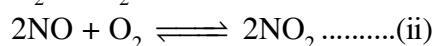
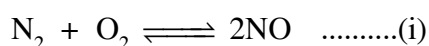
The two constants are related as $K'_p = K_p^2$.

In general, $nH_2(g) + nI_2(g) \rightleftharpoons 2nHI(g)$

$$K' = K^n$$

When the coefficient of a balanced equation are multiplied by a common factor 'n', the equilibrium constant must be raised to the respective factor, K_{eq}^n .

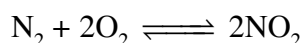
(C) Addition the reaction :



For the 1st step, $K_1 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$

For the 2nd step, $K_2 = \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]}$

from eq. (i) + (ii)



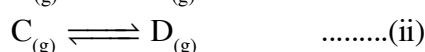
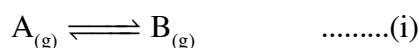
For this reaction, $K = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2}$

the above reaction is related as

$$\therefore K_1 \times K_2 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]} \times \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]} = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2} = K$$

If there is addition of two reaction then the equilibrium constant is multiplied.

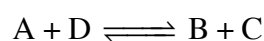
(D) Subtract the reaction :



For the (i) step, $K_1 = \frac{[\text{B}]}{[\text{A}]}$

For the (ii) step, $K_2 = \frac{[\text{D}]}{[\text{C}]}$

from eq. (i) – (ii)



For this reaction, $K = \frac{[\text{B}][\text{C}]}{[\text{A}][\text{D}]}$

the above reaction is related as, $K = \frac{K_1}{K_2} = \frac{[\text{B}]}{[\text{A}]} \times \frac{[\text{C}]}{[\text{D}]} = K$

If we subtract of two reaction, then the equilibrium constant will be divided.

Ex.11 What should be the equilibrium constant for the reaction $\text{HI} \rightleftharpoons \frac{1}{2}\text{H}_2 + \frac{1}{2}\text{I}_2$ if the equilibrium constant for the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ at 444° C is 64 ?

(A) 64

(B) 8

(C) 0.12

(D) 0.81

Sol. $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ $\text{HI} \rightleftharpoons \frac{1}{2}\text{H}_2 + \frac{1}{2}\text{I}_2$

$$K_1 = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

$$K_2 = \frac{[\text{H}_2]^{1/2}[\text{I}_2]^{1/2}}{[\text{HI}]}$$

Therefore, $K_2 = \frac{1}{\sqrt{K_1}} = \frac{1}{\sqrt{64}} = \frac{1}{8} = 0.12$

(E) Equilibrium constant depends upon the temperature.

The equilibrium constant of a particularly balanced reaction depends only on temperature. It is independent from all other factor like amount of components, concentration, pressure, volume etc.

$$K_{eq.} = \frac{K_f}{K_b}$$

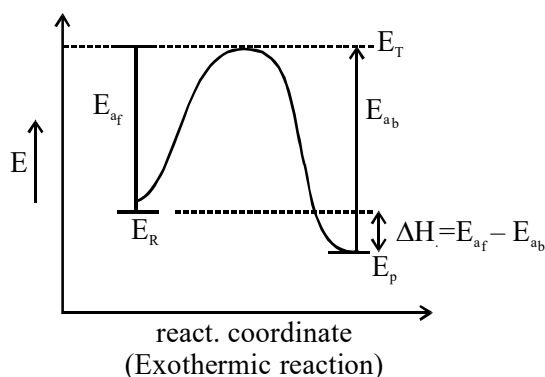
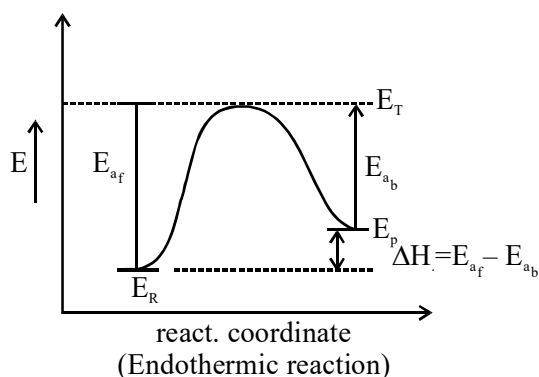
And rate constant, $K = A \cdot e^{-E_a/RT}$

$$\text{Now, } K_{eq.} = \frac{K_f}{K_b} = \frac{A_f \cdot e^{-E_{af}/RT}}{A_b \cdot e^{-E_{ab}/RT}} = \frac{A_f}{A_b} \cdot e^{-(E_{af}-E_{ab})/RT}$$

$$K_{eq.} = A \cdot e^{-\Delta H/RT} \quad (\text{Van't Hoff's equation})$$

where $A = \frac{A_f}{A_b} = \text{constant}$

and $\Delta H = E_{af} - E_{ab}$



If K_1 and K_2 be the equilibrium constants of a reaction at absolute temperatures T_1 and T_2 , then

$$\ln \frac{K_2}{K_1} = \frac{\Delta H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

On increasing the temperature the rate constant of reaction increases. But the reaction with higher activation energy is more sensitive towards temperature change. For the reaction of higher activation energy the value of rate constant increases largely on increasing temperature as well as the rate constant decreases largely as decrease in temperature.

For endothermic reaction

$$\Delta H = E_{af} - E_{ab} = +ve$$

$$\Rightarrow T \uparrow ; K_{eq.} = \frac{K_f \uparrow}{K_b \uparrow} \quad (K_f \text{ larger increasing and hence } K_{eq.} \text{ increase})$$

For exothermic reaction

$$\Delta H = E_{af} - E_{ab} = -ve$$

$$\Rightarrow T \uparrow ; K_{eq.} = \frac{K_f \uparrow}{K_b \uparrow} \quad (K_b \text{ larger increasing and hence } K_{eq.} \text{ decreases})$$

Ex.12 The equilibrium constant for the reaction $H_2(g) + S(s) \rightleftharpoons H_2S(g)$; is 18.5 at 925 K and 9.25 at 1000 K respectively. The enthalpy of the reaction will be :

- (A) $-68000.05 \text{ J mol}^{-1}$ (B) $-71080.57 \text{ J mol}^{-1}$
(C) $-80071.75 \text{ J mol}^{-1}$ (D) $57080.75 \text{ J mol}^{-1}$

Sol. (B)

Using the relation,

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\log \frac{9.25}{18.5} = \frac{\Delta H}{2.303 \times 8.314} \times \frac{75}{925 \times 1000}$$

$$-0.301 = \frac{\Delta H \times 75}{2.303 \times 8.314 \times 925 \times 1000} \Rightarrow \Delta H = -71080.57 \text{ J mol}^{-1}.$$

Ex.13. The equilibrium $2A + B_2 \rightleftharpoons 2AB$ involves the two component equilibria,

$A + B_2 \rightleftharpoons AB + B$ and $A + B \rightleftharpoons AB$ with respective equilibrium constants K_1 and K_2 . How is the equilibrium constant, K , for the overall equilibrium related to K_1 and K_2 ?

- (A) $K = K_1 + K_2$ (B) $K = K_1 K_2$ (C) $K = K_1 / K_2$ (D) $K/2 = K_1 K_2$

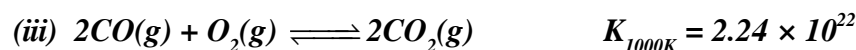
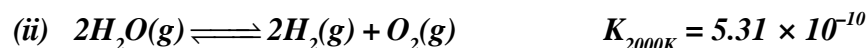
Sol. (B)

Ex.14. For the reactions :



Sol. 9.04×10^{-16}

Ex.15. From the following data :



State whether the reaction (iii) is exothermic or endothermic?

Sol. Equation (iii) = $-[2 \times (i) + (ii)]$

$$\therefore K_{2000(iii)} = \frac{1}{K_1^2 K_2} = \frac{1}{(4.4)^2 \times 5.31 \times 10^{-10}} = 9.7 \times 10^7$$

$$\therefore T \uparrow K \downarrow \Rightarrow \text{reaction is exothermic.}$$

7. APPLICATION OF EQUILIBRIUM CONSTANT :

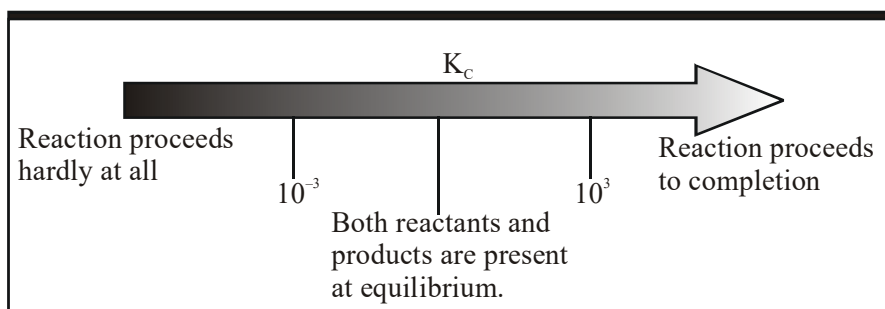
- (I) Predicting the extent of a reaction
- (II) Predicting the direction of the reaction and
- (III) Calculating equilibrium composition.

7.1 Predicting the extent of reaction :

The numerical value of the equilibrium constant for a reaction indicates the extent of the reaction. But it is important to note that an equilibrium does not give any information about the rate at which the equilibrium is reached. The magnitude of K_c or K_p is directly proportional to the concentrations of products (as these appear in the numerator of equilibrium constant expression) and inversely proportional to the concentrations of the reactants (these appear in the denominator). This implies that a high value of K is suggestive of a high concentration of products and vice-versa.

We can make the following generalisations concerning the composition of equilibrium mixtures:

- (i) If $K_c > 10^3$, products predominate over reactants. If K_c is very large, the reaction proceeds almost all the way to completion.
- (ii) If $K_c < 10^{-3}$, reactants predominate over products. If K_c is very small, the reaction proceeds hardly at all.
- (iii) If K_c is in the range 10^{-3} to 10^3 , appreciable concentration of both reactants and products are present.



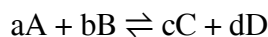
Ex.16 In the following cases which is predominant (reactant or product) for the given value of K_c .

- (a) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$, at 300K has $K_c = 4.0 \times 10^{31}$.
- (b) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, at 298 K has $K_c = 4.8 \times 10^{-31}$.
- (c) $N_2O_4 \rightleftharpoons 2NO_2$, at 298 K has $K_c = 4.64 \times 10^{-3}$.

- Sol.**
- (a) If $K_c > 10^3$, products predominate over reactants. If K_c is very large, the reaction proceeds almost all the way to completion.
 - (b) If $K_c < 10^{-3}$, reactants predominate over products. If K_c is very small, the reaction proceeds hardly at all.
 - (c) Also, gas phase decomposition of N_2O_4 to NO_2 is another reaction with a value of $K_c = 4.64 \times 10^{-3}$ at 25°C which is neither too small nor too large. Hence, equilibrium mixtures contain appreciable concentrations of both N_2O_4 and NO_2 .

7.2 Predicting the direction of the Reaction :

The equilibrium constant is also used to find in which direction, the reaction mixture of reactants and products will proceed. For this purpose, we calculate the reaction quotient, Q . The reaction quotient is defined in the same way as the equilibrium constant (with molar concentrations to give Q_C , or with partial pressure to give Q_P) at any stage of reaction. For a general reaction:



$$Q_C = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Then, if $Q_C > K_C$, the reaction will proceed in the direction of reactants (reverse reaction).

if $Q_C < K_C$, the reaction will move in the direction of the products

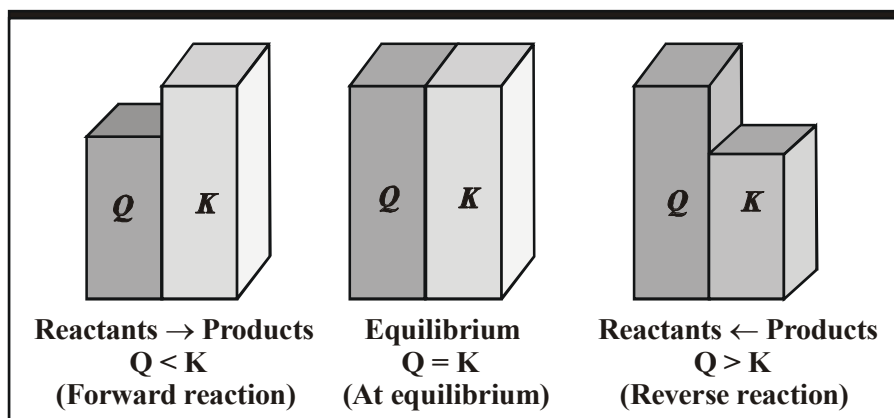
if $Q_C = K_C$, the reaction mixture is already at equilibrium.

In the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, if the molar concentrations of H_2 , I_2 and HI are 0.1 M, 0.2 M and 0.4 M, respectively at 783 K, then reaction quotient at this stage of the reaction is

$$Q_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{(0.4)^2}{(0.1)(0.2)} = 8$$

K_C for this reaction at 783 K is 46 and we find that $Q_C < K_C$. The reaction, therefore, will move to right i.e. more $H_2(g)$ and $I_2(g)$ will react to form more $HI(g)$ and their concentration will decrease till $Q_C = K_C$.

Thus, a reaction has a tendency to form products if $Q < K$ and to form reactants if $Q > K$.



Ex.17. For the reaction $NOBr(g) \rightleftharpoons NO(g) + \frac{1}{2} Br_2(g)$

$K_p = 0.15$ atm at $90^\circ C$. If $NOBr$, NO and Br_2 are mixed at this temperature having partial pressures 0.5 atm, 0.4 atm and 2.0 atm respectively, will Br_2 be consumed or formed ?

Sol.
$$Q_P = \frac{[P_{Br_2}]^{1/2} [P_{NO}]}{[P_{NOBr}]} = \frac{[0.20]^{1/2} [0.4]}{[0.50]} = 0.36$$

$$\therefore K_p = 0.15$$

Hence, reaction will shift in backward direction

$\therefore Br_2$ will be consumed

- (ii) If A(s), B(s) and C(s) are placed in a room where $P_{H_2O}(g) = 0.025 \text{ atm}$, which will act as dehydrating agent.
 (A) A(s) (B) B(s) & C(s) (C) B(s) (D) All
- (iii) Which of the following will be efflorescent at 7.5% R.H. if aqueous $\text{terim} = 0.04 \text{ atm}$.
 (a) Only A. $2H_2O(s)$ (b) Only B. $3H_2O(s)$
 (c) Only C. $4H_2O(s)$ (d) Only A. $2H_2O(s)$ & B. $3H_2O(s)$
 (e) Only B. $3H_2O(s)$ & C. $4H_2O(s)$
- Sol.** (i) D (ii) A (iii) C

7.3 Calculating equilibrium concentrations :

In case of a problem in which we know the initial concentrations but do not know any of the equilibrium concentrations, the following three steps shall be followed :

Step 1. Write the balanced equation for the reaction.

Step 2. Under the balanced equation, make a table that lists for each substance involved in the reaction :

- (a) the initial concentration,
- (b) the change in concentration on going to equilibrium, and
- (c) the equilibrium concentration.

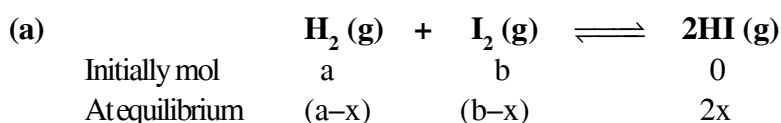
In constructing the table, define x as the concentration (mol/L) of one of the substances that reacts on going to equilibrium, then use the stoichiometry of the reaction to determine the concentrations of the other substances in terms of x .

Step 3. Substitute the equilibrium concentrations into the equilibrium equation for the reaction and solve for x . If you are to solve a quadratic equation choose the mathematical solution that makes chemical sense.

Step 4. Calculate the equilibrium concentrations from the calculated value of x .

Step 5. Check your results by substituting them into the equilibrium equation.

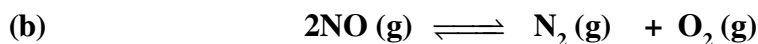
7.3.1 Some General Equilibrium Expressions :



$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)} = \frac{4x^2}{(a-x)(b-x)}$$

$$K_p = \frac{(P_{HI})^2}{P_{H_2} \times P_{I_2}} = \frac{\left(\frac{2x}{a+b} \cdot P\right)^2}{\left(\frac{a-x}{a+b} \cdot P\right)\left(\frac{b-x}{a+b} \cdot P\right)} = \frac{4x^2}{(a-x)(b-x)}$$

So $K_c = K_p \quad (\Delta n_g = 0)$



Initially mol	a	0	0
At equilibrium	(a-x)	x/2	x/2

$$K_c = \frac{[\text{N}_2][\text{O}_2]}{[\text{NO}]^2} = \frac{x/2 \times x/2}{(a-x)^2} = \frac{x^2}{4(a-x)^2} = K_p \quad (\Delta n_g = 0)$$



Initially mol	a	b	0	0
At equilibrium	(a-x)	(b-x)	x	x

$$K_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}]} = \frac{x^2}{(a-x)(b-x)}$$

K_p should not be given for this reaction,



Initially mol	a	0	0
At equilibrium	(a-x)	x	x
Active mass	$\frac{(a-x)}{v}$	$\frac{x}{v}$	$\frac{x}{v}$

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{\frac{x}{v} \times \frac{x}{v}}{\frac{(a-x)}{v}} = \frac{x^2}{(a-x)v}$$

$$K_p = \frac{p_{\text{PCl}_3} \times p_{\text{Cl}_2}}{p_{\text{PCl}_5}} = \frac{\left(\frac{x}{a+x} \cdot P\right) \times \left(\frac{x}{a+x} \cdot P\right)}{\left(\frac{a-x}{a+x}\right) P} = \frac{x^2 P}{(a+x)(a-x)} = \frac{x^2 P}{a^2 - x^2}$$



Initially mol	a	b	0
At equilibrium	(a-x)	(b-3x)	2x
Active mass	$\frac{(a-x)}{v}$	$\left(\frac{b-3x}{v}\right)$	$\left(\frac{2x}{v}\right)$

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{\left(\frac{2x}{v}\right)^2}{\left(\frac{a-x}{v}\right)\left(\frac{b-3x}{v}\right)^3} = \frac{4x^2 v^2}{(a-x)(b-3x)^3}$$

$$K_p = \frac{(p_{\text{NH}_3})^2}{p_{\text{N}_2} \times (p_{\text{H}_2})^3} = \frac{\left[\frac{2xP}{a+b-2x}\right]^2}{\left[\frac{(a-x)P}{a+b-2x}\right]\left[\frac{(b-3x)P}{a+b-2x}\right]^3} = \frac{4x^2(a+b-2x)^2}{(a-x)(b-3x)^3 P^2}$$

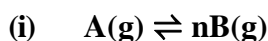
(g) Equilibrium constant expressions in term of ' α '



Initial moles	1	—	—
At. Eq.	$1-\alpha$	α	α

$$K_C = \frac{\alpha^2}{1-\alpha} \left(\frac{1}{V} \right) \quad \text{and} \quad K_P = \frac{\alpha^2}{1-\alpha^2} P_T$$

(h) Degree of dissociation in terms of molar mass and vapour density



$$\alpha = \frac{M_T - M_O}{M_O(n-1)} \quad \text{or} \quad \alpha = \frac{D_T - D_O}{D_O(n-1)}$$

M_T = Theoretical molar mass of reactant

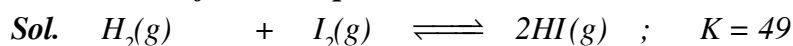
M_O = Observed molar mass of mixture

$$D_T = \frac{M_T}{2} \quad \text{and} \quad D_O = \frac{M_O}{2}$$



$$\alpha = \frac{M_T - M_O}{M_O \left(\frac{1}{n} - 1 \right)}$$

Ex.22 5 moles H_2 gas and 5 moles iodine-vapours are taken in vessel of 10 L capacity. Determine the moles of each at equilibrium.



$$49 = \frac{(2x)^2 v^2}{(5-x)(5-x)v^2}$$

$$49 = \frac{(2x)^2}{(5-x)^2}$$

$$7 = \frac{2x}{5-x}$$

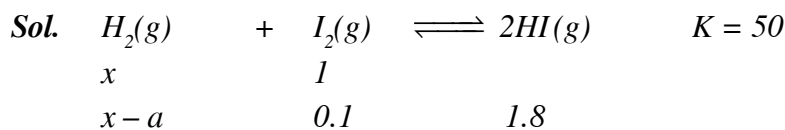
$$35 - 7x = 2x$$

$$35 = 9x$$

$$x = \frac{35}{9}$$

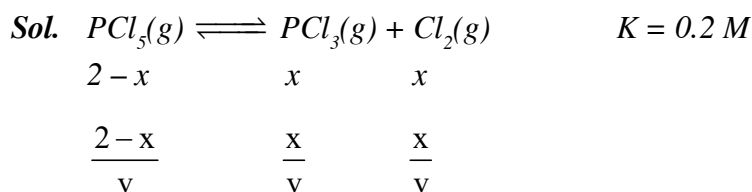
$$\text{Moles at equilibrium} = \frac{10}{9}, \frac{10}{9}, \frac{70}{9}$$

Ex.23 How many moles of $H_2(g)$ should be mixed with each mole I_2 vapours in order to convert 90% of it into HI .



$$50 = \frac{3.24}{(x-0.9)(0.1)} \Rightarrow x = 1.548$$

Ex.24 2 moles of $PCl_5(g)$ is taken in 10 ℓ vessel. Calculate its concentration at equilibrium.



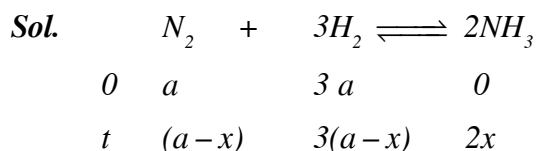
$$0.2 = \frac{\left(\frac{x}{v}\right)^2}{\left(\frac{2-x}{v}\right)} \Rightarrow x = 1.24$$

$$[PCl_5] = 0.076 M$$

$$[PCl_3] = 0.124 M$$

$$[Cl_2] = 0.124 M$$

Ex.25 1 : 3 molar ratio mixture of N_2 and H_2 yields 20% by mole NH_3 at 30 atm; calculate K_p for the equilibrium represented by



$$\text{Total moles} = 4a - 2x$$

$$X_{NH_3} = 0.2 = \frac{2x}{4a-2x} \Rightarrow 2x = 0.8a - 0.4x \Rightarrow x = a/3$$

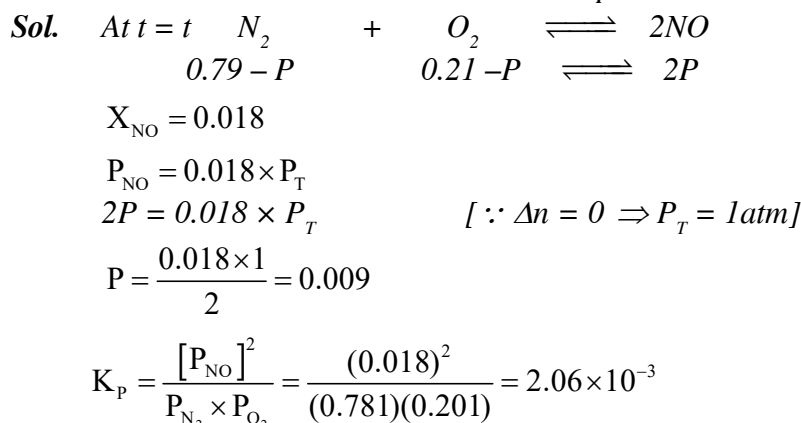
$$P_{NH_3} = 0.2 \times 30 = 6 \text{ atm}$$

$$P_{NH_3} = X_{N_2} \times P_T = \frac{a-x}{4a-2x} \times 30 = 0.2 \times 30 = 6 \text{ atm}$$

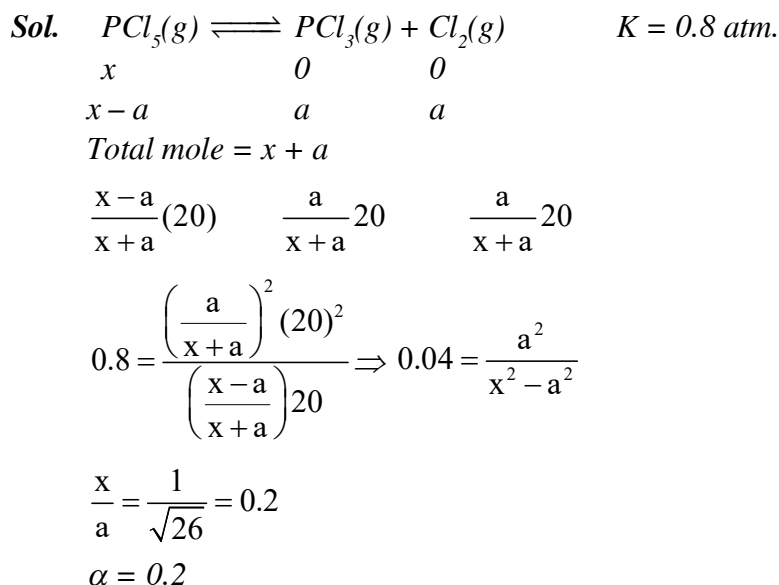
$$P_{H_2} = \frac{3(a-x)}{4a-2x} = 0.6 \times 30 = 18 \text{ atm}$$

$$\therefore K_p = \frac{[P_{NH_3}]^2}{[P_{N_2}][P_{H_2}]^3} = \frac{6}{6 \times (18)^2} = 1.03 \times 10^{-3} \text{ atm}^{-2}$$

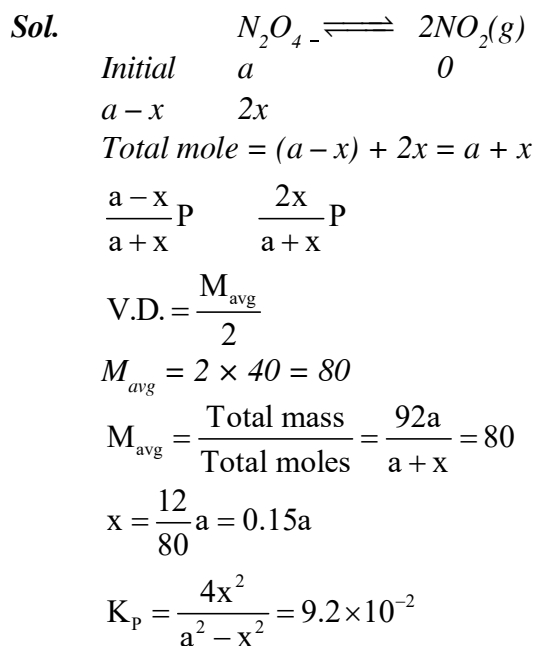
Ex.26 An air sample containing 21:79 (mole ratio) of O_2 & N_2 is heated to $2400^\circ C$. If the mole percent of NO at equilibrium is 1.8%, calculate K_p for the reaction $N_2 + O_2 \rightleftharpoons 2NO$.



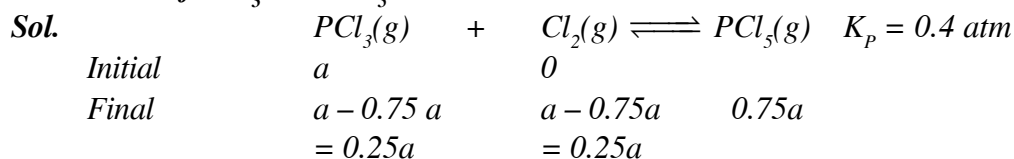
Ex.27 Calculate the degree of dissociation of $PCl_5(g)$ at 20 atm.



Ex.28 The vapour density a sample of $N_2O_4(g)$ is 40 at 10 atm. Calculate K_p for the reaction.



Ex.29 At what pressure an equimolar mixture of $\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$ should be taken in order to convert 75% of PCl_3 into PCl_5 .



Total mole = $1.25a$

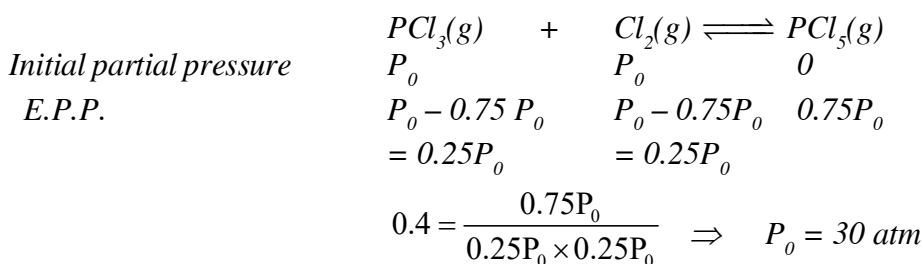
Equ. par. pr.	$\frac{0.25a \times P}{1.25a}$	$\frac{0.25a \times P}{1.25a}$	$\frac{0.75a \times P}{1.25}$
	$\frac{P}{5}$	$\frac{P}{5}$	$\frac{3P}{5}$

$$K_p = \frac{P_{\text{PCl}_5}}{P_{\text{PCl}_3} P_{\text{Cl}_2}} \Rightarrow 0.4 = \frac{\frac{3P}{5}}{\frac{P}{5} \times \frac{P}{5}} \Rightarrow P = 37.5$$

But this is pressure at equilibrium but we have to find initial pressure
Now, from $PV = nRT$

$$\frac{P_1}{P_2} = \frac{n_1}{n_2} \Rightarrow \frac{P_1}{37.5} = \frac{2}{1.25} \Rightarrow P = 60 \text{ atm}$$

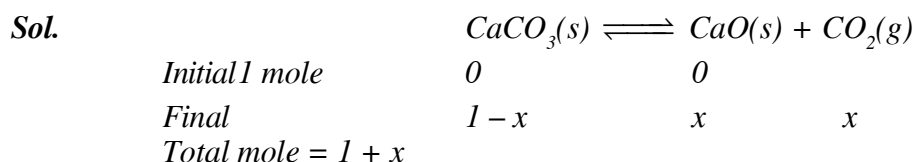
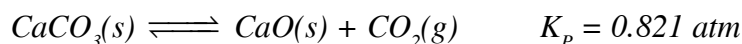
Alternate method :



$\therefore$ Initial total pressure of mixture = $P_0 + P_0 = 60 \text{ atm}$

Note : In any problem related with relative extent of reaction, initial mole may be taken as 1 but the initial pressure or concentration should not be taken as 1.

Ex.30 100 gm CaCO_3 is taken in 30 l empty vessel and the vessel is sealed and the sample is heated to 627°C . Calculate the mass % of CaCO_3 decomposed.

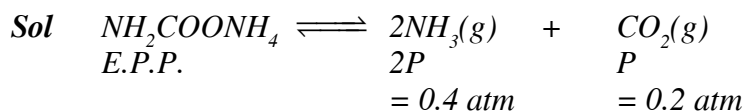


$$K_p = \frac{P_{\text{CO}_2}}{1} = P_{\text{CO}_2} \Rightarrow P_{\text{CO}_2} = \frac{nRT}{V}$$

$$0.821 = \frac{x \times 0.0821 \times 900}{30} \Rightarrow x = \frac{1}{3}$$

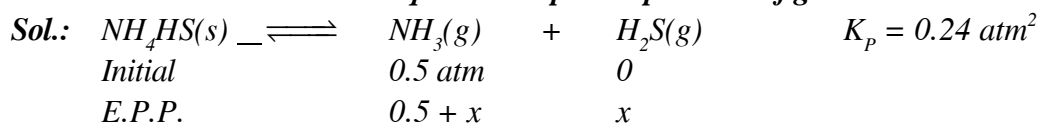
$$\% \text{ of } \text{CaCO}_3 \text{ decomposed} = \frac{1}{3} \times 100 = \frac{100}{3} \%$$

Ex.31 Some solid ammonium carbamate $\text{NH}_4\text{COONH}_4$ is taken in an evacuated vessel and the vessel is sealed after a very long time a constant pressure of 0.6 atm is observed due to dissociation of the solid into NH_3 and CO_2 gas. Calculate the dissociation constant of solid.



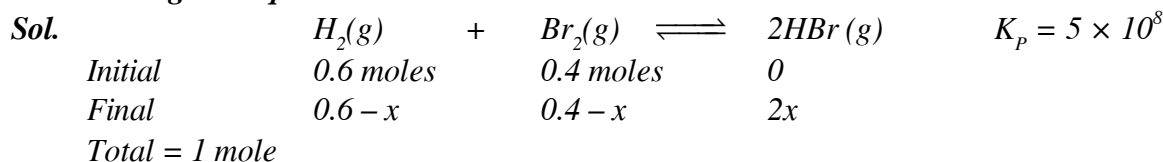
$$K_p = \frac{[0.4]^2 [0.2]}{1} = 0.032 \text{ atm}^3$$

Ex.32 Some solid ammonium hydrogen sulphide NH_4HS is in a vessel containing ammonium gas at 0.5 atm. Calculate the equilibrium partial pressure of gas.



$$K_p = \frac{P_{\text{NH}_3}}{1} \cdot P_{\text{H}_2\text{S}} \Rightarrow 0.24 = (0.5 + x)x \quad \Rightarrow \quad x = 0.3 \text{ atm.}$$

Ex.33 0.6 moles of $H_2(g)$ and 0.4 moles of Br_2 vapour are allowed to reactant. Calculate the moles of each gas at equilibrium.



$$K = \frac{(2x)^2}{(0.6-x)(0.4-x)} = \frac{4x^2}{x^2 - x + 0.24}$$

$$(5 \times 10^8 - 4)x^2 - 5 \times 10^8 x + 1.2 \times 10^8 = 0$$

$$5 \times 10^8 x^2 - 5 \times 10^8 x + 1.2 \times 10^8 = 0$$

$$8x^2 - x + 0.24 = 0$$

$x = 0.4, 0.6$ (Not acceptable)

As K_p is much larger than 1000 the reaction almost completes.

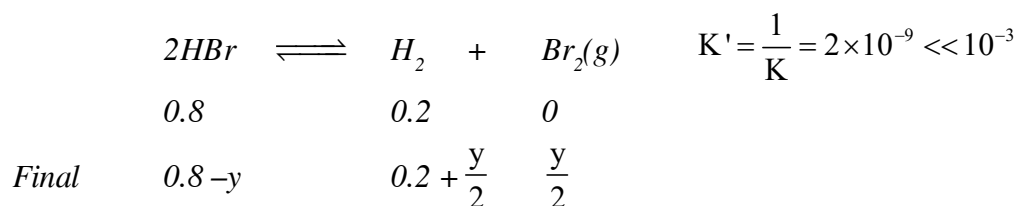
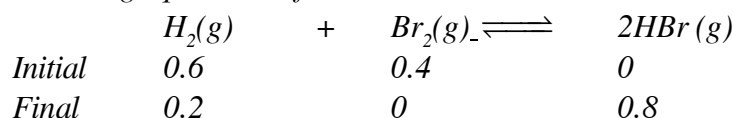
$$\text{Final moles of HBr} = 0.8$$

Final moles of $H_2 = 0.2$

But Final moles of $Br_2 \neq 0$

Note :

The final moles of Br_2 may be calculated by first completing the reaction in forwards direction and then achieving equilibrium from backward direction.



$$K' = \frac{[H_2][Br_2]}{[HBr]^2} = 2 \times 10^{-9} = \frac{\left(0.2 + \frac{y}{2}\right) \frac{y}{2}}{(0.8 - y)^2}$$

As K' is very small we may assume that $0.8 - y \simeq 0.8$

$$\text{and, } 0.2 + \frac{y}{2} \simeq 0.2$$

$$\therefore 2 \times 10^{-9} = \frac{0.2 \times \frac{y}{2}}{0.64}$$

$$y = 1.28 \times 10^{-8}$$

$$\text{Final moles HBr} = 0.8 - y \simeq 0.8$$

$$H_2 = 0.2 + \frac{y}{2} \simeq 0.2$$

$$Br_2 = \frac{y}{2} = 6.4 \times 10^{-9}$$

Ex.34 Calculate concentration of A at equilibrium for a reaction, $A(g) \rightleftharpoons B(g)$; $K_C = 10^{50}$, if the initial concentration of A is 2 M.

Sol.	$A(g) \rightleftharpoons B(g)$	
initial conc.	2	0
Change in conc.	-x	x
equib. conc.	2-x	0+x

K_C is very-very high so maximum reactant converted into product.

$$\therefore \frac{K_C}{\text{initial conc.}} \geq 10^3 \text{ then } \frac{10^{50}}{2} \geq 10^3.$$

So $2-x \simeq 0 \Rightarrow x \simeq 2$ but reactant never zero.

$$\text{equib. conc.} \quad 2-x \simeq y \quad x \simeq 2$$

$$K_C = \frac{[B]}{[A]}$$

$$10^{50} = \frac{x}{2-x}$$

$$10^{50} = \frac{2}{y} \Rightarrow y = 2 \times 10^{-50} \text{ M}$$

$$[A] = 2 \times 10^{-50} \text{ M}$$

8. LE CHATELIER'S PRINCIPLE :

Le Chatelier's principle is applicable for any system in equilibrium which states as :

When a system in equilibrium is disturbed by external agency, the system tends to attain again an equilibrium by adjusting itself. These are these some factors by which a system in equilibrium can be disturbed.

- Changing the concentration of reactant and product.
- Changing the pressure (or volume) of the system.

(iii) Changing the temperature.

For a chemical reaction in equilibrium, Le Chatelier's principle can be stated as,

If we change concentration, pressure or temperature of a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimise the change.

8.1 Effect of a change in concentration :

- If more reactant is added or some product is removed from an equilibrium mixture having equilibrium constant K then the reaction moves in the forward direction (as $Q_c < K_c$) to give a new equilibrium and more products are produced.
- If more product are added to or some reactant are removed from an equilibrium mixture, the reaction moves in the reverse direction (as $Q_c > K_c$) to give a new equilibrium and more reactant are produced.

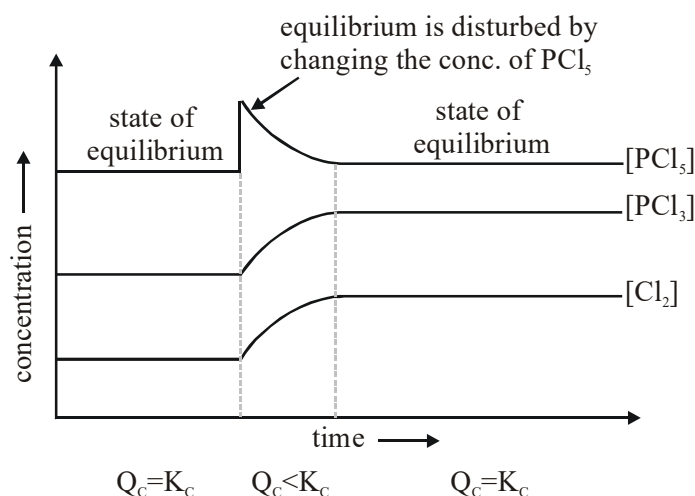
Ex. The decomposition of gaseous PCl_5 is a reversible reaction, $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$. Let the equilibrium concentrations of PCl_5 , PCl_3 and Cl_2 are respectively $[\text{PCl}_5]$, $[\text{PCl}_3]$ and $[\text{Cl}_2]$. The K_c for this reaction can be written as :

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} \text{ Also we know that at equilibrium } K_c = Q_c.$$

Case-I : Let the concentration of PCl_5 is doubled to disturb the equilibrium. This will change the reaction quotient, Q_c to :

$$Q_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[2\text{PCl}_5]} = \frac{1}{2} \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{K_c}{2}$$

After disturbing the equilibrium, the value of Q_c becomes less than K_c . In order to restore the Q_c value to K_c , the concentration of PCl_5 must be decreased while the concentrations of PCl_3 and Cl_2 are to be increased. This is achieved by favoring the forward reaction.

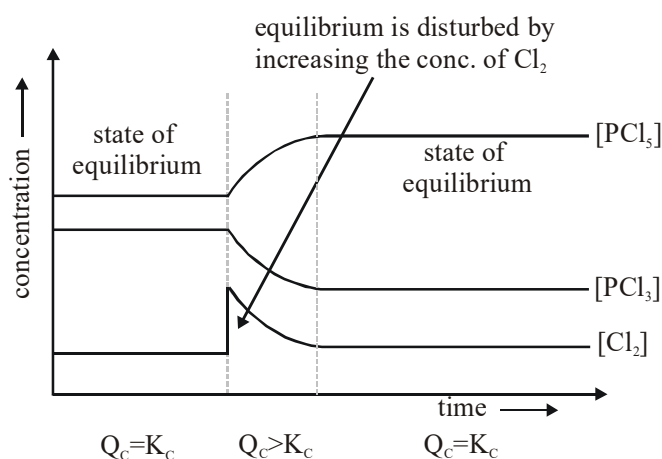


The forward reaction is also favored by removing the products from the reaction mixture (decrease in the concentration of products). Upon removal of products, the rate of forward reaction becomes greater than that of backward reaction momentarily. This will also decrease the reaction quotient. Hence the system tries to reestablish the equilibrium by converting more reactants to products so as to make the rates of both forward and backward reactions become equal again.

Case-II : For example, in case of the decomposition of PCl_5 , if the concentration of Cl_2 is increased by two times at equilibrium, the Q_c value becomes greater than the K_c value.

$$Q_c = \frac{[\text{PCl}_3][2\text{Cl}_2]}{[\text{PCl}_5]} = 2 \times \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = 2K_c$$

Hence the system tries to restore the value of Q_c to K_c again. The backward reaction is favored to decrease the concentration of Cl_2 . However the concentration of PCl_3 also decreases automatically while the concentration of PCl_5 increases while doing so.



8.2 Effect of a change in pressure :

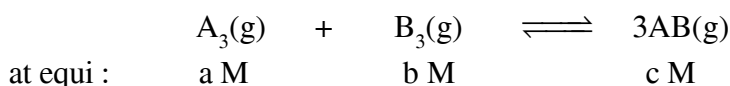
We know that, if we increase the volume of container then the pressure of the container decreases and if we decrease the volume of container then the pressure of the container increases.

$$\text{i.e., Pressure (P)} \propto \frac{1}{\text{Volume (V)}}$$

Hence, if the pressure is decreased by increasing the volume of a reaction mixture, the reaction shifts in the direction of more moles of gas while if the pressure is increased by decreasing the volume of a reaction mixture, the reaction shifts in the direction of fewer moles of gas.

Here moles of reactant or product means co-efficients of reactant or product.

(a) Suppose a general equation



For this reaction, moles of product > moles of reactant i.e., $[\Delta n = (3 - 2) > 0]$

$$K_c = \frac{(c)^3}{a \times b}$$

If we increase the pressure of the system from 0.5 atm pressure to 1 atm pressure by decreasing the volume of container from 5 litre to 2.5 litre. Hence, concentration of all the reactant and product will change and it will be greater than the initial value. Hence we can say that,

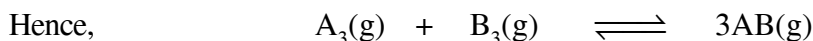
If pressure becomes double then volume becomes half and hence concentration becomes double as

$$(\text{Concentration} \propto \frac{\text{Moles}}{\text{Volume}})$$

∴ Concentration $\propto$ Pressure

$$Q_c = 2 \times \frac{c^3}{a \times b} > K_c$$

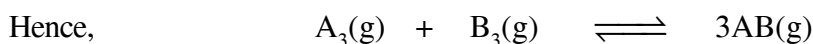
Hence reaction will move from right to left i.e., in the backward direction.



$$\text{Moles at equi.} \quad 2a+x \quad 2b+x \quad 2c-3x$$

It can be simply said that, reaction will move from right to left by increasing the pressure of the system as moles of product > moles of reactant i.e., move in the direction of fewer moles.

- (b) If pressure decreases from 1 atm to 0.5 atm. Volume will increase by factor 2 concentration will decrease by factor 1/2



$$\text{changed conc.} \quad a/2 \quad b/2 \quad c/2$$

$$\text{at equi. conc.} \quad a/2-x \quad b/2-x \quad c/2+3x$$

$$Q_c = \frac{(c/2)^3}{a/2 \times b/2} = \frac{1}{2} \frac{c^3}{a \times b} < K_c$$

i.e., $Q_c < K_c$

Therefore, reaction will move from left to right and as moles of product > moles of reactants, therefore reaction will move from left to right by decreasing the pressure of the system i.e. move in the direction of greater moles.

- (c) If moles of reactant = moles of product i.e., $\Delta n = 0$ then change in pressure of equilibrium mixture has no effect i.e. at this position $Q_c = K_c$ will always exist.

8.3 Effect of a change in temperature :

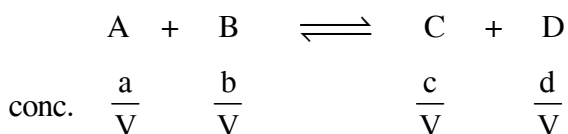
On increasing the temperature equilibrium shifts in the direction of absorption of heat (endothermic direction)

8.4 EFFECT OF INERT GAS AT THE EQUILIBRIUM :

(i) Effect of inert gas at constant volume :

When the addition of inert gas (non-reacting gas) is carried out at constant volume (V), the equilibrium remains unaffected for reactions whether they have $\Delta n = 0$ or $\Delta n \neq 0$.

Let us consider a general equation at temperature T K, and a, b, c and d are the moles of A, B, C and D respectively,



Since, V = constant, so addition of inert gas has no effect on equilibrium

∴ $\frac{a}{V}$, $\frac{b}{V}$, $\frac{c}{V}$ and $\frac{d}{V}$ will not change

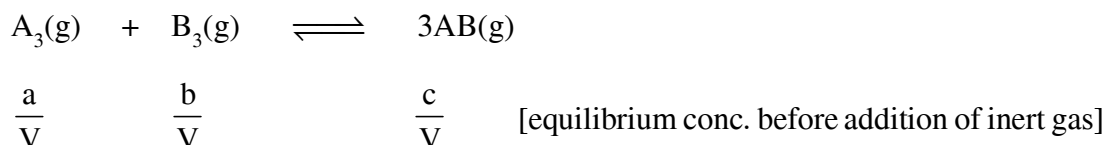
$$Q_c = K_c \quad \text{or} \quad Q_p = K_p$$

(ii) Effect of inert gas at constant pressure :

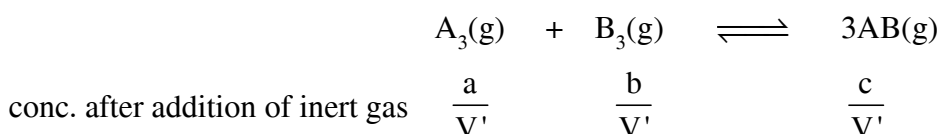
- (a) The addition of an inert gas at constant pressure to an equilibrium, the equilibrium shifts in the direction of greater number of moles.

At constant pressure, addition of inert gas or non-reacting gas means increase in volume of the system.

Suppose a reaction having $\Delta n > 0$ and a, b, c and d are the moles of A, B, C and D respectively and V be the volume before addition of inert gas.



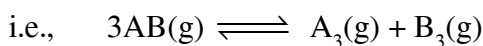
Let us consider V' be the total volume of the system after addition of inert gas So, $V' > V$



$$\therefore \frac{a}{V'} < \frac{a}{V}, \frac{b}{V'} < \frac{b}{V} \text{ and } \frac{c}{V'} < \frac{c}{V}$$

$$\text{Now, } K_c = \frac{\left(\frac{c}{V}\right)^3}{\left(\frac{a}{V}\right)\left(\frac{b}{V}\right)} = \frac{1}{V} \times \frac{c^3}{a \times b} \text{ and } Q_c = \frac{\left(\frac{c}{V'}\right)^3}{\left(\frac{a}{V'}\right)\left(\frac{b}{V'}\right)} = \frac{1}{V'} \times \frac{c^3}{a \times b}$$

$K_c > Q_c$, hence reaction will move from left to right (forward direction) as $\Delta n > 0$ and similarly we can prove that for a reaction having $\Delta n < 0$

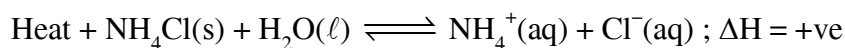


For this reaction $K_c < Q_c$, after addition of inert gas. Hence, reaction will move from right to left (reverse direction) as $\Delta n < 0$.

8.5 APPLICATION OF LE-CHATELIER'S PRINCIPLE :

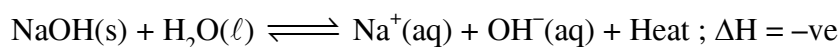
The principle is applicable not only to chemical equilibria but also to physical equilibria in similar way.

- (a) Dissolution of ammonium chloride in water



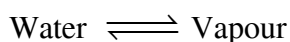
With rise in temperature, equilibrium will shift in direction which will lower the temperature counteract the effect. So, equilibrium shifts in forward direction which is endotherm reaction. Hence, solubility of NH_4Cl increases with rise in temperature.

- (b) Dissolution of sodium hydroxide in water



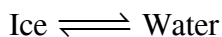
Sodium hydroxide dissolve in water with evolution of heat. Therefore, a rise in temperature will decrease its solubility. On the other hand, a decrease in temperature will increase the solubility of sodium hydroxide and reaction will shift in forward direction.

- (c) Effect of pressure on boiling point :



An increase in pressure will favour backward reaction i.e., the reaction in which volume decreases ($V_{\text{vap.}} > V_{\text{w.}}$). thus more water will exist at equilibrium (B.pt of solvent increase with increase in pressure). So decrease in pressure will shift the reaction forward.

- (d) Effect of pressure on melting point :

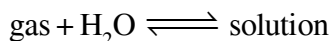


increase in pressure will favour forward reaction because $V_{\text{ice}} > V_{\text{water}}$. Thus more ice melt or the m.pt of ice is lowered with pressure.

For, $\text{Solid} \rightleftharpoons \text{Liquid}$

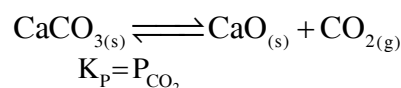
An increase in pressure will favour backward reaction because volume of liquid is more than solid thus more solid will exist at equilibrium (m.pt of solid increases with pressure).

- (e) Effect of pressure on solubility of gases :



increase in pressure favour forward reaction. [Henry's law]

- (f) In solid reactants equilibrium will not shift to the right side even if the more reactants are added.

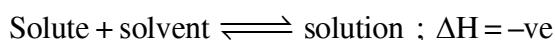


Equilibrium will not be affected by adding CaO or CaCO₃ at that temperature. But if volume is increased the equilibrium will shift to the right side to keep the pressure of CO₂ constant.

- (g) Effect of temperature on solubility



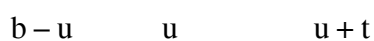
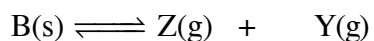
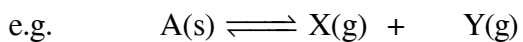
An increase in temperature favour endothermic or forward reaction i.e., solubility increases (urea, glucose).



In exothermic dissolution solubility decreases with temperature i.e., reaction will be in backward direction.

9. SIMULTANEOUS EQUILIBRIUM

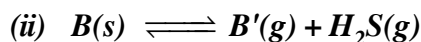
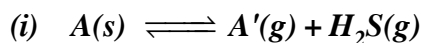
If in any container there are two or more equilibria existing simultaneously involving one or more than one common species, then in both/all the equilibrium, the concentration of common species is the total concentration of that species due to all the equilibria under consideration.



$$K_{C_1} = t(u + t)$$

$$K_{C_2} = (u + t)u$$

Ex.35 Two solid compounds A and B dissociate into gaseous products at 20°C as



At 20°C, pressure over excess solid A is 50 mm and that over excess solid B is 68 mm find

(a) The dissociation constant of A and B

(b) Relative no. of moles of A and B in the vapour phase over a mixture of solid A and B.

(c) Show that the total pressure of the gas over the solid mixture would be 84.4 mm



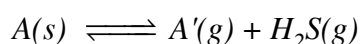
$$A(s) \rightleftharpoons A'(g) + H_2S(g) \quad P = 50 \text{ mm} \quad \Rightarrow \quad P_{H_2S} = P_{A'} = \frac{P}{2} = 25 \text{ mm}$$

$$B(s) \rightleftharpoons B'(g) + H_2S(g) \quad P = 68 \text{ mm} \quad \Rightarrow \quad P_{H_2S} = P_{B'} = \frac{P}{2} = 34 \text{ mm}$$

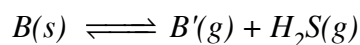
$$(i) \quad K_{p_1} = (25)^2 = 625 \text{ mm}^2$$

$$K_{p_2} = (34)^2 = 1156 \text{ mm}^2$$

(ii) Ratio of moles is same as that of partial pressure so,



$$P_1 \quad P_1 + P_2$$



$$P_2 \quad P_2 + P_1$$

$$K_{p_1} = P_{A'} \times P_{H_2S} = P_1(P_1 + P_2) \quad \dots(i)$$

$$K_{p_2} = P_{B'} \times P_{H_2S} = P_2(P_1 + P_2) \quad \dots(ii)$$

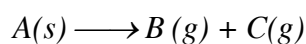
$$\frac{K_{p_1}}{K_{p_2}} = \frac{P_1}{P_2} = \frac{625}{1156}$$

$$(iii) \text{ total pressure} = P_1 + P_2 + (P_1 + P_2) = 2(P_1 + P_2)$$

$$(i) + (ii) = (P_1 + P_2)^2$$

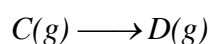
$$\sqrt{K_{p_1} + K_{p_2}} = P_1 + P_2 \quad \Rightarrow \quad P_T = \sqrt{K_{p_1} + K_{p_2}} = 84.4 \text{ mm}$$

10. SEQUENTIAL EQUILIBRIUM



$$a-x \quad x \quad x-y$$

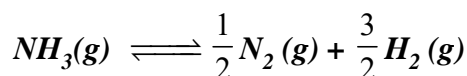
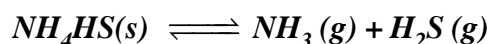
$$K_{C_1} = (x)(x-y)$$



$$x-y \quad y$$

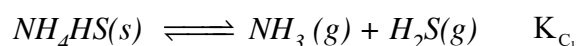
$$K_{C_2} = \frac{y}{x-y}$$

Ex36. 102 g of solid NH_4HS is taken in the 2L evacuated flask at 57°C . Following two equilibrium exist simultaneously

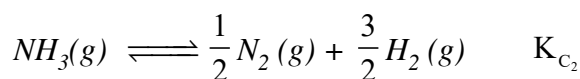


one mole of the solid decomposes to maintain both the equilibrium and 0.75 mole of H_2 was found at the equilibrium then find the equilibrium concentration of all the species and K_C for both the reaction.

Sol. Moles of $\text{NH}_4\text{HS} = \frac{102}{51} = 2$



$$\begin{array}{ccc} 2 & 1 & 0 \\ 1-x & 0 & 1 \end{array}$$



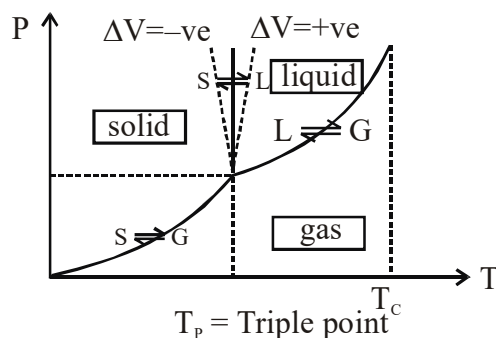
$$\begin{array}{ccc} 1-x & \frac{x}{2} & \frac{3x}{3} \end{array}$$

Given that moles of $\text{H}_2 = \frac{3x}{3} = 0.75 \Rightarrow x = \frac{1}{2}$

$$K_{C_1} = \frac{1}{2} \frac{(1-x)}{2} = \frac{1}{8} \quad [\text{Since } V = 2 \text{ L}]$$

$$K_{C_3} = \frac{\left(\frac{3x}{4}\right)^{3/2} \left(\frac{x}{4}\right)^{1/2}}{\left(\frac{1-x}{2}\right)} = \frac{\left(\frac{3}{8}\right)^{3/2} \left(\frac{1}{4}\right)^{1/2}}{\left(\frac{1}{4}\right)} = (3)^{3/2} \frac{1}{64} \times \frac{4}{1} = \frac{(3)^{3/2}}{16}$$

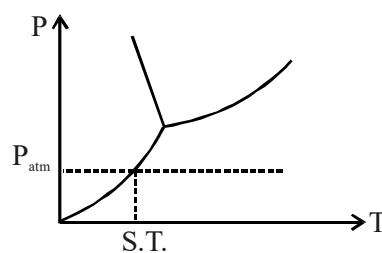
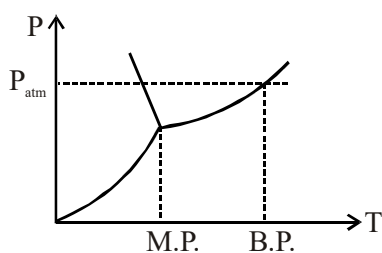
11. PHYSICAL EQUILIBRIUM



- The combination of temperature and pressure at which all the three physical state of matter co-exist is called triple point of that matter.
- The vapour pressure solid is minimum (0) at absolute zero and maximum at triple point.
- The vapour pressure of liquid is minimum at triple point and maximum at critical temperature.

- (iv) On each line in the graph, the matter exist in two physical states is equilibrium.
- (v) In between two lines, the matter exist in only one physical state.
- (vi) The melting point of solid and the triple point of matter differs slightly.
- (vii) Any solid may be directly converted into gaseous state or through liquid state by adjusting the external pressure relative to triple point pressure.

(viii)



MISSLENIUS PREVIOUS YEARS QUESTION

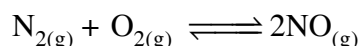
1. For the reaction $\text{CO}_{(g)} + 2\text{H}_{2(g)} \rightleftharpoons \text{CH}_3\text{OH}_{(g)}$; H_2 is introduced into a five litre flask at 327°C , containing 0.2 mole of $\text{CO}_{(g)}$ and a catalyst till the pressure is 4.92 atmosphere. At this point 0.1 mole of $\text{CH}_3\text{OH}_{(g)}$ is formed. Calculate K_c and K_p . [JEE 1990]

Ans. $K_c = 279.64 \text{ litre}^2 \text{ mole}^{-2}$; $K_p = 0.115 \text{ atm}^{-2}$

2. At temperature T, a compound $\text{AB}_{2(g)}$ dissociates according to the reaction $2\text{AB}_{2(g)} \rightleftharpoons 2\text{AB}_{(g)} + \text{B}_{2(g)}$; with a degree of dissociation 'x', which is small compared to unity. Deduce the expression for 'x' in terms of the equilibrium constant K_p and the total pressure P. [JEE 1994]

Ans. $x = \sqrt[3]{\frac{2K_p}{P}}$

3. A sample of air consisting of N_2 and O_2 was heated to 2500 K until the equilibrium



was established with an equilibrium constant $K_c = 2.1 \times 10^{-3}$. At equilibrium, the mole% of NO was 1.8. Estimate the initial composition of air in mole fraction of N_2 and O_2 . [JEE 1997]

Ans. 79%, 21%

4. The degree of dissociation is 0.4 at 400K & 1.0 atm for the gasoeus reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2(g)$. Assuming ideal behaviour of all gases. Calculate the density of equilibrium mixture at 400K & 1.0 atm pressure. [JEE 1999]

Ans. 4.54 g dm^{-3}

Sol. $\text{PCl}_5 = \text{PCl}_3 + \text{Cl}_2$
 $\begin{matrix} 1 & - & - \\ 1-0.4 & 0.4 & 0.4 \end{matrix}$

$$M_{av} = \frac{208.5}{1.4}$$

$$PM = dRT$$

$$1 \times \frac{208.5}{1.4} = d \times 0.0821 \times 400$$

5. For the reversible reaction : [JEE 2000]

$\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$ at 500°C . The value of K_p is 1.44×10^{-5} , when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mol L^{-1} is :

(A) $1.44 \times 10^{-5} / (0.082 \times 500)^2$

(B) $1.44 \times 10^{-5} / (8.314 \times 773)^2$

(C) $1.44 \times 10^{-5} / (0.082 \times 500)^2$

(D) $1.44 \times 10^{-5} / (0.082 \times 773)^{-2}$

Ans. Ans.(D)

Sol. $K_p = K_c (RT)^{-2}$

6. When two reactants A and B are mixed to give products C and D, the reaction quotient Q , at the initial stages of the reaction : [JEE 2000]
- (A) is zero (B) decrease with time
 (C) independent of time (D) increases with time

Ans. (D)

Sol. Q increases at initial steps

7. When 3.06g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C , 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide. [JEE 2000]
- (i) Calculate K_c & K_p for the reaction at 27°C .
 (ii) What would happen to the equilibrium when more solid NH_4HS is introduced into the flask?

Ans. (i) $K_c = 8.1 \times 10^{-5} \text{ mol}^2 \text{ L}^2$; $K_p = 4.91 \times 10^{-2} \text{ atm}^2$ (ii) No effect;

Sol. (i) $\text{NH}_4\text{HS(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}$

$$\frac{0.06}{2} \text{ mole} \quad - \quad -$$

$$\frac{0.042}{2} \text{ mole} \quad \frac{0.018}{2} \quad \frac{0.018}{2}$$

$$K_c = \frac{0.018}{2} \times \frac{0.018}{2}$$

$$K_c = 8.1 \times 10^{-5}$$

$$K_p = (8.1 \times 10^{-5}) \times (RT)^2$$

(ii) When more of $\text{NH}_4\text{HS(s)}$ is added then no-effect will occur on equilibrium

8. At constant temperature, the equilibrium constant (K_p) for the decomposition reaction. $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ is expressed by $K_p = 4x^2P/(1-x^2)$ where P is pressure, x is extent of decomposition. Which of the following statement is true ? [JEE 2001]
- (A) K_p increases with increase of P
 (B) K_p increases with increase of x
 (C) K_p increases with decrease of x
 (D) K_p remains constant with change in P or x

Ans. (D)

Sol. K_p depends only on temperature.

9. Consider the following equilibrium in a closed container : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$.

At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α) : [JEE 2002]

- (A) Neither K_p nor α changes
- (B) Both K_p and α change
- (C) K_p changes, but α does not change
- (D) K_p does not change, but α changes

Ans. (D)

Sol. K_p is independent of volume of container

10. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :-

[AIEEE-2011]

- (1) 0.3 atm
- (2) 0.18 atm
- (3) 1.8 atm
- (4) 3 atm

Ans. (3)

CE0108

11. The equilibrium constant (K_C) for the reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g})$ at temperature T is 4×10^{-4} . The value of K_C for the reaction. $\text{NO}(\text{g}) \longrightarrow \frac{1}{2}\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$ at the same temperature is :- [AIEEE-2012]

- (1) 50.0
- (2) 0.02
- (3) 2.5×10^2
- (4) 4×10^{-4}

CE0109

Ans. (1)

12. 8 mol of $\text{AB}_3(\text{g})$ are introduced into a 1.0 dm^3 vessel. If it dissociates as $2\text{AB}_3(\text{g}) \rightleftharpoons \text{A}_2(\text{g}) + 3 \text{B}_2(\text{g})$

At equilibrium, 2mol of A_2 are found to be present. The equilibrium constant of this reaction is :-

[JEE-MAINS(online)-12]

- (1) 36
- (2) 3
- (3) 27
- (4) 2

CE0110

Ans. (3)

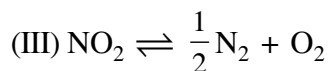
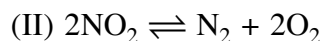
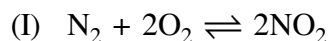
13. The value of K_p for the equilibrium reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ is 2. The percentage dissociation of $\text{N}_2\text{O}_4(\text{g})$ at a pressure of 0.5 atm is [JEE-MAINS(online)-12]

- (1) 71
- (2) 50
- (3) 88
- (4) 25

CE0111

Ans. (1)

14. K_1 , K_2 and K_3 are the equilibrium constants of the following reactions (I), (II) and (III), respectively



The correct relation from the following is :

[JEE-MAINS(online)-12]

(1) $K_1 = \sqrt{K_2} = K_3$ (2) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$ (3) $K_1 = \frac{1}{K_2} = K_3$ (4) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

CE0112

Ans. (4)

15. One mole of $\text{O}_2(\text{g})$ and two moles of $\text{SO}_2(\text{g})$ were heated in a closed vessel of one litre capacity at 1098 K. At equilibrium 1.6 moles of $\text{SO}_3(\text{g})$ were found. The equilibrium constant K_C of the reaction would be :-

[JEE-MAINS(online)-12]

(1) 60 (2) 80 (3) 30 (4) 40

CE0113

Ans. (2)

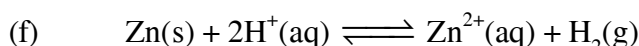
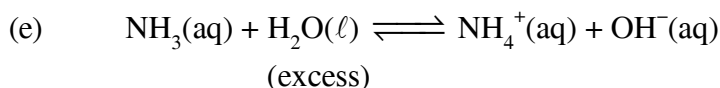
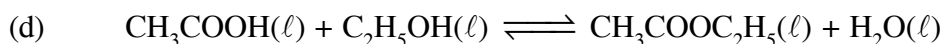
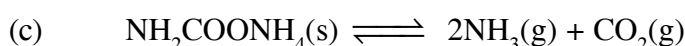
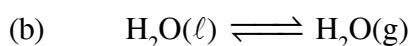
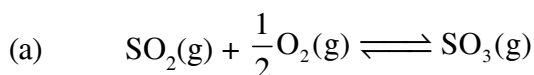
EXERCISE # S-I

1. Which of the following reactions goes almost all the way to completion, and which proceeds hardly at all?

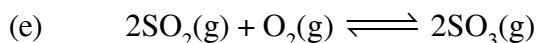
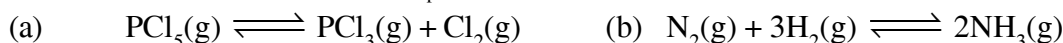


CE0001

2. Write down the expression for equilibrium constant K_c and K_p for the following reactions :

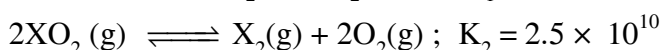
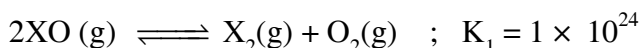
**CE0002**

- 3.** Write the relationship between K_n and K_c for the following reactions :



CE0003

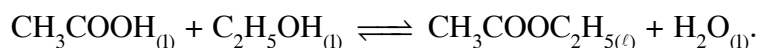
4. Which reactant is more stable from the given equilibrium reaction :

**CE0004**

5. The value of K_c for the reaction $2A \rightleftharpoons B + C$ is 2×10^{-3} . At a given time, the composition of reaction mixture is $[A] = [B] = [C] = 3 \times 10^{-4}$ M. In which direction the reaction will proceed?

CE0005

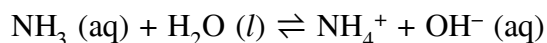
6. The ester or ethyl acetate is formed by the reaction between ethanol and acetic acid and equilibrium is represented as :



- Write the concentration ratio (reaction quotient), Q_c for this reaction. (Note that water is not in excess and is not a solvent in this reaction).
- At 293 K, if starts with 1.00 mole of acetic acid and 0.180 moles of ethanol, there is 0.17 moles of ethyl acetate in the final equilibrium mixture. Calculate the equilibrium constant.
- Starting with 0.500 moles of ethanol and 1.000 mole of acetic acid and maintaining it at 293 K, 0.214 moles of ethyl acetate is found after some time. Has equilibrium been reached ?

CE0006

33. Ammonia is a weak base that reacts with water according to the equation

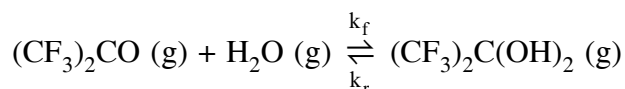


Will any of the following increase the percent of ammonia that is converted to the ammonium ion in water?

- (a) Addition of NaOH. (b) Addition of HCl. (c) Addition of NH_4Cl .

CE0033

34. Consider the gas-phase hydration of hexafluoroacetone, $(\text{CF}_3)_2\text{CO}$:



At 76°C , the forward and reverse rate constants are $k_f = 0.15 \text{ M}^{-1}\text{s}^{-1}$ and $k_r = 6 \times 10^{-4}\text{s}^{-1}$. What is the value of the equilibrium constant K_c (in M^{-1}) ?

CE0034

35. Listed in the table are forward and reverse rate constants for the reaction $2\text{NO} (\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + \text{O}_2(\text{g})$

Temperature (K)	$k_f (\text{M}^{-1}\text{s}^{-1})$	$k_r (\text{M}^{-1}\text{s}^{-1})$
1400	0.29	1.1×10^{-6}
1500	1.3	1.4×10^{-5}

Is the reaction endothermic or exothermic? Explain in terms of kinetics.

CE0035

EXERCISE # S-II

1. One mole of $\text{N}_2\text{O}_4(\text{g})$ at 300 K is left in a closed container under one atm. It is heated to 600 K when 20 % by mass of $\text{N}_2\text{O}_4(\text{g})$ decomposes to $\text{NO}_2(\text{g})$. Calculate resultant pressure (in atm).

CE0036

2. Solid Ammonium carbamate dissociates as: $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. The ratio of total pressure at new equilibrium to that of original total pressure is $x : 3$, the value of 'x' is.

CE0037

3. A definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. NH_4HS decomposes to give NH_3 and H_2S and at equilibrium total pressure in flask is 0.84 atm. Calculate equilibrium constant for the reaction (in atm^2) :

CE0038

4. For the reaction $A(g) + 2B(g) \rightleftharpoons C(g) + D(g)$; $K_c = 10^{12}$.
If the initial moles of A,B,C and D are 0.5, 1, 0.5 and 3.5 moles respectively in a one litre vessel.
What is the equilibrium concentration of B (in millimole/litre) ?

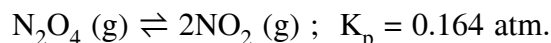
CE0039

5. The ratio of equilibrium molar concentration of PCl_3 and PCl_5 at 300 K if initially 1 mol of PCl_5 is taken in a closed container of having volume 10 litre, is $x : 1.00$. The value of 'x' is.

[Given : $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$; $K_c = 10^{-2}$ at 300 K, $\sqrt{41} = 6.4$]

CE0040

6. When 36.8g $\text{N}_2\text{O}_4(\text{g})$ is introduced into a 1.0-litre flask at 27°C . The following equilibrium reaction occurs :



- Calculate K_c of the equilibrium reaction. (in 10^{-3} M)
- What are the number of millimoles of N_2O_4 at equilibrium?
- What is the total pressure of gases (in atm) in the flask at equilibrium?
- What is the percent dissociation of N_2O_4 ?

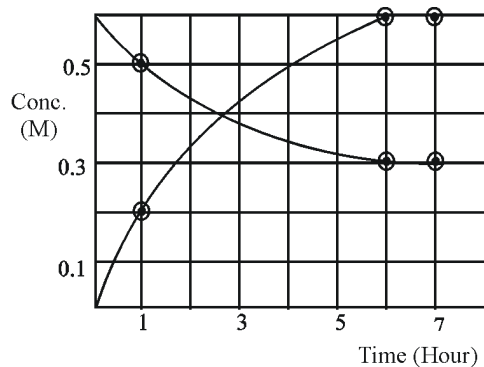
CE0041

- ## 7. The progress of the reaction

A $\rightleftharpoons$ nB with time, is presented in figure. Determine

- (i) the value of n .

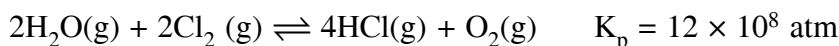
- (ii) the equilibrium constant, K (in M^{n-1}).

**CE0042**

8. A mixture of hydrogen & iodine in the mole ratio 1.5 : 1 is maintained at 450°C. After the attainment of equilibrium $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, it is found on analysis that the mole ratio of I_2 to HI is 1 : 18. Calculate the equilibrium constant, if initially, 127 grams of iodine were taken. ($I = 127$)

CE0043

9. At certain temperature (T) for the gas phase reaction



If Cl_2 , HCl & O_2 are mixed in such a manner that the partial pressure of each is 2 atm and the mixture is brought into contact with excess of liquid water. What would be approximate partial pressure (in 10^{-3} atm) of Cl_2 when equilibrium is attained at temperature (T)?

[Given : Vapour pressure of water is 380 mm Hg at temperature (T)]

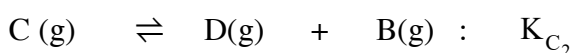
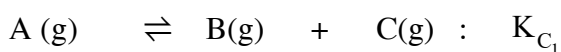
CE0044

10. Two solids X and Y dissociate into gaseous products at a certain temperature as follows:
 $\text{X}(\text{s}) \rightleftharpoons \text{A}(\text{g}) + \text{C}(\text{g})$, and $\text{Y}(\text{s}) \rightleftharpoons \text{B}(\text{g}) + \text{C}(\text{g})$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. When they are preset in separate containers. Calculate

- the values of K_p for two reactions (in mm^2)
- the ratio of moles of A and B in the vapour state over a mixture of X and Y.
- the total pressure of gases (in mm) over a mixture of X and Y.

CE0045

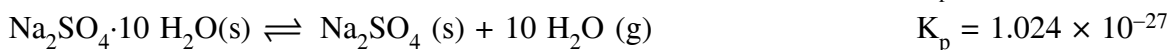
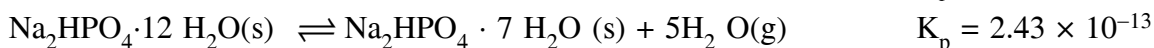
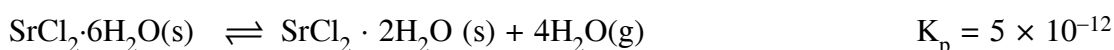
11. When 1 mole of $\text{A}(\text{g})$ is introduced in a closed rigid 1 litre vessel maintained at constant temperature the following equilibria are established.



The pressure at equilibrium is twice the initial pressure. Calculate the value of $\frac{K_{C_2}}{K_{C_1}}$ if $\frac{[\text{C}]_{\text{eq}}}{[\text{B}]_{\text{eq}}} = \frac{1}{5}$

CE0046

12. Equilibrium constants are given (in atm) for the following reactions at 0°C:



$$(\% \text{ Relative humidity} = \frac{P_{\text{H}_2\text{O}}}{\text{Vapour pressure of H}_2\text{O}} \times 100)$$

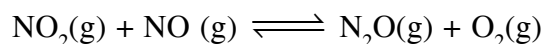
The vapor pressure of water at 0°C is 4.56 torr.

Out of $\text{SrCl}_2 \cdot 2\text{H}_2\text{O}(\text{s})$, $\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}(\text{s})$ and $\text{Na}_2\text{SO}_4(\text{s})$,

Which is the most effective drying agent at 0°C?

CE0047

13. At certain temperature, $K_c = 1.0$ for reaction :



equal moles of NO and NO_2 are to be placed in 5 litre container until N_2O concentration at equilibrium is 0.5M. How many mole of $(\text{NO} + \text{NO}_2)$ must be placed in the container.

CE0048

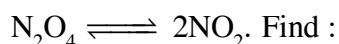
14. In a vessel, two equilibrium are simultaneously established at same temperature as follows,



Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is $7 P_0$ in which pressure due to ammonia is P_0 and due to hydrogen is $2P_0$. Find the values of equilibrium constants (K_p 's) for both the reactions.

CE0049

15. If vapour density of equilibrium mixture NO_2 and N_2O_4 is found to be 40 for the reaction,



- (a) Molar mass of equilibrium mixture (in gm/mol)
- (b) Degree of dissociation
- (c) Percentage of NO_2 in the mixture

CE0050

EXERCISE # O-1

ONLY ONE IS CORRECT :

- 1.** $x \rightleftharpoons y$ reaction is said to be in equilibrium, when :-

(A) Only 10% conversion of x to y takes place

(B) Complete conversion of x to y has taken place

(C) Conversion of x to y is only 50% complete

(D) The rate of change of x to y is just equal to the rate of change of y to x in the system

CE0051

2. For which reaction is $K_p = K_c$:-

$$(A) \quad 2\text{NOCl}(g) \rightleftharpoons 2\text{NO}(g) + \text{Cl}_2(g)$$

(B) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

(C) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

$$(D) \quad 2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$
CE0052

- 3.** For the reaction



Which one is correct representation :-

$$(A) K_p = (P_{(H_2O)})^2$$

(B) $K_c = [\text{H}_2\text{O}]^2$

(C) $K_p = K_c(RT)^2$

(D) All

CE0053

4. $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-

(A) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$

$$(B) \quad 2SO_2 + O_2 \rightleftharpoons 2SO_3$$

(C) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

(D) (B) and (C) both

CE0054

- 5.** For a reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, the value of K_c does not depends upon :-

(a) Initial concentration of the reactants

(b) Pressure

(c) Temperature

(d) Catalyst

(A) Only c

(B) a, b, c

(C) a, b, d

(D) a, b, c, d

CE0055

6. If some He gas is introduced into the equilibrium $\text{PCl}_{5(g)} \rightleftharpoons \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$ at constant pressure and temperature then equilibrium constant of reaction:

(A) Increase

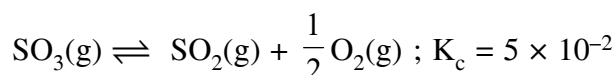
(B) Decrease

(C) Unchange

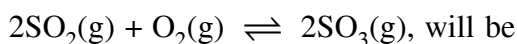
(D) Nothing can be said

CE0056

7. The equilibrium constant for the given reaction :



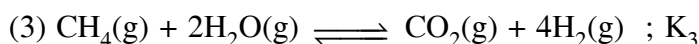
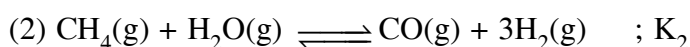
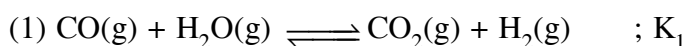
The value of K_c for the reaction :



- (A) 400 (B) 2.40×10^{-3} (C) 9.8×10^{-2} (D) 4.9×10^{-2}

CE0057

8. For the following three reactions, 1, 2 and 3 equilibrium, constants are given :



Which of the following relations is correct ?

- (A) $K_1\sqrt{K_2} = K_3$ (B) $K_2K_3 = K_1$ (C) $K_3 = K_1K_2$ (D) $K_3 = K_2^3 K_1^2$

CE0058

9. Sulfide ion in alkaline solution reacts with solid sulfur to form polysulfide ions having formulas S_2^{2-} , S_3^{2-} , S_4^{2-} and so on. The equilibrium constant for the formation of S_2^{2-} is 12 (K_1) & for the formation of S_3^{2-} is 132 (K_2), both from S and S^{2-} . What is the equilibrium constant for the formation of S_3^{2-} from S_2^{2-} and S?

- (A) 11 (B) 12 (C) 132 (D) None of these

CE0059

10. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a ten-litre container at 444°C and at same temperature value of equilibrium constant K_c is 49, the ratio of $[\text{HI}]$ and $[\text{I}_2]$ will be :-

- (A) 7 (B) $\frac{1}{7}$ (C) $\sqrt{\frac{1}{7}}$ (D) 49

CE0060

11. In the reaction, $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$, the moles/litre of N_2 , O_2 and NO respectively 0.25, 0.05 and 1.0 at equilibrium. The initial concentrations of N_2 and O_2 will respectively be

- (A) 0.75 mole/lit., 0.55 mole/lit. (B) 0.50 mole/lit., 0.75 mole/lit.
(C) 0.25 mole/lit., 0.50 mole/lit. (D) 0.25 mole/lit., 1.0 mole/lit.

CE0061

12. How many moles per litre of PCl_5 has to be taken to obtain 0.1 mole Cl_2 , if the value of equilibrium constant K_c is 0.04 ?

- (A) 0.15 (B) 0.25 (C) 0.35 (D) 0.05

CE0062

13. In the beginning of the reaction, $A \rightleftharpoons B + C$, 2 moles of A are taken, out of which 0.5 mole gets dissociated. What is the degree of dissociation of A ?
 (A) 0.5 (B) 1 (C) 0.25 (D) 4.2

CE0063

14. In the reaction, $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the amount of each PCl_5 , PCl_3 and Cl_2 is 2 mole at equilibrium and total pressure is 3 atmosphere. The value of K_p will be
 (A) 1.0 atm. (B) 3.0 atm. (C) 2.9 atm. (D) 6.0 atm.

CE0064

15. 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium :-
 (A) 4.5 (B) 6 (C) 3 (D) 4

CE0065

16. For the reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equilibrium. What should be the concentration of B at equilibrium ?
 (A) $\frac{a}{4}$ (B) $\frac{a}{2}$ (C) $\frac{3a}{4}$ (D) All of the above.

CE0066

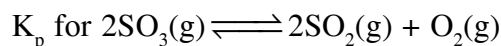
17. For the reaction $3A(g) + B(g) \rightleftharpoons 2C(g)$ at a given temperature, $K_c = 9.0$. What must be the volume of the flask, if a mixture of 2.0 mol each of A, B and C exist in equilibrium?
 (A) 6L (B) 9L (C) 36 L (D) None of these

CE0067

18. For the following gases equilibrium. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
 K_p is found to be equal to K_c . This is attained when temperature is
 (A) 0°C (B) 273 K (C) 1 K (D) 12.19 K

CE0068

19. The degree of dissociation of SO_3 is α at equilibrium pressure p^0 .



- (A) $\frac{p^0 \alpha^3}{2(1-\alpha)^3}$ (B) $\frac{p^0 \alpha^3}{(2+\alpha)(1-\alpha)^2}$ (C) $\frac{p^0 \alpha^2}{2(1-\alpha)^2}$ (D) None of these

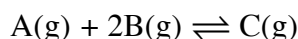
CE0069

20. For the reaction : $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociated (α) of $HI(g)$ is related to equilibrium constant K_p by the expression

- (A) $\frac{1+2\sqrt{K_p}}{2}$ (B) $\sqrt{\frac{1+2K_p}{2}}$ (C) $\sqrt{\frac{2K_p}{1+2K_p}}$ (D) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

CE0070

21. The equilibrium constant for the reaction



is $0.25 \text{ dm}^6 \text{ mol}^{-2}$. In a volume of 5 dm^3 , what amount of A must be mixed with 4 mol of B to yield 1 mol of C at equilibrium.

- (A) 3 moles (B) 24 moles (C) 26 moles (D) None of these

CE0071

22. A 20.0 litre vessel initially contains 0.50 mole each of H_2 and I_2 gases. These substances react and finally reach an equilibrium condition. Calculate the equilibrium concentration of HI if $K_{eq} = 49$ for the reaction $H_2 + I_2 \rightleftharpoons 2HI$.

- (A) 0.78 M (B) 0.039 M (C) 0.033 M (D) 0.021 M

CE0072

23. At 675 K, $H_2(g)$ and $CO_2(g)$ react to form $CO(g)$ and $H_2O(g)$, K_p for the reaction is 0.16. If a mixture of 0.25 mole of $H_2(g)$ and 0.25 mol of CO_2 is heated at 675 K, mole % of $CO(g)$ in equilibrium mixture is :

- (A) 7.14 (B) 14.28 (C) 28.57 (D) 33.33

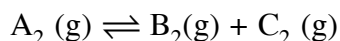
CE0073

24. The vapour density of N_2O_4 at a certain temperature is 30. What is the % dissociation of N_2O_4 at this temperature?

- (A) 53.3% (B) 106.6% (C) 26.7% (D) None

CE0074

25. The equilibrium constant K_p (in atm) for the reaction is 9 at 7 atm and 300 K.



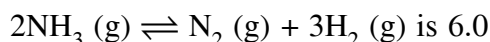
Calculate the average molar mass (in gm/mol) of an equilibrium mixture.

Given : Molar mass of A_2 , B_2 and C_2 are 70, 49 & 21 gm/mol respectively.

- (A) 50 (B) 45 (C) 40 (D) 37.5

CE0075

26. Vapour density of the equilibrium mixture of the reaction



Percent dissociation of ammonia gas is:

- (A) 13.88 (B) 58.82 (C) 41.66 (D) None of these

CE0076

27. The equilibrium concentration of B $[(B)_e]$ for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:-

- (A) $K_C[A]_e^{-1}$ (B) $\frac{k_f}{k_b}[A]_e^{-1}$ (C) $k_f k_b^{-1} [A]_e$ (D) $k_f k_b [A]^{-1}$

CE0077

34. The equilibrium $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$ is attained at 25°C in a closed rigid container and an inert gas, helium is introduced. Which of the following statements is/are correct.

- (A) concentrations of SO_2 , Cl_2 and SO_2Cl_2 do not change
- (B) more chlorine is formed
- (C) concentration of SO_2 is reduced
- (D) more SO_2Cl_2 is formed

CE0084

35. The yield of product in the reaction



would be lower at :

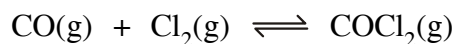
- (A) low temperature and low pressure
- (B) high temperature & high pressure
- (C) low temperature and to high pressure
- (D) high temperature & low pressure

CE0085

EXERCISE # O-II

MORE THAN ONE MAY BE CORRECT :

- 1.** Following two equilibrium is simultaneously established in a container



If some Ni(s) is introduced in the container forming $\text{Ni}(\text{CO})_4(\text{g})$ then at new equilibrium

- (A) PCl_3 concentration will increase (B) PCl_3 concentration will decrease
(C) Cl_2 concentration will remain same (D) CO concentration will remain same **CE0086**

2. For the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, the forward reaction at constant temperature is favoured by

- (A) introducing an inert gas at constant volume (B) introducing chlorine gas at constant volume
(C) introducing an inert gas at constant pressure (D) introducing PCl_5 at constant volume.

CE0087

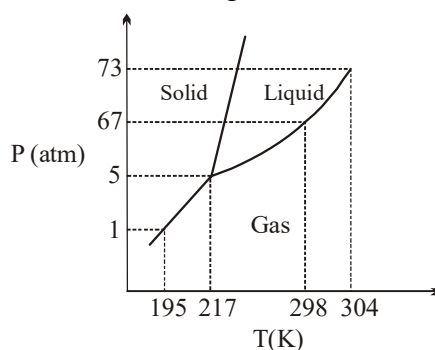
3. When NaNO_3 is heated in a closed vessel, oxygen is liberated and NaNO_2 is left. At equilibrium
- (A) addition of NaNO_2 favours reverse reaction
 - (B) addition of NaNO_3 favours forward reaction
 - (C) increasing temperature favours forward reaction
 - (D) increasing pressure favours reverse reaction

CE0088

4. For the gas phase reaction, $\text{C}_2\text{H}_4 + \text{H}_2 \rightleftharpoons \text{C}_2\text{H}_6$ ($\Delta H = -32.7 \text{ kcal}$), carried out in a closed vessel, the equilibrium moles of C_2H_4 can be increased by
- (A) increasing the temperature (B) decreasing the pressure
- (C) removing some H_2 (D) adding some C_2H_6

CE0089

- 5.** Phase diagram of CO_2 is shown as following



Based on above find the correct statement(s)

- (A) 298K is the normal boiling point of liquid CO₂
(B) At 1 atm & 190 K CO₂ will exist as gas.
(C) CO₂(s) will sublime above 195K under normal atmospheric pressure
(D) Melting point & boiling point of CO₂ will increase on increasing pressure
- CE0090**

6. The equilibrium between, gaseous isomers A, B and C can be represented as

Reaction	Equilibrium constant
$A(g) \rightleftharpoons B(g)$	$K_1 = ?$
$B(g) \rightleftharpoons C(g)$	$K_2 = 0.4$
$C(g) \rightleftharpoons A(g)$	$K_3 = 0.6$

If one mole of A is taken in a closed vessel of volume 1 litre, then

- (A) $[A] + [B] + [C] = 1$ M at any time of the reactions
 (B) Concentration of C is 4.1 M at the attainment equilibrium in all the reactions
 (C) The value of K_1 is $\frac{1}{0.24}$
 (D) Isomer [A] is least stable as per thermodynamics.

CE0091

7. For the gas phase exothermic reaction, $A_2 + B_2 \rightleftharpoons C_2$, carried out in a closed vessel, the equilibrium moles of A_2 can be increased by
 (A) increasing the temperature (B) decreasing the pressure
 (C) adding inert gas at constant pressure (D) removing some C_2

CE0092

8. Consider the equilibrium $HgO(s) + 4I^-(aq) + H_2O(l) \rightleftharpoons HgI_4^{2-}(aq) + 2OH^-(aq)$, which changes will decrease the equilibrium concentration of HgI_4^{2-}
 (A) Addition of 0.1 M HI (aq) (B) Addition of HgO (s)
 (C) Addition of $H_2O(l)$ (D) Addition of KOH (aq)

CE0093

9. Decrease in the pressure for the following equilibria : $H_2O(s) \rightleftharpoons H_2O(l)$ result in the :
 (A) formation of more $H_2O(s)$ (B) formation of more $H_2O(l)$
 (C) increase in melting point of $H_2O(s)$ (D) decrease in melting point of $H_2O(s)$

CE0094

ASSERTION REASON :

10. **Statement -1 :** Total number of moles in a closed system at new equilibrium is less than the old equilibrium if some amount of a substance is removed from a system
 (consider a reaction $A(g) \rightleftharpoons B(g)$) at equilibrium.

Statement -2 : The number of moles of the substance which is removed, is partially compensated as the system reached to new equilibrium.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is false, statement-2 is true.
 (D) Statement-1 is true, statement-2 is false.

CE0095

11. **Statement-1** : Ammonia at a pressure of 10 atm and CO₂ at a pressure of 20 atm are introduced into an evacuated chamber. If K_p for the reaction

$$\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$$
 is 2020 atm³, the total pressure after a long time is less than 30 atm.
Statement-2 : Equilibrium can be attained from both directions.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true. CE0096
12. **Statement-1** : Catalyst does not alter the equilibrium constant
Statement-2 : Because for the catalysed reaction and uncatalysed reaction ΔH remain same an equilibrium constant depend on ΔH.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true. CE0097
13. **Statement-1** : Equilibrium constant may show higher or lower values with increase in temperature
Statement-2 : The change depend on heat of reaction.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true. CE0098

COMPREHENSION :

Paragraph for Question Nos. 14 to 17

In a 7.0 L evacuated chamber, 0.50 mol H₂ and 0.50 mol I₂ react at 427°C .

$$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$$
 At the given temperature, K_C = 49 for the reaction.

14. What is the value of K_p ?
 (A) 7 (B) 49 (C) 24.5 (D) None CE0099
15. What is the total pressure (atm) in the chamber?
 (A) 83.14 (B) 831.4 (C) 8.21 (D) None CE0099
16. How many moles of the iodine remain unreacted at equilibrium?
 (A) 0.388 (B) 0.112 (C) 0.25 (D) 0.125 CE0099
17. What is the partial pressure (atm) of HI in the equilibrium mixture?
 (A) 6.385 (B) 12.77 (C) 40.768 (D) 646.58 CE0099

Paragraph for Question Nos. 18 to 20

Influence of pressure, temperature, concentration and addition of inert gas on a reversible chemical reaction in equilibrium can be explained by formulating the expression for equilibrium constant K_c or K_p for the equilibrium. On the other hand Le Chatelier principle can be theoretically used to explain the effect of P, T or concentration on the physical or chemical equilibrium both.

18. For the reaction : $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ increase of pressure shows :

- (A) An increase in degree of dissociation and a decrease in K_c
- (B) A decrease in degree of dissociation and a decrease in K_c
- (C) An increase in degree of dissociation but K_c remains constant
- (D) A decrease in degree of dissociation but K_c remains constant

CE0100

19. For the reaction : $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$; $\Delta H = -ve$, An increase in temperature shows :

- (A) More dissociation of SO_3 and a decrease in K_c
- (B) Less dissociation of SO_3 and an increase in K_c
- (C) More dissociation of SO_3 and an increase in K_c
- (D) Less dissociation of SO_3 and an decrease in K_c

CE0101

20. For the reaction : $\text{Fe}^{3+}(\text{aq.}) + \text{SCN}^-(\text{aq.}) \rightleftharpoons [\text{Fe}(\text{SCN})]^{2+}(\text{aq.})$; in equilibrium if little more aqueous

solution of FeCl_3 is added, than :

- (A) Forward reaction becomes more and red colours is deepened
- (B) Backward reaction becomes more and red colour faintens
- (C) Solution becomes colourless
- (D) No change will occur

CE0102

MATCH THE COLUMN :

21.	Column I	Column II
(A)	$K_p < K_c$	(P) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
(B)	Introduction of inert gas at constant pressure will decrease the concentration of reactants	(Q) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
(C)	K_p° is dimensionless	(R) $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$
(D)	Temperature increase will shift the reaction on product side.	(S) $\text{NH}_3(\text{g}) + \text{HI}(\text{g}) \rightleftharpoons \text{NH}_4\text{I}(\text{s})$

CE0103

22.	Column-I (Reactions)	Column-II (Favourable conditions)
(A)	Oxidation of nitrogen $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) + 180.5 \text{ kJ} \rightleftharpoons 2\text{NO}(\text{g})$	(p) Addition of inert gas at constant pressure
(B)	Dissociation of $\text{N}_2\text{O}_4(\text{g})$ $\text{N}_2\text{O}_4(\text{g}) + 57.2 \text{ kJ} \rightleftharpoons 2\text{NO}_2(\text{g})$	(q) Decrease in pressure
(C)	Oxidation of $\text{NH}_3(\text{g})$ $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g}) + 905.6 \text{ kJ}$	(r) Decrease in temperature
(D)	Formation of $\text{NO}_2(\text{g})$ $\text{NO}(\text{g}) + \text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + 200 \text{ kJ}$	(s) Increase in temperature

CE0104

MATCHING LIST TYPE :

Answer Q.No. 23, 24 and 25 by appropriately matching the information given in the three columns of the following table.

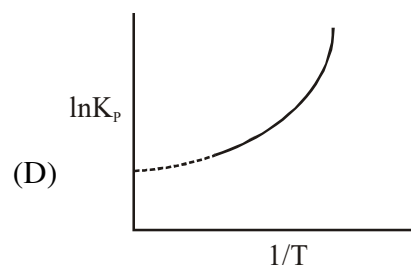
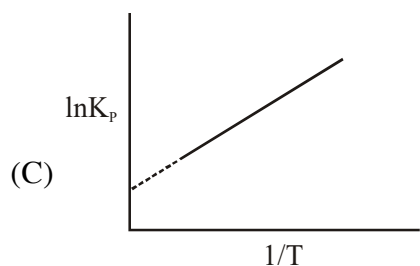
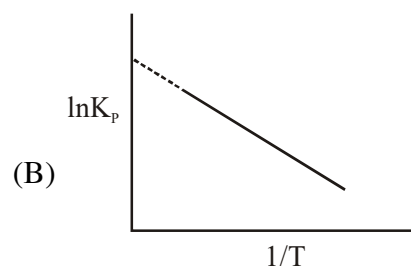
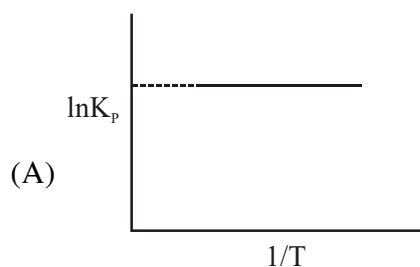
Column-I (Reaction)	Column-II (If α is negligible w.r.t. 1)	Column-III
(A) $2\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g}), \Delta H = -\text{ve}$	(P) $\alpha = 2 \times \sqrt{K_p}$	(I) K increase with increase in temperature
(B) $\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g}), \Delta H = +\text{ve}$	(Q) $\alpha = 3 \times \sqrt{K_p \cdot P}$	(II) K decrease with increase in temperature
(C) $3\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g}), \Delta H = +\text{ve}$	(R) $\alpha = \left(\frac{2K_p}{P} \right)^{1/3}$	(III) Pressure has no effect
(D) $2\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + 2\text{Z}(\text{g}), \Delta H = +\text{ve}$	(S) $\alpha = \sqrt{K_p / P}$	(IV) Addition of inert gas at constant pressure shift equilibrium in forward direction

23. Which of the following option is correctly matched ?

- (A) A-Q-III (B) B-S-II (C) C-Q-II (D) D-R-IV

CE0105

24. Correct representation of reaction in option (A) is represented by graph



CE0106

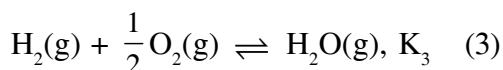
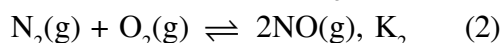
25. For the reaction of option (B) if degree of dissociation varies inversely as square root of pressure of the system. Suppose at constant temperature volume is increased 16 times of its initial volume the degree of dissociation (α) for this reaction will become :

(A) 4 times (B) 2 times (C) 1/4 times (D) 1/2 times

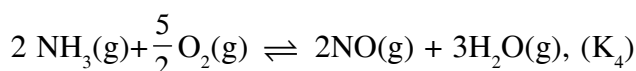
CE0107

EXERCISE # J-MAIN

1. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}), K_1$ (1) [JEE-MAINS(online)-13]



The equation for the equilibrium constant of the reaction



in terms of K_1 , K_2 and K_3 is :

- (1) $\frac{K_1 K_3^2}{K_2}$ (2) $\frac{K_2 K_3^3}{K_1}$ (3) $\frac{K_1 K_2}{K_3}$ (4) $K_1 K_2 K_3$

CE0114

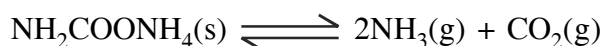
2. In reaction $\text{A} + 2\text{B} \rightleftharpoons 2\text{C} + \text{D}$, initial concentration of B was 1.5 times of concentration of A, but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is :

[JEE-MAINS(online)-13]

- (1) 4 (2) 6 (3) 12 (4) 8

CE0115

3. For the decomposition of the compound, represented as [JEE-MAINS(online)-14]



the $K_p = 2.9 \times 10^{-5} \text{ atm}^3$.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be

- (1) $38.8 \times 10^{-2} \text{ atm}$ (2) $1.94 \times 10^{-2} \text{ atm}$
(3) $5.82 \times 10^{-2} \text{ atm}$ (4) $7.66 \times 10^{-2} \text{ atm}$

CE0116

4. For the reaction $\text{SO}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} \rightleftharpoons \text{SO}_{3(\text{g})}$, if $K_p = K_C (\text{RT})^x$ where the symbols have usual meaning then the value of x is : (assuming ideality) [JEE-MAINS-14]

- (1) $\frac{1}{2}$ (2) 1 (3) -1 (4) $-\frac{1}{2}$

CE0117

5. The equilibrium constants at 298 K for a reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is 100. If the initial concentration of all the four species were 1 M each, then equilibrium concentration of D (in mol L^{-1}) will be :

[JEE-MAINS-16]

- (1) 1.182 (2) 0.182 (3) 0.818 (4) 1.818

CE0118

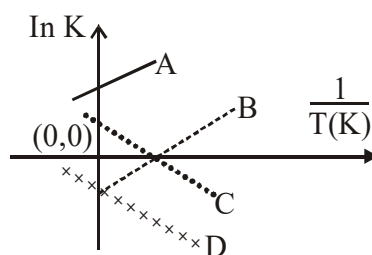
6. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal:
 $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightleftharpoons 2\text{Fe}(\text{l}) + 3\text{CO}_2(\text{g})$ [JEE-MAINS(online)-17]

Using the Le Chatelier's principle, predict which one of the following will **not** disturb the equilibrium?

- (1) Removal of CO_2 (2) Addition of Fe_2O_3
 (3) Addition of CO_2 (4) Removal of CO

CE0119

7. Which of the following lines correctly show the temperature dependence of equilibrium constant, K , for an exothermic reaction ? [JEE-MAINS(offline)-18]



- (1) B and C (2) C and D (3) A and D (4) A and B

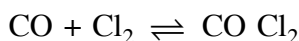
CE0120

8. In which of the following reactions, an increase in the volume of the container will favour the formation of products ? [JEE-MAINS(online)-18]

- (1) $2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$
 (2) $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$
 (3) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 (4) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$

CE0121

9. At a certain temperature in a 5 L vessel, 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction,



At equilibrium, if one mole of CO is present then equilibrium constant (K_c) for the reaction is :-

- (1) 4 (2) 3 (3) 2 (4) 2.5 [JEE-MAINS(online)-18]

CE0122

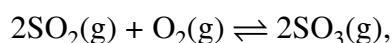
10. The gas phase reaction $2\text{NO}_2(\text{g}) \rightarrow \text{N}_2\text{O}_4(\text{g})$ is an exothermic reaction. The decomposition of N_2O_4 , in equilibrium mixture of $\text{NO}_2(\text{g})$ and $\text{N}_2\text{O}_4(\text{g})$, can be increased by :-

- (1) Addition of an inert gas at constant volume. [JEE-MAINS(online)-18]
 (2) Increasing the pressure.
 (3) Lowering the temperature.
 (4) Addition of an inert gas at constant pressure.

CE0124

11. For the reaction,

[JEE-MAINS(online)-19]



$$\Delta H = -57.2 \text{ kJ mol}^{-1} \text{ and}$$

$$K_c = 1.7 \times 10^{16}.$$

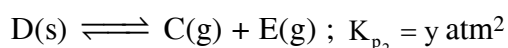
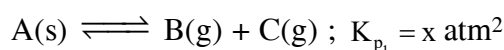
Which of the following statement is INCORRECT?

- (1) The equilibrium constant is large suggestive of reaction going to completion and so no catalyst is required.
- (2) The equilibrium will shift in forward direction as the pressure increase.
- (3) The equilibrium constant decreases as the temperature increases.
- (4) The addition of inert gas at constant volume will not affect the equilibrium constant.

CE0126

12. Two solids dissociate as follows

[JEE-MAINS(online)-19]



The total pressure when both the solids dissociate simultaneously is :-

- (1) $(x + y) \text{ atm}$
- (2) $x^2 + y^2 \text{ atm}$
- (3) $2(\sqrt{x + y}) \text{ atm}$
- (4) $\sqrt{x + y} \text{ atm}$

CE0128

13. In a chemical reaction, $\text{A} + 2\text{B} \xrightleftharpoons{K} 2\text{C} + \text{D}$, the initial concentration of B was 1.5 times of the concentration of A, but the equilibrium concentrations of A and B were found to be equal. The equilibrium constant(K) for the aforesaid chemical reaction is :

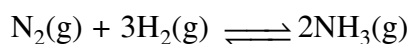
[JEE-MAINS(online)-19]

- (1) 16
- (2) 4
- (3) 1
- (4) $\frac{1}{4}$

CE0129

14. Consider the reaction,

[JEE-MAINS(online)-19]



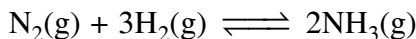
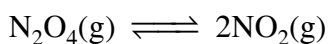
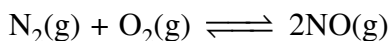
The equilibrium constant of the above reaction is K_p . If pure ammonia is left to dissociate, the partial pressure of ammonia at equilibrium is given by (Assume that $P_{\text{NH}_3} \ll P_{\text{total}}$ at equilibrium)

- (1) $\frac{3^{\frac{3}{2}} K_p^{\frac{1}{2}} P^2}{4}$
- (2) $\frac{3^{\frac{3}{2}} K_p^{\frac{1}{2}} P^2}{16}$
- (3) $\frac{K_p^{\frac{1}{2}} P^2}{16}$
- (4) $\frac{K_p^{\frac{1}{2}} P^2}{4}$

CE0130

15. The value of K_p/K_C for the following reactions at 300K are, respectively :

(At 300K, $RT = 24.62 \text{ dm}^3\text{atm mol}^{-1}$)



[JEE-MAINS(online)-19]

(1) 1, $24.62 \text{ dm}^3\text{atm mol}^{-1}$, $606.0 \text{ dm}^6\text{atm}^2\text{mol}^{-2}$

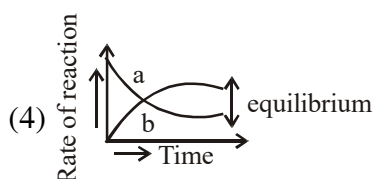
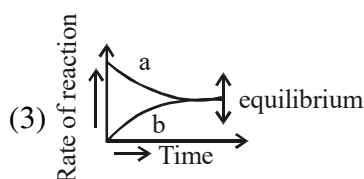
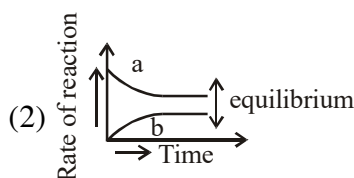
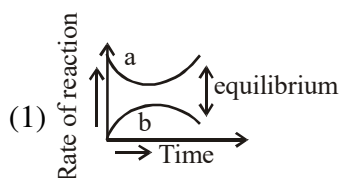
(2) 1, $4.1 \times 10^{-2} \text{ dm}^3\text{atm}^{-1} \text{ mol}^{-1}$, $606.0 \text{ dm}^6 \text{atm}^2 \text{mol}^{-2}$

(3) $606.0 \text{ dm}^6\text{atm}^2\text{mol}^{-2}$, $1.65 \times 10^{-3} \text{ dm}^3\text{atm}^{-2} \text{mol}^{-1}$

(4) 1, $24.62 \text{ dm}^3\text{atm mol}^{-1}$, $1.65 \times 10^{-3} \text{ dm}^{-6}\text{atm}^{-2} \text{mol}^2$

CE0131

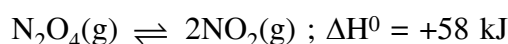
16. For the equilibrium $\text{A} \rightleftharpoons \text{B}$, the variation of the rate of the forward (a) and reverse (b) reaction with time is given by [JEE-MAINS(online)-2020]



CE0151

17. Consider the following reaction

[JEE-MAINS(online)-2020]



For each of the following cases (a, b), the direction in which the equilibrium shifts is:

(a) Temperature is decreased

(b) Pressure is increased by adding N_2 at constant T

(1) (a) towards reactant, (b) no change

(2) (a) towards product, (b) towards reactant

(3) (a) towards product, (b) no change

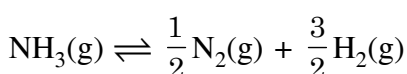
(4) (a) towards reactant, (b) towards product

CE0152

18. The value of K_C is 64 at 800 K for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

The value of K_C for the following reaction is :

[JEE-MAINS(online)-2020]



(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) 8

(4) $\frac{1}{64}$

CE0153

19. At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value K_P for the reaction $\text{Cl}_{2(g)} \rightleftharpoons 2\text{Cl}_{(g)}$ under the above conditions is $x \times 10^{-1}$. The value of x is _____. (Rounded of to the nearest integer)

[JEE-MAINS(online)-2021]

CE0154

- 20.** A homogeneous ideal gaseous reaction $\text{AB}_{2(\text{g})} \rightleftharpoons \text{A}_{(\text{g})} + 2\text{B}_{(\text{g})}$ is carried out in a 25 litre flask at 27°C . The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is _____. (Integer answer) **[JEE-MAINS(online)-2021]**

[JEE-MAINS(online)-2021]

CE0155

- 21.** Consider the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$. The temperature at which $K_C = 20.4$ and $K_P = 600.1$, is _____ K. (Round off to the Nearest Integer).

[Assume all gases are ideal and $R = 0.0831 \text{ L bar K}^{-1} \text{ mol}^{-1}$] [JEE-MAINS(online)-2021]

CE0156

22. For the reaction, $A + B \rightleftharpoons 2C$

the value of equilibrium constant is 100 at 298 K. If the initial concentration of all the three species is 1 M each, then the equilibrium concentration of C is $x \times 10^{-1}$ M. The value of x is _____.

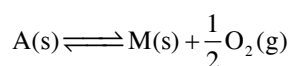
(Nearest integer)

[JEE-MAINS(online)-2021]

CE0157

- 23.** The equilibrium constant for the reaction

[JEE-MAINS(online)-2021]



is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. **(Round off to the nearest integer)**

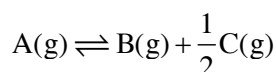
CE0158

24. When 5.1 g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C , 20% of the solid decomposes into gaseous ammonia and hydrogen sulphide. The K_p for the reaction at 27°C is $x \times 10^{-2}$. The value of x is _____. (Integer answer) [Given $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

[JEE-MAINS(online)-2021]

CE0159

- 25.** For a reaction at equilibrium



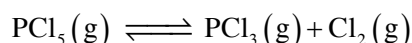
the relation between dissociation constant (K), degree of dissociation (α) and equilibrium pressure (p) is given by : **[JEE-MAINS(online)-2022]**

[JEE-MAINS(online)-2022]

$$(A) \quad K = \frac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{\left(1 + \frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)} \quad (B) \quad K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)} \quad (C) \quad K = \frac{(\alpha p)^{\frac{3}{2}}}{\left(1 + \frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)} \quad (D) \quad K = \frac{(\alpha p)^{\frac{3}{2}}}{(1+\alpha)(1-\alpha)^{\frac{1}{2}}}$$

CE0160

26. PCl_5 dissociates as



5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is $\times 10^{-3}$. (nearest integer)

(Given: $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$: Assume ideal gas behaviour)

[JEE-MAINS(online)-2022]

CE0161

27. $2\text{NOCl}(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{Cl}_2(\text{g})$

In an experiment, 2.0 moles of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be 0.4 mol/L. The equilibrium constant at 30°C is $\times 10^{-4}$.

[JEE-MAINS(online)-2022]

CE0162

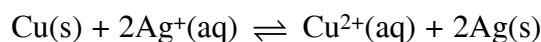
28. 4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is [Given : $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

[JEE-MAINS(online)-2022]

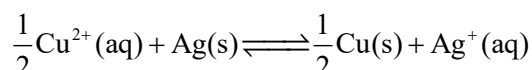
- (A) 2.25 (B) 6.24 (C) 12.13 (D) 15.24

CE0163

29. At 298 K, the equilibrium constant is 2×10^{15} for the reaction :



The equilibrium constant for the reaction

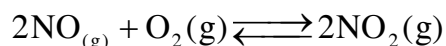


is $x \times 10^{-8}$. The value of x is _____. (Nearest Integer)

[JEE-MAINS(online)-2022]

CE0164

30. At 600K, 2 mol of NO are mixed with 1 mol of O_2 .



The reaction occurring as above comes to equilibrium under a total pressure of 1 atm. Analysis of the system shows that 0.6 mol of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer).

[JEE-MAINS(online)-2022]

CE0165

EXERCISE # J-ADVANCED

1. The thermal dissociation equilibrium of $\text{CaCO}_3(\text{s})$ is studied under different conditions.



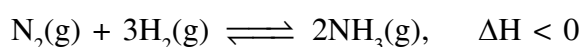
[JEE 2013]

For this equilibrium, the correct statement(s) is(are)

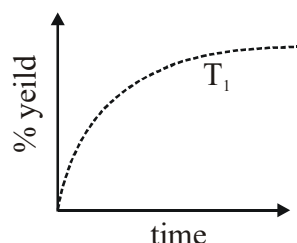
- (A) ΔH is dependent on T
- (B) K is independent of the initial amount of CaCO_3
- (C) K is dependent on the pressure of CO_2 at a given T
- (D) ΔH is independent of the catalyst, if any

CE0134

2. The % yield of ammonia as a function of time in the reaction

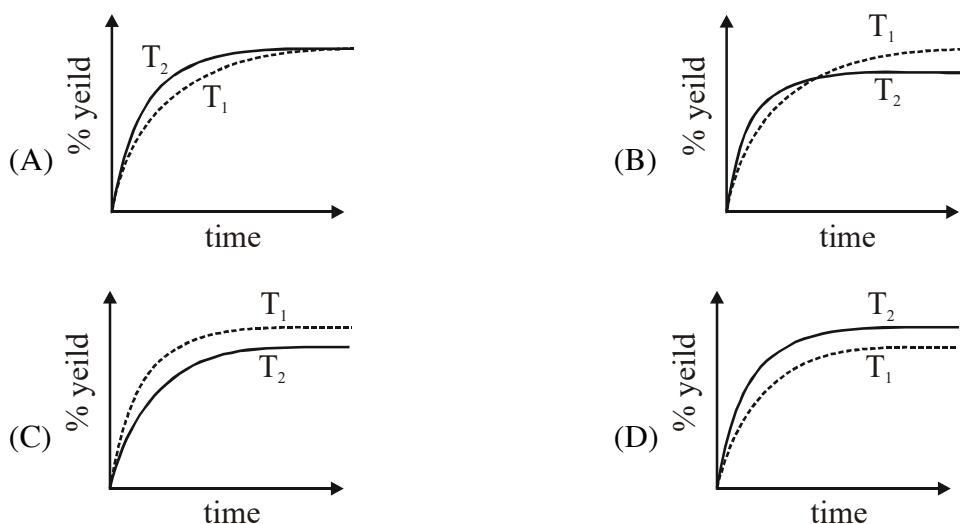


at (P, T_1) is given below -



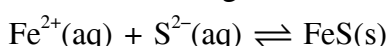
If this reaction is conducted at (P, T_2) , with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by -

[JEE 2015]



CE0166

3. For the following reaction, the equilibrium constant K_c at 298 K is 1.6×10^{17} .



When equal volumes of 0.06 M $\text{Fe}^{2+}(\text{aq})$ and 0.2 M $\text{S}^{2-}(\text{aq})$ solutions are mixed, the equilibrium concentration of $\text{Fe}^{2+}(\text{aq})$ is found to be $Y \times 10^{-17}$ M. The value of Y is (upto two decimal places.)

[JEE 2019]

CE0135

ANSWER KEY**EXERCISE # S-I**

1. (a) negligible (b) almost complete

$$2. \quad (a) \quad K_p = \frac{(P_{SO_3})}{(P_{SO_2})(P_{O_2})^{\frac{1}{2}}} ; K_c = \frac{[SO_3]}{[SO_2][O_2]^{\frac{1}{2}}}$$

$$(b) \quad K_c = [H_2O]$$

$$K_p = P_{H_2O}$$

$$(c) \quad K_c = [NH_3]^2 [CO_2]$$

$$K_p = (P_{NH_3})^2 P_{CO_2}$$

$$(d) \quad K_c = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]} \quad [\text{here } H_2O \text{ is not in excess}]$$

K_p is not defined for liquid phase.

$$(e) \quad K_c = \frac{[NH_4^+][OH^-]}{[NH_3]} \quad [\text{here } H_2O(l) \text{ is in excess (solvent) so its concentration is 1}]$$

$$(f) \quad K_{PC} = \frac{P_{H_2} \times [Zn^{2+}]}{[H^+]^2}$$

3. (a) Here, $\Delta n_g = 1 + 1 - 1 = 1$

$$K_p = K_c (RT)^1 = K_c RT$$

- (b) Here, $\Delta n_g = 2 - (3+1) = -2$

$$K_p = K_c (RT)^{-2}$$

- (c) Here, $\Delta n_g = 2 - (1+1) = 0$

$$K_p = K_c (RT)^0 = K_c$$

- (d) Here, $\Delta n_g = 2 + 1 - 2 = 1$

$$K_p = K_c (RT)^1 = K_c RT$$

- (e) Here, $\Delta n_g = 2 - (2 + 1) = -1$

$$K_p = K_c (RT)^{-1} = K_c / RT$$

4. XO_2
5. $Q_c > K_c$ so the reaction will proceed in the reverse direction.
6. (a) $Q_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}]}$ (b) 3.48 (c) No
7. 2.00 8. 0.90 9. 100.00 10. 0.24, 0.01
11. No, 6.40×10^{-31} M 12. 0.44 13. 1.20
14. 0.33 15. (a) 0.10, (b) 0.40 16. 1.00
17. 4.00 18. $a = \sqrt{\frac{K_p}{4P + K_p}}$ 19. 16.00 20. 2.00
21. 0.50 22. 0.05 23. 0.09 or 0.10 24. 50.00
25. 0.01 26. 15.00 27. 4.00 28. 1.80
29. 22.40 30. 5.00 31. 0.82
32. (a) $K = [\text{CH}_3\text{OH}]/[\text{H}_2]^2[\text{CO}]$,
 (b) 1. $[\text{H}_2]$ increase, $[\text{CO}]$ decrease, $[\text{CH}_3\text{OH}]$ increase ; 2. $[\text{H}_2]$ increase, $[\text{CO}]$ decrease, $[\text{CH}_3\text{OH}]$ decrease ; 3. $[\text{H}_2]$ increase, $[\text{CO}]$ increase, $[\text{CH}_3\text{OH}]$ increase ; 4. $[\text{H}_2]$ increase, $[\text{CO}]$ increase, $[\text{CH}_3\text{OH}]$ increase ; 5. $[\text{H}_2]$ increase, $[\text{CO}]$ increase, $[\text{CH}_3\text{OH}]$ decrease ; 6. no change
33. 6
34. 250.00
35. k_r increase more than k_f , this means that E_a (reverse) is greater than E_a (forward). Hence exothermic reaction.

EXERCISE # S-II

1. (2.40) 2. (3.44)
3. (0.11) 4. (0.20)
5. (0.36 or 0.37)
6. (a) 6.66 or 0.67; (b) 375.00; (c) 10.45 or 10.46; (d) 6.25
7. (i) 2; (ii) 1.20 8. 54.00
9. 3.60 10. (a) 400, 900, (b) 4: 9, (c) 72.15
11. (4.00) 12. $(\text{SrCl}_2 \cdot 2\text{H}_2\text{O})$
13. (10.00) 14. $K_{p_1} = \frac{1}{20P_0^2}$; $K_{p_2} = \frac{3}{20P_0^2}$
15. (a) 80 ; (b) 0.15 ; (c) 26.08 or 26.09

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	D	B	C	C	A	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	C	A	B	B	A	D	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	A	C	C	C	B	C	D
Que.	31	32	33	34	35					
Ans.	A	A	C	A	D					

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C,D	C,D	A,B,C,D	C,D	A,C,D	A,B,C	C,D	A,C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	A	B	C	B	A	D	A	A
Que.	21				22				23	24
Ans.	A-P,R,S	B-P,Q,R	C-P,Q,R,S	D-Q	A-s	B-p,q,s	C-p,q,r	D-r	D	C
Que.	25									
Ans.	A									

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	3	4	4	2	4	1	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	2	2	4	3	1	2	5	73
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	354	25	16	6	B	1107	125	A	2	2

EXERCISE # J-ADVANCED

Que.	1	2	3							
Ans.	A,B,D	B	8.92 or 8.93							

Important Notes